1.0 INTRODUCTION AND GENERAL DESCRIPTION OF PLANT

1.1 INTRODUCTION

This Final Safety Analysis Report (FSAR) was originally submitted to support the application of Georgia Power Company (GPC), Oglethorpe Power Corporation (OPC), the Municipal Electric Authority of Georgia (MEAG), and the City of Dalton, Georgia, for an operating license for a nuclear power plant designated as the Vogtle Electric Generating Plant, herein referred to as the VEGP. Pursuant to an application dated September 18, 1992, the NRC issued an operating license amendment on March 17, 1997, which was effective March 22, 1997, designating Southern Nuclear Operating Company, Inc. (SNC) as the exclusive operating licensee of VEGP. SNC has no ownership interest in VEGP.

1.1.1 PLANT UNITS

The application is for two units, each with a reactor core rated at a power level of 3411 MWt under section 103(b) of the Atomic Energy Act of 1954, as amended, and the regulations of the Nuclear Regulatory Commission (NRC) set forth in Part 50 of Title 10 of the Code of Federal Regulations (10 CFR 50). The VEGP power uprate increased the licensed reactor core power level from 3411 MWt to 3565 MWt. The VEGP Measurement Uncertainty Recapture Power Uprate (MUR-PU) increased the licensed reactor core power level from 3565 MWt to 3625.6 MWt.

The plant was constructed as a two-unit plant with each unit essentially the same. Descriptions of one unit shall be interpreted as applying to both units.

Differences between Units 1 and 2 and, particularly, structures, systems, and components that are shared between the two units are specified in the appropriate location in the FSAR.

Units 3 and 4 will be constructed to the west of Units 1 and 2 as shown on drawing CX2D45V002. The Units 1 and 2 support facilities and systems, which are located in areas affected by Units 3 and 4 preconstruction and limited work authorization (LWA) related activities, are being modified, relocated or demolished. The figures incorporated by reference into the FSAR that show the Unit 3 and 4 construction impact area have been annotated to show the impact area; however, the figures will not be updated until construction is complete.

The FSAR text, tables, and figures that describe Unit 1 and 2 facilities and systems in the construction impact area will be updated as necessary to reflect the modifications to, relocation of, or demolition of the facility and/or system upon completion of the action.

The FSAR will also be updated to reflect the presence of Units 3 and 4, identify the facilities and systems that are shared by all four units, and, prior to Unit 3 fuel load, to address any impacts on Units 1 and 2 due to Units 3 and 4 operations.

1.1.2 PLANT LOCATION

The location of the VEGP is on the southwest side of the Savannah River approximately 23 river miles upstream from the intersection of the Savannah River and U.S. Highway 301, as shown on drawing CX2D45V002. The site is in the eastern sector of Burke County, Georgia, and across the river from Barnwell County, South Carolina. The VEGP site is directly across the Savannah River from the Department of Energy Savannah River Site. Refer to paragraph 1.2.1.1.

1.1.3 CONTAINMENT TYPE

The containment for each of the VEGP units is a steel-lined, prestressed, post-tensioned concrete cylinder with a hemispherical dome. The containment was designed by the Los Angeles Regional Office of the Bechtel Power Corporation.

1.1.4 NUCLEAR STEAM SUPPLY SYSTEM

1.1.4.1 Reactor Type and Supplier

The nuclear steam supply system (NSSS) for each of the VEGP units is a pressurized water reactor. Westinghouse Electric Corporation is the designer and supplier of these units for the VEGP.

1.1.4.2 Power Output

Each NSSS unit has a design net core power output of 3636 MWt plus a maximum of 17 MWt from nonreactor sources, primarily pump heat, for a total of 3653 MWt.

The turbine-generator unit of the steam and power conversion system has the capability of generating with valves wide open a gross electrical output of 1249.7 MWe (at 3.5 in. HgA condenser vacuum). This corresponds to an NSSS output of 3636 MWt.

All safety systems, including the containment and engineered safety features, are designed and evaluated for operation at a power level of 3636 MWt net core power. This power rating is used in the analysis of all postulated accidents bearing significantly on the acceptability of the site.

1.1.5 SCHEDULE FOR COMPLETION AND COMMERCIAL OPERATION

The two VEGP units are scheduled to be completed and begin commercial operation as tabulated below:

Unit	Completion of Construction Fuel Loading	Commercial Power Operation
1	January 1987	June 1987
2	February 1989	June 1989

1.1.6 FORMAT AND CONTENT

1.1.6.1 Regulatory Guide 1.70

This FSAR has used as a guide the format and content recommendations of Regulatory Guide 1.70, Revision 3, Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants - LWR Edition, November 1978.

This FSAR generally uses the same chapter, section, subsection, and paragraph headings used in the standard format. Where appropriate, the FSAR is subdivided beyond the extent of the standard format to isolate all information specifically requested in that document. Information presented that is not specifically requested by the standard format and is identified numerically (chapter, section, subsection, or paragraph), is presented under the appropriate general

heading as a subdivision following all subdivisions containing information specifically requested by the standard format.

1.1.6.2 Standard Review Plan (SRP)

Because the NRC utilizes NUREG-0800, Standard Review Plan for the Review of Safety Analysis Reports for Nuclear Power Plants - LWR Edition, July 1981, for guidance in the determination of the acceptability of an application, the SRP is addressed in this FSAR as required in 10 CFR 50.34. In addition, the applicants, with the assistance of their agents and contractors identified in section 1.4, have reviewed the SRP in accordance with 10 CFR 50.34 and have determined that this FSAR contains no unacceptable deviations from the acceptance criteria given in the applicable portions of the SRP. A table of SRP differences appears in section 1.8.

1.1.6.3 Tables and Figures

Tabulations of data are designated "tables." Each is identified by the section or subsection number followed by a number according to its order of mention in the section or subsection. (Table 3.3-5 is the fifth table of section 3.3 and table 3.3.1-5 is the fifth table of subsection 3.3.1.) Tables are located at the end of the section or subsection immediately following the text. Drawings, pictures, sketches, curves, graphs, and engineering diagrams identified as "figures" are numbered according to the order of mention in the section or subsection. (Figure 3.4-2 is the second figure of section 3.4 and figure 3.4.1-2 is the second figure in subsection 3.4.1.) Figures are located at the end of the applicable section or subsection, following tables.

Some drawings and engineering diagrams are included, in conjunction with specific system descriptions, by reference to their drawing identification number in lieu of inclusion as a figure.

1.1.6.4 Numbering of Pages

Text pages are numbered sequentially within each section or subsection. (Page 1.1-4 is the fourth page of section 1.1 and 1.1.1-4 is the fourth page of subsection 1.1.1.) Tables and figures are not paginated. When it becomes necessary during revision of this FSAR to insert additional information, impacted text material will be moved to subsequent pages as required, and these pages will be renumbered.

1.1.6.5 Amending the FSAR

When it becomes necessary to submit additional information or revise information presently contained in the FSAR, the following procedures will be followed:

A. When a change is made to the FSAR text, those pages affected will be marked with the amendment number and date in the lower right corner of the margin. A vertical line in the same margin will indicate the material affected.

B. Figures will be revised by indicating the amendment number and date in the lower right corner. Notation will be made of the revisions made in that amendment

1.2 GENERAL PLANT DESCRIPTION

1.2.1 SITE CHARACTERISTICS

1.2.1.1 Location

The site is located in southeast Georgia on the west side of the Savannah River about 26 miles southeast of Augusta, Georgia, and 15 miles east-northeast of Waynesboro, Georgia. It is in Burke County, Georgia, just across the river from the Savannah River Site as shown in figure 2.1.1-2. The Universal Transverse Mercator Grid Coordinates for the center of the Unit 1 containment are Zone 17S, east 428,900 m, north 3,666,900 m. For the center of the Unit 2 containment, the coordinates are Zone 17S, east 428,800 m, north 3,666,900 m.

1.2.1.2 Site Ownership

The 3169-acre site is owned by Georgia Power Company and the co-owners.

1.2.1.3 Access to the Site

All activities on the site are under the control of Southern Nuclear Operating Company, Inc. (SNC). Access to the plant's protected area is controlled by a security fence and guard force.

1.2.1.4 Site Environs

There are no people living on the site. Due to the remoteness of the VEGP site, there are few human activities within a 5-mile radius. A survey by the Central Savannah River Area Planning and Development Commission shows no public or private schools, hospitals, commercial plants, sports facilities, residential development parks, or recreational areas within 5 miles of the site on the Georgia side of the river. In South Carolina the restricted nature of the adjacent Savannah River Site excludes any such uses of the land. The surrounding terrain is characterized by gently rolling land of which approximately 30 percent is farmland; the remainder is primarily wooded.

The above paragraph was written prior to receipt of an operating license. It is presently considered historical information. However, subsequent to receipt of license, some sports facilities/recreational areas within 5 miles of the site have been constructed. These are a concrete, public, fishing-boat ramp on the Georgia side of the Savannah River near the VEGP site training center, and a VEGP employees' recreational facility (picnic tables, pavilion-type shed, restrooms/showers, tennis courts, fishing area, softball field) across from the site's GPC land/surveying office building. In addition, adjacent to the owner controlled area (OCA) is a Georgia Wildlife Game Preserve where state-controlled, wild-game hunting is permitted.

1.2.1.5 Geology

The topography of the site consists of gently rolling hills carved from a terrace ranging in elevation from approximately 300 ft above sea level to the lowest point, approximately 80 ft above sea level, and lying at the edge of the Savannah River. Shallow solution basins are present where leaching of dispersed calcareous sands and shells has resulted in local near-surface subsidence.

At the site, the upper 80 to 110 ft consists of red to yellow sands and clays of Eocene and Miocene age. The lower portion of the yellow sands is generally calcareous, often locally becoming a coquina, which has leached out to form shallow near-surface depressions. A hard impervious clayey marl approximately 70 ft thick underlies the calcareous sands and shells. Beneath this are increasingly dense sands composed of deposits of the Cretaceous Age.

The local structure in the sediments is primarily homoclinal with a dip towards the sea varying from about 30 ft/mile at the basement contact to 6 ft/mile at the contact between the red and yellow sands. A monoclinal flexure occurs at the northwest boundary of the site and has a maximum slope of approximately 5 percent. This feature is not tectonic in origin but appears related to existing conditions at the time of deposition.

There are no indications of major or active faulting at or near the site, nor is there any record of nearby seismic epicenters. Solution features observed on the site are clearly related to materials overlying the bearing stratum and therefore do not constitute a safety hazard for the proposed plant.

1.2.1.6 Seismology

The region of the United States in which the site lies is one of considerable earthquake activity, but this is of low to moderate intensity. The exception to this was the 1886 Charleston, South Carolina, earthquake whose epicenter was 104 miles from the site. Reports from nearby towns indicate that this strongest historic shock within the region probably would have had an intensity at the site of VI and no greater than VII on the Modified Mercalli Intensity Scale of 1931 (MM). The New Madrid, Missouri, earthquakes of 1811-1812, whose epicenters were approximately 500 miles to the west, would have had an intensity of no greater than VI (MM) at the site.

The Seismic Risk Map of the United States (Algermissen, 1969) shows the site in Zone 2, described as "moderate damage corresponds to intensity VII on the Modified Mercalli Intensity Scale of 1931." No earthquake epicenters have been recorded within 50 miles of the site and none with an intensity greater than VI (MM) within a 100-mile radius. Evaluation of the site indicates that the largest shock intensities felt were from great, distant earthquakes.

Earthquake frequency data for the period in which records have been kept indicate that in the last 300 years the only shocks of intensity V (MM) or greater felt at the site were those of the 1886 Charleston earthquake and the 1811-1812 New Madrid earthquake.

1.2.1.7 Hydrology

The dominant hydrological feature of the site is the Savannah River, which empties into the Atlantic Ocean some 151 miles downstream. This region is not prone to flooding. The area of the drainage basin affecting the Savannah River at the site is about 8015 mi^2. The average flow in the river during 72 years of record at Augusta, Georgia, is 10,300 ft^3/s, and the minimum flow is 5800 ft^3/s. The maximum historical flow was 360,000 ft^3/s during the flood of 1796. This

corresponds to an estimated maximum flood stage at the site of about 116 ft msl. No flooding has been noted in the region since the completion of the Clark Hill Dam in 1954.

The probable maximum flood for the Savannah River at the VEGP site has been calculated to be 708,969 ft^3/s, which corresponds to an estimated maximum flood stage of 165.0 ft msl; the plant site grade is approximately 220 ft msl.

1.2.1.8 Meteorology

The general climate of this region is characterized by long periods of mild sunny weather in the autumn, mild short winters, somewhat more windy but mild weather in the spring, and long hot summers. The site is far enough inland that the strong winds due to the occasional tropical storms and hurricanes that affect this area are greatly reduced, although these storms do cause heavy precipitation at the site in late summer. The site lies in a region of relatively low tornado activity.

In April 1972 an onsite meteorological measurement program began. Data for the program is collected by instrumentation placed on a 150-ft tower located in a cleared area south-southwest of the plant on the site.

1.2.1.9 Environmental Radiation Monitoring

The purpose of the environmental radiation monitoring program is to measure radioactive material in the environment which may be released from the plant. The measurements made will allow evaluation of possible significant modes of human exposure. Background radiation reference measurements to assess radiation sources in the environment not attributable to the plant will be made as well as measurements of radiation in the environment due to plant operation. Background measurements at locations least affected by the plant will be made concurrently with measurements at locations that are more likely to be affected by releases from the plant. These samples will include air, water, and selected biota. The monitoring program will be initiated a few months before startup of Unit 1.

1.2.2 FACILITY ARRANGEMENT

The principal buildings and structures associated with the plant include the containments, the equipment buildings, the turbine building, the auxiliary building, the control building, the diesel generator buildings, the auxiliary feedwater pumphouses, the fuel handling building, portions of the radwaste transfer building, the nuclear service cooling water towers, and the circulating water cooling towers. Ancillary structures include the administration building, warehouse building and receiving facility, service building, maintenance building, plant entry and security building, vehicle maintenance facility, intake and outfall structures, boathouse, chlorination facilities, fire pumphouse, demineralizer building, field support building, the nuclear training facility, alternate radwaste building, radwaste processing facility, alternate plant entry and security building, outage support building, outage storage building, and low-level waste storage facility.

These buildings and structures are founded upon suitable material for their intended application. Structures essential to the safe operation and shutdown of the plant are designed to withstand more extreme loading conditions than normally considered in conventional nonnuclear design practice. The safety-related buildings and their internal structures are designed to provide protection as required from floods, tornadoes, earthquakes, and the failure of equipment

producing flooding, missiles, and pipe breaks. Additional discussion of design considerations is provided in chapter 3.

Location and orientation of the buildings on the site are shown on drawings AX1D45A01, CX2D45V002 and CX2D45V003. The general arrangement of the power block buildings is shown in drawings AX4DD301, AX4DD302, AX4DD303, AX4DD304, AX4DD305, AX4DD306-1, AX4DD306-2, AX4DD307, AX4DD308, AX4DD309, AX4DD310, and AX4DD311.

Equipment locations for Unit 1 are also shown in drawings 1X4DE312, 1X4DE313, 1X4DE314, 1X4DE317, 1X4DE320, 1X4DE322, 1X4DE315, 1X4DE318, 1X4DE321, 1X4DE323, 1X4DE324, 1X4DE325, 1X4DE316, 1X4DE327, 1X4DE330, 1X4DE301, 1X4DE302, 1X4DE303, 1X4DE304, 1X2D05E001, 1X2D48E007, 1X2D48E008, AX1D65A14-1, and AX2D65A400.

The containment, shown schematically in drawings 1X2D48E007 and 1X2D48E008, encloses the reactor coolant system, the steam generators, some of the engineered safety features systems, and supporting systems. The functional design basis of the Seismic Category 1 containment, including its penetrations and isolation valves, is to contain with adequate design margin the energy released from a design basis, high energy line break accident and to provide a leaktight barrier against the uncontrolled release of radioactivity to the environment, even assuming the loss of one of the two trains of engineered safety features. The containment is a prestressed, post-tensioned, reinforced concrete, right circular cylinder with a hemispherical dome.

The equipment building, shown in drawings 1X4DE314 and 1X4DE317, provides protection from the weather for equipment located within the building. The equipment building consists of portions of the control and fuel handling buildings and is designed to Seismic Category 1 requirements.

The turbine building, shown in drawings 1X4DE301, 1X4DE302, 1X4DE303 and 1X4DE304, houses all equipment associated with the main turbine-generator. Other auxiliary equipment, such as air compressors, is also located in this building.

The auxiliary building, shown in drawings 1X4DE315, 1X4DE318, 1X4DE321, 1X4DE323, 1X4DE324 and 1X4DE325, is a multistory, Seismic Category 1, reinforced concrete structure which contains safety systems and necessary auxiliary support systems. Redundant safety trains in the auxiliary building and other areas of the plant are separated and protected so that a loss of function of one train will not prevent the other train from performing its safety function.

The control building, shown in drawings 1X4DE312, 1X4DE313, 1X4DE314, 1X4DE317, 1X4DE320 and 1X4DE322, is a Seismic Category 1, multistory, reinforced concrete structure, in which many of the control and electrical systems, including required support systems directly related to safety or necessary for plant operations, are located.

The auxiliary feedwater pumphouse and condensate storage facility are shown in drawing 1X4DE316. These structures are designed to Seismic Category 1 requirements. The pumphouse contains the train A and B motor-driven pumps as well as the train C turbine-driven pump. The condensate storage facility consists of two concrete condensate storage tanks and associated equipment and structures.

The diesel generator building, shown in drawings 1X4DE327 and 1X4DE330, is a Seismic Category 1 reinforced concrete structure. The building contains two diesel generators, two fuel oil day tanks, four starting air receiver-compressor units, air intake vents and filters, silencers, and controls. Each diesel generator and its associated equipment is in an individual room within the diesel generator building. The building interior and exterior walls which separate the

diesel generators and associated equipment constitute a fire barrier wall having a 3-h fire resistance rating.

The fuel handling building, shown in drawings 1X4DE312, 1X4DE313, 1X4DE314, 1X4DE317, 1X4DE320 and 1X4DE322, is a Seismic Category 1 reinforced concrete structure consisting mainly of spent fuel pools, a fuel receiving and storage area, and a new fuel inspection area. The handling, loading, and shipping of spent fuel casks and the handling, storage, and transfer to the core of new fuel assemblies is provided for in this building.

The radwaste transfer building, adjacent to the auxiliary building (drawing AX4DE350) was formerly intended as a radwaste collection and transfer facility. The structure and its contents have been abandoned with the exception that certain areas associated with HVAC and piping to the alternate radwaste building have been retained. The underground transfer tunnel routes the process piping from the auxiliary building to the radwaste processing facility.

The alternate radwaste building which formerly housed the liquid radwaste systems, shown in drawing AX4DE350, is a metal siding building supported on a base slab with a "lean-to" structure on the north side for storage. The building basemat is designed with curbing to retain radioactive liquid in the event of an operating basis earthquake per Regulatory Guide 1.143. It contains a demineralizer vault, high integrity container (HIC) system storage vault, laydown area, and a truck-trailer loading bay. Areas have been allotted to stage process shields and process skids. The radwaste processing facility, located between the solidification building and the field support building, is designed in accordance with Regulatory Guide 1.143 and applicable codes, standards, calculations, and specifications. It is a concrete building supported on a slab to house process equipment for handling radioactive liquids, resins, and filters. The radwaste processing facility, shown on drawing AX4DE357, contains a subterranean demineralizer vault, subterranean HIC storage vaults, a rollup door for a truck bay, and a 40-ton bridge crane to service equipment. The slab and shield walls inside the building are designed for the operational basis earthquake (OBE) condition and are capable of retaining radioactive liquids during and after an OBE event. A low-level radioactive waste storage facility is provided as shown on drawing AX2D47A469 to provide temporary storage of low-level radioactive waste pending NRC approval of an appropriate disposal facility.

The circulating water cooling tower is a concrete, natural draft, hyperbolic structure. The tower is designed to dissipate all excess heat removed from the main condensers and accomplishes this function by the use of the spray network, the tower basin, and circulating water pumps, piping, and valves. The intake structure houses the circulating water pumps, turbine plant cooling water pumps, and associated auxiliary equipment and piping.

The nuclear service cooling water towers, shown in drawing 1X2D05E001, are Seismic Category 1 concrete mechanical draft structures. The towers house the equipment required to cool the heated nuclear service cooling water, and the basins provide a cooling water storage supply for the ultimate heat sink.

The plant is arranged so that Unit 1 can be placed in commercial operation before the completion of Unit 2. To minimize the exposure of construction personnel to radiation, to prevent unauthorized construction personnel from entering the Unit 1 protected area, and to ensure that no construction condition for Unit 2 affects operation of Unit 1, the following measures are taken:

A. Physical separation (temporary barrier), to the extent practical between Unit 1 and Unit 2, separates Unit 1 from major construction and testing activities on Unit 2.

B. Since Unit 2 is outside the Unit 1 protected area (PA) access to those Unit 2 systems/components inside the PA is through the plant security system.

C. Unit 2 construction personnel access into the PA is minimized.

D. Prior to the start of operation of Unit 1, electrical, ventilation, and process systems within the Protected Area of Unit 1 were segregated to the extent practical from similar systems of Unit 2.

E. A temporary barrier exists in the control room to separate the Unit 1 portion from the under-construction Unit 2 portion.

F. Common systems and interfaces between units are initially isolated from Unit 2. Control of the operating portions of common systems and interfaces, isolation between units, and/or the change of state of these systems and interfaces are governed by Unit 1 procedures.

G. Unit 2 interface activities with Unit 1, including construction work within the protected area, is conducted in accordance with the administrative controls set forth in the Plant Administration Procedures Manual.

- Construction work within the protected area is handled under the controls of the Unit 1 maintenance program established in the plant administration procedures.
- The initial test program preoperational test phase activities are controlled by procedures in the Startup Manual based on the requirements set forth in the plant administration procedures.

1.2.2.1 Units 1 and 2 Shared Facilities

The following facilities are shared by Units 1 and 2; the locations are noted in parentheses:

- Plant access control (plant entry and security building and alternate plant entry and security building).
- Power block entry (control building).
- Control room (control building).
- Control building support facilities (locker rooms, showers, health physics office, laundry, first aid station, etc.).
- Radioactive laboratories (control building).
- Communication room (service building).
- Technical support center (control building).
- Hot machine shop (auxiliary building).
- Hot instrument decontamination shop (auxiliary building).
- Drum storage area (auxiliary building).
- New fuel pit (fuel handling building).

- Spent fuel cask handling areas (auxiliary and fuel handling building).
- Portions of radwaste transfer building.
- Alternate radwaste building.
- Radwaste processing facility.
- Water analysis room (turbine building).
- Nitrogen storage area (outdoors).
- Hydrogen storage area (outdoors).
- Sewage treatment plant.
- River intake structure (outdoors).
- Water treatment building (outdoors).
- Waste water effluent facility (outdoors).
- Low-level waste storage facility (outdoors).
- Outage storage building.

1.2.2.2 Units 1 and 2 Shared Systems

The following systems or portions of systems are shared by Units 1 and 2:

- Boron recycle.
- Waste processing - liquid.
- Waste processing - gaseous.
- Radwaste solidification - portable vendor supplied.
- Fire protection.
- Instrument and service air.
- Auxiliary gas.
- Plant makeup water treatment.
- Plant makeup water treatment waste neutralization.
- Plant demineralized water.
- Auxiliary steam.

- Turbine-generator hydrogen gas cooling.
- Potable water.
- Laboratory gas and liquid supply.
- Fuel handling building heating, ventilation, and air-conditioning (HVAC).
- Control building normal HVAC.
- Control room HVAC.
- Control room lighting.
- Spent fuel cask bridge crane.
- Plant communications.
- Process and effluent monitoring system.
- Fuel handling machine.
- Condensate filter demineralizer spent resin dewatering system.
- CVCS chiller.
- 4160-V nonsafety-related electrical distribution system (outage loads only).

1.2.3 NUCLEAR STEAM SUPPLY SYSTEM (NSSS)

Each NSSS consists of a pressurized water reactor and a four-loop reactor coolant system (RCS). The mechanical, thermal-hydraulic, and nuclear design of the reactor core is similar to the design of other Westinghouse units. The maximum expected thermal output rating of each NSSS unit is 3653 MWt. Heat balances for the major parameters of the plant are shown for the turbine valves wide open condition in drawing 1X4DC101, and for the as-built heat balance in drawings 1X4DC103 and 2X4DC103.

1.2.3.1 Reactor Core

The core is cooled and moderated by light water in the RCS. The moderator/coolant contains boron as a neutron poison. The concentration of the boron in the coolant is varied as required to control relatively slow reactivity changes including the effects of fuel burnup. Additional boron, in the form of burnable poison rods, is employed in the first core to establish the desired initial reactivity.

The fuel rods consist of slightly enriched uranium dioxide cylindrical pellets contained in slightly cold-worked, Zircaloy-4 tubing which is plugged and seal welded at the ends to encapsulate the fuel. All fuel rods are pressurized with helium during fabrication to reduce stresses and strains and to increase fatigue life.

Additional reactivity control is provided by control rod assemblies which consist of a group of individual absorber rods fastened at the top end to a common hub or spider assembly. The control rod drive mechanisms for the control rod assemblies are of the magnetic latch type. The latches are controlled by three magnetic coils. They are so designed that upon a loss of power to the coils, the control rod assembly is released and falls by gravity to shut down the reactor.

The components of the reactor internals are divided into three structures consisting of the lower core support structure (including the entire core barrel and neutron shield pad assembly), the upper core support structure, and the incore instrumentation support structure. The reactor internals support the core, maintain fuel alignment, limit fuel assembly movement, maintain alignment between fuel assemblies and control rod drive mechanisms, direct coolant flow past the fuel elements and to the pressure vessel head, provide gamma and neutron shielding, and provide guides for the incore instrumentation.

Instrumentation is provided in and out of the core to monitor the nuclear, thermal-hydraulic, and mechanical performance of the reactor and to provide inputs to automatic control functions.

The reactor core design together with corrective actions of the reactor control, protection, and emergency core cooling systems ensure that attained peak local power densities do not:

- Lead to fuel damage during normal operation or faults of moderate frequency.
- Cause failure of more than a small fraction of fuel rods due to infrequent fault.
- Prevent acceptable heat transfer during transients associated with limiting faults.

1.2.3.2 Reactor Coolant System

The RCS is arranged as four closed loops connected in parallel to the reactor vessel. Each loop consists of one 29-in. inside diameter (ID) outlet pipe (hot leg), one steam generator, and one 27.5-in. ID inlet pipe (cold leg), with one reactor coolant pump in each cold leg. The piping between the steam generator and the reactor coolant pump suction has an inside diameter of 31 in. to reduce pressure drop and improve flow conditions to the pump suction. An electrically heated pressurizer is connected to one of the loops. A safety injection line is connected to each of the four cold legs. The RCS operates at a nominal pressure of 2235 psig.

The reactor coolant enters the reactor vessel through four inlet nozzles, turns and flows downward between the reactor vessel shell and the core support barrel, and enters the lower plenum through the flow skirt. The coolant then turns and flows upward through the core barrel lower support structure, through the core support plate flow holes, and continues parallel to the axis of the fuel assemblies to remove the heat generated within the fuel. The coolant continues its upward flow through the upper guide structure, then turns and leaves the reactor vessel through the four outlet nozzles and the hot leg pipes, which lead to the steam generators. The coolant flows through the tube side of the four steam generators, where heat is transferred to the secondary system. Reactor coolant pumps return the reactor coolant to the reactor vessel.

The pressure in the RCS is controlled by regulating the temperature of the coolant in the pressurizer, where steam and water are held in thermal equilibrium. Steam is formed by the pressurizer heaters or condensed by the pressurizer spray to reduce pressure variations caused by expansion and contraction of the reactor coolant due to system temperature changes. The pressurizer limits pressure variations during plant load transients and keeps system pressure within design limits during abnormal conditions.

1.2.4 STEAM AND POWER CONVERSION SYSTEM

The turbine-generator is furnished by the General Electric Company. The gross generator output at turbine guaranteed rating is 1,156,622 kW output at 1800 rpm, using a tandem compound 6-stage flow machine with 38-in. last-stage buckets. The throttle steam conditions are 940 psia, 1191.4 Btu/lb, with one stage of reheat. The exhaust pressure is 3.5-in. Hg abs; the makeup is 0.0 percent. Extraction steam is used for normal feedwater heating and for steam generator feed pump turbine operation.

General features of the generator include a 1,350,000-kVA output at 1800 rpm, four poles direct connected, 3-phase, 60-Hz, 25,000-V conductor cooled synchronous generator rated at 0.90 power factor, 0.5 short circuit ratio at maximum hydrogen pressure of 75 psig.

The auxiliary feedwater system contains two motor-driven pumps and one turbine-driven pump. This system is designed to provide emergency heat removal capacity.

1.2.5 CONTAINMENT

The containment completely encloses the reactor and reactor coolant system. It is a vertical, right-cylindrical, prestressed, post-tensioned concrete structure with a dome and flat base with a depressed center for a reactor cavity and instrumentation tunnel. The interior is lined with carbon steel plate for leaktightness. Approximate dimensions of the containment are as follows: inside diameter, 140 ft; inside height, 228 ft; vertical wall thickness, 3 ft 9 in.; dome thickness, 3 ft 9 in.; foundation thickness, 10 ft 6 in., with a minimum thickness of the concrete basemat of 8 ft 3 in.; and the minimum thickness of the concrete instrumentation cavity of 8 ft.

Inside the containment, the reactor and other nuclear steam supply system components are shielded with concrete. A vent stack is attached to the outside of the containment and extends a short distance above the top of the containment dome. Access to portions of the containment during power operation is permissible.

The containment, along with the safety features, is designed to withstand the internal pressure and coincident temperature resulting from the energy release of the high energy line break accident. The containment design internal pressure is based upon the design basis loss-of-coolant accident, and the internal temperature is based upon design basis main steam line break.

1.2.6 SAFETY FEATURES

The safety features limit the potential radiation exposure to the public and to plant personnel following an accidental release of radioactive fission products from the reactor system, particularly as the result of a loss-of-coolant accident (LOCA). These safety features function to localize, control, mitigate, and terminate such accidents, ensuring that 10 CFR 100 guidelines are not exceeded. The safety features include the following systems:

- Emergency core cooling system (ECCS).
- Containment spray system.
- Containment cooling system.
- Penetration room filtration systems.

- Hydrogen recombiners.

1.2.6.1 Emergency Core Cooling System

The ECCS injects borated water into the reactor coolant system following a LOCA. This provides cooling to limit core damage, metal-water reactions, and fission product release and ensures adequate shutdown margin regardless of temperature. The ECCS also provides continuous long term, post-accident cooling of the core by recirculating borated water between the containment sump and the reactor core.

1.2.6.2 Containment Spray System

The containment spray system is composed of two redundant, full-capacity trains which are designed to reduce the post-accident containment building iodine concentrations so that offsite doses are less than 10 CFR 100 guidelines.

The containment spray system supplies borated water during injection and borated water mixed with trisodium phosphate during recirculation to the containment atmosphere. The spray system in combination with four of the eight containment air coolers (operating at reduced speed) is sized to provide adequate cooling with either or both of the two containment spray pumps in service. These pumps take suction from the refueling water storage tank. When the RWST empty alarm is received, suction of the containment spray pumps is aligned to pump water from the containment sump directly into the containment during the recirculation mode of operation.

1.2.6.3 Containment Cooling System

The containment cooling system consists of two independent, redundant, full-capacity trains containing the equipment necessary for safe shutdown of the plant following an accident. The system is designed to reduce the post-accident containment pressure to 50 percent or less of the peak containment pressure within 24 h or less following the accident.

1.2.6.4 Penetration Room Filtration Systems

The penetration room filtration systems for each unit collect and process potential airborne contamination due to leakage from process components during the recirculation mode of ECCS operation. This filtration limits the environmental activity levels following an accident.

1.2.6.5 Hydrogen Recombiners

Fully redundant electrical hydrogen recombiners inside the containment reduce the percentage of hydrogen in the post- accident containment atmosphere to below combustible levels.

1.2.7 UNIT CONTROL

The reactor is controlled by control rod movement and regulation of the boric acid concentration in the reactor coolant. During steady-state operation, the reactor control system maintains a programmed average reactor coolant temperature that rises in proportion to the load.

The solid state protection logic system automatically initiates appropriate action whenever the parameters monitored by this system reach preestablished setpoints. This system acts to trip the reactor, actuate emergency core cooling, close containment isolation valves, and initiate the operation of other safety feature systems.

1.2.8 PLANT ELECTRICAL POWER

The two main turbine-generators are each rated at 1350 MVA, 0.90 power factor, 25,000 V; they are 3-phase, 60-Hz, 1800 rpm hydrogen- and water-cooled units. The power from these units is delivered to the Georgia Power Company 230-kV and 500-kV transmission lines. Termination points of these lines are described in section 8.2.

Each unit has three separate sources of power for its auxiliaries. The three sources of power and associated electrical equipment ensure the functioning of both units without undue risk to the health and safety of the public and provide reliable power sources for startup, normal operation, safe shutdown, and emergency situations.

The three sources for each unit are:

A. The main turbine-generator supplies normal auxiliary loads during plant operation.

B. The two reserve auxiliary transformers supply the two safety feature buses from the 230-kV switchyard. The standby auxiliary transformer (SAT) may supply either of the two safety feature buses from Plant Wilson.

C. The two standby diesel generators provide emergency power to two safety feature buses. Upon loss of all offsite power, either diesel generator and its associated bus has the capacity to power the equipment required to safely shutdown the reactor and mitigate the consequences of the design bases accidents.

Plant batteries ensure a constant supply of power to vital instruments and controls.

Plant power is distributed through buses at 13,800 V, 4160 V, and 480 V, and 120 V ac. The safety-related dc loads in each unit are powered from four 125-V dc buses.

1.2.9 PLANT INSTRUMENTATION AND CONTROL SYSTEM

To avoid undue risk to the health and safety of the public, instrumentation and controls monitor and maintain neutron flux, primary coolant pressure, temperature, and control rod positions within prescribed operating ranges.

The regulating, process, and containment instrumentation measure temperatures, pressures, flows, and levels in the reactor coolant system, steam systems, containment, and auxiliary systems. Process variables required on a continuous basis for startup, operation, and shutdown of the unit are indicated, recorded, and controlled from the control room. The quality and types of process instrumentation provided ensure safe and orderly operation of all systems and processes over the full operating range of the plant.

Startup and shutdown of the reactor and adjustment of reactor power in response to turbine load demand are provided by the reactor control system. The reactor is controlled by control rod motion for startup, shutdown, and load-follow transients; it is also controlled by a soluble neutron absorber (boron, injected as boric acid into the primary coolant) which is inserted during cold shutdown and refueling, partially removed during reactor startup, and adjusted in concentration during core lifetime to compensate for such effects as fuel consumption and accumulation of fission products which reduce the levels of desirable nuclear reactions. The control system permits each unit to accept step load increases or decreases of 10 percent and ramp load increases or decreases of 5 percent/min over the load range from 15 percent to, but not exceeding, 100-percent power under normal operating conditions, subject to xenon limitations.

Under normal conditions both the reactor and the turbine-generator are controlled from the control room by Nuclear Regulatory Commission-licensed personnel.

1.2.10 AUXILIARY SYSTEMS

Nuclear auxiliary systems perform the following functions:

- Supply reactor coolant system (RCS) water requirements.
- Purify reactor coolant water.
- Introduce chemicals for corrosion inhibition.
- Introduce and remove chemicals for reactivity control.
- Cool system components.
- Remove residual heat during a portion of the reactor cooling period and when the reactor is shut down.
- Cool the spent fuel pool water.
- Permit sampling of reactor coolant water.
- Provide for safety injection.
- Vent and drain the RCS and the auxiliary systems.
- Provide containment ventilation and cooling.
- Process liquid and gaseous wastes and dispose of solid wastes.

These functions are performed by the systems discussed below.

1.2.10.1 Chemical and Volume Control System (CVCS)

The CVCS performs the following functions:

- Reactivity control.

- Regulation of reactor coolant inventory.
- Reactor coolant purification.
- Chemical additions for corrosion control.
- Seal water injection to reactor coolant pump seals.

The purity level in the RCS is controlled by continuous purification of a bypass stream of reactor coolant. Water removed from the RCS is cooled in the regenerative heat exchanger. From there, the coolant flows to a letdown heat exchanger and through a demineralizer where corrosion and fission products are removed. The coolant then passes through a filter and is sprayed into the volume control tank.

The CVCS automatically adjusts the amount of reactor coolant to compensate for changes in specific volume due to coolant temperature changes and reactor coolant pump shaft seal leakage in order to maintain a constant level in the pressurizer.

1.2.10.2 Residual Heat Removal (RHR) System

The RHR system is used to reduce the temperature of the reactor coolant at a controlled rate from 350°F to 140°F and to maintain the proper reactor coolant temperature during refueling.

The RHR pumps are used to circulate the reactor coolant through two RHR heat exchangers, returning it to the RCS through the low-pressure injection header.

1.2.10.3 Auxiliary Feedwater (AFW) System

The AFW system provides feedwater to the steam generators for removal of reactor core decay heat following a loss of main feedwater. The AFW system is also used to cool down the reactor to the temperature and pressure conditions required for initiation of the RHR system.

1.2.10.4 Component Cooling Water (CCW) System

The CCW system consists of two independent trains of pumps and heat exchangers to remove heat from the various auxiliary systems handling the reactor coolant. Corrosion-inhibited demineralized water is circulated by the system through the spent fuel pool heat exchangers, the residual heat exchangers, and the RHR pump seal coolers.

The CCW system provides an intermediate barrier between the RCS and the nuclear service cooling water (NSCW) system described in paragraph 1.2.10.8.

1.2.10.5 Auxiliary Component Cooling Water (ACCW) System

The ACCW system consists of pumps and heat exchangers to remove heat from various auxiliary systems handling reactor coolant. Corrosion-inhibited demineralized water is circulated by the system through the letdown heat exchanger, the reactor coolant pump coolers, sample heat exchangers, waste gas compressors, and other nonsafety-related heat loads.

The ACCW system provides an intermediate barrier between the RCS and the NSCW system described in paragraph 1.2.10.8.

1.2.10.6 Fuel Handling and Storage System

The reactor is refueled with equipment designed to handle spent fuel underwater from the time it leaves the reactor vessel until it is placed in a cask for temporary storage pending shipment offsite. Transfer of spent fuel underwater enables the use of an optically transparent radiation shield and provides a reliable source of coolant for removal of residual heat.

The fuel handling system also provides for the safe handling of rod cluster control assemblies under all foreseeable conditions and for the required assembly, disassembly, and storage of reactor internals. This system includes a manipulator crane located inside the containment above the refueling pool, fuel transfer carriage, upending devices, fuel transfer tube, spent fuel cask bridge crane in the spent fuel pool area, and various devices used for handling the reactor vessel head and internals.

The new fuel storage area is shared by both units and is sized to accommodate storage of 162 fuel assemblies and control rods. The new fuel assemblies are stored in racks in parallel rows having a center-to-center distance of approximately 21 in. to preclude criticality.

Together, the two stainless steel-lined, reinforced concrete spent fuel pools provide storage capacity for 3574 fuel assemblies. Spent fuel assemblies are stored in vertical racks spaced to preclude criticality with no credit taken for the borated pool water. See drawings 1X6AN10B-66 and AX6AN10A-6 for the layout of the pools. The layout of the pools is also shown in figures 9.1.2-3 and 9.1.2-5.

Purification and redundant cooling equipment is provided for each spent fuel pool. This equipment may also be used for cleanup of refueling water after each fuel change in the reactor.

Additional spent fuel storage capacity is provided by the independent spent fuel storage installation (ISFSI) operated in accordance with the general license provisions of Part 72, Subpart K. The ISFSI consists of dry spent fuel storage casks, two storage pads, and applicable radiological and security provisions. The ISFSI is designed to hold a total of 36 HI-STORM 100 spent fuel cask systems, each containing 32 pressurized water reactor fuel assemblies, for a total storage capacity of 1,152 assemblies as described in subsection 9.1.6.

1.2.10.7 Sampling Systems

Three sampling systems are provided: nuclear sampling system - liquid, nuclear sampling system - gaseous, and turbine plant sampling system. These systems are used for determining both chemical and radiochemical conditions of the various fluids used in the plant.

1.2.10.8 Cooling Water Systems

The turbine-generator condenser is cooled by the circulating water system, which rejects heat to a cooling tower.

The cooling water requirements for various nuclear components, CCW system, ACCW system, and diesel generators are supplied by pumps taking suction from the NSCW tower basins. Water makeup to the storage basins is supplied by the normal plant makeup wells or the Savannah River. In addition, a standby nuclear service makeup well can supply makeup flow to the NSCW system as a backup source of water.

1.2.10.9 Plant Ventilation Systems

Separate ventilation systems are provided for the containment, penetration rooms, auxiliary building, fuel handling building, control room, control building, turbine building, and emergency diesel generator building. In addition, a purge system, mini- purge system, post-loss-of-coolant accident purge system, and containment preaccess filtration system are provided for the containment.

The auxiliary building, fuel handling building, and control building penetration rooms are ventilated by the penetration room filtration systems, which include filters for control of any leakage from process components during the recirculation mode of emergency core cooling system operation.

1.2.10.10 Plant Fire Protection System

The major fire protection system contains both diesel- and electric-driven fire pumps which supply the various hydrants, hose stations, sprinklers, and deluge systems. Hydrants and hose stations are manually operated; the sprinkler and deluge systems are a combination of automatic and manually actuated systems. Supplementary to these facilities, chemical fire-extinguishing equipment is provided to accommodate special requirements for various classes of hazards. Noncombustible and fire-resistant materials are selected for use wherever practical throughout the facility, particularly in critical portions of the plant such as the containment, control room, and components of the safety features system.

1.2.10.11 Compressed Air Systems

One reciprocating and two rotary air compressors, with separate aftercoolers and separators, discharge compressed air to two separate headers. These headers provide a continuous supply of filtered, dried, oil-free, compressed air for pneumatic instrument operation and control of pneumatic actuations. The system also provides service air at outlets throughout the plant. An additional reciprocating compressor train can be aligned to either unit.

1.2.10.12 Radioactive Waste Disposal System

The waste disposal system provides controlled handling and disposal of liquid, gaseous, and solid wastes. The waste processing system provides all equipment necessary for controlled treatment and preparation for retention or disposal of all liquid, gaseous, and solid wastes produced as a result of reactor operation.

The liquid waste processing system collects, processes, and recycles reactor grade water; removes or concentrates radioactive constituents; and processes them until suitable for release or shipment offsite. Liquid wastes are sampled and activity levels verified and recorded prior to release. Processed liquid effluent from the RCS will have been subjected to the CVCS purification ion exchanger and the components of the waste processing system and will be within the limits established by the Offsite Dose Calculation Manual.

The gaseous waste processing system functions to remove fission product gases from the reactor coolant and to contain these gases during normal plant operation. Waste gases are collected in the vent header. These gases are withdrawn from the vent header by one of two compressors and discharged to a waste gas decay tank. The tank contents will be released to the environment in accordance with the program requirements of Technical Specifications, and

releases will be within the limits set forth in the Offsite Dose Calculation Manual. Seven waste gas decay tanks and two shared waste gas decay shutdown tanks are provided. Gaseous wastes are discharged through an absolute particle filter to the vent stack.

Solid wastes are stored in suitable containers for eventual disposal. A low-level waste storage facility provides temporary storage of dry radioactive waste pending disposal in an NRC-licensed facility. The low-level waste facility also provides temporary storage for dewatered resins pending collection of a sufficient quantity for processing and/or reduction to a form suitable for disposal.

1.2.11 SHARED FACILITIES AND EQUIPMENT

An integrated, twin-unit station enables certain components, systems, and facilities to be designed and arranged so that they are common to both units without impairing the safety or reliability of either unit.

Separate and similar systems and equipment are provided for each unit except as noted in paragraphs 1.2.2.1 and 1.2.2.2.

1.2.12 PRINCIPAL DESIGN CRITERIA

The VEGP is designed to comply with the intent of the General Design Criteria for Nuclear Power Plants contained in Appendix A to 10 CFR 50. The details of compliance to the general design criteria are provided in section 3.1.

The principal design criteria are presented in two ways--power generation function or safety function--and grouped according to system. Although the distinctions between power generation or safety functions are not always clear and occasionally overlap, the functional classification facilitates safety analyses, while the grouping by system facilitates the understanding of both the system function and design.

1.2.12.1 General Design Criteria

1.2.12.1.1 Power Generation Design Criteria

A. The plant is designed to produce steam for direct use in a turbine-generator unit.

B. Heat removal systems are provided with sufficient capacity and operational adequacy to remove heat generated in the reactor core for the full range of normal operational conditions and abnormal operational transients.

C. Backup heat removal systems are provided to remove decay heat generated in the core under circumstances wherein the normally operational heat removal systems become inoperative. The capacity of such systems is adequate to prevent fuel cladding damage.

D. The fuel cladding, in conjunction with other plant systems, is designed to retain integrity through the range of normal operational conditions and abnormal operational transients.

E. The fuel cladding accommodates, without loss of integrity, the pressures generated by fission gases released from fuel material throughout the design life of the fuel.

F. Control equipment is provided to allow the reactor to respond automatically to minor load changes, major load changes, and abnormal operational transients.

G. Reactor power level is manually controllable.

H. Control of the reactor is possible from a single location.

I. Reactor controls and related indicators and alarms are arranged to allow the operator to rapidly assess the condition of the reactor system and locate system malfunctions.

J. Interlocks, or other automatic equipment, are provided as backup to procedural controls to avoid conditions requiring the functioning of nuclear safety systems or engineered safety features (ESF).

1.2.12.1.2 Safety Design Criteria

A. The plant design conforms to applicable regulations as discussed in sections 1.9 and 3.1.

B. The plant is designed, fabricated, erected, and operated in such a way that the release of radioactive materials to the environment does not exceed the limits and guideline values of applicable government regulations pertaining to the release of radioactive materials for normal operations and for abnormal transients and accidents.

C. The reactor core is designed so its nuclear characteristics do not contribute to a divergent power transient.

D. The reactor is designed so that there is no tendency for divergent oscillation of any operating characteristic, considering the interaction of the reactor with other appropriate plant systems.

E. Gaseous, liquid, and solid waste disposal facilities are designed so that the discharge of radioactive effluents and offsite shipment of radioactive materials can be made in accordance with applicable regulations.

F. The design provides means by which plant operators are alerted when limits on the release of radioactive material are approached.

G. Sufficient indications are provided to allow determination that the reactor is operating within the envelope of conditions considered by plant safety analysis.

H. Radiation shielding is provided and access control patterns are established to allow a properly trained operating staff to control radiation doses within the limits of applicable regulations in any mode of normal plant operations.

I. Those portions of the nuclear steam supply system (NSSS) that form part of the reactor coolant pressure boundary (RCPB) are designed to retain integrity as a radioactive material containing barrier following abnormal operational transients and accidents.

J. Nuclear safety systems and ESF function to ensure that no damage to the RCPB results from internal pressures caused by abnormal operational transients and accidents.

K. Where positive, precise action is immediately required in response to abnormal operational transients and accidents, such action is automatic and requires no decision or manipulation of controls by plant operations personnel.

L. Essential safety actions are provided by equipment of sufficient redundance and independence so that no single failure of active components can prevent the required actions. For systems or components to which Institute of Electrical and Electronics Engineers (IEEE) 279-1971, Criteria for Protection Systems for Nuclear Power Generating Stations, applies, single failures of both active and passive electrical components are considered in recognition of the higher anticipated failure rates of passive electrical components relative to passive mechanical components.

M. Provisions are made for control of active components of nuclear safety systems and ESF from the control room.

N. Nuclear safety systems and ESF are designed to permit demonstration of their functional performance requirements.

O. The design of nuclear safety systems and ESF includes allowances for natural environmental disturbances such as earthquakes, floods, and storms at the station site.

P. Standby electrical power sources have sufficient capacity to power all nuclear safety systems and ESF requiring electrical power.

Q. Standby electrical power sources are provided to allow prompt reactor shutdown and removal of decay heat under circumstances where normal auxiliary power is not available.

R. A containment is provided which completely encloses the reactor system.

S. The containment is designed to allow periodic integrity and leaktightness testing.

T. The containment, in conjunction with other ESF, limits to less than the prescribed acceptable limits the radiological effects of accidents resulting in the release of radioactive material to the containment volume.

U. Provisions are made for removing energy from the containment as necessary to maintain the integrity of the containment system following accidents that release energy to the containment.

V. Piping that penetrates the containment and could serve as a path for the uncontrolled release of radioactive material to the environs is automatically isolated whenever such uncontrolled radioactive material release is threatened. Such isolation is effected in time to limit radiological effects to less than the specified acceptable limits.

W. Emergency core cooling systems (ECCS) are provided to limit fuel cladding temperature to less than the limits established by 10 CFR 50.46 in the event of a loss-of-coolant accident (LOCA).

X. The ECCS provides for continuity of core cooling over the complete range of postulated break sizes in RCPB.

Y. Actuation of the ECCS occurs automatically when required, regardless of the availability of offsite power supplies and the normal generating system of the station.

Z. The control room is shielded against radiation so that continued occupancy under accident conditions is possible.

AA. In the event that the control room becomes uninhabitable, it is possible to bring the reactor from power range operation to cold shutdown conditions by utilizing the shutdown panels and other equipment available outside the control room.

BB. Backup reactor shutdown capability is provided independent of normal reactivity control provisions. This backup system has the capability to shut down the reactor from any normal operating condition and subsequently to maintain the shutdown condition.

CC. The fuel handling and storage facility is designed to prevent inadvertent criticality and to maintain shielding and cooling of spent fuel.

1.2.12.2 System Criteria

The principal design criteria for particular systems are discussed below.

1.2.12.2.1 Nuclear System Criteria

A. The fuel cladding is designed to retain integrity as a radioactive material barrier throughout the design power range. The fuel cladding is designed to accommodate, without loss of integrity, the pressures generated by the fission gases released from the fuel material throughout the design life of the fuel.

B. The fuel cladding, in conjunction with other plant systems, is designed to retain integrity throughout any abnormal operational transient.

C. Those portions of the nuclear system that form part of the RCPB are designed to retain integrity as a radioactive material barrier during normal operation and following abnormal operational transients and accidents.

D. Heat removal systems are provided in sufficient capacity and operational adequacy to remove heat generated in the reactor core for the full range of normal operational transients. The capacity of these systems is adequate to prevent fuel cladding damage.

E. Heat removal systems are provided to remove decay heat generated in the core under circumstances wherein the normal operational heat removal systems become inoperative. The capacity of these systems is adequate to prevent fuel cladding damage. The reactor is capable of being shut down automatically in sufficient time to permit decay heat removal systems to become effective following loss of normal heat removal systems.

F. The reactor core and reactivity control system is designed so that control rod action is capable of bringing the core to subcriticality and maintaining it subcritical even with the rod of highest reactivity worth fully withdrawn and unavailable for insertion into the core.

G. The reactor core is designed so that its nuclear characteristics do not contribute to a divergent power transient.

H. The nuclear system is designed so there is no tendency for divergent oscillation of any operating characteristic, considering the interaction of the nuclear system with other appropriate plant systems.

1.2.12.2.2 Power Conversion Systems Criteria

Components of the power conversion systems are designed to produce electrical power from the steam coming from the steam generator, condense the steam into water, and return the water to the steam generator as heated feedwater.

1.2.12.2.3 Electrical Power System Criteria

Sufficient normal and standby auxiliary sources of electrical power are provided to facilitate prompt shutdown and continued maintenance of the plant in a safe condition under all credible circumstances. The power sources are adequate to accomplish all required essential safety actions under postulated design bases accident conditions.

1.2.12.2.4 Radwaste System Criteria

A. The gaseous and liquid radwaste systems are designed to minimize the release of radioactive effluents from the station to the environs. Such releases as may be necessary during normal operations are limited to values that meet the requirements of applicable regulations.

B. The solid radwaste disposal systems are designed so that in plant processing and offsite shipments are in accordance with applicable regulations.

C. The system design provides means by which station operations personnel are alerted whenever specified limits on the release of radioactive material are approached.

1.2.12.2.5 Auxiliary Systems Criteria

A. Fuel handling and storage facilities are designed to prevent criticality and to maintain adequate shielding and cooling for spent fuel. Provisions are made for maintaining the cleanliness of spent fuel cooling and shielding water.

B. Other auxiliary systems or portions of these systems, such as service water, cooling water, fire protection, heating and ventilating, communications, and lighting, are designed to function during normal and/or accident conditions.

C. Auxiliary systems which are not required to effect safe shutdown of the reactor or maintain it in a safe condition are designed so that a failure of these systems does not prevent the essential auxiliary systems from performing their design functions.

1.2.12.2.6 Shielding and Access Control Criteria

Shielding and access control criteria are as follows:

A. Radiation shielding is provided, and access control patterns are established to allow a properly trained operating staff to control radiation doses within the limits of applicable regulations in any normal mode of plant operation.

B. The control room is shielded against radiation so that occupancy is possible under accident conditions.

1.2.12.2.7 Nuclear Safety Systems and ESF Criteria

Principal design criteria for nuclear safety systems and ESF are as follows:

A. Nuclear safety systems and ESF function to ensure that the safety design criteria J through Q, X, Y, and AB of paragraph 1.2.12.1.2 are satisfied.

B. ECCSs are provided to preclude core damage in excess of limits specified in 10 CFR 50, Appendix K.

C. If the control room is uninhabitable, it is possible to bring the reactor from power range operation to a hot shutdown condition by manipulating the local controls and equipment that are available outside the control room. Furthermore, the plant design includes the capability to bring the reactor to a cold shutdown condition from the hot shutdown condition using controls and indications located outside the control room.

1.2.12.2.8 Process Control Systems Criteria

The principal design criteria for the process control systems are given in the following paragraphs.

1.2.12.2.8.1 NSSS Process Control Criteria.

A. Control equipment is provided to allow the reactor to respond automatically to main load changes within design limits.

B. It is possible to control the reactor power level manually.

C. Control of the NSSS is possible from a central location.

D. The NSSS process controls and alarms are arranged to allow the operator to rapidly assess the condition of the NSSS and to locate process system malfunctions.

E. Interlocks or other automatic equipment are provided as a backup to procedural controls to avoid conditions requiring the actuation of ESF.

1.2.12.2.8.2 Power Conversion Systems Process Control Criteria.

A. Control equipment is provided to control the reactor pressure throughout its operating range.

B. The turbine is capable of responding automatically to minor changes in load.

C. Control of the power conversion equipment is possible from a central location.

D. Interlocks or other pieces of automatic equipment are provided in addition to procedural controls to avoid conditions requiring the actuation of ESF.

1.2.12.2.8.3 Electrical Power System Process Control Criteria.

A. The Class 1E power system is designed as two safety feature buses, with either bus being capable of bringing the unit to a safe shutdown condition.

B. Protective devices are used to detect and isolate faulted equipment from the system with a minimum of disturbance in the event of equipment failure.

C. Voltage detection devices are used to isolate the safety feature buses from the normal electrical system in the event of loss of offsite power and to initiate starting of the standby emergency power system diesel generators.

D. The standby diesel generators are started manually or automatically. The generators are sequentially loaded by a load sequencing system to meet the existing emergency condition.

E. Electrically operated breakers are generally controllable from the control room.

F. Metering for essential generators, transformers, and circuits is provided in the control room.

1.2.13 PARTICULARLY DIFFICULT ENGINEERING PROBLEMS

In general, particularly difficult engineering problems can be defined as those requiring development work or vendor testing to finalize the design. Such areas are discussed in section 1.5.

1.2.14 EXTRAPOLATION OF TECHNOLOGY

There are no significant extrapolations of design technology incorporated in the VEGP.

1.3 COMPARISON TABLES (HISTORICAL)

1.3.1 COMPARISON WITH SIMILAR FACILITY DESIGNS

Table 1.3.1-1 presents a design comparison of the major parameters or features of the VEGP units at the time of initial licensing with the SNUPPS units, Comanche Peak, Units 1 and 2, and the Trojan unit. Changes made to VEGP or other referenced plants following initial licensing are not reflected in table 1.3.1-1.

Table 1.3.1-2 presents a design comparison of the major balance of plant parameters or features of the VEGP design at the time of initial licensing with data for San Onofre, Units 2 and 3; Farley, Units 1 and 2; and Calvert Cliffs, Units 1 and 2. Changes made to VEGP or other referenced plants following initial licensing are not reflected in table 1.3.1-2.

1.3.2 COMPARISON OF FINAL AND PRELIMINARY INFORMATION

Table 1.3.2-1 identifies the significant changes that have been made to the power block since submittal of the VEGP Preliminary Safety Analysis Report (PSAR) at the time of initial licensing. Only items not reported in the PSAR and its subsequent amendments and supplements are listed in table 1.3.2-1. Changes made following initial licensing are not reflected in table 1.3.2-1.

Table 1.3.2-2 provides a listing of the Three Mile Island (TMI) action plan tasks approved for incorporation in NUREG-0737 and indicates the subsections or paragraphs in this Final Safety Analysis Report (FSAR) which describe the implementation of the tasks for VEGP.

TABLE 1.3.1-1 (SHEET 1 OF 5) (HISTORICAL)

DESIGN COMPARISON

Parameter or Feature	*VEGP-FSAR Chapter/Section*	*VEGP[c] (Original Design)*	*SNUPPS*	*Comanche Peak*	*Trojan*
Reactor core heat output (MWt)	*4.0, 5.0, 15.0*	*3411*	*3411*	*3411*	*3411*
Minimum departure from nucleate boiling ratio for design transients	*4.1, 4.4, 15.0*	*1.30*	*1.30*	*1.30*	*1.30*
Total thermal flowrate 10^6 (lb/h)	*4.1, 4.4, 5.1*	*142.1*	*142.1*	*140.3*	*132.7*
Reactor coolant temperatures (°F)	*4.1, 4.4*				
Core outlet		*621.4*	*621.4*	*620.8*	*619.5*
Vessel outlet		*618.2*	*618.2*	*618.2*	*616.8*
Core average		*591.8*	*591.8*	*589.4*	*585.9*
Vessel average		*588.5*	*588.5*	*588.2*	*584.7*
Core inlet		*558.8*	*558.8*	*558.1*	*552.5*
Vessel inlet		*558.8*	*558.8*	*558.1*	*552.5*
Average linear power(kW/ft)	*4.1, 4.4*	*5.44*	*5.44*	*5.44*	*5.44*
Peak linear power for normal operation (kW/ft)	*4.1, 4.4*	*12.6*	*12.6*	*12.6*	*13.6*
Heat flux hot channel factor (F_Q)	*4.1, 4.4, 15.0*	*2.30*	*2.32*	*2.32*	*2.50*
Fuel assembly array	*4.1, 4.3*	*17 x 17*	*17 x 17*	*17 x 17*	*17 x 17*
Number of fuel assemblies	*4.1, 4.3*	*193*	*193*	*193*	*193*
Uranium dioxide rods per assembly	*4.1, 4.3*	*264*	*264*	*264*	*264*
Fuel weight as uranium dioxide (lb)	*4.1, 4.3*	*222,739*	*222,739*	*222,739*	*222,739*
Number of grids per assembly	*4.1, 4.3*	*8, type R*	*8, type R*	*8, type R*	*8, type R*

TABLE 1.3.1-1 (SHEET 2 OF 5) (HISTORICAL)

Parameter or Feature	VEGP-FSAR Chapter/Section	VEGP[c] (Original Design)	SNUPPS		Comanche Peak		Trojan
Rod cluster control assemblies	4.1, 4.3						
Number of full-length rods absorber material		53 hafnium	53 Ag-In-Cd		53 Ag-In-Cd[a] and B_4C		53 Ag-In-Cd
Clad material		SS	SS		SS		SS
Clad thickness		0.0185	0.0185		0.0185		0.0185
Equivalent core diameter (in.)	4.1, 4.3	132.7	132.7		132.7		132.7
Active fuel length (in.)	4.1, 4.3	143.7	143.7		143.7		143.7
Fuel enrichment (wt%)	4.1, 4.3		Core A	Core BB	Unit 1	Unit 2	
Region 1		2.10	2.10	1.40	1.60	1.40	2.10
Region 2		2.60	2.60	2.10	2.40	2.10	2.60
Region 3		3.10	3.10	2.90	3.10	2.90	3.10
Number of coolant loops	5.0	4	4		4		4
Total steam flow, 10^6 (lb/h)	5.1	15.14	15.14		15.14		15.14
Reactor vessel	5.3						
Inside diameter (in.)		173	173		173		173
Inlet nozzle inside diameter(in.)		27 1/2	27 1/2		27 1/2		27 1/2
Outlet nozzle inside diameter (in.)		29	29		29		29
Number of reactor closure head studs		54	54		54		54

TABLE 1.3.1-1 (SHEET 3 OF 5) (HISTORICAL)

Parameter or Feature	VEGP-FSAR Chapter/Section	VEGP[(c)] (Original Design)	SNUPPS	Comanche Peak	Trojan
Reactor coolant pumps	5.4.1				
Horsepower		7000	7000	7000	6000
Capacity (gal/min)		100,600	100,600	99,000	88,500
Steam generators	5.4.2				
Model		F	F	D	51
Heat transfer area (ft^2)		55,000	55,000	48,000	51,500
Number of U-tubes		5626	5626	4578	3388
Residual heat removal	5.4.7				
Initiation pressure (psig)		≈425	≈425	≈425	≈400
Initiation/completion temperature (°F)		≈350/140	≈350/140	≈350/140	≈350/140
Component cooling water design temperature (°F)		105	105	105	95
Cooldown time after initiation (h)		≈16	≈16	≈16	≈16
Heat exchanger removal capacity, 10^6 (Btu/h)		32.8	39.0	39.1	34.2
Pressurizer	5.4.10				
Heatup rate using heaters (F°h)		55	55	55	55
Internal volume (ft^3)		1800	1800	1800	
Pressurizer safety valves	5.4.13				
Number		3	3	3	
Maximum relieving capacity (lb/h)		420,000	420,000	420,000	

TABLE 1.3.1-1 (SHEET 4 OF 5) (HISTORICAL)

Parameter or Feature	*VEGP-FSAR Chapter/Section*	*VEGP[(c)] (Original Design)*	*SNUPPS*	*Comanche Peak*	*Trojan*
Accumulators	*6.3*				
Number		*4*	*4*	*4*	*4*
Operating pressure, minimum (psig)		*600*	*600*	*600*	*600*
Minimum operating water volume, each(ft^3)		*950*	*950*	*950*	*870*
Centrifugal charging pumps	*6.3*				
Number		*2*	*2*	*2*	*2*
Design flow (gal/min)		*150*	*150*	*150*	*150*
Design head (ft)		*5800*	*5800*	*5800*	*5800*
Safety injection pumps	*6.3*				
Number		*2*	*2*	*2*	*2*
Design flow (gal/min)		*425*	*440*	*425*	*425*
Design head (ft)		*2680*	*2780*	*2680*	*2500*
Residual heat removal pumps	*5.4.7, 6.3*				
Number		*2*	*2*	*2*	*2*
Design flow (gal/min)		*3700*	*3800*	*3800*	*3000*
Design head (ft)		*375*	*350*	*350*	*375*
Instrumentation and controls	*7.0*	*(b)*	*(b)*	*(b)*	*(b)*
New fuel storage racks center-to-center spacing (in.)	*9.1.1*	*21*	*21*	*21*	*21*

TABLE 1.3.1-1 (SHEET 5 OF 5) (HISTORICAL)

Parameter or Feature	*VEGP-FSAR Chapter/Section*	*VEGP[c] (Original Design)*	*SNUPPS*	*Comanche Peak*	*Trojan*
Chemical and volume control	*9.3.4*				
Total seal water supply flowrate, nominal (gal/min)		*32*	*32*	*32*	*32*
Total seal water return flowrate, nominal (gal/min)		*12*	*12*	*12*	*12*
Letdown flow normal/ maximum (gal/min)		*75/120*	*75/120*	*75/120*	*75/120*
Charging flow, normal/ maximum (gal/min)		*55/100*	*55/100*	*55/100*	*55/100*

a. The Comanche Peak FSAR presents the absorber material as Ag-In-Cd or Ag-In-Cd and B_4C. The second entry applies to Ag-In-Cd and B_4C.

b. The instrumentation and control systems discussed in chapter 7 of VEGP are functionally similar to those systems implemented in SNUPPS, Comanche Peak, and Trojan.

c. The VEGP power uprate increases the licensed reactor core power level from 3411 MWt to 3565 Mwt.

TABLE 1.3.1-2 (SHEET 1 OF 5) (HISTORICAL)

DESIGN COMPARISON OF MAJOR BALANCE OF PLANT DESIGN FEATURES

Item	*Vogtle Units 1 and 2 (FSAR) (Original Design)*	*San Onofre Units 2 and 3 (FSAR)*	*Farley Units 1 and 2 (FSAR)*	*Calvert Cliffs Units 1 and 2 (FSAR)*
Containment System Parameters				
Type	*Steel-lined, prestressed, post-tensioned, concrete cylinder, hemispherical dome roof*	*Steel-lined, prestressed, post-tensioned, concrete cylinder, hemispherical dome roof*	*Steel-lined, prestressed, post-tensioned, concrete cylinder, curved dome roof*	*Steel-lined, prestressed, post-tensioned, concrete cylinder, curved dome roof*
Design parameters				
Inside diameter (ft)	*140*	*150*	*130*	*130*
Inside height (ft)	*228*	*172*	*183*	*182*
Nominal free volume (ft^3)	*2,750,000*	*2,335,000*	*2,024,900*	*2,000,000*
Design pressure(psig)	*52*	*60*	*54*	*50*
Concrete thickness (ft)				
Vertical wall	*3 3/4*	*4 1/3*	*3 3/4*	*3 3/4*
Dome	*3 3/4*	*3 3/4*	*3 1/4*	*3 1/4*
Containment leak prevention and mitigation systems	*Leaktight penetrations and continuous steel liner. Automatic isolation where required. The exhaust from penetration rooms to vent stack.*	*Leaktight penetrations and continuous steel liner. Automatic isolation where required. The exhaust from penetration rooms to vent stack.*	*Leaktight penetrations and continuous steel liner. Automatic isolation where required. The exhaust from penetration room to vent stack.*	*Leaktight penetrations and continuous steel liner. Automatic isolation where required. The exhaust from penetration room to vent stack.*
Gaseous effluent purge stack	*Discharge through stack*	*Discharge through stack*	*Discharge through stack*	*Discharge through stack*
Engineered Safety Features				
Safety injection system				
High head pumps	*2*	*3*	*3*	*3*
Low head pumps	*2*	*2*	*2*	*2*
Accumulator tanks	*4*	*4*	*3*	*4*

TABLE 1.3.1-2 (SHEET 2 OF 5) (HISTORICAL)

Item	*Vogtle Units 1 and 2 (FSAR) (Original Design)*	*San Onofre Units 2 and 3 (FSAR)*	*Farley Units 1 and 2 (FSAR)*	*Calvert Cliffs Units 1 and 2 (FSAR)*
Containment fan coolers				
No. of units	*8*	*4*	*4*	*4*
Airflow capacity, each at emergency conditions (ft^3/min)	*43,500*	*31,000*	*60,000*	*60,000*
Auxiliary feedwater system				
No. of pumps	*2 electric-driven* *1 turbine-driven*	*1 electric-driven* *1 turbine-driven*	*2 electric-driven* *1 turbine-driven*	*2 turbine-driven*
Initiation	*Automatic*	*Automatic*	*Automatic*	*Manual*
Condensate storage tanks	*2*	*2*	*1*	*1*
Capacity, each (gal)	*480,000* *500,000 (Cat. 2)*	*480,000*	*150,000 (Cat. 1)*	*500,000; 350,000*
Post-accident filters	*Piping penetration area, control room, fuel handling bldg.*	*Fuel handling bldg.*	*Penetration room*	*Penetration room*
Containment spray				
No. of pumps	*2*	*2*	*2*	*2*
Spray additive	*NaOH*	*NaOH*	*NaOH*	*None*
Combustible gas control	*Electric H; recombiners inside containment; post-accident manual vent*	*Electric H; recombiners inside containment; post-accident manual vent*	*Electric H; recombiners inside containment; post-accident manual vent*	*Electric H; recombiners inside containment; post-accident manual vent*
Electrical Components				
Standby power system	*Total of 4 diesels; supply each unit. Diesels are connected to 4160-V buses. No capability for sharing.*	*Total of 4 diesels; 2 supply each unit. Diesels are connected to 4160-V buses. No capability for sharing.*	*Total of 5 diesels; 3 are shared between Units 1 and 2. Diesels are connected to 4160-V buses.*	*3 diesels connected to 4-kV buses and shared between Units 1 and 2.*
Engineered safety feature buses	*Two 4160-V buses/units divided into 2 separate and redundant systems*	*Two 4160-V buses/units divided into 2 separate and redundant systems*	*Six 4160-V buses/units divided into 2 separate and redundant systems*	*Two 4-kV buses/units divided into separate and redundant systems*

TABLE 1.3.1-2 (SHEET 3 OF 5) (HISTORICAL)

Item	*Vogtle Units 1 and 2 (FSAR) (Original Design)*	*San Onofre Units 2 and 3 (FSAR)*	*Farley Units 1 and 2 (FSAR)*	*Calvert Cliffs Units 1 and 2 (FSAR)*
dc systems	*4 separate and independent 125-V dc subsystems for each unit provide reliable power for safety-related dc control, instrumentation, and motor loads. A separate 125-V dc system serves nonsafety-related loads.*	*Separate and redundant 125-V dc systems for ESF loads. Separate 125-V dc and 250-V dc systems for non-ESF loads.*	*Separate and redundant 125-V dc systems for ESF loads. Separate dc systems for loads in auxiliary building, turbine building, cooling tower area, diesel generator building and switchyard.*	*4 batteries between 2 units divided to give 2 separate and redundant 125-V dc systems. Separate dc systems for turbine building and the switchyard.*
Vital instrumentation systems	*6 inverters arranged to give 4 separate and redundant channels; separate systems serve nonsafety-related loads*	*4 inverters arranged to give 4 separate and redundant channels*	*4 inverters arranged to give 4 separate and redundant channels*	*4 inverters between 2 units to give 4 separate and redundant channels per unit*
Offsite power system	*Units 1 and 2 connected to the 230-kV switchyard. Each unit has 2 unit auxiliary and 2 reserve auxiliary transformers. The ESF buses are normally supplied from reserve aux. transformers.*	*One 230-kV switchyard common to Units 2 and 3. Each unit is provided with 2 unit auxiliary and 3 startup transformers supplied from the common switchyard.*	*Unit 1, 230-kV switchyard. Unit 2, 500-kV switchyard. Each unit has 2 startup transformers and 2 unit auxiliary transformers with the ESF buses supplied from startup transformers.*	*500-kV switchyard. 2 startup transformers shared between 2 units*
Radioactive Waste Management System				
Liquid radwaste system				
Miscellaneous liquid waste system	*Shared: disposable waste subsystem Separate: resin handling subsystem recyclable waste subsystem*	*Shared*	*Shared*	*Shared*

TABLE 1.3.1-2 (SHEET 4 OF 5) (HISTORICAL)

Item		*Vogtle Units 1 and 2 (FSAR) (Original Design)*	*San Onofre Units 2 and 3 (FSAR)*	*Farley Units 1 and 2 (FSAR)*	*Calvert Cliffs Units 1 and 2 (FSAR)*
Discharge:	*Evaporator distillate (waste evaporator abandoned in place)*	*Waste evaporator condensate tank*	*Circulating water outfall*	*Circulating water outfall*	*Circulating water outfall*
	Evaporator bottoms (waste evaporator abandoned in place)	*Reactor makeup water storage tank (RMWST) solid radwaste system*	*Solid radwaste system*	*Solid radwaste system*	*Solid radwaste system*
Recycle capability		*Yes*	*Yes*	*Yes*	*Yes*
Total reprocessing storage capacity (holdup tanks)		*41,600 gal (both units)*	*1 at 6000 gal 2 at 25,000 gal*	*40,000 gal*	*8000 gal*
Filter type and backflushable		*Disposable cartridge and backflushable*	*Disposable cartridge*	*Disposable cartridge*	*Disposable cartridge*
Evaporator capacity (waste evaporator abandoned in place)		*15 gal/min*	*50 gal/min*	*35 gal/min 20 gal/min*	
Coolant and boric acid recycle system		*Shared BRS (part of CVCS for each unit)*	*Shared*	*Shared*	*Shared (reactor coolant waste processing system)*
	Concentrator bottoms	*Recycled to boric acid storage tank*	*No; recycled to boric acid makeup and batching tanks*	*Liquid radwaste system*	*Solid radwaste system*
Discharge:	*Concentrator condensate*	*Recycled to RMWST*	*No; recycled to CVCS discharge*	*No; recycled to CVCS*	*Circulating water*
Concentrator capacity		*15 gal/min*	*50 gal/min*	*30 gal/min*	*2 at 20 gal/min*
Concentrated boric acid storage tanks		*2 at 46,000 gal each*	*2 at 25,000 gal each*	*2 at 21,000 gal each*	*2 at 10,000 gal each*
Radwaste receiver tanks		*2 at 112,000 gal each*	*2 primary at 120,000 (recycle holdup tank)*	*3 at 28,000 gal each gal each and 2 secondary at 120,000 gal each*	*2 waste receiver tanks at 90,000 gal each*

TABLE 1.3.1-2 (SHEET 5 OF 5) (HISTORICAL)

Item	*Vogtle Units 1 and 2 (FSAR) (Original Design)*	*San Onofre Units 2 and 3 (FSAR)*	*Farley Units 1 and 2 (FSAR)*	*Calvert Cliffs Units 1 and 2 (FSAR)*
Waste gas system	*Separate system for each unit*	*Shared*	*Shared*	*Shared*
Number of decay tanks	*7*	*6, Seismic Cat. 1*	*8*	*3*
Tank size (each)	*600 ft^3*	*500 ft^3*	*600 ft^3*	*610 ft^3*
Design pressure	*150 psig*	*350 psig*	*150 psig*	*150 psig*
Discharge point	*Plant vent*	*Plant vent stack*	*Plant vent*	*Plant vent*
Holdup time available	*30 days*	*30 days (minimum)*	*30 days (minimum)*	*60 days*
Surge tank	*Yes, 2 shutdown/startup (shared)*	*Yes, Seismic Cat. 1*	*No*	*Yes*
Surge tank size	*600 ft^3 at 150 psig*	*500 ft^3 at 150 psig*	*-*	*610 ft^3 at 50 psig*
Compressor capacity	*2 per unit 40 sf^3/min*	*2 at 5 sf^3/min*	*2 at 40 sf^3/min*	*2 at 4.7 sf^3/min*
Radwaste solidification system	*Portable vendor supplied-shared*	*Shared*	*Shared*	*Shared*
Solidification agent	*-*	*(a)*	*Vermiculite - cement*	*(a)*
Onsite storage:				
Intermediate level	*High integrity containers*	*20 50-ft^3 drums[(a)]*	*175 55-gal drums*	*(a)*
Low level		*25 55-gal drums[(a)]*	*400 55-gal drums*	*(a)*
Shipping containers used	*High integrity containers*	*55-gal drums and 50-ft^3 drums[(a)]*	*55-gal drums*	*(a)*

a. System presently not being used.

TABLE 1.3.2-1 (SHEET 1 OF 6)

SIGNIFICANT DESIGN CHANGES FROM THE PSAR

Item	FSAR Chapter/Section	Reason for Change
Deletion of two units from license application	1.1	Lower than expected power demand growth and continued construction of four units monetarily prohibitive
Deletion of reactor loop stop valves	1.2, 5.1	No plans to operate with an isolated loop
In the area north of the turbine building, the backfill is compacted to an average of 95 percent of the maximum dry density	2.5	The tested static and dynamic properties of the 95-percent backfill meet the design static and dynamic properties of the 97-percent backfill
Nonsafety-related chemical or membrane waterproofing utilized on exterior surfaces of Category 1 structures. No waterproofing provided for interior surfaces.	3.4	The thick concrete walls of Category 1 structures obviates the need for safety-related membranes on the exterior building surfaces and any type of waterproofing on interior surfaces, e.g., nuclear service cooling water (NSCW) basins
Revision of tornado missile spectrum to include missiles C and F of Standard Review Plan (SRP) 3.5.1.4 and corresponding velocities	3.5	To comply with the criteria provided in SRP 3.5.1.4 (11/24/75)
Part of the roof and related walls of level 4 of the control building are not designed to resist tornado missiles. The level 4 floor slab is missile resistant and openings in the slab are provided with local missile protection.	3.5.1.4	There is no safety-related equipment located in the affected area of level 4 of the control building
Revision of pipe break location criteria and analysis method	3.6	Overall update of pipe break criteria to meet current requirements
Unaffected Category 2 equipment is used to mitigate consequences of high energy line breaks in seismically supported lines located outside of the containment, when the radiological consequences are insignificant in comparison to 10 CFR 100 dose guidelines.	3.6	This approach is consistent with APCSB 3-1

TABLE 1.3.2-1 (SHEET 2 OF 6)

Item	FSAR Chapter/Section	Reason for Change
The Summer 1979 Addenda to the American Society of Mechanical Engineers (ASME) Code, Section III, Division 1, Subsection NB, subarticles NB-3650 and NB-3680, have been utilized in the Class 1 piping analysis	3.6, 3.9	The addenda makes the piping stress analysis more consistent with the method of vessel analysis
Damping values as high as 15 percent have been used in the design of cable tray supports for both the safe shutdown earthquake (SSE) and operating basis earthquake (OBE) condition	3.7.1	Documented test data supports the use of these damping values
The equipment building dimensions are such that equipment is properly enclosed and height relationships with other buildings are maintained	3.8.4	The equipment building is being constructed as an extension of the fuel handling and control buildings.
Deletion of part length control rods incorporation of T_{avg} 9.5 control D bank changes from nine to five rods	4.0, 7.7	To provide improved power distribution control during load follow operations
Baffle-to-barrel region configuration has been changed from downflow to upflow	5.3, 15.6.5	To reduce baffle plate and baffle slot loading and to minimize the potential for excessive baffle joint jetting
Steam generators changed from model D to model F	5.4	To increase reliability of the steam generators
Added a check valve in series with encapsulated, residual heat removal (RHR) containment sump, motor-operated isolation valve	5.4	Prevents draining of the refueling water storage tank (RWST) to containment sump via RHR sump suction lines
The number of containment building air cooling units has been increased to four units per train	6.2	Increased reliability of air cooling system
Steam generator subcompartment differential pressure analysis performed utilizing restrained breakflow area mass and energy release data	6.2.1.2	Greater than expected reduction in vent area necessitated more realistic calculational assumption

TABLE 1.3.2-1 (SHEET 3 OF 6)

Item	FSAR Chapter/Section	Reason for Change
Through design evolution, the closure times for some containment isolation valves have been changed from those presented in the PSAR	6.2.4	The valve closure times conform to Regulatory Guide 1.141 and American National Standards Institute (ANSI) N271-1976
Incorporation of semiautomatic switch-over from injection to recirculation mode of emergency core cooling system (ECCS) operation	6.3	To reduce number of operator actions required during ECCS operation
Revision of containment emergency sump design	6.3	To meet requirements of Regulatory Guide 1.82
Increased volume of RWST to 715,000 gal	6.3	Consideration of instrument accuracy and single most limiting failure
Alteration of control room heating, ventilation, and air-conditioning (HVAC) systems to provide four 100-percent redundant engineered safety features (ESF) trains and two 100-percent normal units	6.4	To increase system reliability
Deletion of reactor trip following turbine trip below 50-percent power	7.2	To increase plant availability
Elimination of low feedwater flow reactor trip	7.2, 15.0	To increase plant availability
An improved steam line break protection system has been incorporated	7.2, 15.0	To improve plant availability by preventing spurious safety injection actuation
Modification of fluid systems and shutdown panels to allow safety-grade cold shutdown from control room and safe shutdown from outside the control room	7.4	To meet the requirements of General Design Criterion (GDC) 19, Appendix R and Regulatory Guide 1.139
Revision of post-accident monitoring parameters	7.5	To meet new regulatory requirements
Automatic feedwater control at low power has been incorporated	7.7	To provide the capability of automatically monitoring steam generator water level from 0- to 25-percent power
Revised Class 1E dc system to provide four 125-V dc batteries	8.3	Increases system reliability

TABLE 1.3.2-1 (SHEET 4 OF 6)

Item	FSAR Chapter/Section	Reason for Change
Battery sizing revised to provide power at 55°F for 2 3/4 h	8.3	Increases system reliability
Revision to cable, tray, and raceway designs	8.3	Increases system reliability
Increased spent fuel storage capacity from 1 1/3 cores to 4 2/3 cores	9.1	Increases onsite storage capacity of spent fuel
Decrease in spent fuel pool accident temperature from 212°F to 180°F	9.1	Provide margin to boiling of pool
Integrated head, missile, shield, and control rod drive mechanism (CRDM) cooling system have been adopted.	9.1.4	Reduces man-hours of effort required to remove and replace reactor vessel head with subsequent reduction of man-rem radiation exposures
Improved pressure vessel head closure system.	9.1.4	Reduces man-hours of effort required to remove and replace reactor vessel head with subsequent reduction of man-rem radiation exposures
The ultimate heat sink design based upon two-train operation during first 24 h and one-train operation for subsequent 29 days	9.2	Changed mode of operation to ensure 30 days of ultimate heat sink operation without external makeup
The auxiliary component cooling water (ACCW) system is classified as nonsafety related although components are procured in accordance with the ASME code.	9.2.8	The ACCW system cools various nonsafety-related loads, none of which are required for safe shutdown or accident mitigation
The principal design codes and standards for the liquid portions of the boron recycle system have been revised.	9.3.4	This portion of the system serves no safety function
The number of plant vents has been reduced and classified as nonsafety related. Ducting in equipment building associated with safety-related systems is Category 1.	9.4	The functioning of the plant vent and ducting in the equipment building is not required for safe shutdown or accident mitigation
The carbon dioxide fire suppression system has been deleted.	9.5.1	The water fire suppression systems fulfill the suppression system requirements of Appendix A to APCSB 9.5-1

TABLE 1.3.2-1 (SHEET 5 OF 6)

Item	FSAR Chapter/Section	Reason for Change
The principal design codes and standards for the diesel generator fuel oil storage tank flame arrestors have been revised	9.5.4	The flame arrestors serve no safety function
The principal design codes and standards for the diesel generator starting air compressor and air dryers have been revised	9.5.8	The starting system air compressor and air dryers serve no safety function
Sizing of steam generator atmosphere relief valves increased from 10 percent to 15 percent	10.3	Increased load shed capabilities and plant reliability
Incorporated second main steam isolation valve (MSIV) into each main steam line	10.3	Ensures isolation of steam generator in the event of a main steam line break (MSLB) downstream of isolation valves
Deleted phosphate injection system and incorporated all volatile chemical treatment for steam generators	10.3	To decrease corrosion of steam generators
Incorporated low T_{avg}, high-high steam generator level, and reactor trip signals for initiation of feedwater isolation	10.4	Increases responsiveness of feedwater isolation to abnormal conditions
Revision of auxiliary feedwater system to consist of two 100-percent electric-driven pumps and one 200-percent steam pump where control power for steam-driven pump and associated valves is provided from the station batteries	10.4.9	Increases reliability of automatic initiation and subsequent operation of the auxiliary feedwater system
The principal design codes and standards for the liquid radwaste have been revised	11.2	To conform to Regulatory Guide 1.143
Revision of solid radwaste handling system	11.4	Upgraded system to meet regulatory requirements
Revise reactor makeup water system to provide a seismic category and flow driving path from the reactor makeup water storage tank (RMWST) to the spent fuel pool and other miscellaneous safety-related surge tanks	9.2.7	Provide Category 1 backup source of water for spent fuel pool. Gravity provides head from RMWST to spent fuel pool

TABLE 1.3.2-1 (SHEET 6 OF 6)

Item	FSAR Chapter/Section	Reason for Change
Deleted position switches on drain line isolation valves and incorporated locked closed valves	9.3.3	Consideration of locked closed valves and administrative control insures their proper positioning
The applicable piping codes or standards have been used for the hydrostatic testing of embedded piping in lieu of American Concrete Institute (ACI) 318-71	3.8	To consistently test piping in accordance with the piping codes or standards
Revised the auxiliary feedwater lines to feed directly into each steam generator	10.4.9	To reduce or eliminate the potential for cracking problems
Reduced the volume of the diesel fuel oil storage tanks	9.5.4	To provide diesel fuel oil supply consistent with the quantity required for design basis accident safety-related loads
Added an alternate centrifugal charging pump miniflow system	6.3	To prevent pump deadheading should reactor coolant system (RCS) pressure rise following isolation of the normal miniflow lines
Addition of AMSAC	7.7/15.8	The anticipated transient without scram (ATWS) mitigation system actuation circuitry (AMSAC) provides a backup to the reactor trip system (RTS) and ESF actuation system (ESFAS) for initiating turbine trip and auxiliary feedwater, and isolating the steam generator blowdown and sample lines.

TABLE 1.3.2-2 (SHEET 1 OF 3)

TMI ACTION PLAN TASKS AND LOCATION OF FSAR DESCRIPTION

Action Plan Task	Title	FSAR Section
I.A.1.1	Shift Technical Advisor	13.1.2
I.A.1.2	Shift Supervisor Administrative Duties	13.5.1
I.A.1.3	Shift Manning	13.5.1
I.A.2.1	Reactor Operator and Senior Reactor Operator Training and Qualifications	13.2.1/13.1.2 13.1.3/13.2.2
I.A.2.3	Administration of Training Programs	13.2.1/13.1.3 13.2.2
I.A.3.1	Scope and Criteria for License Exams	13.2.1
I.B.1.2	Independent Safety Engineering Group	13.4.3/17.2.1
I.C.1	Evaluation and Development of Procedures for Transients and Accidents	13.5.1/13.5.2 1.9.33
I.C.2	Shift and Relief Turnover Procedures	13.5.1
I.C.3	Shift Supervisor Responsibilities	13.5.1
I.C.4	Control Room Access	13.5.1
I.C.5	Feedback of Operating Experience	13.5.1
I.C.6	Verify Correct Performance of Operating Activities	13.5.1/13.5.2
I.C.7	Nuclear Steam Supply System (NSSS) Vendor Review of Procedures	13.5
I.D.1	Control Room Design Review	18.1.1
I.D.2	Safety Parameter Display System (SPDS)	7.5.4/18.3.5
I.G.1	Training During Low Power Testing	14.2.5
II.B.1	RCS Vents	5.1.1/5.4.15
II.B.2	Plant Shielding	12.2.1/12.3.1
II.B.3	Post-Accident Sampling	9.3.2/7.5.2
II.B.4	Training for Mitigating Core Damage	13.2.2
II.D.1	Relief and Safety Valve Test Requirements	5.4.13
II.D.3	Valve Position Indication	5.4.13
II.E.1.1	Auxiliary Feedwater System Reliability Evalua	10.4.9/10A.1
II.E.1.2	Auxiliary Feedwater System Initiation and Flow	10.4.9/7.5.4
II.E.3.1	Emergency Power for Pressurizer Heaters	5.4.10
II.E.4.1	Containment Dedicated Penetrations	NA
II.E.4.2	Containment Isolation Dependability	6.2.4

TABLE 1.3.2-2 (SHEET 2 OF 3)

Action Plan Task	Title	FSAR Section
II.F.1	Additional Accident Monitoring Instrumentation	7.5.4/6.2.5
II.F.2	Inadequate Core Cooling Instruments	7.5.4/7.7.2 Appendix 4A
II.G.1	Emergency Power for Pressurizer Equipment	5.4.11
II.K.1	IE Bulletins	
	5. Review ESF Valves	6.3.1/7.5.4
	10. Operability Status	13.5.1
	17. Trip per Low Level B/S	5.4.10
II.K.2.13	Thermal Mechanical Report	5.3.3
II.K.2.17	Voiding in RCS	15.0.1
II.K.2.19	Benchmark Analysis Seq. AFW Flow	
II.K.3	Final Recommendation of B & O Task Force	
	1. Auto Power-Operated Relief Valves (PORVs) Isolation	5.4.11
	2. Report on PORV Failures	NA
	3. Reporting Relief Valve and Safety Valve Failures and Challenges	5.4.1
	5. Auto Trip of Reactor Coolant Pumps	5.4.1
	9. PID Controller	PLS[(a)]/5.4.10
	10. Anticipatory Trip at High Power	7.2.1
	12. Confirm Anticipatory Trip	7.2.1
	17. ECCS Outages	13.5.1
	25. Power to Pump Seals	5.4.1
	30. Small Break Loss-of-Coolant Accident (LOCA) Methods	15.0.1
	31. Plant Specific Analysis	15.0.1
III.A.1.1	Upgrade Emergency Preparedness	13.3
III.A.1.2	Upgrade Emergency Support Facilities	9.5.10
III.A.2	Upgrade Emergency Plans, App. E, 10 CFR 50	13.3
	Meteorological Data	2.3.3

TABLE 1.3.2-2 (SHEET 3 OF 3)

Action Plan Task	Title	FSAR Section
III.D.1.1	Primary Coolant Sources Outside Containment	12.1.3/9.3.4
III.D.3.3	Inplant Radiation Monitoring	12.5.2
III.D.3.4	Control Room Habitability	6.4

a. The precautions, limitations, and setpoints document states that derivative control shall be set at zero on the pressure PID controller.

1.4 IDENTIFICATION OF AGENTS AND CONTRACTORS

1.4.1 APPLICANT - CONSTRUCTION MANAGER AND OPERATOR

Georgia Power (GPC), a co-owner of the Vogtle Electric Generating Plant (VEGP), was responsible for the design, construction, and operation of the plant through March 21, 1997. Since March 22, 1997, as the exclusive operating licensee, Southern Nuclear Operating Company, Inc. (SNC) is responsible for the planning, design, licensing, operation, maintenance, repair, modification, addition, renewal, retirement, and decommissioning of VEGP pursuant to a nuclear operating agreement between SNC and GPC. GPC will act as agent for the co-owners identified as follows:

A. Oglethorpe Power Corporation - An electric membership corporation organized and operating under Title 34B of the Georgia Code Annotated.

B. Municipal Electric Authority of Georgia - A public body corporate and politic, being an instrumentality of the State of Georgia.

C. City of Dalton, Georgia - An incorporated municipality in the State of Georgia acting by and through its Board of Water, Light, and Sinking Fund Commissioners.

1.4.1.1 Description of Business

GPC is a public utility engaged in the generation, distribution, and sale of electricity at retail in 646 cities and communities and at wholesale to 50 municipalities and 39 U.S. Rural Electrification Administration cooperatives. GPC's electric generating facilities consist of 11 steam-electric generating plants, 18 hydroelectric generating plants, and 7 generating plants with gas turbines having peak hour capability of approximately 13,000 MW as of January 1, 1982. Approximately 87.5 percent of GPC's total capacity is provided by steam-electric installations. The steam-electric generating plants are listed below:

- Plant Arkwright near Macon.
- Plant Atkinson near Atlanta.
- Plant Bowen near Cartersville.
- Plant Harlee Branch near Milledgeville.
- Plant Hammond near Rome.
- Plant McDonough near Atlanta.
- Plant McManus near Brunswick.
- Plant Mitchell near Albany.
- Plant Yates near Newnan.

- Plant Wansley near Carrolton.
- Plant Hatch near Baxley.

GPC has constructed 38 thermal units (not including combustion turbines) and is presently constructing six additional thermal units which include Plant Scherer Units 1 through 4, near Macon, and VEGP Units 1 and 2.

A brief description of the utility electric grid is given in section 8.1.

1.4.1.2 Description of Corporate Organization

The GPC is a public utility incorporated under the Laws of the State of Georgia with its principal offices located in Atlanta, Georgia. SNC performs nuclear operating services for GPC with respect to the VEGP, and SNC's principal offices are located in Birmingham, Alabama. GPC and SNC are wholly-owned subsidiaries of Southern Company.

1.4.1.3 Technical Qualifications

GPC has participated in the development of the nuclear power industry since 1953. Early activities included financial contributions to the Dow Chemical-Detroit Edison study group. GPC's continuing involvement in industry development has included charter memberships in the Institute of Nuclear Power Operations founded in 1980 and the Electric Power Research Institute founded in 1973. Since March 22, 1997, as the exclusive operating licensee, SNC has replaced GPC in the nuclear related aspects of these organizations.

In September 1969, GPC received a provisional construction permit for its first nuclear unit, the Edwin I. Hatch Nuclear Plant. On August 6, 1974, and June 13, 1978, GPC received operating licenses (Atomic Energy Commission Docket Nos. 50-321 and 50-366) for Edwin I. Hatch Units 1 and 2, respectively. SNC's personnel are presently operating these facilities.

In addition to experience at Plant Hatch, many GPC staff members have undergone training and observation assignments at various nuclear power plants. An aggressive training program, including a 22-week operational technology course, is being pursued to further improve the technical competency of GPC personnel. The operations technology course consists of 16 weeks of classroom lecture and plant walkthroughs followed by a 6-week simulator course. Since March 22, 1997, GPC's nuclear staff has been transferred to SNC as the exclusive operating licensee.

Georgia Power Company has obtained considerable knowledge of, and has contributed to, the development of nuclear power through its participation and ownership of the above projects and activities. This knowledge has now been transferred to SNC. The technical qualifications of SNC are further delineated in section 13.1, Organizational Structure of SNC.

1.4.2 SOUTHERN NUCLEAR OPERATING COMPANY, INC.

SNC, a wholly-owned subsidiary of Southern Company, is the exclusive operating licensee of VEGP and is responsible to GPC for the operation of VEGP pursuant to a nuclear operating agreement.

SNC was initially formed from support organizations of Southern Company Services, Georgia Power Company, and Alabama Power Company, whose individuals have many years of experience in the area of nuclear operations. On March 22, 1997, the remainder of GPC's nuclear personnel were transferred to SNC as the exclusive operating licensee for VEGP and Edwin I. Hatch Nuclear Plant (HNP). SNC is currently responsible for the operation of the six nuclear units within Southern Company located at Joseph M. Farley Nuclear Plant, HNP, and VEGP. Pursuant to the nuclear operating agreement, SNC is responsible for nuclear operating services, including fuel services, new investment services, and operation and maintenance services.

Fuel services include work relating to supplying and managing the nuclear fuel for VEGP including, but not limited to, planning, procurement, contract administration, fuel cycle design, fuel core and assembly design, fuel quality assurance, nuclear materials management, and all activities relating to procurement, conversion, enrichment, fabrication, transportation, installation, monitoring, repairing, storage, reprocessing and disposal of uranium, nuclear fuel, related materials and waste products.

New investment services include work undertaken with respect to the planning, design, licensing, acquisition, construction, completion, renewal, improvement, addition, repair, replacement, enlargement, or modification of VEGP.

Operation and maintenance services include work for the VEGP co-owners relating to the possession, management, control, startup, operation, availability, production of energy, maintenance, modification, shutdown, retirements, and decommissioning, including, but not limited to, any planning, design, engineering, labor, procurement of materials and supplies, materials management, quality assurance, training, security, environmental protection, and handling of any source material, special nuclear material or by-product material together with maintaining or obtaining licenses and regulatory approvals related thereto, governmental affairs or regulatory relationships, and all other activity that is not included in or performed as new investment services or fuel services, but which is required for the operation and maintenance of VEGP or that may be required to comply with legal requirements.

SNC is also responsible for the following coordination of the licensing activity:

A. Assurance, through quality assurance audits, of the proper implementation and compliance of the quality program.

B. Assurance of the proper implementation and execution of the supplier inspection program.

SNC will also participate in engineering and design activities of certain systems and structures.

1.4.3 BECHTEL POWER CORPORATION

The Bechtel Corporation, the parent of Bechtel Power Corporation, has been continuously engaged in construction and engineering activities since 1898. Since the close of World War II, Bechtel has placed strong emphasis on electric power generation projects. During this period, Bechtel has been responsible for the design of over 204 thermal generating units, representing more than 126,860 MW of new generating capacity. Of this number, a nuclear capacity of more than 65,800 MW has been engineered by the company itself.

The ratings of thermal generating plants designed by Bechtel range to 1470 MW per unit and include most types of station designs and arrangements, such as reheat and nonreheat, indoor and outdoor stations, single and multiple units, and wide ranges of steam conditions up to 3500 psig, 1050°F. Also, some of the larger units were fully automated and computer controlled. The majority of contracts for these facilities provide Bechtel with complete responsibility for both engineering and construction, although several contracts have been engineering design assignments only.

The Los Angeles Regional Office of Bechtel Power Corporation (Bechtel) was retained by GPC to provide A/E services during the design and construction phase, including procurement, for the balance of plant portions of the nuclear electric generating facilities and administration of the nuclear steam supply system (NSSS) contract.

The Gaithersburg Regional Office of Bechtel Power Corporation (Bechtel) is under contract to SNC to provide A/E services during the operational phase for structures, systems, and components assigned by SNC.

1.4.3.1 Nuclear Experience

For over 25 years Bechtel Power Corporation has been actively working on nuclear projects involving power plants, as well as such facilities as nuclear accelerators, research laboratories, hot cells, experimental reactors, and nuclear fuel processing plants. Its responsibilities have covered design, construction, site surveys, license applications, feasibility studies, and equipment procurement.

1.4.4 NUCLEAR STEAM SUPPLY SYSTEM MANUFACTURER

Westinghouse Electric Corporation (Westinghouse) was responsible for supplying the NSSS and first fuel load for each VEGP unit.

Westinghouse has designed, developed, and manufactured nuclear facilities since the 1950s, beginning with the world's first large central station nuclear plant (Shippingport), which has produced power since 1957.

More than 100 commercial nuclear power plants with a combined electrical generating capacity in excess of 90,000 MW have been completed, including the two VEGP units.

Westinghouse pioneered new nuclear design concepts, such as chemical shim control of reactivity and the rod cluster control concept, throughout the last two decades. Among the company's own related manufacturing facilities are the commercial nuclear fuel fabrication

facility at Columbia, South Carolina, and nuclear component manufacturing facilities at Tampa, Florida; Pensacola, Florida; Blairsville, Pennsylvania; and Cheswick, Pennsylvania.

1.4.5 DIVISION OF RESPONSIBILITY

1.4.5.1 Design Stage

Westinghouse Electric Corporation and Bechtel Power Corporation have been delegated the responsibility for design of the NSSS and the balance of the plant, respectively. For preparation of the Final Safety Analysis Report, all parties (GPC, SCS, Bechtel, and Westinghouse) have participated in the preparation and review of design bases and philosophies of both systems and structures by a review of the plant design.

1.4.5.2 Utility Company

The ultimate responsibility for the proper design, construction, and operation for the entire spectrum of safety of each unit rests with the applicants. Since March 22, 1997, SNC is the exclusive operating licensee of VEGP.

1.4.5.3 Architect/Engineer

The A/E is responsible for the design, engineering, and procurement of the standard power block, which includes the following:

- Turbine building.
- Containment building.
- Auxiliary building.
- Fuel handling building.
- Radwaste buildings.
- Diesel generator building.
- Control building.
- Auxiliary feedwater pumphouse.
- Nuclear service cooling water towers.
- Tanks.

All systems, equipment, and structures within the standard power block are designed or specified by the A/E. The NSSS portion of the facility was procured by individual contract between GPC and the NSSS supplier. Similarly, the turbine-generator was obtained by direct

contract between the turbine-generator supplier and GPC. However, the A/E (acting as representative) retains responsibility for monitoring the design and integrating the systems into the power block to ensure that the NSSS and turbine-generator components being supplied are consistent with the needs of the facility. Other equipment and material for areas within their scope are procured by the A/E.

1.4.5.4 Procurement of Safety-Related Equipment

1.4.5.4.1 Westinghouse Scope of Supply

Westinghouse is responsible for the manufacture and/or procurement of all items within the Westinghouse scope of supply.

1.4.5.4.2 Bechtel Scope of Supply

For the safety-related equipment under the Bechtel scope of supply, procurement procedures have been established to require participation by both SNC, on behalf of GPC, and Bechtel. Bechtel prepares the inquiries and transmits them to the SNC for review to ascertain that sufficient information is contained therein to inform the bidders of all requirements for the supplied equipment, including, but not limited to, materials, documentation, and shipping requirements. From this point, Bechtel has the responsibility of sending the inquiry out for bids in accordance with an established bidders list. After review of the bids, preparation of the requisition by Bechtel, and approval by SNC, the purchase order is prepared by SNC.

1.4.5.5 Construction

As of March 22, 1997, all construction activities at the site are under the management of SNC with independent testing agencies being contracted as necessary to perform special testing and to provide expertise in the interpretation of results.

1.4.5.6 Operation

During the design, construction, and testing phases, lines of communication were established between the plant site, GPC's general office, Southern Company Services, Inc. (SCS), and various vendors and consultants. These lines are maintained and others were established as necessary during operation to enable the plant staff to receive technical support from the most authoritative source. Since March 22, 1997, lines of communication have been established between the plant site, SNC's general office, SCS, and various vendors and consultants as a result of SNC becoming the exclusive operating licensee. As a result of the consolidation of SCS and SNC nuclear expertise, and in addition to being the licensee, SNC also serves as its own A/E and performs the functions previously performed by SCS.

1.5 REQUIREMENTS FOR FURTHER TECHNICAL INFORMATION

In accordance with 10 CFR 50.34(b)(5), this section describes and provides references to the evaluations of programs, including research and development, if any, to demonstrate that any safety questions identified at the construction permit stage have been resolved. Westinghouse has conducted and completed the research and development programs relating to the requirements of this section. The core stability evaluation program is considered necessary to establish design adequacy. The other programs addressed below are considered as demonstrating the margin of conservatism of the design. The programs that apply to the VEGP are listed below.

1.5.1 CORE STABILITY EVALUATION PROGRAM

The core stability evaluation program is complete. Refer to section 1 of WCAP-8768 for a discussion of this program.

1.5.2 FUEL DEVELOPMENT PROGRAM FOR OPERATION AT HIGH-POWER DENSITIES

The fuel development program for operation at high-power densities is complete. Refer to section 8 of WCAP-8485 for a discussion of this program.

1.5.3 INCORE DETECTOR PROGRAM

The incore detector program is complete. Refer to section 9 of WCAP-8353 for a discussion of this program.

1.5.4 FULL-LENGTH EMERGENCY CORE COOLING HEAT TRANSFER TEST

The full-length emergency core cooling heat transfer test is complete. See section 12 of WCAP-8768, Revision 2, for a discussion of this program.

1.5.5 BLOWDOWN FORCES PROGRAM

The blowdown forces program is complete. See section 15 of WCAP-8004 for a discussion of this program.

1.5.6 REACTOR VESSEL THERMAL SHOCK PROGRAM

The reactor vessel thermal shock program is complete. See section 16 of WCAP-8768, Revision 2, for a discussion of this program. [a]

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Reactor vessel neutron embrittlement was evaluated as a time-limited aging analysis (TLAA) for license renewal in accordance with 10 CFR Part 54. The results of this analysis are provided in subsection 19.4.1.

1.5.7 ENVIRONMENTAL TESTING OF ENGINEERED SAFETY FEATURES RELATED EQUIPMENT

The environmental testing of engineered safety features related equipment is complete. See WCAP-7856 for a discussion of this program.

1.5.8 GENERIC PROGRAMS OF BECHTEL

GPC has contributed, with other utilities, to tests of prototypical cable trays under seismically induced loads. A primary objective of the tests has been evaluation of damping coefficients under safe shutdown earthquake conditions. Design of cable tray supports in VEGP is verified by the results of this test program.

1.6 MATERIAL INCORPORATED BY REFERENCE

The VEGP Final Safety Analysis Report incorporates, by reference, various topical reports as part of the applications. Bechtel topical reports are listed in table 1.6-1, and Westinghouse topical reports are listed in table 1.6-2. Other topical reports are listed in table 1.6-3. The referenced topical reports have been filed separately in support of this and similar applications.

TABLE 1.6-1 (SHEET 1 OF 2)

BECHTEL TOPICAL REPORTS INCORPORATED BY REFERENCE

Bechtel Topical Report No.	Title	Revision Number	FSAR Section Reference	Report Submitted to the NRC	Review Status[(a)]
BC-TOP-1	Containment Building Liner Plate Design Report	Rev 1	3.8	12/72	A 2/7/74
BC-TOP-3-A	Tornado and Extreme Wind Design Criteria for Nuclear Power Plants	Rev 3	3.3	8/74	A 10/4/74
BC-TOP-4-A	Seismic Analyses of Structures and Equipment for Nuclear Power Plants	Rev 3	1.9, 3.7.B	11/74	A 10/31/74
BC-TOP-5-A	Prestressed Concrete Nuclear Reactor Containment Structures	Rev 3	3.8	2/75	A 3/28/75
BC-TOP-7	Full Scale Buttress Test for Prestressed Nuclear Containment Structures	Rev 0	3.8	9/72	A 8/24/73
BC-TOP-8	Tendon End Anchor Reinforcement Test	Rev 0	3.8	9/72	A 8/24/73
BN-TOP-1	Test Criteria for Integrated Leak Rate Testing of Primary Containment Structures for Nuclear Power Plants	Rev 1	6.2	11/72	A 2/1/73

TABLE 1.6-1 (SHEET 2 OF 2)

Bechtel Topical Report No.	Title	Revision Number	FSAR Section Reference	Report Submitted to the NRC	Review Status[a]
BN-TOP-2	Design for Pipe Break Effects	Rev 2	3.6, 3F, 3.9.B, 3.8	5/74	A 6/17/74
BN-TOP-4	Subcompartment Pressure and Temperature Transient Analysis	Rev 1	6.2, 3F, 3.6	10/77	A 2/23/79
BP-TOP-1	Seismic Analysis of Piping Systems	Rev 3	1.9, 3.7.B	1/76	A 9/29/76

TABLE 1.6-2 (SHEET 1 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-2048	The Doppler Effect for a Non-Uniform Temperature Distribution in Reactor Fuel Elements	Rev 0	4.3	7/62	O
WCAP-2850-L(P) WCAP-7916	Single-Phase Local Boiling and Bulk Boiling Pressure Drop Correlations	Rev 0	4.4	5/66	O
WCAP-2923	In-Pile Measurement of UO_2 Thermal Conductivity	Rev 0	4.4	3/66	O
WCAP-3269-8	Hydraulic Tests of the San Onofre Reactor Model	Rev 0	4.4	6/64	O
WCAP-3269-26	LEOPARD - A Spectrum Dependent Non-Spatial Depletion Code for the IBM-7094	Rev 0	4.3, 15.0, 15.4	9/63	O
WCAP-3385-56	Saxton Core II Fuel Performance Evaluation WCAP-3385-56, Part II, Evaluation of Mass Spectrometric and Radiochemical Materials Saxton Plutonium Fuel	Rev 0	4.3, 4.4	7/70	O
WCAP-3680-20	Xenon-Induced Spatial Instabilities in Large Pressurized Water Reactors (EURAEC-1974)	Rev 0	4.3	3/68	O
WCAP-3680-21	Control Procedures for Xenon-Induced X-Y Instabilities in Large Pressurized Water Reactors (EURAEC-2111)	Rev 0	4.3	2/69	O

TABLE 1.6-2 (SHEET 2 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-3680-22	Xenon-Induced Spatial Instabilities in Three Dimensions (EURAEC-2116)	Rev 0	4.3	9/69	O
WCAP-3696-8	Pressurized Water Effect Final Report (EURAEC-2074)	Rev 0	4.3	10/68	O
WCAP-3726-1	PUO_2 -UO_2 Fueled Critical Experiments	Rev 0	4.3	7/67	O
WCAP-6065	Melting Point of - Irradiated UO_2	Rev 0	4.4	2/65	O
WCAP-6069	Burnup Physics of Heterogeneous Reactor Lattices	Rev 0	4.4	6/65	O
WCAP-6073	LASER – Depletion Program for Lattice Calculations Based on MUFT and THERMOS	Rev 0	4.3	4/66	O
WCAP-6086	Supplementary Report on Evaluation of Mass Spectrometric and Radiochemical Analyses of Yankee Core I Spent Fuel, Including Isotopes of Elements Thorium through Curium	Rev 0	4.3	8/69	O
WCAP-7015	Subchannel Thermal Analysis of Rod Bundle Cores	Rev 1	4.4	2/14/69	O
WCAP-7048-P-A(P) WCAP-7757-A	PANDA Code	Rev 0	4.3	1/9/75	A
WCAP-7198-L(P) WCAP-7825	Evaluation of Protective Coatings for Use in Reactor Containment	Rev 0	6.1	4/23/69 12/16/71	O

TABLE 1.6-2 (SHEET 3 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-7213-P-A(P) WCAP-7758-A	TURTLE 24.0 Diffusion Depletion Code	Rev 0	4.3, 15.0, 15.4	1/9/75	A
WCAP-7308-L(P) WCAP-7810	Evaluation of Nuclear Hot Channel Factor Uncertainties	Rev 0	4.3	7/9/70 12/16/71	U
WCAP-7359-L(P) WCAP-7838	Application of THINC Program to PWR Design	Rev 0	4.4	9/8/69 1/17/72	O
WCAP-7397-L(P) WCAP-7817	Seismic Testing of Electrical and Control Equipment	Rev 0	3.10.N	2/6/70 12/16/71	U
WCAP-7397-L(P) WCAP-7817	Seismic Testing of Electrical and Control Equipment (WCID Process Control Equipment)	Supplement 1	3.10.N	1/27/71 12/16/71	U
WCAP-7427	Effective Structural Damping of the KEP L105 CRDM	Rev 0	3.7.N	1/70	0
WCAP-7427	Effective Structural Damping of the KEP L105 CRDM	Addendum 1	3.7.N	12/70	0
WCAP-7477-L(P) WCAP-7735	Sensitized Stainless Steel in Westinghouse PWR Nuclear Steam Supply Systems	Rev 0	5.2	3/26/70 8/12/71	A
WCAP-7488-L(P) WCAP-7672	Solid State Logic Protection System Description	Rev 0	7.2, 7.3	3/24/71 5/27/71	B
WCAP-7536-L(P) WCAP-7821	Seismic Testing of Electrical and Control Equipment (High Seismic Plants)	Rev 0	3.10.N	11/70	U
WCAP-7558	Seismic Vibration Testing with Sine Beats	Rev 0	3.10.N	9/25/72	U

TABLE 1.6-2 (SHEET 4 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-7588	Evaluation of Rod Ejection Accident in Westinghouse Pressurized Water Reactors Using Spatial Kinetics Methods	Rev 1A	15.4	1/7/75	A
WCAP-7667-P-A(P) WCAP-7755-A	Interchannel Thermal Mixing with Mixing Vane Grids	Rev 0	4.4	1/27/75	A
WCAP-7695-P-A(P) WCAP-7958-A	DNB Tests Results for New Mixing Vane Grids (R)	Rev 0	4.4	1/21/75	A
WCAP-7705	Testing of Engineered Safety Features Actuation System	Rev 2	7.3	5/5/76	B
WCAP-7706-L(P) WCAP-7706	An Evaluation of Solid State Logic Reactor Protection in Anticipated Transients	Rev 0	4.6, 7.1, 7.2	9/2/71	U
WCAP-7709-L(P) WCAP-7820	Electrical Hydrogen Recombiner for Water Reactor Containments	Rev 0	6.2	7/14/71 12/16/71	A
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner for PWR Containments-Final Development Report	Supplement 1	6.2	5/23/72 5/31/72	A
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner for PWR Containments Equipment Qualification Report	Supplement 2	6.2	9/24/73 11/2/73	A
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner for PWR Containments Long-Term Tests	Supplement 3	6.2	1/23/74 3/22/74	A
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner for PWR Containments	Supplement 4	6.2	4/21/74	A
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner Special Tests	Supplement 5	6.2	1/7/76	A

TABLE 1.6-2 (SHEET 5 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner IEEE 323-1974 Qualification	Supplement 6	6.2	11/5/76	A
WCAP-7709-L(P) WCAP-7820	Electric Hydrogen Recombiner LWR Containments Supplemental Test No. 2	Supplement 7	6.2	9/21/77	A
WCAP-7769	Overpressure Protection for Westinghouse Pressurized Water Reactors	Rev 1	5.2, 15.2	7/5/72	U
WCAP-7798-L(P) WCAP-7803	Behavior of Austenitic Stainless Steel in Post Hypothetical Loss-of-Coolant Environment	Rev 0	6.1	12/6/71 1/4/72	O
WCAP-7800	Nuclear Fuel Division Quality Assurance Program Plan	Rev 4A	4.2, 17B	4/28/75	A
WCAP-7806	Nuclear Design of Westinghouse Pressurized Water Reactors with Burnable Poison Rods	Rev 0	4.3	12/16/71	B
WCAP-7811	Power Distribution Control of Westinghouse Pressurized Water Reactors	Rev 0	4.3	12/16/71	O
WCAP-7817	Seismic Testing of Electrical and Control Equipment (Low Seismic Plants)	Supplement 2	3.10.N	1/17/72	A
WCAP-7817	Seismic Testing of Electrical and Control Equipment (Westinghouse Solid-State Protection System) (Low Seismic Plants)	Supplement 3	3.10.N	1/17/72	A
WCAP-7817	Seismic Testing of Electrical and Control Equipment (WCID NUCANA 7300 Series) (Low Seismic Plants)	Supplement 4	3.10.N	12/14/72	A

TABLE 1.6-2 (SHEET 6 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-7817	Seismic Testing of Electrical and Control Equipment (Instrument Bus Distribution Panel) (Low Seismic Plants)	Supplement 5	3.10.N	3/19/75	A
WCAP-7817	Seismic Testing of Electrical and Control Equipment - Type DB Reactor Trip Switchgear	Supplement 6	3.10.N	8/74	U
WCAP-7817	Seismic Testing of Electrical and Control Equipment for Low Seismic Plants	Supplement 7	3.10.N	9/76	U
WCAP-7817	Seismic Testing of Electrical and Control Equipment – Low Seismic Plants	Supplement 8	3.10.N	6/75	A
WCAP-7821	Seismic Testing of Electrical and Control Equipment (High Seismic Plants)	Supplement 1, 2	3.10.N	12/71	U
WCAP-7821	Seismic Testing of Electrical and Control Equipment (WICD-NUCANA 7300 Series) (High Seismic Plants)	Supplement 3	3.10.N	9/72	U
WCAP-7821	Seismic Testing of Electrical and Control Equipment (Type DB Reactor Trip switchgear)	Supplement 4	3.10.N	8/74	U
WCAP-7821	Seismic Testing of Electrical and Control Equipment for High Seismic Plants	Supplement 5	3.10.N	9/76	U
WCAP-7821	Seismic Testing of Electrical and Control Equipment (High Seismic Plants)	Supplement 6	3.10.N	6/75	U

TABLE 1.6-2 (SHEET 7 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-7832	Evaluation of Steam Generator Tube, Tube Sheet, and Divider Plate Under Combined LOCA Plus SSE Conditions	Rev 0	5.4	12/26/73	A
WCAP-7836	Inlet Orificing of Open PWR Cores	Rev 0	4.4	1/17/72	B
WCAP-7856	Safety-Related Research and Development for Westinghouse Pressurized Water Reactors Program Summaries, Fall 1971 through Spring 1972	Rev 0	1.5	1/72	B
WCAP-7870	Neutron Shielding Pads	Rev 0	3.9.N	7/17/72	A
WCAP-7907	LOFTRAN Code Description	Rev 0	5.2, 6.2, 15.0, 15.1, 15.2, 15.3, 15.4, 15.5, 15.6	10/11/72	U
WCAP-7908	FACTRAN - A FORTRAN-IV Code for Thermal Transients in a UO_2 Fuel Rod	Rev 0	15.0, 15.2, 15.3, 15.4	9/20/72	U
WCAP-7912-P-A(P) WPAC-7912-A	Power Peaking Factors	Rev 0	4.3, 4.4	1/16/75	A
WCAP-7913	Process Instrumentation for Westinghouse Nuclear Steam Supply Systems (Four-Loop Plant Using WCID-7300 Series Process Instrumentation)	Rev 0	7.2, 7.3	3/9/73	B
WCAP-7921-AR	Damping Values of Nuclear Power Plant Components	Rev 0	3.7.N	7/11/74	A
WCAP-7941-P-A(P) WCAP-7959-A	Effect of Axial Spacing On Interchannel Thermal Mixing with the R Mixing Vane Grid	Rev 0	4.4	1/27/75	A

TABLE 1.6-2 (SHEET 8 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-7956	THINC-IV - An Improved Program for Thermal-Hydraulic Analysis of Rod Bundle Cores	Rev 0	4.4	10/22/73	A
WCAP-7964	Axial Xenon Transient Tests at Rochester Gas and Electric Reactor	Rev 0	4.3	6/15/71	O
WCAP-7979-P-A(P) WCAP-8028-A	TWINKLE - A Multidimensional Neutron Kinetics Computer Code	Rev 0	15.0, 15.4	1/7/75	A
WCAP-7988-P-A(P) WCAP-8030-A	Application of Modified Spacer Factor to L-Grid Typical and Cold Wall Cell DNB	Rev 0	4.4	1/75	A
WCAP-8004	Safety Related Research and Development for Westinghouse Pressurized Water Reactors Program Summaries - Fall 1972	Rev 0	1.5	2/73	B
WCAP-8054(P) WCAP-8195	Application of THINC-IV Program to PWR Design	Rev 0	4.4	12/7/73 1/11/74	A
WCAP-8082-P-A(P) WCAP-8172-A	Pipe Breaks for the LOCA Analysis of Westinghouse Primary Coolant Loop	Rev 0	3.6	1/16/75	A
WCAP-8163	Reactor Coolant Pump Integrity in LOCA	Rev 0	1.9, 5.4	9/20/73	U
WCAP-8170(P) WCAP-8171	Calculational Model for Core Reflooding After a Loss-of-Coolant Accident (WREFLOOD Code)	Rev 0	15.6	7/3/74	AE
WCAP-8174-P-A(P) WCAP-8202-A	Effect of Local Heat Flux on DNB in Nonuniformly Heated Rod Bundles	Rev 0	4.4	2/75	A
WCAP-8183	Operational Experience with Westinghouse Cores (up to December 31, 1977)	Rev 7	4.2	4/20/78	B

TABLE 1.6-2 (SHEET 9 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-8200(P) WCAP-8261	WFLASH - A FORTRAN-IV Computer Program for Simulation of Transients in a Multiloop PWR	Rev 2 Rev 1	15.6	7/3/74	AE
WCAP-8218-P-A(P) WCAP-8219-A	Fuel Densification Experimental Results and Model for Reactor Application	Rev 0	4.1, 4.2, 4.3, 4.4	3/6/75	A
WCAP-8236(P) WCAP-8288	Safety Analysis of 17 x 17 Fuel Assembly for Combined Seismic and Loss-of-Coolant Accident	Rev 0	4.2, 3.7.N	2/28/74 3/1/74	U
WCAP-8236(P) WCAP-8288	Safety Analysis of Eight-Grid 17 x 17 Fuel Assembly for Combined Seismic Loss-of-Coolant Accident	Addendum 1	3.7.N	4/15/74	A
WCAP-8252	Documentation of Selected Westinghouse Structural Analysis Computer Codes	Rev 1	3.6, 3.9.N	7/19/77	U
WCAP-8253	Source Term Data for Westinghouse Pressurized Water Reactors	Amendment 1	11.1	2/13/76	B
WCAP-8255	Nuclear Instrumentation System	Rev 0	7.2, 7.7	4/9/74	B
WCAP-8258	Spraco Model 1713A Nozzle Spray Drop Size Distribution	Rev 1	6.5	5/75	B
WCAP-8278(P) WCAP-8279	Hydraulic Flow Test of 17 x 17 Fuel Assembly	Rev 0	4.2, 4.4	2/28/74 3/1/74	U
WCAP-8296-P-A(P) WCAP-8297-A	Effect of 17 x 17 Fuel Assembly Geometry on DNB	Rev 0	4.4	2/6/75	A
WCAP-8298-P-A(P) WCAP-8299-A	The Effect of 17 x 17 Fuel Assembly Geometry on Interchannel Thermal Mixing	Rev 0	4.4	1/28/75	A
WCAP-8301(P) WCAP-8305	LOCA-IV Program: Loss-of-Coolant Transient Analysis	Rev 0	15.0, 15.6	7/12/74	AE

TABLE 1.6-2 (SHEET 10 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-8302(P) WCAP-8306	SATAN-IV Program: Comprehensive Space-Time-Dependent Analysis of Loss-of-Coolant	Rev 0	15.0, 15.6	7/12/74	AE
WCAP-8303-P-A(P) WCAP-8317-A	Prediction of the Flow-Induced Vibration of Reactor Internals by Scale Model Tests	Rev 0	3.9.N	7/18/75	A
WCAP-8324-A	Control of Delta Ferrite in Austenitic Stainless Steel Weldments	Rev 0	5.2	6/23/75	A
WCAP-8327(P) WCAP-8326	Containment Pressure Analysis Code (COCO)	Rev 0	15.6, 6.2	7/3/74	AE
WCAP-8330	Westinghouse Anticipated Transients Without Trip Analysis	Rev 0	4.3, 4.6, 15.8	9/25/74	U
WCAP-8339	Westinghouse Emergency Core Cooling System Evaluation Model – Summary	Rev 0	15.6	7/3/74	AE
WCAP-8340(P) WCAP-8356	Westinghouse Emergency Core Cooling System - Plant Sensitivity Studies	Rev 0	15.6	8/1/74	AE
WCAP-8341(P) WCAP-8342	Westinghouse Emergency Core Cooling System Evaluation Model-Sensitivity Studies	Rev 0	15.6	7/3/74	AE
WCAP-8353	Safety-Related Research and Development for Westinghouse Pressurized Water Reactor Program Summaries - Spring 1974	Rev 0	1.5	8/74	B
WCAP-8359	Effects of Fuel Densification Power Spikes on Clad Thermal Transients	Rev 0	4.3	7/2/74	AE
WCAP-8370	Westinghouse Water Reactor Divisions Quality Assurance Plan	Rev 9A	1.9, 17B	11/14/77	A

TABLE 1.6-2 (SHEET 11 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-8376	Iodine Removal by Spray in the Joseph M. Farley Station Containment	Rev 0	6.5	7/74	B
WCAP-8377(P) WCAP-8381	Revised Clad Flattening Model	Rev 0	4.2	8/7/74 8/6/74	A
WCAP-8385(P) WCAP-8403	Power Distribution Control and Load Following Procedures	Rev 0	4.3, 4.4	10/9/74	A
WCAP-8424	Evaluation of Loss-of-Flow Accidents Caused by Power System Frequency Transients - in Westinghouse PWRs	Rev 1	15.3	5/30/75	U
WCAP-8446(P) WCAP-8449	17 x 17 Drive Line Components Tests Phase IB, II, III, D-Loop Drop, and Deflection	Rev 0	3.9.N	12/31/74	A
WCAP-8453-A	Analysis of Data from Zion (Unit 1) THINC Verification Test	Rev 0	4.4	5/10/76	A
WCAP-8471(P) WCAP-8472	Westinghouse ECCS - Evaluation Model – Supplementary Information	Rev 0	15.6	2/10/75 2/11/75	AE
WCAP-8485	Safety-Related Research and Development for Westinghouse Pressurized Water Reactors, Program Summaries - Fall 1974	Rev 0	1.5	4/2/75	B
WCAP-8498	Incore Power Distribution Determination in Westinghouse Pressurized Water Reactors, Program - Summaries - Fall 1974	Rev 0	4.3	7/22/75	U
WCAP-8510	Method for Fracture Mechanics Analysis of Nuclear Reactor Vessels Under Severe Thermal Transients	Rev 0	5.3	7/76	U

TABLE 1.6-2 (SHEET 12 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-8516-P(P) WCAP-8517	UHI Plant Internals Vibration Measurement Program and Pre- and Post-Hot Functional Examinations	Rev 0	3.9.N	4/11/75	A
WCAP-8536(P) WCAP-8537	Critical Heat Flux Testing of 17 x 17 Fuel Assembly Geometry with 22-in. Grid Spacing	Rev 0	4.4	5/30/75	A
WCAP-8565-P-A(P) WCAP-8566-A	Westinghouse ECCS-Four-Loop Plant (17 x 17) Sensitivity Studies	Rev 0	15.6	7/17/75	A
WCAP-8577	Application of Preheat Temperatures after Welding Pressure Vessel Steels	Rev 0	1.9	2/3/76	A
WCAP-8584(P) WCAP-8760	Failure Mode and Effects Analysis (FMEA) of Engineered Safeguard Features Actuation System	Rev 0 Rev 1	4.6, 7.3	4/23/76 2/80	U
WCAP-8587	Equipment Qualification Data Packages	Rev 1	3.10.N, 3.11.N	4/17/78 11/78	U
WCAP-8587	Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety-Related Electrical Equipment	Rev 2 Rev 3	1.9, 3.10.N, 3.11.N	3/5/79 5/80	U
WCAP-8622(P) WCAP-8623	Westinghouse ECCS - Evaluation Model - October 1975 Version	Rev 0	15.6	11/20/75	AE
WCAP-8624(P) WCAP-8695	General Method of Developing Multifrequency-Biaxial Test Inputs for Bistables	Rev 0	3.10.N	9/75	U
WCAP-8682(P) WCAP-8683	Experimental Verification of Wet Fuel Storage Criticality Analyses	Rev 0	4.3	3/18/76	B
WCAP-8691(P) WCAP-8692	Fuel Rod Bowing	Rev 0	4.2	1/9/76	U

TABLE 1.6-2 (SHEET 13 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-8693	Delta Ferrite in Production Austenitic Stainless Steel Weldments	Rev 0	5.2	3/16/76	B
WCAP-8708-P-A(P), Volumes I and II WCAP-8709-A, Volumes I and II	MULTIFLEX - FORTRAN-IV Computer Program for Analyzing Thermal-Hydraulic Structure System Dynamics	Rev 0	3.6, 3.9.N	9/16/77	A
WCAP-8720(P) WCAP-8785	Improved Analytical Models Used in Westinghouse Fuel Rod Design Computations	Rev 0	4.2, 15.6	11/2/76	A
WCAP-8768	Safety-Related Research and Development for Westinghouse Pressurized Water Reactors, Program Summaries - Spring 1976	Rev 0	1.5	6/17/76	B
WCAP-8768	Safety-Related Research and Development for Westinghouse Pressurized Water Reactors, Program Summaries - Winter 1977 through Summer 1978	Rev 2 Rev 1	1.5, 4.2, 4.3, 5.4	9/28/78	B
WCAP-8766(P) WCAP-8780	Verification of Neutron Pad and 17 x 17 Guide Tube Designs by Preoperational Tests on the Trojan 1 Power Plant	Rev 0	3.9.N	5/21/76	A
WCAP-8768	Safety-Related Research and Development for Westinghouse Pressurized Water Reactors Program Summaries - Winter 1977 through Summer 1978	Rev 2	5.4	10/78	B
WCAP-8776	Corrosion Study for Determining Hydrogen Generation from Aluminum and Zinc During Post-Accident Conditions	Rev 0	6.2	10/76	B

TABLE 1.6-2 (SHEET 14 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-8822 (P) WCAP-8860	Mass and Energy Releases Steam Line Rupture	Rev 0	6.2	9/76	U
WCAP-8892-A	Westinghouse 7300 Series Process Control System Noise Tests	Rev 0	7.1	6/15/77	A
WCAP-8929	Benchmark Problem Solution Employed for Verification of WECAN Computer Program	Rev 0	3.9.N	5/26/77	U
WCAP-8963(P) WCAP-8964	Safety Analysis for the Revised Fuel Rod Internal Pressure Design Basis	Rev 0	4.2	3/31/71 8/11/77	A
WCAP-8976	Failure Mode and Effects Analysis (FMEA) of Solid State Full-Length Rod Control System	Rev 0	4.6	10/26/77	U
WCAP-9168(P) WCAP-9150	Westinghouse Emergency Core Cooling System Evaluation Model - Modified October 1975 Version	Rev 0	15.6	9/27/77	U
WCAP-9179(P) WCAP-9224	Properties of Fuel and Core Component Materials	Rev 1	4.2	8/2/78	U
WCAP-9207(P) WCAP-8966	Evaluation of Mispositioned ECCS Valves	Rev 0	6.3	3/21/78	U
WCAP-9220-P-A(P) WCAP-9221-A	Westinghouse ECCS Evaluation Model - February 1978 Version	Rev 0	15.6	2/78	A
WCAP-9226 WCAP-9227	Reactor Core Response to Excessive Secondary Steam Releases	Rev 1	15.1	9/78	U
WCAP-9230(P) WCAP-9231	Report on the Consequences of a Postulated Main Feedline Rupture	Rev 0	15.2	1/27/78	U
WCAP-9283	Integrity of Primary Piping Systems of Westinghouse Nuclear Power Plants during Postulated Seismic Events	Rev 0	4.2	3/21/78	U

TABLE 1.6-2 (SHEET 15 OF 15)

Westinghouse Topical Report No.[a]	Title	Revision Number	FSAR Section Reference	Submitted to the NRC	Review Status[b]
WCAP-9292	Dynamic Fracture Toughness of ASME SA508 Class 2a, ASME SA533 Grade A Class 2 Base, Heat Affected Zone Material, and Applicable Weld Metals	Rev 0	5.2	3/17/78	U
WCAP-9401-P-A(P) WCAP-9402	Verification Testing and Analyses of the 17 x 17 Optimized Fuel Assembly	Rev 0	4.2	3/79	A
WCAP-9600(P) WCAP-9601	Report on Small Break Accidents for Westinghouse NSSS System	Rev 0	5.4	6/79	A
WCAP-11731(P) WCAP-11731	LOFTTR2 Analysis for a Steam Generator Tube Rupture Event for the Vogtle Electric Generating Plant Units 1 and 2	Rev 0	15.6	3/01/88	U
WCAP-10858P-A	AMSAC Generic Design Package	Rev 1	7.7/15.8	6/85	A

a. (P) - Proprietary

b. Legend for the review status code letters:

A - NRC review complete; NRC acceptance letter issued.

AE - NRC accepted as part of the Westinghouse ECCS evaluation model only; does not constitute acceptance for any purpose other than for ECCS analyses.

B - Submitted to NRC as background information; not undergoing formal NRC review.

O - On file with NRC; older generation report with current validity; not actively under formal NRC review.

U - Actively under formal NRC review.

TABLE 1.6-3

OTHER TOPICAL REPORTS INCORPORATED BY REFERENCE

Topical Report No.	Prepared By	Title	Revision No.	FSAR Section Reference	Report Submitted to the NRC	Review Status
AECC-3-P	Aerojet Energy Conversion Company	Topical Report Radioactive Waste Volume Reduction System Combined Incinerator/Dryer	0	11.4	12/81	Submitted to NRC; acceptance review to be concurrent with review of the first FSAR to reference the topical report.
OFC-1	Owens-Corning Fiberglass Co.	Nuclear Containment Insulation System	-	6.1	9/1/77	NRC acceptance 12/8/78.

1.7 DRAWINGS AND OTHER DETAILED INFORMATION

1.7.1 ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS

This section originally contained lists of electrical, instrumentation, and control drawings that were considered to be necessary to evaluate the safety-related features pertaining to VEGP. Three uncontrolled copies of each drawing were provided to the NRC in 1983 with the original FSAR submittal in conjunction with the application for operating licenses for use in reviewing the design. These lists are not included in this Updated FSAR.

Table 1.7.1-1 is a cross reference between the applicable electrical, instrumentation, and control drawings as well as the piping and instrumentation flow diagrams and the section where they are referenced in the Updated FSAR. This cross reference is an aid in locating engineering drawings, which are included by reference throughout the Updated FSAR in conjunction with specific system descriptions.

Drawings 1X3D-AA-A00E, 1X3D-AA-A00M, 1X5DN002-1, 1X5DN002-2, 1X5DN002-3 and 1X5DV002 are the legends for electrical and control drawings. Drawing AX3DG030 is the legend for communications and lighting drawings.

1.7.2 PIPING AND INSTRUMENTATION DIAGRAMS

This section originally contained a list of each piping and instrumentation diagram (P&ID) and the corresponding Final Safety Analysis Report (FSAR) figure number (if applicable) as it appeared at the end of the respective text section. Two uncontrolled, large-scale copies of each P&ID were provided to the NRC in 1983 with the original FSAR submittal in conjunction with the application for operating licenses. That list is not included in this Updated FSAR.

A cross reference between the Updated FSAR figures and the applicable piping and instrumentation flow diagrams contained in the Updated FSAR is included in table 1.7.1-1. This cross reference is an aid in locating engineering drawings that are included throughout the Updated FSAR in conjunction with specific system descriptions. Those engineering drawings identified as "modified for FSAR" have been either partially used to produce the Updated FSAR figure or enhanced by the addition of notations or information with respect to the specific system description. The appropriate revision number of each engineering drawing is reflected on the Updated FSAR figure. Drawings 1X4DB101 and 1X4DB102 provide an explanation of symbols and characters used in these FSAR figures. Drawing 1X4DB100 is a P&ID flow diagram legend.

1.7.3 OTHER DETAILED INFORMATION

The geologic and geophysical logs obtained during the geology and site foundation investigations and examinations have been submitted under separate cover. The results and conclusions are summarized in section 2.5 and appendix 2B.

TABLE 1.7.1-1 (Sheet 1 of 43)

CROSS REFERENCE OF ENGINEERING DRAWINGS TO UFSAR DISCUSSION SECTION

Drawing Number	Section	Title
1K5-1305-058-01	10.4.7	Feedwater System Isometric Containment and Valve Room
1K5-1305-062-01	10.4.7	Feedwater System Isometric Containment and Valve Room
1P01C016	3.8.1	Containment Basemat Hoop and Dowel Reinforcement (Unit 1)
1P01C017	3.8.1	Containment Basemat Reinforcement Typical E-W Layout (Unit 1)
1P01C018	3.8.1	Containment Basemat Reinforcement Typical N-S Layout (Unit 1)
1P01W058	3.8.1	Shell Wall Reinforcement Containment Building Outside Face at Buttress 1 (Unit 1)
1P01W059	3.8.1	Shell Wall Reinforcement Containment Building Buttress 1 Plan and Sections (Unit 1)
1P01W072	3.8.1	Shell Wall Reinforcement Containment Building Inside Face Vertical and Horizontal at Equipment Hatch (Unit 1)
1P01W077	3.8.1	Shell Wall Reinforcement Containment Building Outside Face Vertical and Horizontal Buttress 1 to 2 (Unit 1)
1P01W081	3.8.1	Shell Wall Reinforcement Containment Building Inside Face Vertical and Horizontal at Equipment Hatch (Unit 1)
1X2D01A001	3.8.1	Containment Section and Plan View (Unit 1)
1X2D01J004	3.8.3	Primary and Secondary Shield Wall Anchorage to Basemat
1X2D01J008	3.8.3	NSSS Support Anchorage to Basemat (Unit 1)
1X2D01J015	3.8.1	Containment Liner Plate Plan, Sections, Details (Unit 1)
1X2D01J017	3.8.1	Containment - Wall Liner Plate Hatch and Lock Details Sh. 1
1X2D01J018	3.8.1	Containment - Wall Liner Plate Hatch and Lock Details Sh. 2

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Drawing Number	Section	Title
1X2D01J019	3.8.2	Containment - Wall Liner Plate Penetrations Details Sh. 1
1X2D01J020	3.8.2	Containment - Wall Liner Plate Penetrations Details Sh. 2
1X2D01J022	3.8.1	Polar Crane Bracket Details
1X2D01K002	3.8.1	Containment Tendon Arrangement (Unit 1)
1X2D01K003	3.8.1	Containment Tendon Arrangement (Unit 1)
1X2D01K005	3.8.1	Containment - Prestressing Requirements - Wall and Dome Cross Section
1X2D01K006	3.8.1	Containment - Prestressing Requirements - Hatch and Lock Details Sh. 1
1X2D01K007	3.8.1	Containment - Prestressing Requirements - Hatch and Lock Details Sh. 2
1X2D01K008	3.8.1	Containment - Prestressing Requirements - Sections and Details Sh. 1
1X2D05E001	1.2.2	NSCW Cooling Towers 1A and 1B General Arrangement Plan, Elevation, and Sections
1X2D28A001	3.8.4	Reactor Make-up Water Storage Tank - Forming Plan - Section and Details
1X2D29A001	3.8.4	Refueling Water Storage Tank - Forming Plan - Sections and Details
1X2D30A001	3.8.4	Condensate Storage Tank - Forming Plan and Sections
1X2D44A005	3.8.4	Outside Area Tunnels - Forming Plan - Bottom Electrical Tunnel 1T4A & 1T4B
1X2D48A018	3.8.3	Primary Shield Wall and Reactor Cavity Plan Views (Unit 1)
1X2D48A019	3.8.3	Primary Shield Wall and Reactor Cavity Sections Views (Unit 1)
1X2D48A050	12.3.2	Fuel Transfer Tube Shielding Details
1X2D48E003	3.8.3	Containment Plan View (Unit 1)
1X2D48E004	3.8.3	Containment Plan View (Unit 1)

TABLE 1.7.1-1 (Sheet 3 of 43)

Drawing Number	Section	Title
1X2D48E005	3.8.3	Containment Plan View (Unit 1)
1X2D48E007	3.8.3	Containment Sections (Unit 1)
1X2D48E007	1.2.2	Containment Internals General Arrangement Section Looking North
1X2D48E008	3.8.3	Containment Sections (Unit 1)
1X2D48E008	1.2.2	Containment Internals General Arrangement Section Looking West
1X2D48J006	12.3.2	Fuel Transfer Tube Shielding Details
1X2D48J006	3.8.2	Fuel Transfer Tube Penetration Assembly
1X3AE01-20	18.1	Control Room Panels, QEAB, Section 1A, Part 1 (Unit 1)
1X3AE01-21	18.1	Control Room Panels, QEAB, Section 1A, Part 2 (Unit 1)
1X3AE01-24	18.1	Control Room Panels, QEAB, Section 1B (Unit 1)
1X3D-AA-A00E	1.7.1	Legend for Electrical Drawings
1X3D-AA-A00M	1.7.1	Legend for Electrical Drawings
1X3D-AA-A01A	8.3.1	Main One Line (Unit 1)
1X3D-AA-G01A	7.6.1	Main One-Line Class 1E 125-V dc and 120-V Vital ac Systems
1X3D-AA-G01B	8.3.1	Non-Class 1E Essential ac Power System and dc Power System
1X3D-AA-G02A	8.3.1	120-volt Vital ac Instrument Distribution Panels
1X3D-AA-G02C	8.3.1	120-volt Vital ac Instrument Distribution Panels
1X3D-AA-K02A	8.3.1	Safety Load Sequencing Table (Train A, Unit 1)
1X3D-AA-K02B	8.3.1	Safety Load Sequencing Table (Train A, Unit 1)
1X3D-BD-JO1A	7.3.11	Containment Spray System Elementary Diagram
1X3D-BD-JO1B	7.3.11	Containment Spray System Elementary Diagram
1X3D-BD-JO2A	7.3.11	Containment Spray System Elementary Diagram
1X3D-BD-JO2B	7.3.11	Containment Spray System Elementary Diagram

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Drawing Number	Section	Title
1X3D-BD-JO2C	7.3.11	Containment Spray System Elementary Diagram
1X3D-BD-JO2D	7.3.11	Containment Spray System Elementary Diagram
1X3DG001	9.5.2	Telephone/Page System Riser Diagram
1X3DG002	9.5.2	Sound-Powered Phone System Riser Diagram
1X3DG004	9.5.2	PABX System Riser Diagram
1X3DG020	9.5.3	Lighting Systems Schematic
1X3DG021	9.5.3	Lighting Systems Schematic
1X3DG031	9.5.2	Telephone/Page System Riser Diagram
1X3DG032	9.5.2	Telephone/Page System Riser Diagram
1X3DG033	9.5.2	Telephone/Page System Riser Diagram
1X3DG040	9.5.2	Sound-Powered Phone System Riser Diagram
1X3DG041	9.5.2	Sound-Powered Phone System Riser Diagram
1X3DG042	9.5.2	Sound-Powered Phone System Riser Diagram
1X3DG043	9.5.2	PABX System Riser Diagram
1X3DG045	9.5.2	Sound-Powered Phone System Riser Diagram
1X4AH04-25	3.8.2	Containment Spray Isolation Valve Encapsulation Vessel
1X4AK01-30	8.3.1	Diesel Generator Initiating Circuit Logic Diagrams
1X4AK01-31	8.3.1	Diesel Generator Initiating Circuit Logic Diagrams
1X4DB100	1.7.2	P&ID Flow Diagram Legend
1X4DB101	1.7.2	P&ID Instrumentation Identification and Symbols
1X4DB102	1.7.2	P&ID Instrumentation Identification and Symbols
1X4DB110	9.3.2	Post-Accident Sampling System
1X4DB111	5.1.2	Reactor Coolant System P&ID (Unit 1)
1X4DB112	5.1.2	Reactor Coolant System P&ID (Unit 1)
1X4DB113	5.1.2	Reactor Coolant System P&ID (Unit 1)

TABLE 1.7.1-1 (Sheet 5 of 43)

Drawing Number	Section	Title
1X4DB114	9.3.4	Chemical and Volume Control System
1X4DB115	9.3.4	Chemical and Volume Control System
1X4DB116-1	9.3.4	Chemical and Volume Control System
1X4DB116-2	9.3.4	Chemical and Volume Control System
1X4DB117	9.3.4	Chemical and Volume Control System
1X4DB118	9.3.4	Chemical and Volume Control System
1X4DB119	6.3.2	Safety Injection System (Unit 1)
1X4DB120	6.3.2	Safety Injection System P&ID
1X4DB121	6.3.2	Safety Injection System P&ID
1X4DB122	5.4.7	Residual Heat Removal System P&ID (Unit 1)
1X4DB124	11.2.1	Waste Processing System - Liquid
1X4DB125	11.2.1	Waste Processing System - Liquid
1X4DB126	11.2.1	Waste Processing System - Liquid
1X4DB127	11.2.1	Waste Processing System - Liquid
1X4DB128	11.3.2	Waste Processing System - Gas
1X4DB129	11.3.2	Waste Processing System - Gas
1X4DB130	9.1.3	Spent Fuel Pool Cooling and Purification System P&ID
1X4DB131	6.2.2	Containment Spray System
1X4DB132	6.2.6	Containment ILRT System P&ID
1X4DB133-1	9.2.1	P&ID Nuclear Service Cooling Water System System No. 1202
1X4DB133-2	9.2.1	P&ID Nuclear Service Cooling Water System System No. 1202
1X4DB134	9.2.1	P&ID Nuclear Service Cooling Water System System No. 1202
1X4DB135-1	9.2.1	P&ID Nuclear Service Cooling Water System System No. 1202

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Drawing Number	Section	Title
1X4DB135-2	9.2.1	P&ID Nuclear Service Cooling Water System System No. 1202
1X4DB136	9.2.2	Component Cooling Water System
1X4DB137	9.2.2	Component Cooling Water System
1X4DB138-1	9.2.8	Auxiliary Component Cooling Water System
1X4DB138-2	9.2.8	Auxiliary Component Cooling Water System
1X4DB139	9.2.8	Auxiliary Component Cooling Water System
1X4DB140	9.3.2	Nuclear Sampling System - Liquid (Unit 1)
1X4DB141	9.3.2	Nuclear Sampling System - Gaseous
1X4DB142-1	9.3.3	Control Building Drain System
1X4DB142-2	9.3.3	Control Building Drain System
1X4DB143	9.3.3	Containment and Auxiliary Building Drains - Radioactive
1X4DB144-1	9.3.3	Containment and Auxiliary Building Drains - Radioactive
1X4DB144-2	9.3.3	Containment and Auxiliary Building Drains - Radioactive
1X4DB145-1	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB145-2	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB145-3	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB145-4	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB145-5	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB145-6	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB145-7	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB146-1	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB146-2	9.3.3	Auxiliary Building and Miscellaneous Drains
1X4DB146-3	9.3.3	Auxiliary Building and Miscellaneous Drains

TABLE 1.7.1-1 (Sheet 7 of 43)

Drawing Number	Section	Title
1X4DB147-1	9.3.3	Auxiliary Building Flood Retaining Rooms Alarms and Drains
1X4DB147-2	9.3.3	Auxiliary Building Flood Retaining Rooms Alarms and Drains
1X4DB148	11.4.2	Backflushable Filter System P&ID
1X4DB148-1	11.4.2	RCS Filter System
1X4DB148-2	11.4.2	Seal Water Return Filter System
1X4DB148-3	11.4.2	Seal Injection Filter System
1X4DB148-4	11.4.2	Seal Injection Filter System
AX4DB148-5	11.4.2	Recycle Evaporator Feed Filter System
AX4DB148-6	11.4.2	Recycle Evaporator Feed Filter System
AX4DB148-7	11.4.2	Recycle Evaporator Concentrate Filter System
1X4DB148-8	11.4.2	Waste Evaporator Feed Filter System
1X4DB148-9	11.4.2	Resin Sluice Filter System
1X4DB148-10	11.4.2	Floor Drain Tank Filter System
1X4DB148-11	11.4.2	Waste Monitor Tank Filter System
1X4DB148-12	11.4.2	Spent Fuel Pit Filter System
1X4DB148-13	11.4.2	Steam Generator Blowdown Filter System
1X4DB149-1*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 1)
1X4DB149-2*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 1)
1X4DB149-3*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 1)
1X4DB149-4*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 1)
1X4DB150	10.4.5	Circulating Water System
1X4DB150-1	10.4.5	Circulating Water System
1X4DB151-1	9.2.11	Turbine Plant Cooling Water System

TABLE 1.7.1-1 (Sheet 8 of 43)

Drawing Number	Section	Title
1X4DB151-2	9.2.11	Turbine Plant Cooling Water System
1X4DB151-3	9.2.11	Turbine Plant Cooling Water System
1X4DB154-1	9.2.10	Turbine Plant Closed Cooling Water System
1X4DB154-2	9.2.10	Turbine Plant Closed Cooling Water System
1X4DB157	10.4.10	Condensate Chemical Injection System
1X4DB159-1	10.3.2	Main Steam Supply System (Unit 1)
1X4DB159-2	10.3.2	Main Steam Supply System (Unit 1)
1X4DB159-3	10.3.2	Main Steam Supply System (Unit 1)
1X4DB160-1	10.2.2	General Electric Turbine-Generator (Main Steam System)
1X4DB160-2	10.2.2	General Electric Turbine-Generator (Main Steam System)
1X4DB160-3	10.2.2	General Electric Turbine-Generator (Turbine Steam Seal System)
1X4DB160-3	10.4.3	Turbine Steam Sealing System
1X4DB161-1	9.2.6	Condensate Storage and Degasifier System
1X4DB161-2	10.4.9	Auxiliary Feedwater P&ID
1X4DB161-3	10.4.9	Auxiliary Feedwater P&ID
1X4DB166	10.2.2	General Electric Turbine-Generator (Turbine Drive Steam System)
1X4DB167-1	10.2.2	General Electric Turbine-Generator (Turbine Lube Oil Storage and Filtration System)
1X4DB167-2	10.2.2	General Electric Turbine-Generator (Turbine Lube Oil Storage and Filtration System)
1X4DB168-1	10.4.1	Condensate and Feedwater System
1X4DB168-2	10.4.1	Condensate and Feedwater System
1X4DB168-3	10.4.1	Condensate and Feedwater System
1X4DB170-1	9.5.4	Diesel Generator System

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Drawing Number	Section	Title
1X4DB170-2	9.5.4	Diesel Generator System
1X4DB171-1	9.3.2	Turbine Plant Sampling System
1X4DB171-2	9.3.2	Turbine Plant Sampling System
1X4DB171-3	9.3.2	Turbine Plant Sampling System
1X4DB171-4	9.3.2	Turbine Plant Sampling System
1X4DB171-5	9.3.2	Turbine Plant Sampling System
1X4DB171-6	9.3.2	Turbine Plant Sampling System
1X4DB171-7	9.3.2	Turbine Plant Sampling System
1X4DB171-8	9.3.2	Turbine Plant Sampling System
1X4DB171-9	9.3.2	Turbine Plant Sampling System
1X4DB173	9.5.1	Fire Protection Water Systems
1X4DB174-1	9.5.1	Fire Protection Halon Systems
1X4DB174-2	9.5.1	Fire Protection Water Systems (Unit 1)
1X4DB174-3	9.5.1	Fire Protection Water Systems
1X4DB174-4	9.5.1	Fire Protection Water Systems (Unit 1)
1X4DB174-5	9.5.1	Fire Protection Water Systems
1X4DB174-6	9.5.1	Fire Protection Water System System No. 2303
1X4DB175-1	9.3.1	Instrument and Service Air System
1X4DB175-2	9.3.1	Instrument and Service Air System
1X4DB175-3	9.3.1	Instrument and Service Air System
1X4DB175-4	9.3.1	Instrument and Service Air System
1X4DB176-3	9.3.5	Auxiliary Gas System P&ID
1X4DB179-1	10.4.8	Steam Generator Blowdown Processing System
1X4DB179-2	10.4.8	Steam Generator Blowdown Processing System
1X4DB180-1	9.3.3	Turbine Building Drain System
1X4DB180-2	9.3.3	Turbine Building Drain System

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Drawing Number	Section	Title
1X4DB180-3	9.3.3	Turbine Building Drain System
1X4DB180-4	9.3.3	Turbine Building Drain System
1X4DB182	10.4.2	Condenser Air Ejection System P&ID
1X4DB183	9.3.3	Fuel Handling Building and Electrical Chase Tunnel Drains
1X4DB184	9.2.7-1	Reactor Makeup Water System
1X4DB185-1	10.4.6	Condensate Filter Demineralizer System
1X4DB185-2	10.4.6	Condensate Filter Demineralizer System
1X4DB185-3	10.4.6	Condensate Filter Demineralizer System
1X4DB185-5	10.4.6	Condensate Filter Demineralizer System
1X4DB185-6	9.3.1	Instrument Air System
1X4DB186-1	9.3.1	Service Air System
1X4DB186-4	9.3.1	Instrument Air System
1X4DB186-5	9.3.1	Instrument Air System
1X4DB186-6	9.3.1	Instrument Air System
1X4DB186-7	9.3.1	Instrument Air System
1X4DB186-8	9.3.1	Instrument Air System
1X4DB186-9	9.3.1	Instrument Air System
1X4DB188	9.3.1	Service Air System
1X4DB191	10.2.2	General Electric Turbine-Generator (Turbine-Generator Hydrogen Seal Oil System)
1X4DB192	9.3.7	NSCW Chemical Injection and Chlorine Analysis System
1X4DB193	10.2.2	General Electric Turbine-Generator (Turbine-Generator Stator Cooling System) (Unit 1)
1X4DB194	10.2.2	EHC System
1X4DB196	10.2.2	General Electric Turbine-Generator (Turbine-Generator Gas System)

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Drawing Number	Section	Title
1X4DB203	9.4.9	P&ID - Equipment Building HVAC System (Unit 1)
1X4DB205-1	9.4.3	Piping Penetration Area Filtration and Exhaust System (Unit 1)
1X4DB205-2	9.4.3	Piping Penetration Area Filtration and Exhaust System
1X4DB207-1	9.4.5	Control Building ESF Safety Feature Equipment Room Level B
1X4DB208-1	9.4.3	Auxiliary Building Normal Ventilation System (Unit 1)
1X4DB208-2	9.4.3	Auxiliary Building Normal Ventilation System
1X4DB208-3	9.4.3	Auxiliary Building Normal Ventilation System
1X4DB210	9.4.1	Control Building Wing Area Levels A and B Normal HVAC System
1X4DB211	9.4.1	Control Building Level B Normal HVAC System
1X4DB212	6.2.2	Containment Cooling System P&ID
1X4DB212	9.4.6	Containment Normal, Auxiliary, and Lower Level Heat Removal Systems
1X4DB213-1	9.4.6	Containment Purge and Preaccess Filter Systems (Unit 1)
1X4DB213-2	9.4.6	Containment Purge and Preaccess Filter Systems
1X4DB214-1	9.4.6	Reactor Cavity, Reactor Support, and CRDM Cooling Systems
1X4DB217	9.4.7	Diesel Generator Building Ventilation System
1X4DB221	9.2.9	Essential Chilled Water System
1X4DB227	9.4.8	Auxiliary Feedwater Pumphouse HVAC System
1X4DB228	9.4.3	ESF Room Coolers System
1X4DB229-1	9.4.4	P&ID Turbine Building HVAC
1X4DB229-2	9.4.4	P&ID Turbine Building HVAC
1X4DB229-3	9.4.4	Turbine Building HVAC P&ID
1X4DB233	9.2.9	Essential Chilled Water System

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Drawing Number	Section	Title
1X4DB234	9.2.9	Essential Chilled Water System
1X4DB238	9.4.9	P&ID Electrical Tunnel Ventilation System
1X4DB245	9.4.9	P&ID - Piping Penetration Ventilation System (Unit 1)
1X4DB250-1	9.4.9	Miscellaneous HVAC Systems Flow Diagrams (Unit 1)
1X4DB251-1	9.4.6	Containment Building HVAC Flow Diagrams
1X4DB252	9.4.6	Containment Building HVAC Flow Diagrams
1X4DB253-1	9.4.6	Containment Building HVAC Flow Diagrams
1X4DB257-1	9.4.1	Control Building Levels A and B Normal HVAC Flow Diagrams
1X4DB258-1	9.4.5	Control Building ESF Safety Feature Electrical Equipment Room
1X4DB259-1	9.4.5	Control Building Electrical and Pipe Penetration Room Filter and Exhaust System (Unit 1)
1X4DB261-1	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-2	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-3	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-4	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-5	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-6	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-7	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-8	9.4.3	Auxiliary Building Flow Diagrams
1X4DB261-9	9.4.3	Auxiliary Building Flow Diagrams
1X4DB267	9.4.7	Diesel Generator Building HVAC Flow Diagram
1X4DB268	9.4.8	Auxiliary Feedwater Pumphouse HVAC Flow Diagram
1X4DB301	9.4.4	Airflow Diagram Turbine Building HVAC Systems
1X4DB301-1	9.4.4	Airflow Diagram Turbine Building 1 HVAC Systems

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Drawing Number	Section	Title
1X4DB301-2	9.4.4	Airflow Diagram Turbine Building HVAC Systems
1X4DC101	10.1	Turbine Cycle Heat Balance VWO 105 Percent of Manufacturer's Guaranteed Rating
1X4DC103	10.1	As Built Heat Balance Diagram (100 Percent Reactor Power)
1X4DE301	1.2.2	Turbine Building Equipment Layout
1X4DE302	1.2.2	Turbine Building Equipment Layout
1X4DE303	1.2.2	Turbine Building Equipment Layout
1X4DE304	1.2.2	Turbine Building Equipment Layout
1X4DE312	1.2.2	Equipment Location Layout Containment, Control, and Fuel Handling Buildings Plan Level 4 el. 280 ft 0 in. (Unit 1)
1X4DE313	1.2.2	Equipment Location Layout Containment, Control, and Fuel Handling Buildings Plan Level 3 el. 260 ft 0 in. (Unit 1)
1X4DE314	1.2.2	Equipment Location Layout Containment, Control, and Fuel Handling Buildings Plan Level 2 el. 240 ft 0 in. (Unit 1)
1X4DE315	1.2.2	Auxiliary Building Equipment Location Layout Level 2 el. 240 ft 0 in. and Above (Unit 1)
1X4DE316	1.2.2	AFW Pumphouse and Condensate Storage Tank Equipment Location Layout
1X4DE317	1.2.2	Fuel Handling Building, Control Building, and Equipment Locations
1X4DE318	1.2.2	Equipment Location Layout Auxiliary Building Plan Level 1 el. 220 ft 0 in. to el. 240 ft 0 in. (Unit 1)
1X4DE320	1.2.2	Equipment Location Layout Containment, Control, and Fuel Handling Buildings Plan Level A el. 200 ft 0 in. (Unit 1)
1X4DE321	1.2.2	Equipment Location Layout Auxiliary Building Plan Level A el. 195 ft 0 in. to el. 220 ft 0 in. (Unit 1)
1X4DE322	1.2.2	Equipment Location Layout Containment, Control,

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Drawing Number	Section	Title
		and Fuel Handling Buildings Plan Level B el. 180 ft 0 in. (Unit 1)
1X4DE323	1.2.2	Equipment Location Layout Auxiliary Building Plan Level B el. 170 ft 6 in. to el. 195 ft 0 in. (Unit 1)
1X4DE324	1.2.2	Equipment Location Layout Auxiliary Building Plan Level C el. 143 ft 6 in. to el. 170 ft 6 in. (Unit 1)
1X4DE325	1.2.2	Equipment Location Layout Auxiliary Building Plan Level D el. 119 ft 3 in. to el. 143 ft 6 in. (Unit 1)
1X4DE327	1.2.2	Equipment Location Layout Diesel Generator Building (Unit 1)
1X4DE330	1.2.2	Equipment Location Layout Diesel Generator Building (Unit 1)
1X4DE502	9.1.5	Heavy Load Handling - Auxiliary Building Level B, el. 170 ft 6 in. to 195 ft 0 in. (Unit 1)
1X4DE503	9.1.5	Heavy Load Handling - Auxiliary Building Level A, el. 195 ft 0 in. to 220 ft 0 in. (Unit 1)
1X4DE504	9.1.5	Heavy Load Handling - Auxiliary Building Level 1, el. 220 ft 0 in. to 240 ft 0 in. (Unit 1)
1X4DE505	9.1.5	Heavy Load Handling - Auxiliary Building Level 2, el. 240 ft 0 in. and Above (Unit 1)
1X4DE506	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level A, el. 200 ft 0 in. (Unit 1)
1X4DE507	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level 1, el. 220 ft 0 in. (Unit 1)
1X4DE508	9.1.5	Heavy Load Handling - Containment Internal Section Looking North
1X4DE509	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level 2, el. 240 ft 0 in. (Unit 1)
1X4DE510	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level 3, el. 260 ft 0 in. (Unit 1)
1X4DE511	9.1.5	Heavy Load Handling - Diesel Generator Building (Unit 1)
1X4DE512	9.1.5	Heavy Load Handling - Auxiliary Feedwater Pump House and Condensate Storage Tank

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Drawing Number	Section	Title
1X4DE513	9.1.5	Heavy Load Handling - Control Building Piping Area 2B Level 4, el. 280 ft 0 in. and Above
1X4DE514	9.1.5	Heavy Load Handling - Control Building Piping Area 2B Level 3, el. 260 ft 0 in. to el. 280 ft 0 in.
1X4DE515	9.1.5	Heavy Load Handling - North Main Steam Valve Room Piping Area 2E Level 1, el. 220 ft 0 in. and Above
1X4DE516	9.1.5	Heavy Load Handling - Control Building Piping Area 2F Level 1, el. 220 ft 0 in. and Above
1X4DE517	9.1.5	Heavy Load Handling - North Feedwater Valve Room Piping Area 2E Level A, el. 200 ft 0 in. to el. 220 ft 0 in.
1X4DE518	9.1.5	Heavy Load Handling - Train A Nuclear Service Cooling Water Tower Level 1, el. 220 ft 0 in. (Unit 1)
1X4DE519	9.1.5	Heavy Load Handling - Train B Nuclear Service Cooling Water Tower Level 1, el. 220 ft 0 in. (Unit 1)
1X4DE520	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level B, el. 180 ft 0 in. (Unit 1)
1X4DE521	9.1.5	Heavy Load Handling - Auxiliary Building Level D, el. 119 ft 3 in. to 143 ft 3 in. (Unit 1)
1X4DE522	9.1.5	Heavy Load Handling - Auxiliary Building Level C, el. 143 ft 6 in. to 170 ft 6 in. (Unit 1)
1X4DE600	9.1.5	Safe Load Path for Reactor Upper Internal Containment Building 2101-R4-001 (Unit 1)
1X4DE601	9.1.5	Safe Load Path for Reactor Head Containment Building 2101-R4-001 (Unit 1)
1X4DE602	9.1.5	Safe Load Path for Reactor Coolant Pump and Motor Containment Building 2101-R4-001 (Unit 1)
1X4DE603	9.1.5	Safe Load Path for Movement of New Fuel Containers Utilizing Spent Fuel Cask Bridge Crane A-2109-R4-001
1X4DE604	9.1.5	Safe Load Path for Reactor Internals Lifting Rig Containment Building 2101-R4-001 (Unit 1)

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Drawing Number	Section	Title
1X4DE605	9.1.5	Safe Load Path for Backflushable Filter Monorail Auxiliary Building Level B Monorail 2108-R4-012 (Unit 1)
1X4DE606	9.1.5	Safe Load Path for Cartridge Filter Monorail Auxiliary Building Level D Monorail 2108-R4-064 (Unit 1)
1X4DE607	9.1.5	Safe Load Path for Auxiliary Feedwater Pumphouse Sump Pump Hatches (Unit 1)
1X4DE608	9.1.5	Safe Load Path for Containment Building, Carbon Bed Containers
1X4DJ4103	6.2.2	Containment Cooling System Equipment Location
1X4DJ4113	6.2.2	Containment Cooling System Equipment Location
1X4DJ4123	6.2.2	Containment Cooling System Equipment Location
1X4DJ4133	6.2.2	Containment Cooling System Equipment Location
1X4DL4A01	6.5.2	Spray Nozzle Locations and Orientations
1X4DL4A14	3.8.2	Containment Building - Containment Wall Piping Penetration Design List (Unit 1)
1X4DL4A17	5.1.2	Reactor Coolant Loops Plan and Section (Unit 1)
1X4DL4D01	6.5.2	Spray Nozzle Locations and Orientations
1X4DL4D09	6.5.2	Spray Nozzle Locations and Orientations
1X5AB01-66	18.1	Control Room Panels, QPCP, Section 1 (Unit 1)
1X5AB01-67	18.1	Control Room Panels, QPCP, Section 2 (Unit 1)
1X5AB01-557	18.1	Control Room Panels, QHVC, Section 1 (Unit 1)
1X5AB01-560	18.1	Control Room Panels, QHVC, Sections 2 and 3 (Unit 1)
1X5AB01-563	18.1	Control Room Panels, QPCP, Section 3 (Unit 1)
1X5DN002-1	1.7.1	Legend for Control Logic Drawings
1X5DN002-2	1.7.1	Legend for Control Logic Drawings
1X5DN002-3	1.7.1	Legend for Control Logic Drawings
1X5DN013-1	7.3.11	Containment Coolers

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Drawing Number	Section	Title
1X5DN013-2	7.3.11	Containment Coolers
1X5DN013-4	7.3.3	Containment Combustible Gas Control System
1X5DN013-4	7.3.11	Containment Coolers
1X5DN015-1	7.3.3	Containment Combustible Gas Control System
1X5DN017-2	7.3.3	Containment Combustible Gas Control System
1X5DN019-1	7.3.4	Containment Purge Isolation Logic Diagram
1X5DN019-2	7.3.4	Containment Purge Isolation Logic Diagram
1X5DN030-1	7.3.13	Auxiliary Building ESF HVAC System Actuation Logic Diagram
1X5DN030-3	7.3.13	Auxiliary Building ESF HVAC System Actuation Logic Diagram
1X5DN030-4	7.3.13	Auxiliary Building ESF HVAC System Actuation Logic Diagram
1X5DN030-5	7.3.13	Auxiliary Building ESF HVAC System Actuation Logic Diagram
1X5DN044-1	7.3.12	Control Building ESF HVAC System Actuation Logic Diagram
1X5DN045-1	7.3.12	Control Building ESF HVAC System Actuation Logic Diagram
1X5DN058-1	7.3.15	Diesel Generator Building ESF HVAC System Actuation Logic Diagram
1X5DN058-3	7.3.15	Diesel Generator Building ESF HVAC System Actuation Logic Diagram
1X5DN058-4	7.3.15	Diesel Generator Building ESF HVAC System Actuation Logic Diagram
1X5DN058-5	7.3.15	Diesel Generator Building ESF HVAC System Actuation Logic Diagram
1X5DN065-1	7.3.13	Auxiliary Building ESF HVAC System Actuation Logic Diagram
1X5DN068-1	7.3.14	Auxiliary Feedwater Pumphouse ESF HVAC System Actuation Logic Diagram

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Drawing Number	Section	Title
1X5DN068-3	7.3.14	Auxiliary Feedwater Pumphouse ESF HVAC System Actuation Logic Diagram
1X5DN069-1	7.3.16	Electrical Tunnel ESF HVAC System Actuation Logic Diagram
1X5DN086-1	7.3.9	NSCW System Logic Diagram
1X5DN087-1	7.3.9	NSCW System Logic Diagram
1X5DN087-2	7.3.9	NSCW System Logic Diagram
1X5DN087-3	7.3.9	NSCW System Logic Diagram
1X5DN087-4	7.3.9	NSCW System Logic Diagram
1X5DN087-5	7.3.9	Control Logic Diagram Nuclear Service Cooling Water Pumps
1X5DN089-1	7.3.9	NSCW System Logic Diagram
1X5DN089-2	7.3.9	NSCW System Logic Diagram
1X5DN089-3	7.3.9	NSCW System Logic Diagram
1X5DN090-1	7.3.9	NSCW System Logic Diagram
1X5DN090-2	7.3.9	NSCW System Logic Diagram
1X5DN090-3	7.3.9	NSCW System Logic Diagram
1X5DN090-6	7.3.9	Control Logic Diagram Nuclear Service Cooling Water System Auxiliaries and Alarms
1X5DN091-1	7.3.10	Component Cooling Water System Logic Diagram
1X5DN091-2	7.3.10	Component Cooling Water System Logic Diagram
1X5DN091-3	7.3.10	Component Cooling Water System Logic Diagram
1X5DN092-1	7.3.10	Component Cooling Water System Logic Diagram
1X5DN092-2	7.3.10	Component Cooling Water System Logic Diagram
1X5DN094-3	7.6.6	RCP Thermal Barrier Cooling Water Isolation Logic Diagram
1X5DN107-1	7.3.17	Diesel Generator Fuel Oil System Actuation Logic Diagram

TABLE 1.7.1-1 (Sheet 19 of 43)

Drawing Number	Section	Title
1X5DN114-4	7.6.6	Refueling Water Storage Tank Isolation Logic Diagram
1X5DN117-1	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN117-2	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN117-3	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN120-1	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN120-2	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN120-3	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN120-5	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN120-6	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN121-1	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN121-2	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN122-1	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN122-2	7.3.7	Auxiliary Feedwater Actuation Logic Diagram
1X5DN149-1	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN149-2	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN149-3	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN149-4	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN150-1	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN150-2	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN150-3	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN150-4	7.3.8	Main Steam and Feedwater Isolation Logic Diagram
1X5DN151-1	7.6.6	CVCS Letdown Line Isolation Logic Diagram
1X5DN151-2	7.6.6	CVCS Letdown Line Isolation Logic Diagram
1X5DN152-1	7.6.6	Steam Generator Blowdown Line Isolation Logic Diagram

TABLE 1.7.1-1 (Sheet 20 of 43)

Drawing Number	Section	Title
1X5DN152-2	7.6.6	Steam Generator Blowdown Line Isolation Logic Diagram
1X5DN203-1	10.2.2	Control Logic Diagram Main Turbine Tripping System
1X5DV002	1.7.1	Legend for Control Loop Drawings
1X6AA02-225	7.2.1	Index and Symbols
1X6AA02-226	7.2.1	Reactor Trip Signals
1X6AA02-227	7.2.1	Nuclear Instrument and Manual Trip Signals
1X6AA02-228	7.2.1	Nuclear Instrument Permissives and Blocks
1X6AA02-229	7.2.1	Primary Coolant System Trip Signals
1X6AA02-230	7.2.1	Pressurizer Trip Signals
1X6AA02-231	7.2.1	Steam Generator Trip Signals
1X6AA02-232	7.2.1	Safeguard Actuation System
1X6AA02-233	7.2.1	Rod Controls and Rod Stops
1X6AA02-234	7.2.1	Steam Dump Control
1X6AA02-235	7.2.1	Pressurizer Pressure Control
1X6AA02-236	7.2.1	Pressurizer Heater Control
1X6AA02-237	7.2.1	Feedwater Control and Isolation
1X6AA02-238	7.2.1	Feedwater Control and Isolation
1X6AA02-239	7.2.1	Auxiliary Feedwater Pumps Startup
1X6AA02-240	7.2.1	Turbine Trips, Runback, and Other Signals
1X6AA02-494	7.2.1	Pressurizer Pressure Relief System (Train A)
1X6AA02-495	7.2.1	Pressurizer Pressure Relief System (Train B)
1X6AA02-496	7.2.1	Pressurizer Level Control
1X6AA02-519	7.2.1	Safeguard Actuation System
1X6AH02-30000	6.3.2	Charging Pump Orifice Assembly (Unit 1)
1X6AN10B-66	1.2.10	Spent Fuel Pool-Spent Fuel Rack Layout (Unit 1)

TABLE 1.7.1-1 (Sheet 21 of 43)

Drawing Number	Section	Title
1X6AVO2-46	18.1	Control Room Panels, QMCB, Section A1 (Unit 1)
1X6AVO2-47	18.1	Control Room Panels, QMCB, Section A2 (Unit 1)
1X6AVO2-48	18.1	Control Room Panels, QMCB, Section C (Unit 1)
1X6AVO2-49	18.1	Control Room Panels, QMCB, Section B1 (Unit 1)
1X6AVO2-50	18.1	Control Room Panels, QMCB, Section B2 (Unit 1)
1X6AVO2-60	18.1	Control Room Panels QMCB, Section D (Unit 1)
1X6DD001	12.3.1	Radiation Zones - Accessibility During Operations - Containment Building - Floor Plan el. 171 ft 9 in.
1X6DD002	12.3.1	Radiation Zones - Accessibility During Operations - Containment Building - Floor Plan el. 195 ft 0 in.
1X6DD003	12.3.1	Radiation Zones - Accessibility During Operations - Containment Building and Equipment Building - Floor Plan el. 220 ft 0 in.
1X6DD004	12.3.1	Radiation Zones - Accessibility During Operations - Diesel Generator Building - Floor Plan
1X6DD100	12.3.1	Radiation Zone Map Post TMI 24 Hours - Diesel Generator Building Floor Plans
1X6DD101	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Feedwater Pump House - Floor Plan
1X6DD200	12.3.1	Personnel Flow - Containment Building - Floor Plan el. 171 ft 9 in. (Unit 1)
1X6DD201	12.3.1	Personnel Flow - Containment Building - Floor Plan el. 195 ft 0 in. (Unit 1)
1X6DD202	12.3.1	Personnel Flow - Containment Building and Equipment Building-Floor Plan el. 220 ft 0 in. (Unit 1)
1X6DD203	12.3.1	Personnel Flow - Diesel Generator Building - Floor Plan
1X6DD300	3F	Auxiliary Feedwater Pump House Area Nodal Boundary
1X6DD301	3F	Nodal Boundary MSIV/MFIV Area – Control Building
1X6DD302	3F	Nodal Boundary MSIV/MFIV Area – Control Building

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Drawing Number	Section	Title
2X2D01J019	3.8.2	Containment - Wall Liner Plate Penetrations Details Sh. 1
2X3D-AA-A01A	8.3.1	Main One Line (Unit 2)
2X3D-AA-G01A	8.3.1	Class 1E dc and 120-Volt ac Power Supply
2X3D-AA-K02A	8.3.1	Safety Load Sequencer Table (Train A, Unit 2)
2X3D-AA-K02B	8.3.1	Safety Load Sequencing Table (Train A, Unit 2)
2X3DG030	9.5.3	Lighting Systems Schematic
2X4DB111	5.1.2	Reactor Coolant System P&ID (Unit 2)
2X4DB112	5.1.2	Reactor Coolant System P&ID (Unit 2)
2X4DB116-1	6.3.2	Chemical and Volume Control System
2X4DB116-2	9.3.4	Chemical and Volume Control System
2X4DB119	6.3.2	Safety Injection System (Unit 2)
2X4DB120	6.3.2	Safety Injection System P&ID
2X4DB121	6.3.2	Safety Injection System P&ID
2X4DB122	5.4.7	Residual Heat Removal System P&ID (Unit 2)
2X4DB140	9.3.2	Nuclear Sampling System - Liquid (Unit 2)
2X4DB149-1*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 2)
2X4DB149-2*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 2)
2X4DB149-3*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 2)
2X4DB149-4*	9.2.1	Flow Diagram for NSCW, CCW, and ACCW Systems (Unit 2)
2X4DB159-1	10.3.2	Main Steam Supply System (Unit 2)
2X4DB170-1	9.5.4	Diesel Generator System
2X4DB170-2	9.5.4	Diesel Generator System
2X4DB174-2	9.5.1	Fire Protection Water Systems (Unit 2)

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Drawing Number	Section	Title
2X4DB174-4	9.5.1	Fire Protection Water Systems (Unit 2)
2X4DB203	9.4.9	P&ID - Equipment Building HVAC System (Unit 2)
2X4DB205-1	9.4.3	Piping Penetration Area Filtration and Exhaust System (Unit 2)
2X4DB208-1	9.4.3	Auxiliary Building Normal Ventilation System (Unit 2)
2X4DB228	9.4.2	P&ID - ESF Room Coolers System
2X4DB245	9.4.9	P&ID - Piping Penetration Ventilation System (Unit 2)
2X4DB250-1	9.4.9	Miscellaneous HVAC Systems Flow Diagrams (Unit 2)
2X4DB259-1	9.4.5	Control Building Electrical and Pipe Penetration Room Filter and Exhaust System (Unit 2)
2X4DB266-1	9.4.1	Control Building Level B HVAC Flow Diagrams
2X4DC103	1.2.3	As-Built Heat Balance Diagram (100 Percent Reactor Power)
2X4DE327	9.5.8	Equipment Location Layout, Diesel Generator Building (Unit 2)
2X4DE330	9.5.8	Equipment Location Layout, Diesel Generator Building (Unit 2)
2X4DE502	9.1.5	Heavy Load Handling - Auxiliary Building Level B, el. 170 ft 6 in. to 195 ft 0 in. (Unit 2)
2X4DE503	9.1.5	Heavy Load Handling - Auxiliary Building Level A, el. 195 ft 0 in. (Unit 2)
2X4DE504	9.1.5	Heavy Load Handling - Auxiliary Building Level 1, el. 220 ft 0 in. to 240 ft 0 in. (Unit 2)
2X4DE505	9.1.5	Heavy Load Handling - Auxiliary Building Level 2, el. 240 ft 0 in. and Above (Unit 2)
2X4DE506	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level A, el. 200 ft 0 in. (Unit 2)
2X4DE507	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level 1, el. 220 ft 0 in. (Unit 2)
2X4DE508	9.1.5	Heavy Load Handling - Containment Internal Section Looking North

TABLE 1.7.1-1 (Sheet 24 of 43)

Drawing Number	Section	Title
2X4DE509	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level 2, el. 240 ft 0 in. (Unit 2)
2X4DE510	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level 3, el. 260 ft 0 in. (Unit 2)
2X4DE511	9.1.5	Heavy Load Handling - Diesel Generator Building (Unit 2)
2X4DE512	9.1.5	Heavy Load Handling - Auxiliary Feedwater Pump House and Condensate Storage Tank
2X4DE514	9.1.5	Heavy Load Handling - Control Building Piping Area 2B Level 3, el. 260 ft 0 in. to 280 ft 0 in.
2X4DE515	9.1.5	Heavy Load Handling - North Main Steam Valve Room Piping Area 2E Level 1, el. 220 ft 0 in. and Above
2X4DE516	9.1.5	Heavy Load Handling - Control Building Piping Area 2F Level 1, el. 220 ft 0 in. and Above
2X4DE517	9.1.5	Heavy Load Handling - North Feedwater Valve Room Piping Area 2E Level A, el. 200 ft 0 in. to el. 220 ft 0 in.
2X4DE518	9.1.5	Heavy Load Handling - Train A Nuclear Service Cooling Water Tower Level 1, el. 220 ft 0 in. (Unit 2)
2X4DE519	9.1.5	Heavy Load Handling - Train B Nuclear Service Cooling Water Tower Level 1, el. 220 ft 0 in. (Unit 2)
2X4DE520	9.1.5	Heavy Load Handling - Containment, Control, and Fuel Handling Building Level B, el. 180 ft 0 in. (Unit 2)
2X4DE521	9.1.5	Heavy Load Handling - Auxiliary Building Level D, el. 119 ft 3 in. to 143 ft 3 in. (Unit 2)
2X4DE522	9.1.5	Heavy Load Handling - Auxiliary Building Level C, el. 143 ft 6 in. to 170 ft 6 in. (Unit 2)
2X4DE600	9.1.5	Safe Load Path for Buttress No. 3 Cranes Control Building, Level 3 el. 260 ft 0 in. 2-2101-R4-016 and 2-2101-R4-015 (Unit 2)
2X4DE601	9.1.5	Safe Load Path for Cartridge Filter Monorail Auxiliary Building Level D Monorail 2108-R4-064 (Unit 2)

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Drawing Number	Section	Title
2X4DL4A14	3.8.2	Containment Building - Containment Wall Piping Penetration Design List (Unit 2)
2X4DL4A17	5.1.2	Reactor Coolant Loops Plan and Section (Unit 2)
2X6AH02-30000	6.3.2	Charging Pump Orifice Assembly (Unit 2)
2X6DD001	12.3.1	Radiation Zones – Accessibility During Operations – Containment Building – Floor Plan, el. 171 ft 9 in.
2X6DD002	12.3.1	Radiation Zones – Accessibility During Operations – Containment Building – Floor Plan, el. 195 ft 0 in.
2X6DD003	12.3.1	Radiation Zones – Accessibility During Operations – Containment Building and Equipment Building – Floor Plan, el. 220 ft. 0 in.
2X6DD200	12.3.1	Personnel Flow – Containment Building – Floor Plan, el. 171 ft 9 in. (Unit 2)
2X6DD201	12.3.1	Personnel Flow – Containment Building – Floor Plan, el. 195 ft 0 in. (Unit 2)
2X6DD202	12.3.1	Personnel Flow –Containment Building and Equipment Building – Floor Plan, el. 220 ft 0 in. (Unit 2)
AX1D08A02-2	3F	Auxiliary Building Level D (el. 119 ft 3 in.) Hazards Analysis Room Locations
AX1D08A02-3	3F	Auxiliary Building Level D (el. 119 ft 3 in.) Hazards Analysis Room Locations
AX1D08A03-3	3F	Auxiliary Building Level C (el. 143 ft 6 in.) Hazards Analysis Room Locations
AX1D08A03-4	3F	Auxiliary Building Level C (el. 143 ft 6 in.) Hazards Analysis Room Locations
AX1D08A04-4	3F	Auxiliary Building Level B (el. 170 ft 6 in.) Hazards Analysis Room Locations
AX1D08A31	3F	Auxiliary Building Level C (el. 143 ft 6 in.) Hazards Analysis Room Locations
AX1D11A04	6.4.2	Control Room Envelope
AX1D45A01	1.2.2	Location and Orientation of Buildings

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Drawing Number	Section	Title
AX2D09A017	12.3.2	Fuel Transfer Tube Shielding Details
AX2D11A048	3.8.4	Control Building Electrical Tunnel - Forming Plan at el. 200 ft 0 in. (Unit 1)
AX2D46T001	2.5.4	Excavation Plan Power Block
AX2D46T025	2.5.4	Noncategory Backfill Requirements Yard Area
AX2D55V001	2.5.4	Settlement Observation Markers Location and Details
AX2D94V031	3.8.5	Typical Waterstop Installation at Seismic Separation Joint
AX3D-AA-A01A	8.1	Main One Line
AX3D-AA-A03A	8.2.1	Master One Line
AX3D-AA-G02A	8.3.1	125-V dc High Voltage Substation
AX3D-AA-G02B	8.3.1	Common Non-Class 1E Essential ac Power System and dc Power System
AX3D-AA-G02C	8.3.2	Main One Line, CAS and SAS Distribution System
AX3D-AA-L50A	8.2.1	Switchyard Arrangement
AX3D-CA-L5OA	18.1	Control Room Panels QEAB, Section 1B (Unit 1)
AX3DG030	1.7.1	Legend for Communications and Lighting Drawings
AX3DL060	8.2.1	Ultimate Development - Substation
AX4DB105-4	9.3.3	Auxiliary Building and Miscellaneous Drains
AX4DB123-1	9.3.4	Boron Recycle System
AX4DB123-2	9.3.4	Boron Recycle System
AX4DB124-2	11.2.2	Radwaste Processing Facility Demineralizer System
AX4DB124-3	11.2.2	Radwaste Processing Facility Demineralizer System
AX4DB124-4	11.2.2	Radwaste Processing Facility Dewatering System
AX4DB124-5	11.2.2	Radwaste Processing Facility Filtration System
AX4DB152-2	2.4.13	Waste Water Effluent System

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Drawing Number	Section	Title
AX4DB152-3	2.4.13	Waste Water Effluent System
AX4DB158-1	9.3.7	Circulating Water Chemical Injection System
AX4DB158-3	9.3.7	Circulating Water Chemical Injection System
AX4DB175-5	9.3.1	Instrument and Service Air System
AX4DB176-1	9.3.5	Auxiliary Gas System P&ID
AX4DB176-2	9.3.5	Auxiliary Gas System P&ID
AX4DB177	9.2.3	Demineralized Water System
AX4DB178	9.2.3	Demineralized Water Makeup System
AX4DB186-3	9.3.1	Instrument and Service Air System
AX4DB190-1	9.2.3	Demineralized Water System
AX4DB190-2	9.2.3	Plant Demineralized Water System
AX4DB195-2	9.5.9	Auxiliary Steam System
AX4DB195-3	9.5.9	Auxiliary Steam System Distribution (Unit 1)
AX4DB195-4	9.5.9	Auxiliary Steam System Distribution (Unit 2)
AX4DB195-5	9.5.9	Auxiliary Steam System Fuel Oil
AX4DB198-1	9.2.4	Potable Water System
AX4DB198-2	9.2.4	Potable Water System
AX4DB198-3	9.3.7	River Water Intake Structures System-Chlorine
AX4DB204-1	9.4.2	Fuel Handling Building Ventilation System
AX4DB204-2	9.4.2	Fuel Handling Building Ventilation System
AX4DB206-1	9.4.1	Control Room HVAC System Level 1
AX4DB206-2	9.4.1	Control Room HVAC System Level 1
AX4DB206-3	9.4.1	Control Room HVAC System Level 1
AX4DB215	9.4.1	Control Building Level 1 HVAC System
AX4DB216	9.4.1	Control Building Cable Spreading Room HVAC System Levels 2 and 3

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Drawing Number	Section	Title
AX4DB218	9.2.9	Normal Chilled Water System
AX4DB223	9.4.1	Control Building Normal HVAC System Level 2
AX4DB225	9.4.1	Control Building Cable Spreading Room HVAC System Level A
AX4DB231	9.2.9	Normal Chilled Water System
AX4DB232	9.2.9	Normal Chilled Water System
AX4DB235	9.4.1	Onsite Technical Support Center HVAC System
AX4DB237	9.4.1	Control Building Smoke Exhaust System
AX4DB241	9.4.1	Control Building HVAC Equipment Room Levels 3 and 4
AX4DB242-1	9.4.1	Onsite Technical Support Center HVAC System
AX4DB255-1	6.4.3	Control Room HVAC System, Level 1
AX4DB255-3	6.4.3	Control Room HVAC System, Level 1
AX4DB256-1	9.4.1	Control Building Control Room HVAC Flow Diagrams
AX4DB260-1	9.4.2	Fuel Handling Building HVAC Flow Diagram
AX4DB266-2	9.4.1	Control Building Level B HVAC Flow Diagrams
AX4DB269	9.4.1	Control Building Level A HVAC Flow Diagram
AX4DB270	9.4.1	Control Building Cable Spreading Room Levels 2 and 3 HVAC Flow Diagram
AX4DB353(sh. 2)	9.4.3	Radwaste Building Ventilation System
AX4DB376	9.4.1	Onsite Technical Support Center HVAC Flow Diagram
AX4DB378(sh. 2)	9.4.3	Radwaste Building Flow Diagrams
AX4DB400	9.4.3	Radwaste Processing Facillity HVAC System P&ID
AX4DD300	1.2.2	Plot Plan
AX4DD301	1.2.2	General Arrangement Level 3 Plan - el. 260 ft 0 in.
AX4DD302	1.2.2	General Arrangement Level 2 Plan - el. 240 ft 0 in.
AX4DD303	1.2.2	General Arrangement Level 1 Plan - el. 220 ft 0 in.

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Drawing Number	Section	Title
AX4DD304	1.2.2	General Arrangement Level A Plan - el. 200 ft 0 in.
AX4DD305	1.2.2	General Arrangement Level B Plan - el. 180 ft 0 in.
AX4DD306-1	1.2.2	General Arrangement Level C Plan - el. 160 ft 0 in.
AX4DD306-2	1.2.2	General Arrangement Levels C and D Plan 143 ft 6 in. and 119 ft 3 in.
AX4DD307	1.2.2	General Arrangement Section A Looking North
AX4DD308	1.2.2	General Arrangement Section B Looking West
AX4DD309	1.2.2	General Arrangement Section C Looking West
AX4DD310	1.2.2	General Arrangement Section D Looking North
AX4DD311	1.2.2	General Arrangement Level 4 Plan - el 280 ft 0 in.
AX4DE350	1.2.2	Transfer Building Equipment Location Plan
AX4DE357	1.2.2	Radwaste Processing Facility Equipment Location Plan
AX4DE500	9.1.5	Heavy Load Handling - Alternate Radwaste Transfer Plan Level 1, el. 220 ft 0 in.
AX4DE501	9.1.5	Spent Fuel Cask Bridge Crane, Clearance and Hook Travel Envelope Plan View
AX4DE502	9.1.5	Spent Fuel Cask Bridge Crane, Clearance and Hook Travel Envelope Cross Sections
AX4DJ3101	12.3.3	Typical Layout for Air Cleaning Unit
AX4DJ8004	9A	Fire Areas Radwaste Transfer Building and Alternate Radwaste Building Level 1 Floor Plan el. 220 ft 0 in.
AX4DJ8005	9A	Fire Areas Radwaste Transfer Building Partial Level 1 Floor Plan el. 237 ft 0 in.
AX4DJ8006	9A	Fire Areas Radwaste Transfer Building Level 2 Floor Plan el. 247 ft 0 in.
AX4DJ8007	9A	Fire Areas Auxiliary Building Floor Plan el. 119 ft 3 in., Level D
AX4DJ8008	9A	Fire Areas Auxiliary Building Floor Plan el. 119 ft 3 in., Level D

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Drawing Number	Section	Title
AX4DJ8009	9A	Fire Areas Auxiliary Building Partial Plans Utility Chase, Levels C and D
AX4DJ8010	9A	Fire Areas Auxiliary Building Floor Plan el. 143 ft 6 in., Level C
AX4DJ8011	9A	Fire Areas Auxiliary Building Floor Plan el. 143 ft 6 in., Level C
AX4DJ8012	9A	Fire Areas Auxiliary Building Floor Plan el. 170 ft 6 in., Level B
AX4DJ8013	9A	Fire Areas Auxiliary Building Floor Plan el. 170 ft 6 in., Level B
AX4DJ8014	9A	Fire Areas Auxiliary Building Partial Plans Utility Chase Levels A and B
AX4DJ8015	9A	Fire Areas Auxiliary Building Floor Plan el. 195 ft 0 in., Level A
AX4DJ8016	9A	Fire Areas Auxiliary Building Floor Plan el. 195 ft 0 in., Level A
AX4DJ8017	9A	Fire Areas Auxiliary Building Floor Plan el. 220 ft 0 in., Level 1
AX4DJ8018	9A	Fire Areas Auxiliary Building Floor Plan el. 220 ft 0 in., Level 1
AX4DJ8019	9A	Fire Areas Auxiliary Building Floor Plan el. 240 ft 0 in., Level 2
AX4DJ8020	9A	Fire Areas Auxiliary Building Floor Plan el. 240 ft 0 in., Level 2
AX4DJ8021	9A	Fire Areas Control Building Electrical Tunnels el. 160 ft 0 in., Level C
AX4DJ8022	9A	Fire Areas Control Building Floor Plan el. 180 ft 0 in., Level B
AX4DJ8023	9A	Fire Areas Control Building Floor Plan el. 180 ft 0 in., Level B
AX4DJ8024	9A	Fire Areas Control Building Floor Plan el. 200 ft 0 in., Level A
AX4DJ8025	9A	Fire Areas Control Building Floor Plan el. 200 ft 0 in., Level A

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Drawing Number	Section	Title
AX4DJ8026	9A	Fire Areas Control Building Floor Plan el. 220 ft 0 in., Level 1
AX4DJ8027	9A	Fire Areas Control Building Floor Plan el. 240 ft 0 in., Level 2
AX4DJ8028	9A	Fire Areas Control Building Floor Plan el. 260 ft 0 in., Level 3
AX4DJ8029	9A	Fire Areas Control Building Floor Plan el. 280 ft 0 in. Level 4
AX4DJ8030	9A	Fire Areas Fuel Handling Building Floor Plan el. 160 ft 0 in. and el. 179 ft 1/2 in., Levels C and D
AX4DJ8031	9A	Fire Areas Fuel Handling Building Floor Plan el. 200 ft. 0 in., Level A
AX4DJ8032	9A	Fire Areas Fuel Handling Building Floor Plan el. 220 ft. 0 in. and el. 263 ft 8 in., Levels 1 and 3
AX4DJ8033	9A	Fire Areas Containment Building Floor Plan el. 171 ft 9 in., Level B, Platform el. 184 ft and 185 ft
AX4DJ8034	9A	Fire Areas Containment Building Platform el. 195 ft 2 in. and el. 197 ft 6 in., Level A
AX4DJ8035	9A	Fire Areas Containment Building and Equipment Building Floor Plan el. 220 ft 0 in., Level 1
AX4DJ8036	9A	Fire Areas Containment Building Platform el. 238 ft and 261 ft Equipment Building Roof el. 236 ft, Levels 2 and 3
AX4DJ8037	9A	Fire Areas Diesel Fuel Oil Storage, Pump Room, Valve Room, and Diesel Generator Building Floor Plans
AX4DJ8038	9A	Fire Areas Auxiliary Feedwater Pump House Roof and Floor Plan and Details
AX4DJ8039	9A	Fire Areas NSCW Pump House Unit 1 Plans and Sections - Trains A and B
AX4DJ8040	9A	Fire Areas NSCW & Misc. Category 1 Structures Key Plan - Unit 1
AX4DJ8041	9A	Fire Areas NSCW and Miscellaneous Category 1 Structures Key Plan - Unit 2

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Drawing Number	Section	Title
AX4DJ8042	9A	Fire Areas Containment Building Floor Plan el. 171 ft and 9 in., Platform el. 184 ft and 185 ft, Level B
AX4DJ8043	9A	Fire Areas Containment Building Platform el. 195 ft 2 in. and 197 ft 6 in., Level A
AX4DJ8044	9A	Fire Areas Containment Building and Equipment Building Floor Plan el. 220 ft 0 in., Level 1
AX4DJ8045	9A	Fire Areas Containment Building Platform el. 238 ft and 261 ft, Equipment Building Roof el. 236 ft, Levels 2 and 3
AX4DJ8046	9A	Fire Areas Diesel Fuel Oil Storage Pump Room, Valve Room, and Diesel Generator Building Floor Plans
AX4DJ8047	9A	Fire Areas Auxiliary Feedwater Pump House Roof and Floor Plan and Details
AX4DJ8048	9A	Fire Areas NSCW Pump House Unit 2 Plans and Sections - Trains A and B
AX4DJ8051	9A	Fire Areas Outage Storage Building Floor Plan
AX5DN008-1	7.3.12	Control Building ESF HVAC System Actuation Logic Diagram
AX5DN020-1	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN020-10	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX4DJ8050	9A	Radwaste Processing Facility Floor Plan el. 217 ft 6 in.
AX5DN020-2	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN020-3	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN020-4	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN020-5	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN020-6	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN020-8	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN020-10	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN032-1	7.3.6	Control Room Ventilation Isolation Logic Diagram

TABLE 1.7.1-1 (Sheet 33 of 43)

Drawing Number	Section	Title
AX5DN027-1	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN028-1	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN029-1	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN029-3	7.3.5	FHB Ventilation Isolation Logic Diagram
AX5DN031-1	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN031-2	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN031-4	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN032-1	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN032-3	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN034-2	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN037-1	7.3.6	Control Room Ventilation Isolation Logic Diagram
AX5DN056-2	7.3.12	Control Building ESF HVAC System Actuation Logic Diagram
AX5DN056-6	7.3.12	Control Building Cable Spreading Room HVAC Logic Diagram
AX6AN10A-6	1.2.10	Pool Layout-Spent Fuel Racks (Unit 2)
AX6DD001	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level D, el. 119 ft 3 in.
AX6DD002	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, el. 119 ft 3 in.
AX6DD003	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level C, el. 143 ft 6 in.
AX6DD004	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level C, el. 143 ft 6 in.
AX6DD005	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level B, el. 170 ft 6 in.
AX6DD006	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level B, el. 170 ft 6 in.
AX6DD007	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level A, el. 195 ft 0 in.

TABLE 1.7.1-1 (Sheet 34 of 43)

Drawing Number	Section	Title
AX6DD008	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level A, el. 195 ft 0 in.
AX6DD009	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD010	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD011	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level 2, el. 240 ft 0 in.
AX6DD012	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building Floor Plan, Level 2, el. 240 ft 0 in.
AX6DD013	12.3.1	Radiation Zones - Accessibility During Operations - Partial Plans Utility Chase Levels C & B
AX6DD014	12.3.1	Radiation Zones – Accessibility During Operations – Auxiliary Building – Partial Plans Utility Chase Levels C & D
AX6DD015	12.3.1	Radiation Zones – Accessibility During Operations – Fuel Handling Building – Floor Plan, Levels C & B, el. 160 ft 0 in. and el. 179 ft ½ in.
AX6DD016	12.3.1	Radiation Zones – Accessibility During Operations – Fuel Handling Building – Floor Plan, Level A, el. 200 ft 0 in.
AX6DD017	12.3.1	Radiation Zones - Accessibility During Operations - Fuel Handling Building – Floor Plan Levels 1 & 3, el. 220 ft 0 in. and el. 263 ft 8 in.
AX6DD018	12.3.1	Radiation Zones - Accessibility During Operations - Control Building – Electrical Tunnels Floor Plan Level C, el. 160 ft 0 in.
AX6DD019	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level B, el. 180 ft 0 in.
AX6DD020	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level B, el. 180 ft 0 in.
AX6DD021	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level A, el. 200 ft 0 in.
AX6DD022	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level A, el. 200 ft 0 in.

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Drawing Number	Section	Title
AX6DD023	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level 1, el. 220 ft 0 in.
AX6DD024	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level 2, el. 240 ft 0 in.
AX6DD025	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level 3, el. 260 ft 0 in.
AX6DD026	12.3.1	Radiation Zones - Accessibility During Operations - Control Building - Floor Plan Level 4, el. 280 ft 0 in.
AX6DD027	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Feedwater Pump House - Floor Plan (Turbine Building Floor Plan)
AX6DD028	12.3.1	Radiation Zones - Accessibility During Operations - Radwaste Transfer Building - Floor Plans Levels 1 & 2, el. 220 ft 0 in., 237 ft 0 in., and 247 ft 0 in.
AX6DD029	12.3.1	Radiation Zones - Accessibility During Operations - Auxiliary Building - Partial Plans Utility Chase Levels A & B
AX6DD030	12.3.1	Radiation Zones - Accessibility During Operations - Alternate Radwaste Building - Floor Plans Level 1, el. 220 ft 0 in.
AX6DD031	12.3.1	Radiation Zones – Accessibility During Operations – Radwaste Processing Facility – el. 217 ft 6 in.
AX6DD100	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level D, el. 119 ft 3 in.
AX6DD101	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level D, el. 119 ft 3 in.
AX6DD102	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level C, el. 143 ft 6 in.
AX6DD103	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level C, el. 143 ft 6 in.
AX6DD104	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level B, el. 170 ft 6 in.
AX6DD105	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level B, el. 170 ft 6 in.
AX6DD106	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level A, el. 195 ft 0 in.

TABLE 1.7.1-1 (Sheet 36 of 43)

Drawing Number	Section	Title
AX6DD107	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level A, el. 195 ft 0 in.
AX6DD108	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD109	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD110	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building - Partial Plans Utility Chase Levels C & B
AX6DD111	12.3.1	Radiation Zone Map Post TMI 24 Hours – Fuel Handling Building - Floor Plan, Levels C & B, el. 160 ft 0 in. and 179 ft 1/2 in.
AX6DD112	12.3.1	Radiation Zone Map Post TMI 24 Hours – Fuel Handling Building - Floor Plan, Level A, el. 200 ft 0 in.
AX6DD113	12.3.1	Radiation Zone Map Post TMI 24 Hours - Control Building - Floor Plan Level A, el. 200 ft 0 in.
AX6DD114	12.3.1	Radiation Zone Map Post TMI 24 Hours - Control Building - Floor Plan Level A, el. 200 ft 0 in.
AX6DD115	12.3.1	Radiation Zone Map Post TMI 24 Hours - Control Building - Floor Plan Level B, el. 180 ft 0 in.
AX6DD116	12.3.1	Radiation Zone Map Post TMI 24 Hours - Control Building - Floor Plan Level B, el. 180 ft 0 in.
AX6DD117	12.3.1	Radiation Zone Map Post TMI 24 Hours - Control Building - Floor Plan Level 1, el. 220 ft 0 in.
AX6DD118	12.3.1	Radiation Zone Map Post TMI 24 Hours - Technical Support Center - Floor Plan Level 1, el. 220 ft 0 in.
AX6DD119	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, Level 2, el. 240 ft 0 in.
AX6DD120	12.3.1	Radiation Zone Map Post TMI 24 Hours - Auxiliary Building Floor Plan, el. 240 ft 0 in. and 260 ft 0 in., Levels 2 and 3.
AX6DD200	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level D, el. 119 ft 3 in.
AX6DD201	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level D, el. 119 ft 3 in.

TABLE 1.7.1-1 (Sheet 37 of 43)

Drawing Number	Section	Title
AX6DD202	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level C, el. 143 ft 6 in.
AX6DD203	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level C, el. 143 ft 6 in.
AX6DD204	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level B, el. 170 ft 6 in.
AX6DD205	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level B, el. 170 ft 6 in.
AX6DD206	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level A, el. 195 ft 0 in.
AX6DD207	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level A, el. 195 ft 0 in.
AX6DD208	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD209	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD210	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level 2, el. 240 ft 0 in.
AX6DD211	12.3.1	Personnel Flow - Auxiliary Building Floor Plan, Level 2, el. 240 ft 0 in.
AX6DD212	12.3.1	Personnel Flow - Auxiliary Building - Partial Plans Utility Chase Levels C & B
AX6DD213	12.3.1	Personnel Flow - Fuel Handling Building - Floor Plan, Levels C & B, el. 160 ft 0 in. and el. 179 ft 1/2 in.
AX6DD214	12.3.1	Personnel Flow - Fuel Handling Building - Floor Plan, Level A, el. 200 ft 0 in.
AX6DD215	12.3.1	Personnel Flow - Fuel Handling Building - Floor Plan, Levels 1 & 3, el. 220 ft 0 in. and el. 263 ft 8 in.
AX6DD216	12.3.1	Personnel Flow - Control Building - Electrical Tunnels Floor Plan, Level C, el. 160 ft 0 in.
AX6DD217	12.3.1	Personnel Flow - Control Building - Floor Plan, Level B, el. 180 ft 0 in.
AX6DD218	12.3.1	Personnel Flow - Control Building - Floor Plan, Level B, el. 180 ft 0 in.

TABLE 1.7.1-1 (Sheet 38 of 43)

Drawing Number	Section	Title
AX6DD219	12.3.1	Personnel Flow - Control Building - Floor Plan, Level A, el. 200 ft 0 in.
AX6DD220	12.3.1	Personnel Flow - Control Building - Floor Plan, Level A, el. 200 ft 0 in.
AX6DD221	12.3.1	Personnel Flow - Control Building - Floor Plan, Level 1, el. 220 ft 0 in.
AX6DD222	12.3.1	Personnel Flow - Control Building - Floor Plan, Level 2, el. 240 ft 0 in.
AX6DD223	12.3.1	Personnel Flow - Control Building - Floor Plan, Level 3, el. 260 ft 0 in.
AX6DD224	12.3.1	Personnel Flow - Control Building - Floor Plan, Level 4, el. 280 ft 0 in.
AX6DD225	12.3.1	Personnel Flow - Auxiliary Feedwater Pump House - Floor Plan (Turbine Building Floor Plan)
AX6DD226	12.3.1	Personnel Flow – Radwaste Processing Facility – Floor Plan, el. 217 ft 6 in.
AX6DD300	3F	Nodal Boundary - MSIV/MFIV Area – Auxiliary Building
AX6DD301	3F	Nodal Boundary - MSIV/MFIV Area – Auxiliary Building
AX6DD302	3F	Nodal Boundary - MSIV/MFIV Area – Auxiliary Building
AX6DD303	3.5.1	Plan - Turbine Missile Target Identification
AX6DD304	3.5.1	Power Block - Turbine Missile Plan
AX6DD305	3.5.1	Turbine Missile Targets - Section "A"
AX6DD306	3.5.1	Turbine Missile Targets - Section "B"
AX6DD307	3.5.1	Turbine Missile Targets - Section "C"
AX6DD308	3.5.1	Turbine Missile Targets - Section "D"
AX6DD309	3.6.2	Location of Postulated Breaks in the Reactor Coolant Loop
AX6DD310	6.4.2	Control Room Intake Relationship to Radiological and Toxic Chemical Release Points

TABLE 1.7.1-1 (Sheet 39 of 43)

Drawing Number	Section	Title
AX6DD311	2.1.1	Site Effluent Release Points
AX6DD312	3.8.3	Reactor Coolant System Component Supports and Arrangement (Unit 1)
AX6DD313	3.8.1	Containment Tendon End Anchorage Assembly (Unit 1 Shown)
AX6DD314	3.8.3	Primary and Secondary Shield Wall Anchorage to Basemat
AX6DD315	2.4.3	PMP Isohyets Position 1, Hydromet 51 Savannah River Basin Above Site
AX6DD316	2.4.3	PMP Isohyets Position 2, Hydromet 51 Savannah River Basin Above Site
AX6DD317	2.4.3	PMP Isohyets Position 1, Hydromet 52 Savannah River Basin Above Site
AX6DD318	2.4.3	Probable Maximum Flood Hydrograph (Valley Storage Not Considered) at River Mile 151.1
AX6DD319	2.4.3	Probable Maximum Flood Hydrograph (Valley Storage Considered) at River Mile 151.1
AX6DD320	2.4.3	Location of Savannah River Valley Cross-sections Unsteady, Nonuniform Flow
AX6DD321	2.4.3	Wave Study Fetch Diagrams
AX6DD323	2.4.12	Observation Wells 1971 - 1985
AX6DD324	2.4.12	Construction Dewatering Plan
AX6DD325	2.4.12	Outlying Exploration Holes, Wells and Springs
AX6DD326	2.4.12	Hydrographs of Water Wells-Screven 3; CHA 84
AX6DD327	2.4.12	Piezometric Surface of Tertiary Aquifier October 1971
AX6DD328	2.4.12	Piezometric Surface of Tertiary Aquifier December 1984
AX6DD329	2.4.12	Contours of Water Table Aquifer November 1971
AX6DD330	2.4.12	Contours of Water Table Aquifer December 1984

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Drawing Number	Section	Title
AX6DD331	2.4.12	Stratigraphic Column Showing Lithology and Hydrogeologic Units
AX6DD332	2.4.12	Water Supply Wells
AX6DD333	2.4.12	Observation Wells November 1985
AX6DD334	2.4.12	Observation Wells November 1993
AX6DD335	2.4.12	Observation Wells November 1995
AX6DD336	2.4.12	Well Hydrographs Tertiary Aquifer
AX6DD337	2.4.12	Well Hydrographs Water Table Aquifer
AX6DD338	2.5.1	Regional Generalized Physiographical Map
AX6DD339	2.5.1	Stratigraphic Correlation Chart
AX6DD340	2.5.1	Lithologic Chart
AX6DD341	2.5.1	Regional Geological Map
AX6DD342	2.5.1	Generalized Geologic Section
AX6DD343	2.5.1	Site Map
AX6DD344	2.5.1	Location of Test Holes at River Facility
AX6DD345	2.5.1	Local Geologic Map
AX6DD351	2.5.1	Site Geologic Map
AX6DD352	2.5.1	Geologic Section A-A
AX6DD360	2.5.1	Geologic Map Power Block Area
AX6DD361	2.5.1	Geologic Map Power Block Area
AX6DD362	2.5.1	Geologic Map Power Block Area
AX6DD363	2.5.1	Geologic Map - Cavities in Utley Limestone Power Block Cut Slope
AX6DD364	2.5.1	Geologic Sections Auxiliary Bldg. Basement
AX6DD365	2.5.1	Geologic Sections Auxiliary Bldg. Basement
AX6DD366	2.5.1	Geologic Sections Auxiliary Bldg. Basement
AX6DD367	2.5.1	Geologic Sections Auxiliary Bldg. Basement

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Drawing Number	Section	Title
AX6DD368	2.5.1	Geologic Sections Auxiliary Bldg. Basement
AX6DD369	2.5.1	Geologic Sections Auxiliary Bldg. Basement
AX6DD370	2.5.1	Geologic Sections Auxiliary Bldg. Basement
AX6DD371	2.5.1	Isopachs of the Bearing Horizon (Blue Bluff Marl)
AX6DD372	2.5.1	Upper Surface of the Marl (Lisbon Formation) North and East Slopes
AX6DD373	2.5.1	Upper Surface of the Marl (Lisbon Formation) South and West Slopes
AX6DD374	2.5.1	Geologic Reference Stations - Auxiliary Bldg. Basement Excavation
AX6DD375	2.5.1	Geologic Reference Stations - Auxiliary Bldg. Basement Excavation
AX6DD376	2.5.1	Areal Extent and Isopachs of the Shell Zone (Utley Limestone)
AX6DD377	2.5.1	Top of the Bearing Horizon (Blue Bluff Marl)
AX6DD378	2.5.1	Bottom of the Bearing Horizon (Blue Bluff Marl)
AX6DD379	2.5.1	Geophysical Logs Hole No. 501
AX6DD380	2.5.1	Geophysical Logs Hole No. 502
AX6DD381	2.5.1	Geophysical Logs Hole No. 509
AX6DD382	2.5.1	Geophysical Logs Hole No. 513
AX6DD383	2.5.1	Geophysical Logs Hole No. 516
AX6DD384	2.5.1	Geophysical Logs Hole No. 523
AX6DD385	2.5.2	Significant Seismicity 1776 - 1982
AX6DD386	2.5.4	Power Block Cross Section
AX6DD387	2.5.4	Site Map and Location of Seismic Lines
AX6DD388	2.5.4	Seismic Profiles Lines A, B, and C
AX6DD389	2.5.4	Seismic Profiles Lines D and E
AX6DD390	2.5.4	Seismic Profiles Lines F, G, and H

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Drawing Number	Section	Title
AX6DD391	2.5.4	Seismic Profiles Lines J, K, L, M, N, and P
AX6DD392	2.5.4	Seismic Profiles Line Q
AX6DD393	2.5.4	Seismic Profiles Lines R and S
AX6DD394	2.5.4	Boring and Test Pit Locations in Borrow Areas and Stockpiles Plan
AX6DD395	2.5.4	Results of Cyclic Triaxial Tests (93- and 95-Percent Relative Compaction)
AX6DD396	2.5.4	Results of Cyclic Triaxial Tests (95- and 97-Percent Relative Compaction)
AX6DD397	2.5.4	Laboratory Results of Cyclic Triaxial Tests
AX6DD398	2.5.4	Cyclic Triaxial Tests Data Normalized to 50-Percent Relative Density
AX6DD399	2.5.4	Plot of Heave Versus Time
AX6DD400	2.5.4	Plot of Heave Versus Time
AX6DD401	3.9.N.2	Vibration Checkout Functional Inspection Points
AX6DD402	8.1	The Southern Company Grid System
AX6DD403	10.1	Steam and Power Conversion Systems
AX6DD404	12.3.2	Isometric View of Control Room Shielding
AX6DD406	2.4.3	Location of Savannah River Valley Cross-sections, Unsteady, Nonuniform Flow
AX6DD407	2.4.3	Savannah River Valley Cross-section, Steady-flow Condition
AX6DD408	2.4.12	Well 42 Series (Blue Bluff Marl) Hydrographs and Piezometer Profiles
AX6DD409	2.5.1	Regional Geologic Structures
CX2D45V002	1.1	Location and Vicinity Map
CX2D45V003	1.2.2	Site Plan
CX2D46V003	2.4.1	Site Grading and Drainage Plan
CX2D46V004	2.4.1	Site Grading and Drainage Plan (PMP Storm)

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Drawing Number	Section	Title
CX2D46V005	2.4.1	Site Grading and Drainage Plan (100-year, 6-h Storm)
CX2D46V006	2.4.1	Site Grading and Drainage Plan (PMP Storm)
CX4DB153	9.2.4	Plant Makeup Water Well and Potable Water System
CX4DB173-2	9.5.1	Fire Protection Water Systems
CX4DB173-3	9.5.1	Fire Protection Water Systems
CX4DB173-4	9.5.1	Fire Protection Water Systems
CX4DB173-5	9.5.1	Fire Protection Water Systems
CX4DB173-6	9.5.1	Fire Protection Water Systems

* This drawing provides useful information showing system interactions and configuration in subsection 9.2.1 but does not represent actual operation or accident mitigation parameters. For this reason, it is retained here but represents historical information.

1.8 DIFFERENCES FROM STANDARD REVIEW PLAN ACCEPTANCE CRITERIA

Georgia Power Company (GPC) has reviewed the VEGP design and concludes that it meets the applicable acceptance criteria presented in the Standard Review Plan (SRP) (NUREG-0800), except as noted in table 1.8-1. As the exclusive operating licensee, Southern Nuclear Operating Company, Inc. (SNC) adopts that conclusion. Each cited text reference includes sufficient information to sustain the SNC conclusion that the underlying regulatory requirements have been satisfied. Also, in lieu of including figures as required by Standard Review Plan (SRP)(NUREG 0800) throughout this FSAR, applicable drawings and engineering diagrams are included by reference to their drawing identification number in conjunction with specific system descriptions.

TABLE 1.8-1 (SHEET 1 OF 5)

LIST OF DIFFERENCES WITH THE STANDARD REVIEW PLAN

SRP [a] Section	Specific SRP Acceptance Criteria	Summary Description of Differences	Where Discussed in FSAR
2.5.4 (Rev 2)	II.5, Subsection 2.5.4.8, Stability of Subsurface Materials and Foundations	Liquefaction potential assessments of subsurface soils using both probabilistic and deterministic approaches are urged; the VEGP analysis is based upon a deterministic approach exclusively.	Paragraph 2.5.4.15
3.5.3 (Rev 2)	II, Missiles Generated by Natural Phenomena	The tornado missile spectrum used for VEGP differs from the SRP.	Paragraph 3.5.1.8
3.7.1 (Rev 1)	II.1.b, Seismic Design Parameters	For seismic analysis of structures, the design motion is applied at the foundation level of Seismic Category 1 structures; for VEGP seismic analysis of structures, the design motion is applied at the grade level for deeply embedded Seismic Category 1 structures.	Paragraph 3.7.B.1.5
3.7.2 (Rev 1)	II.4, Seismic System Analysis	For soil-structures interaction, the half-space and finite element methods of analysis are to be used, and the response is determined by: A. Envelop results of two methods. B. Results of one method with conservative design considerations of effects from use of the other method. C. Combination of A and B with provision of adequate conservatism. VEGP soil-structures interaction analysis uses finite element methods for deeply embedded structures and half-space methods for shallowly embedded structures.	Paragraph 3.7.B.2.16
	II.4.c.iii	A soil excitation system interaction is to be selected such that the free-field response spectra generated at the structural foundation level envelop the design response spectra of earthquake motion. Seismic analysis of VEGP deeply embedded structures does not fully conform with this.	
3.7.2 (Rev 1)	II.7, Seismic System Analysis	Closely spaced modes should be combined in accordance with procedures stated in Regulatory Guide 1.92. The VEGP uses the "pi epsilon" method for closely spaced mode combination.	Paragraph 3.7.N.2.16

TABLE 1.8-1 (SHEET 2 OF 5)

SRP [a] Section	Specific SRP Acceptance Criteria	Summary Description of Differences	Where Discussed in FSAR
3.8.1 (Rev 1)	II.2, Concrete Containment	The containment pressure boundary is designed in accordance with American Society of Mechanical Engineers (ASME) Section III, Division 2, Article CC-3000. The extent to which the VEGP design conforms with the regulatory guides is covered in section 1.9 of the FSAR.	Paragraph 3.8.1.8
3.8.2 (Rev 1)	III.5.3, Steel Containment	The defined loads and load combinations in the SRP are not fully consistent with the VEGP design loads and load combinations.	Paragraph 3.8.2.8
3.8.3 (Rev 1)	II.2, Concrete and Steel Internal Structures of Steel or Concrete Containments	The containment internal structure was designed in accordance with American Concrete Institute (ACI) 318-71 whereas the SRP specifies ACI-349 as acceptance criteria for reinforced concrete designs. The VEGP CP was issued using the codes, standards, and guides which require conformance with ACI-318-71. Load combinations in the SRP are not consistent with the VEGP design load combinations. The extent to which the VEGP design conforms with the regulatory guides is covered in section 1.9 of the FSAR.	Paragraph 3.8.3.8
3.8.4 (Rev 1)	II.2, Other Seismic Category 1 Structures	The applicable codes, standards, and guides for these structures include ACI-349. The VEGP CP was issued using the codes, standards, and guides which require conformance with ACI-318-71.	Paragraph 3.8.4.8
3.8.5 (Rev 1)	II.4.b, Foundations	The applicable codes, standards, and guides for these structures include ACI-349. The VEGP CP was issued using the codes, standards, and guides which require conformance with ACI-318-71. The containment base mat is designed per ASME Section III, Division 2, CC-3000.	Paragraph 3.8.5.8
3.9.3 (Rev 1)	II.3.a, ASME Class 1, 2, and 3 Components, Supports, and Core Support Structures	The VEGP FSAR does not fully specify deformation limits of supports where the component supports may affect the operability of the component.	Paragraphs 3.9.B.3.5, 3.9.N.3.5
	II.App.A-1.3.3, Service Limit C	The design basis pipe break is defined by VEGP as a faulted rather than emergency condition.	Paragraph 3.9.B.3.5

TABLE 1.8-1 (SHEET 3 OF 5)

SRP [a] Section	Specific SRP Acceptance Criteria	Summary Description of Differences	Where Discussed in FSAR
3.10 (Rev 2)	II.1.a.(2) Seismic and Dynamic Qualification of Mechanical and Electrical Equipment	The SRP requires that equipment be tested in the operational condition and that loadings simulating normal plant conditions should be superimposed on seismic and dynamic loads. This includes flow-induced loads and degraded flow conditions. VEGP tests for NSSS equipment are made in the operational conditions where practical, simulated as appropriate, or addressed by analysis. Flow loads are not superimposed on seismic loads for valve operability tests.	Paragraphs 3.10.B.4.2, 3.10.N.4.2
	II.1.a.(8)	Evaluation is performed to ensure that test configurations conservatively simulate actual field mountings.	
	II.1.a.(10)	The VEGP does not apply end loadings to active valves during static deflection tests. End loads are evaluated in the active valve analysis.	
	II.1.a.(14).(b).iii	The VEGP does not analyze valve discs for Δp or impact energy resultant from LOCA, except for certain cases where a significant impact from the LOCA is expected.	
	II.1.a.(14).(b).viii	The VEGP does not utilize Regulatory Guide 1.92 guidance for combination of multimodal or multidirectional responses. The VEGP uses a vendor topical report as guidance for combining multinodal response. This report meets the intent of Regulatory Guide 1.92.	
	II.1.C	The VEGP has not performed aging and sequence testing on mechanical safety-related equipment.	
	II.3	The VEGP documentation file conforms to the SRP guidelines with the exception of certain NSSS documentation which remains in an auditable file at the vendor.	
3.11 (Rev 2)	II.1, Environmental Qualification of Mechanical and Electrical Equipment	The VEGP does not provide environmental qualification testing of mechanical equipment to the guidelines of NUREG-0588.	Paragraphs 3.11.N.1.4, 3.11.B.1.1

TABLE 1.8-1 (SHEET 4 OF 5)

SRP [a] Section	Specific SRP Acceptance Criteria	Summary Description of Differences	Where Discussed in FSAR
6.2.1.5 (Rev 2)	II.2, Minimum Containment Pressure Analysis for Emergency Core Cooling System Performance Capability Studies	The VEGP does not employ the heat transfer coefficients supplied in the SRP.	Paragraph 6.2.1.5.9
6.5.1 (Rev 1)	II.E, Engineered Safety Features (ESF) Atmosphere Cleanup Systems	The instrumentation provided for VEGP ESF atmosphere cleanup does not fully conform with the guidance of the SRP.	Paragraph 6.5.1.7
6.5.2 (Rev 1)	II.1.A, Containment Spray as a Fission Product Cleanup System	The VEGP is equipped with a semiautomatic switchover from injection to recirculation modes.	Paragraph 6.5.2.7
8.3.1 (Rev 2)	II.4.F.5, ac Power Systems (Onsite)	The diesel generator controls and monitoring instruments are not mounted on a vibration-free floor area, and vibration isolators have not been provided on the associated control cabinets.	Paragraph 8.3.1.5
9.1.4 (Rev 2)	II.5, Light Load Handling System	Kinetic energy of a dropped fuel handling tool lifted to the maximum height exceeds the kinetic energy of the tool and an assembly lifted to the normal height.	Paragraph 9.1.4.6
9.2.2 (Rev 1)	II.3.e, Reactor Auxiliary Cooling Water System	The VEGP will provide safety-grade instrumentation to detect loss of auxiliary component cooling water to the reactor coolant pump seals, but VEGP does not incorporate an automatic reactor coolant pump trip upon loss of auxiliary component cooling water.	Paragraph 9.2.8.6
9.2.5	II.1, Ultimate Heat Sink	Position C-1 of Regulatory Guide 1.27 requires that the heat sink be capable of providing cooling sufficient for 30 days.	Paragraph 9.2.5.6
9.4.5 (Rev 2)	II.4, ESF Ventilation System	The VEGP is not fully in conformance with Item 2 of Subsection A and Item 1 of Subsection C of NUREG/CR-0660.	Paragraph 9.4.7.6

TABLE 1.8-1 (SHEET 5 OF 5)

SRP [a] Section	Specific SRP Acceptance Criteria	Summary Description of Differences	Where Discussed in FSAR
9.5.1 (Rev 2)	II.2.9, Fire Protection Program	The VEGP is not fully in conformance with BTP CMEB 9.5-1.	Paragraph 9.5.1.6
11.5 (Rev 3)	II.1.A, Process and Effluent Radiological Monitoring Instrumentation and Sampling System	Gaseous and liquid process streams or effluent release points are not monitored and sampled in full accordance with tables 1 and 2 of section 11.5 of the SRP.	Paragraph 11.5.5.2
12.3.4 (Rev 2)	II.0.17, Radiation Protection Design Features	The VEGP did not utilize American National Standards Institute/American Nuclear Society HPSSC-6.8.1-1981.	Paragraph 12.3.4.2
	II.4.e	Criticality monitors as defined in the SRP are not included in the VEGP design.	Paragraph 12.3.4.2
	II.8.11	Dose assessments should be made in accordance with the procedures stated in Regulatory Guide 8.19.	Paragraph 12.4.1.3
13.6	11.7, Regulatory Guide 5.12	Security locks will comply with the requirements of 10 CFR 73.2(m) rather than the recommendations of Regulatory Guide 5.12.	Section 13.6 (Security Plan)

a Throughout this FSAR, in conjunction with specific system descriptions, applicable drawings and engineering diagrams are included by reference to their drawing identification number in lieu of inclusion as a figure as required by NUREG 0800.

1.9 CONFORMANCE TO NRC REGULATORY GUIDES

This section discusses the extent to which the VEGP conforms to the Nuclear Regulatory Commission (NRC) Regulatory Guides. Exceptions to the guides are identified, and justification is presented or referenced. In the discussion of the guides, the sections or tables of the Final Safety Analysis Report (FSAR), where more detailed information is presented, are referenced. Where VEGP conformance during design and construction phase differs from the operation phase, both are discussed.

1.9.1 REGULATORY GUIDE 1.1, NOVEMBER 1970, NET POSITIVE SUCTION HEAD FOR EMERGENCY CORE COOLING AND CONTAINMENT HEAT REMOVAL SYSTEM PUMPS

1.9.1.1 Regulatory Guide 1.1 Position

Emergency core cooling and containment heat removal systems should be designed so that adequate net positive suction head (NPSH) is provided to system pumps assuming maximum expected temperatures of pumped fluids and no increase in containment pressure from that is present prior to postulated loss-of-coolant accidents (LOCA).

1.9.1.2 VEGP Position

Conform, except that a saturated sump model is used to calculate available NPSH for containment sump liquid temperatures of equal to or greater than 211°F at a containment pressure of -0.3 psig or greater. Refer to paragraph 6.2.2.2.2.3.2 and section 6.3.

1.9.2 REGULATORY GUIDE 1.2, NOVEMBER 1970, THERMAL SHOCK TO REACTOR PRESSURE VESSELS

1.9.2.1 Regulatory Guide 1.2 Position

This guide describes a suitable program to assure that the reactor pressure vessel will behave in a nonbrittle manner under loss-of-coolant conditions. Regulatory Guide 1.2 was withdrawn June 17, 1991, and was superseded by 10 CFR 50.61. FSAR subsection 5.3.3 provides a description of compliance with the requirements of 10 CFR 50.61. The information contained in this section is retained for historical purposes.

1.9.2.2 VEGP Position

VEGP design conforms with this guide as discussed below:

Regulatory position C.1 is satisfied by the Westinghouse analytical and experimental programs as well as by the heavy section steel technology (HSST) program at Oak Ridge National Laboratory (ORNL).

Analytical techniques have been developed by Westinghouse to perform fracture evaluations of reactor vessels under thermal shock loadings.

Under the HSST program, a number of 6-in.-thick, 39-in. outside diameter steel pressure vessels containing carefully prepared and sharpened surface cracks are being tested. Test conditions include both hydraulic internal pressure loadings and thermal shock loadings. The objective of this program is to validate analytical fracture mechanics techniques and demonstrate quantitatively the margin of safety inherent in reactor pressure vessels.

A number of vessels have been tested under hydraulic pressure loadings, and results have confirmed the validity of fracture analysis techniques. The results and implications of the hydraulic pressure tests are summarized in ORNL report ORNL-TM-T5090.

Four thermal shock experiments have been completed and are now being evaluated. For representative conditions, flaws are shown to initiate and arrest in a predictable manner.

Westinghouse is continuing to obtain fracture toughness data for reactor pressure vessel steels through internally funded programs as well as HSST sponsored work.

Fracture toughness testing of irradiated compact tension fracture toughness specimens has been completed. The complete post-irradiation data on 0.394-, 2-, and 4-in.-thick specimens are now available from the HSST program. Both static and dynamic post-irradiation fracture toughness data have been obtained. Evaluation of the data obtained to date on material irradiated to fluences between 2.2 and 4.5 x 10^{19} n/cm^2 indicates that the reference toughness curve as contained in the American Society of Mechanical Engineers (ASME) Code, Section III remains a conservative lower bound for toughness values for pressure vessel steels.

Details of progress and results obtained in the HSST program are available in the HSST program progress reports issued by ORNL. Refer to section 5.3 for further discussion.

1.9.3 REGULATORY GUIDE 1.3, REVISION 2, JUNE 1974, ASSUMPTIONS USED FOR EVALUATING THE POTENTIAL RADIOLOGICAL CONSEQUENCES OF A LOSS-OF-COOLANT ACCIDENT FOR BOILING WATER REACTORS

Not applicable to VEGP.

1.9.4 REGULATORY GUIDE 1.4, REVISION 2, JUNE 1974, ASSUMPTIONS USED FOR EVALUATING THE POTENTIAL RADIOLOGICAL CONSEQUENCES OF A LOSS-OF-COOLANT ACCIDENT FOR PRESSURIZED WATER REACTORS

1.9.4.1 Regulatory Guide 1.4 Position

This guide gives acceptable assumptions that may be used in evaluating the radiological consequences of this accident for a pressurized water reactor (PWR).

1.9.4.2 VEGP Position

The extent of conformance with this guide is indicated in table 15.6.5-10.

1.9.5 REGULATORY GUIDE 1.5, MARCH 1971, ASSUMPTIONS USED FOR EVALUATING THE POTENTIAL RADIOLOGICAL CONSEQUENCES OF A STEAM LINE BREAK ACCIDENT FOR BOILING WATER REACTORS

Not applicable to VEGP.

1.9.6 REGULATORY GUIDE 1.6, MARCH 1971, INDEPENDENCE BETWEEN REDUNDANT STANDBY (ONSITE) POWER SOURCES AND BETWEEN THEIR DISTRIBUTION SYSTEMS

1.9.6.1 Regulatory Guide 1.6 Position

This guide describes an acceptable degree of independence between redundant standby (onsite) power sources and between their distribution systems.

1.9.6.2 VEGP Position

Conform, with the exception of regulatory position C.4.d. The VEGP design has provisions for manually connecting redundant load groups together. The connection is accomplished under administrative control by performing a manual, hot-bus transfer between the normal and alternate offsite power sources. This transfer may be made as discussed in paragraph 8.3.1.1.2.D.

Independence between redundant standby power sources and between their distribution systems is discussed further in sections 8.1 and 8.3.

1.9.7 REGULATORY GUIDE 1.7, REVISION 2, NOVEMBER 1978, CONTROL OF COMBUSTIBLE GAS CONCENTRATIONS IN CONTAINMENT FOLLOWING A LOSS-OF-COOLANT ACCIDENT

1.9.7.1 Regulatory Guide 1.7 Position

This guide describes an acceptable method for complying with NRC regulations for controlling combustible gas concentrations inside containment following a LOCA.

1.9.7.2 VEGP Position

The extent of conformance with this guide is as indicated in subsection 6.2.5.

1.9.8 REGULATORY GUIDE 1.8, REVISION 2, APRIL 1987, QUALIFICATION AND TRAINING OF PERSONNEL FOR NUCLEAR POWER PLANTS

1.9.8.1 Regulatory Guide 1.8 Position

Personnel will meet the minimum education and experience recommendations of this guide before they are considered qualified to perform all duties independently. Prior to meeting the recommendations of the guide, personnel may be trained to perform specific tasks and will be qualified to perform those tasks independently. As described in Part D of the guide, personnel who complete an accredited program which has been endorsed by the NRC shall meet the requirements of the accredited program in lieu of other guidance given in the guide.

1.9.8.2 VEGP Position

Conformance is discussed in sections 13.1 and 13.2.

1.9.9 REGULATORY GUIDE 1.9, REVISION 2, DECEMBER 1979, SELECTION, DESIGN, AND QUALIFICATION OF DIESEL GENERATOR UNITS USED AS STANDBY (ONSITE) ELECTRIC POWER SYSTEMS AT NUCLEAR POWER PLANTS

1.9.9.1 Regulatory Guide 1.9 Position

Conformance with the requirements of Institute of Electrical and Electronics Engineers (IEEE) Standard (Std.) 387-1977, IEEE Standard Criteria for Diesel Generator Units Applied as Standby Power Supplies for Nuclear Power Generating Stations, dated June 17, 1977, is acceptable for meeting the requirements of the principal design criteria and qualification testing of diesel generator units as onsite electric power systems for nuclear power plants subject to the following:

1. C.1 When the characteristics of loads are not accurately known, such as during the construction permit stage of design, each diesel generator unit of an onsite power supply system should be selected to have a continuous load rating (as defined in Section 3.7.1 of IEEE Std. 387-1977) equal to or greater than the sum of the conservatively estimated loads needed to be powered by that unit at any one time. In the absence of fully substantiated performance characteristics for mechanical equipment such as pumps, the electric motor drive ratings should be calculated using conservative estimates of these characteristics, e.g., pump runout conditions and motor efficiencies of 90 percent or less and power factors of 85 percent or less.

2. C.2 At the operating license stage of review, the predicted loads should not exceed the short-time rating (as defined in Section 3.7.2 of IEEE Std. 387-1977) of the diesel generator unit.

3. C.3 In Section 5.1.1, General, of IEEE Std. 387-1977, the requirements of IEEE Std. 308-1974 should be used subject to the regulatory position of Regulatory Guide 1.32.

4. C.4 Section 5.1.2, Mechanical and Electrical Capabilities, of IEEE Std. 387-1977 pertains, in part, to the starting and load-accepting capabilities of the diesel generator unit. In conjunction with Section 5.1.2, each diesel generator unit should be capable of starting and accelerating to rated speed, in the required sequence, all the needed engineered safety feature (ESF) emergency shutdown loads. The diesel generator unit design should be such that at no time during the loading sequence should the frequency and voltage decrease to less than 95 percent of nominal and 75 percent of nominal, respectively. (A larger decrease in voltage and frequency may be justified for a diesel generator unit that carries only one large connected load.) Frequency should be restored to within 2 percent of nominal, and voltage should be restored to within 10 percent of nominal within 60 percent of each load-sequence time interval. (A greater percentage of the time interval may be used if it can be justified by analysis. However, the load-sequence time interval should include sufficient margin to account for the accuracy and repeatability of the load-sequencer timer.) During recovery from transients caused by step load increases or resulting from the disconnection of the largest single load, the speed of the diesel generator unit should not exceed the nominal speed plus 75 percent of the difference between nominal speed and the overspeed trip setpoint or 115 percent of nominal, whichever is lower. Further, the transient following the complete loss of load should not cause the speed of the unit to attain the overspeed trip setpoint.

5. C.5 In Section 5.4, Qualification, of IEEE Std. 387-1977, the qualification testing requirements of IEEE Std. 323-1974, IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations, should be used subject to the regulatory position of Regulatory Guide 1.89, Qualification of Class 1E Equipment for Nuclear Power Plants.

6. C.6 Section 5.5, Design and Application Considerations, of IEEE Std. 387-1977 pertains to design features for consideration in diesel generator unit design. In conjunction with Section 5.5, diesel generator units should be designed to be testable during operation of the nuclear power plant as well as while the plant is shut down. The design should include provisions so that the testing of the units will simulate the parameters of operation (outlined in Regulatory Guide 1.108, Periodic Testing of Diesel Generator Units Used as Onsite Electric Power Systems at Nuclear Power Plants) that would be expected if actual demand were to be placed on the system.

   Testability should be considered in the selection and location of instrumentation sensors and critical components (e.g., governor, starting system components), and the overall design should include status indication and alarm features. Instrumentation sensors should be readily accessible and, where practicable, designed so that their inspection and calibration can be verified in place.

7. C.7 Section 5.6.2.2, Automatic Control, of IEEE Std. 387-1977 pertains to automatic startup requirements and their relationship to other operating modes. In conjunction with Section 5.6.2.2, engine-overspeed and generator-differential trips

may be implemented by a single-channel trip. All other diesel generator protective trips should be handled in one of two ways:

a. A trip should be implemented with two or more independent measurements for each trip parameter with coincident logic provisions for trip actuation.

b. A trip may be bypassed under accident conditions, provided the operator has sufficient time to react appropriately to an abnormal diesel generator unit condition. The design of the bypass circuitry should satisfy the requirements of IEEE Std. 279-1971 at the diesel generator system level and should include the capability for:

(1) Testing the status and operability of the bypass circuits.

(2) Alarming in the control room abnormal values of all bypass parameters.

(3) Manual resetting of the trip bypass function. (Capability for automatic reset is not acceptable.)

8. C.8 Section 5.6.3.1, Surveillance Systems, of IEEE Std. 387-1977 pertains to status indication of diesel generator unit conditions. In conjunction with Section 5.6.3.1, in order to facilitate trouble diagnosis, the surveillance system should indicate which of the diesel generator protective trips is activated first.

9. C.9 In Section 6.3, Type Qualification Testing Procedures and Methods, of IEEE Std. 387-1977, the requirements of IEEE Std. 344-1975, Recommended Practice for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations, for seismic analysis or seismic testing by equipment manufacturers should be used subject to the regulatory position of Regulatory Guide 1.100, Seismic Qualification of Electric Equipment for Nuclear Power Plants.

10. C.10 The option indicated by "may" in Section 6.3.2(5)(c) of IEEE Std. 387-1977 should be treated as a requirement.

11. C.11 Section 6.5, Site Acceptance Testing, and Section 6.6, Periodic Testing, of IEEE Std. 387-1977 should be supplemented by Regulatory Guide 1.108.

12. C.12 Section 4, Reference Standards, of IEEE Std. 387-1977 lists additional applicable IEEE standards. The specific applicability or acceptability of these referenced standards has been or will be covered separately in other regulatory guides, where appropriate.

13. C.13 Section 6.3.2, Start and Load Acceptance Qualification, pertains to test requirements for diesel generator unit qualification. In conjunction with Section 6.3.2, fewer successful start-and-load tests and allowed failures than that specified--300 valid tests with no more than 3 failures--may be justified for a diesel generator unit that carries only one large connected load tested under actual conditions, provided an equivalent reliability/confidence level is demonstrated.

14. C.14 In Section 6.3.1, Load Capability Qualification, of IEEE Std. 387-1977, the order of sequence of load tests described in parts (1) and (2) should be as follows: Load equal to the continuous rating should be applied for the time required to reach

engine temperature equilibrium, at which time, the rated short-time load should be applied for a period of 2 h. Immediately following the 2-h short-time load test, load equal to the continuous rating should be applied for 22 h.

1.9.9.2 VEGP Position

1. C.1 Conform.

2. C.2 Conform.

3. C.3 Conform. See Regulatory Guide 1.32 comparison.

4. C.4 The voltage and frequency requirements are met under all conditions except during energizing of the load center transformers (time 0 s on the sequencer). At this time the voltage and frequency may be outside the specified limits but there are no loads on the bus to be affected. The first load sequence step is at time 0.5 s on the sequencer.

5. C.5 Conform. Refer to Regulatory Guide 1.89 comparison.

6. C.6 Conform. See Regulatory Guide 1.108 comparison.

7. C.7 Conform.

8. C.8 Conform.

9. C.9 VEGP diesels are qualified in accordance with IEEE Std. 387-1977 and IEEE Std. 344-1975.

10. C.10 Conform.

11. C.11 Conform. See Regulatory Guide 1.108 comparison.

12. C.12 Conform. Applicable standards are referenced where appropriate.

13. C.13 Conform.

14. C.14 Conform.

Regulatory Guide 1.9, Revision 3, has been used in specific instances to revise diesel generator testing requirements for VEGP as stated in subsection 1.9.108.

Refer to section 8.3 for further discussion.

1.9.10 REGULATORY GUIDE 1.10, REVISION 1, JANUARY 1973, MECHANICAL (CADWELD) SPLICES IN REINFORCING BARS OF CATEGORY I CONCRETE STRUCTURES

1.9.10.1 Regulatory Guide 1.10 Position

Procedures given for testing cadwelds include:

1. C.1 Crew qualification.
2. C.2 Visual inspection.
3. C.3 Tensile testing.
4. C.4 Tensile test frequency.
5. C.5 Procedure for substandard tensile test results.

1.9.10.2 VEGP Position

Procedures for testing cadwelds conform with the requirements of Regulatory Guide 1.10.

Refer to subsection 3.8.1 for discussion on this subject.

1.9.11 REGULATORY GUIDE 1.11, MARCH 1971, INSTRUMENT LINES PENETRATING PRIMARY REACTOR CONTAINMENT

1.9.11.1 Regulatory Guide 1.11 Position

This guide describes an acceptable method for designing instrument lines which penetrate the primary containment and reactor coolant pressure boundary or are connected directly to the containment atmosphere.

1.9.11.2 VEGP Position

VEGP conforms with this guide with the exception of regulatory position C.1.c. Containment pressure sensing lines and reactor vessel level instrumentation system are not equipped with isolation valves, but are provided with sealed capillaries as described in subsection 6.2.4.1.1, paragraph G. This alternative is consistent with Regulatory Guide 1.141.

Two containment hydrogen monitors are provided. Each monitor has a sample line and return line which penetrate the containment. All sample lines penetrating the containment are equipped with remote manual operated valves inside and outside the containment, in accordance with 10 CFR 50, Appendix A, General Design Criteria 55 and 56.

All other sample lines penetrating the containment are similarly equipped.

Refer to subsection 6.2.4 and table 6.2.4-1.

1.9.12 REGULATORY GUIDE 1.12, REVISION 2, MARCH 1997, INSTRUMENTATION FOR EARTHQUAKES

1.9.12.1 Regulatory Guide 1.12 Position

This guide describes seismic instrumentation acceptable to the NRC for meeting Appendix A of 10 CFR 50.

1.9.12.2 VEGP Position

VEGP conforms to this guide to the extent indicated in table 3.7.4-1 and figure 3.7.4-1.

1.9.13 REGULATORY GUIDE 1.13, REVISION 1, DECEMBER 1975, SPENT-FUEL STORAGE FACILITY DESIGN BASIS

1.9.13.1 Regulatory Guide 1.13 Position

This guide describes a method acceptable to the NRC for designing wet spent fuel storage facilities for light-water reactors.

1.9.13.2 VEGP Position

Conform, with exception to regulatory position C.7, as the spent fuel pool level indicator alarms in the control room only, and not locally. Additional information is provided in section 9.1.

1.9.14 REGULATORY GUIDE 1.14, REVISION 1, AUGUST 1975, REACTOR COOLANT PUMP FLYWHEEL INTEGRITY

1.9.14.1 Regulatory Guide 1.14 Position

This guide describes a method acceptable to the NRC for minimizing the potential for failures of the flywheels of reactor coolant pump motors in light-water-cooled reactors.

1.9.14.2 VEGP Position – Prelicense

VEGP uses the standard Westinghouse reactor coolant pump design and takes the following position on this guide:

Since the issuance of Regulatory Guide 1.14, Revision 1, the NRC has provided to Westinghouse a copy of draft 2, Revision 2, of Regulatory Guide 1.14. This draft was formulated from industry and concerned parties' comments. It is significant that the draft 2 version incorporates several of the Westinghouse comments on Revision 1. Since draft 2 has not been formally published as Revision 2 of Regulatory Guide 1.14, the exceptions and clarifications (from the original Westinghouse comments) are provided below:

Cross rolling ratio of 1 to 3

Westinghouse's position is that specification of a cross rolling ratio is unnecessary since past evaluations have shown that ASME SA-533, Grade B, Class 1 materials produced without this requirement have suitable toughness for typical flywheel applications. Proper material selection and specification of minimum material properties in the transverse direction adequately ensure flywheel integrity. An attempt to gain isotropy in the flywheel materials by means of cross rolling is unnecessary since adequate margins of safety are provided by both flywheel material selection (ASME SA-533, Grade B, Class 1) and by specifying minimum yield and tensile levels and toughness test values taken in the direction perpendicular to the maximum working direction of the material.

1. C.1 The requirements for the vacuum melting and degassing process or the electroslag process are not essential in meeting the balance of the regulatory position or do they, in themselves, ensure conformance with the overall regulatory position. The initial Safety Guide 14 (October 27, 1971) stated that the "flywheel material should be produced by a process that minimized flaws in the material and improved its fracture toughness properties." This is accomplished by using ASME SA-533 material including vacuum treatment.

   Specification of a cross rolling ratio is considered unnecessary since past evaluations have shown that ASME SA-533, Grade B, Class 1 materials produced without this requirement have suitable toughness for typical flywheel applications. Proper material selection and specification of minimum material properties in the transverse direction adequately ensure flywheel integrity. An attempt to gain isotropy in the flywheel material by means of cross rolling is unnecessary since adequate margins of safety are provided by both flywheel material selection (ASME SA-533, Grade B, Class 1) and by specifying minimum yield and tensile levels and toughness test values taken in the direction perpendicular to the maximum working direction of the material.

2. C.2 Because the VEGP design specifies a light interference fit between the flywheel and the shaft, at zero speed the hoop stresses and radial stresses at the flywheel bore are negligible. Centering of the flywheel relative to the shaft is accomplished by means of keys and/or centering devices attached to the shaft, and at normal speed, the flywheel is not in contact with the shaft in the sense intended by Revision 1. Hence, the definition of "excessive deformation," as defined in this guide, is not applicable to the design since the enlargement of the bore and subsequent partial separation of the flywheel from the shaft does not cause unbalance of the flywheel. Extensive experience with reactor coolant pump flywheels installed in this fashion has verified the adequacy of the design.

The combined primary stress levels, as defined in Revision 0 of Regulatory Guide 1.14 (regulatory positions C.2.a and C.2.c) are both conservative and proven and no changes to these stress levels are necessary. Westinghouse designs to these stress limits and thus does not have permanent distortion of the flywheel bore at normal or spin test conditions.

Paragraph 2.b is considered as delineated. The interpretation removes the ambiguous reference to an undefined overspeed transient.

3. C.3 Conform.

4. C.4 WCAP-8163 shows that the flywheel would not fail at 290 percent of normal speed for a flywheel flaw of 1.15 in. or less in length. Results for a double-ended guillotine break at the pump discharge with full separation of pipe ends assumed show the maximum overspeed to be less than 110 percent of normal speed. The maximum overspeed was calculated to be about 280 percent of normal speed for the same postulated break, and an assumed instantaneous loss of power to the reactor coolant pump. In comparison with the overspeed presented above, the flywheel is tested at 125 percent of normal speed at the factory. Thus, the flywheel could withstand a speed up to 2.3 times greater than the flywheel spin test speed of 125 percent provided that no flaws greater than 1.15 in. are present. If the maximum speed were 125 percent of normal speed or less, the critical flaw size for failure would exceed 6 in. in length. Nondestructive tests and critical dimension examinations are all performed before the spin tests. The inspection methods employed provide assurance that flaws significantly smaller than the critical flaw size of 1.15 in. for 290 percent of normal speed would be detected. Flaws in the flywheel will be recorded in the prespin inspection program. Flaw growth attributable to the spin test (i.e., from a single reversal of stress, up to speed and back), under the most adverse conditions, is about three orders of magnitude smaller than that which nondestructive inspection techniques are capable of detecting. For these reasons, no post-spin inspections are performed since the prespin test inspections are considered adequate.

   Refer to paragraph 5.4.1.5 for further discussion.

1.9.14.3 VEGP Position – Post License

The VEGP Position – Prelicense applies, except for the flywheel examination requirements of Regulatory Position C.4.b of Regulatory Guide 1.14. Westinghouse WCAP-14535, “Topical Report on Reactor Coolant Pump Flywheel Inspection Elimination,” developed by the Westinghouse Owners Group (WOG), presents the basis for relief from the flywheel inspections required by Regulatory Position C.4.b of Regulatory Guide 1.14. The report was submitted to the NRC for review on January 24, 1996 by Duquesne Light Company for application to Beaver Valley Power Station Units 1 and 2. On September 12, 1996, the NRC issued a Safety Evaluation Report (SER) documenting its approval of WCAP-14535. The SER states that the evaluation methodology described in WCAP-14535 is appropriate and the criteria are in accordance with the design criteria of Regulatory Guide 1.14. The final NRC-accepted version of WCAP-14535 was issued as revision A which compiles all relevant correspondence for this matter. In related NRC correspondence, the Commission stated that other utilities may reference WCAP-14535 and its revision when pursuing the alternative flywheel inspections.

VEGP Technical Specification 5.5.7, “Reactor Coolant Pump Flywheel Inspection Program,” reflects the alternative examination requirements provided by WCAP-14535A.

1.9.15 REGULATORY GUIDE 1.15, REVISION 1, DECEMBER 1972, TESTING OF REINFORCING BARS FOR CATEGORY I CONCRETE STRUCTURES

1.9.15.1 Regulatory Guide 1.15 Position

Yield strength and tensile strength test procedures are required on reinforcing bars, as well as deformation inspections.

1.9.15.2 VEGP Position

The test procedures and deformation inspections conform with Regulatory Guide 1.15, with the clarification that while ASTM A615-72 is referenced in the regulatory guide, later editions of ASTM A615 may be used. Refer to subsection 3.8.1 for discussion on this subject.

1.9.16 REGULATORY GUIDE 1.16, REVISION 4, AUGUST 1975, REPORTING OF OPERATING INFORMATION - APPENDIX A TECHNICAL SPECIFICATIONS

1.9.16.1 Regulatory Guide 1.16 Position

In addition to the applicable reporting requirements of 10 CFR, this guide provides an acceptable basis for meeting the reporting requirements of Appendix A of the Technical Specifications.

1.9.16.2 VEGP Position

VEGP does not conform with this regulatory guide. In addition to the applicable reporting requirements of Title 10 Code of Federal Regulations, the program for reporting of Units 1 and 2, operating information is in accordance with Generic Letter 97-02, "Revised Contents of the Monthly Operating Report," dated May 15, 1997.

1.9.17 REGULATORY GUIDE 1.17, JUNE 1973, PROTECTION OF NUCLEAR POWER PLANTS AGAINST INDUSTRIAL SABOTAGE

1.9.17.1 Regulatory Guide 1.17 Position

The requirements and recommendations contained in the proposed ANSI Standard N18.17, Industrial Security for Nuclear Power Plants, dated March 23, 1973, are generally acceptable

and, with due consideration for the unique characteristics of the plant and its owner organization, provide an adequate basis for a physical security plan for protection of nuclear power plants against industrial sabotage, subject to the qualifications provided in this guide.

1.9.17.2 VEGP Position

VEGP conforms with ANSI/ANS-3.3-1982, as described in the physical security and contingency plan for the plant.

1.9.18 REGULATORY GUIDE 1.18, REVISION 1, DECEMBER 1972, STRUCTURAL ACCEPTANCE TEST FOR CONCRETE PRIMARY REACTOR CONTAINMENTS

1.9.18.1 Regulatory Guide 1.18 Position

This guide describes an acceptable method of implementing the initial structural acceptance test, which demonstrates the capability of a concrete primary reactor containment to withstand postulated pressure loads.

1.9.18.2 VEGP Position

VEGP conforms with this guide, except with regulatory positions C.3 and C.9 as follows:

1. C.3 Measurement of tangential displacements will not be performed for the 12 points on the largest opening (equipment hatch) of the containment. The magnitude of the expected local tangential deformation under the test pressure conditions is so negligibly small that, combined with the difficulty in obtaining fixed reference lines for local measurements, it is impractical to attempt measurement of local tangential deflections.

2. C.9 It is not intended to schedule structural integrity testing for periods when extremely inclement weather is forecast. Should, despite the forecast, snow, heavy rain, or strong winds occur during the test, the test results will be considered valid unless there is evidence to indicate otherwise.

Refer to subsection 3.8.1 for discussion on this subject.

1.9.19 REGULATORY GUIDE 1.19, REVISION 1, AUGUST 1972, NONDESTRUCTIVE EXAMINATION OF PRIMARY CONTAINMENT LINER WELDS

1.9.19.1 Regulatory Guide 1.19 Position

This guide describes acceptable procedures for performing nondestructive examinations (NDEs) of the welds in the liners and penetrations of primary reactor containments of concrete construction.

1. C.1.b Allows magnetic particle or ultrasonic methods for NDE where radiographic examination is not feasible.

2. C.1.c Requires bubble test solution to be checked hourly.

3. C.7 Acceptance standards should be in accordance with ASME Section III NE-5120.

1.9.19.2 VEGP Position

Conform, with the following exceptions:

1. C.1.b Magnetic particle or liquid penetrant testing of seam welds is performed if radiography is not possible due to weld inaccessibility.

2. C.1.c The bubble test solution will be checked every 4 h.

3. C.7 Acceptance standards are in accordance with ASME Section III, NE-5320 and NE-5350.

Refer to subsection 3.8.1 for further discussion.

1.9.20 REGULATORY GUIDE 1.20, REVISION 2, MAY 1976, COMPREHENSIVE VIBRATION ASSESSMENT PROGRAM FOR REACTOR INTERNALS DURING PREOPERATIONAL AND INITIAL STARTUP TESTING

1.9.20.1 Regulatory Guide 1.20 Position

This guide presents a method acceptable for implementing requirements with respect to the internals of light-water-cooled reactors during preoperational and initial startup testing.

1.9.20.2 VEGP Position

The comprehensive vibration assessment program for the VEGP reactor internals during preoperational and initial startup testing conforms with the recommendations of this guide, as discussed in paragraph 3.9.N.2.4.

1.9.21 REGULATORY GUIDE 1.21, REVISION 1, JUNE 1974, MEASURING, EVALUATING, AND REPORTING RADIOACTIVITY IN SOLID WASTES AND RELEASES OF RADIOACTIVE MATERIALS IN LIQUID AND GASEOUS EFFLUENTS FROM LIGHT-WATER-COOLED NUCLEAR POWER PLANTS

1.9.21.1 Regulatory Guide 1.21 Position

This guide describes programs acceptable to the NRC for measuring, reporting, and evaluating releases of radioactive materials in liquid and gaseous effluents and guidelines for classifying and reporting the categories and curie content of solid wastes.

1.9.21.2 VEGP Position

VEGP conforms with this guide to the extent discussed below and in section 11.5.

The requirement for hourly meteorological data for batch releases will be met if the batch releases are made in a non random manner. If the batch releases are made in a random manner, average annual meteorological parameters will be used for these calculations and the hourly meteorological data will not have to be included in the radioactive effluent reports. This methodology is representative for dose calculations and is less susceptible to errors in data management.

This is in accordance with the Technical Specifications, the ODCM, NUREG 0133, and is standard practice in the nuclear industry.

1.9.22 REGULATORY GUIDE 1.22, FEBRUARY 1972, PERIODIC TESTING OF PROTECTION SYSTEM ACTUATION FUNCTIONS

1.9.22.1 Regulatory Guide 1.22 Position

This guide describes an acceptable method for ensuring that the protection system is designed to permit periodic testing of its functioning during reactor operation.

1.9.22.2 VEGP Position

VEGP conforms with this guide to the extent discussed below and in paragraph 7.1.2.5.

1. D.1 The protection system is designed to permit periodic testing to extend to and include actuation devices and actuated equipment. (See position 4.)

   D.1.a. and b. The periodic tests do duplicate, as closely as practicable, the performance that is required of the actuation devices in the event of an accident. The only actuation devices for which the tests do not completely duplicate the performance that is required in the event of an accident are:

- The main steam and feedwater isolation valve actuators - full performance testing of these actuators would result in full closure of the main steam and feedwater isolation valves. The transients that would result under power generating conditions in the plant would include steam generator water level oscillations, or low-low steam generator water level, and would probably result in reactor trip. The actuators will be under power generating conditions to verify proper operation of all components required for fast closure of the valves. Operability of the valves may be checked during operation by performing a partial stroke. This partial stroke closes the valves 10 percent (at a slow speed, but by the same mechanism used for fast closure) and then reopened to verify that the valves are free to close. Positive indications are provided on the main control board to verify that the closure mechanism of the valves has been reset to ensure proper operation after the test. The valve actuators can be fully tested, including full closure at high speed, whenever the plant is not in operation.

- The main turbine trip function - a trip of the main turbine under power generating conditions (unless below approximately 50 percent power) would result in a trip of the reactor. The turbine trip function can be tested whenever the turbine is not in operation.

2. D.2.a. through d. In general, the protection systems can be tested in accordance with method a. The only protection systems that cannot be tested in accordance with method a are the main steam and feedwater isolation system, containment spray system, and the auxiliary feedwater system. The systems not tested in accordance with method a can all be tested in accordance with method b. Methods c and d need not be used.

3. D.3.a. and b. System bypasses are generally not required for testing; in most cases, the actuated equipment actually responds to the test signals. The only exceptions to these criteria are:

   - Bistables - test signals are substituted for the actual plant inputs during bistable tests, and provisions are included for bypassing bistable outputs. The bistables not under test, all digital inputs, and all other portions of the protection system are not affected.

   - Main steam and feedwater isolation valves - the signals to these valves are held in a condition that prevents valve motion during a portion of the test.

   - Each bypass condition is automatically indicated to the reactor operator in the MCR by a separate annunciator for the train in test. Test circuitry does not allow two trains to be tested at the same time so that extension of the bypass condition to the redundant system is prevented.

4. D.4 Where actuated equipment is not tested during reactor operation it has been determined that:

   D.4.a. There is no practicable system design that would permit operation of the equipment without adversely affecting the safety or operability of the plant.

D.4.b. The probability that the protection system will fail to initiate the operation of the equipment is, and can be maintained, acceptably low without testing the equipment during reactor operation.

D.4.c. The equipment can be tested routinely when the reactor is down.

The list of equipment that cannot be tested at power is:

- Manual actuation switches.
- Reactor coolant pump breakers.
- Turbine trip equipment that causes reactor trip.
- Main steam line stop valves (close).
- Main feedwater and feedwater bypass isolation valves (close).
- Feedwater regulating valves (close).
- Main feedwater pump solenoids.
- Reactor coolant pump seal water isolation valves (close).
- Pressurizer power operated relief valves (PORVs) (open).
- Instrument air containment isolation valves (close).

1.9.23 REGULATORY GUIDE 1.23, FEBRUARY 1972, ONSITE METEOROLOGICAL PROGRAMS

1.9.23.1 Regulatory Guide 1.23 Position

This guide describes a suitable onsite meteorological program to provide meteorological data needed to estimate potential radiation doses to the public as a result of the routine or accidental release of radioactive materials to the atmosphere and to assess other environmental effects.

1.9.23.2 VEGP Position

A description of the onsite meteorological program is provided in subsection 2.3.3. Onsite weather instruments are described in table 2.3.2-2. Short- and long-term diffusion estimates are discussed in subsections 2.3.4 and 2.3.5.

1.9.24 REGULATORY GUIDE 1.24, MARCH 1972, ASSUMPTIONS USED FOR EVALUATING THE POTENTIAL CONSEQUENCES OF A PRESSURIZED WATER REACTOR RADIOACTIVE GAS STORAGE TANK FAILURE

1.9.24.1 Regulatory Guide 1.24 Position

This guide lists acceptable assumptions for use in evaluating the radiological consequences of a radioactive gas storage tank failure.

1.9.24.2 VEGP Position

The extent of conformance with this guide is as indicated in table 15.7.1-2.

1.9.25 REGULATORY GUIDE 1.25, MARCH 1972, ASSUMPTIONS USED FOR EVALUATING THE POTENTIAL RADIOLOGICAL CONSEQUENCES OF A FUEL HANDLING ACCIDENT IN THE FUEL HANDLING AND STORAGE FACILITY FOR BOILING AND PRESSURIZED WATER REACTORS

1.9.25.1 Regulatory Guide 1.25 Position

This guide provides acceptable assumptions that may be used in evaluating the radiological consequences of a fuel handling accident in light-water reactors.

1.9.25.2 VEGP Position

The extent of conformance with this guide is as indicated in table 15.7.4-2.

1.9.26 REGULATORY GUIDE 1.26, REVISION 3, FEBRUARY 1976, QUALITY GROUP CLASSIFICATIONS AND STANDARDS FOR WATER-, STEAM-, AND RADIOACTIVE-WASTE-CONTAINING COMPONENTS OF NUCLEAR POWER PLANTS

1.9.26.1 Regulatory Guide 1.26 Position

Defines classification of systems and components.

1.9.26.2 VEGP Position

The quality group classification and standards for water-, steam-, and radioactive-waste-containing components for VEGP are identified in table 3.2.2-1. Westinghouse classifies

components within its scope of supply using the guidelines of ANSI N18.2-1973 and ANSI N18.2a-1975, which is an acceptable alternative to this regulatory guide.

1.9.27 REGULATORY GUIDE 1.27, REVISION 2, JANUARY 1976, ULTIMATE HEAT SINK FOR NUCLEAR POWER PLANTS

1.9.27.1 Regulatory Guide 1.27 Position

This guide describes a method acceptable to the NRC for designing an ultimate heat sink used for dissipation of residual heat following reactor shutdown or an accident.

1.9.27.2 VEGP Position

Conform. Refer to subsection 9.2.5.

1.9.28 REGULATORY GUIDE 1.28, REVISION 2, FEBRUARY 1979, QUALITY ASSURANCE PROGRAM REQUIREMENTS (DESIGN AND CONSTRUCTION) (HISTORICAL)

1.9.28.1 Regulatory Guide 1.28 Position (HISTORICAL)

The overall quality assurance program (QAP) requirements for the design and construction phases that are included in ANSI N45.2-1977 provide an adequate basis for complying with the QAP requirements of Appendix B to 10 CFR 50, as supplemented or modified by this guide.

1.9.28.2 VEGP Position (HISTORICAL)

The VEGP quality assurance program for design and construction phases is in conformance with Regulatory Guide 1.28, formerly Safety Guide 28, June 7, 1972, Quality Assurance Program Requirements (Design and Construction) which invokes ANSI N45.2-1971 as an acceptable standard for nuclear quality assurance programs. The VEGP QAP for design and construction is discussed in chapter 17.1.

1.9.29 REGULATORY GUIDE 1.29, REVISION 3, SEPTEMBER 1978, SEISMIC DESIGN CLASSIFICATION

1.9.29.1 Regulatory Guide 1.29 Position

Defines systems to withstand the safe shutdown earthquake (SSE).

1.9.29.2 VEGP Position

VEGP conforms with this regulatory guide as shown in table 3.2.2-1.

With regard to regulatory position C.1, each nuclear steam supply system (NSSS) component important to safety is classified as Safety Class 1, 2, or 3; these classes are qualified to remain functional in the event of the SSE, except where exempted by meeting all of the below requirements. Portions of systems required to perform the same safety function as required of a safety class component which is part of that system shall be likewise qualified or granted exemption. Conditions to be met for exemption are:

- Failure would not directly cause a Condition III or IV event (as defined in ANSI N18.2-1973).
- There is no safety function to mitigate, nor could failure prevent mitigation of, the consequence of a Condition III or IV event.
- Failure during or following any Condition IV event would result in consequences no more severe than allowed for a Condition III event.
- Routine post-seismic procedures would disclose loss of the safety function.

1.9.30 REGULATORY GUIDE 1.30, AUGUST 1972, QUALITY ASSURANCE REQUIREMENTS FOR THE INSTALLATION, INSPECTION, AND TESTING OF INSTRUMENTATION AND ELECTRIC EQUIPMENT

1.9.30.1 Regulatory Guide 1.30 Position (HISTORICAL)

[The requirements for the installation, inspection, and testing of nuclear power plant instrumentation and electric equipment which are included in ANSI N45.2.4-1972, Installation, Inspection, and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations (also designated IEEE Std. 336-1971) are generally acceptable and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the qualifications in the regulatory guide.]

1.9.30.2 VEGP Position (HISTORICAL)

[(HISTORICAL) The VEGP QAP during design and construction conforms to ANSI N45.2.4-1972 and IEEE 336-1971, as described in section 17.1.]

Regulatory Guide 1.30 provided NRC endorsement of ANSI N45.2.4 (IEEE 336-1971). The SNC Quality Assurance Topical Report (QATR) is based on ASME NQA-1-1994 which incorporates the requirements of the installation, inspection, and testing of instrumentation and electric equipment formerly contained in IEEE 336-1971 are superseded by those contained IEEE 336-1985 as described in the QATR.

1.9.31 REGULATORY GUIDE 1.31, REVISION 3, APRIL 1978, CONTROL OF FERRITE CONTENT IN STAINLESS STEEL WELD METAL

1.9.31.1 Regulatory Guide 1.31 Position

This guide describes a method acceptable to the NRC for implementing requirements for the control of welding in fabricating and joining safety-related austenitic stainless steel components and systems in light-water-cooled nuclear power plants.

1.9.31.2 VEGP Position

Conforms to the basic concept of controlling delta ferrite content except for magnetic measurement of the delta ferrite in procedure qualification samples and in production welds. To meet the intent of the regulatory guide, the control of ferrite content in weld metal is attained by chemical analysis and/or magnetic measurement of the weld metal, as applicable.

Welding materials for welding austenitic stainless steel to austenitic stainless may contain 8- to 25-percent delta ferrite. For welding austenitic stainless steel to carbon, or low alloy, steel welding materials may contain 5- to 15-percent delta ferrite. The use of welding materials with a delta ferrite exceeding the recommended Ferrite Number 20 is done in accordance with the regulatory guide since austenitic stainless steel items are not postweld heat treated above 350°F (except during welding) unless they are given a full solution anneal at the material manufacturer's recommended temperature and holding period, followed by water quenching or spraying from the solution heat treating temperature rapidly enough to prevent carbide precipitation.

Control of ferrite content in stainless steel weld metal for NSSS equipment is discussed in paragraph 5.2.3.4.6.

1.9.32 REGULATORY GUIDE 1.32, REVISION 2, FEBRUARY 1977, CRITERIA FOR SAFETY-RELATED ELECTRIC POWER SYSTEMS FOR NUCLEAR POWER PLANTS

1.9.32.1 Regulatory Guide 1.32 Position

For the portion of safety-related electric power systems within its scope, the criteria, requirements, and recommendations in IEEE Std.-308-1974 are generally acceptable to the NRC staff and provide an adequate basis for complying with General Design Criteria 17 and 18 of Appendix A to 10 CFR 50 with respect to the design, operation, and testing of electric power systems, subject to the qualifications identified in the guide.

1.9.32.2 VEGP Position

Conform except for the periodic test schedule for batteries. Safety-related batteries will be tested periodically in accordance with the Technical Specifications and the version of IEEE 450 as described in the Bases for the Technical Specifications. Refer to comparisons for Regulatory Guides 1.6, 1.9, 1.75, 1.81, 1.93 and 1.129. Further discussion is provided in sections 8.1 and 8.3.

1.9.33 REGULATORY GUIDE 1.33, REVISION 2, FEBRUARY 1978, QUALITY ASSURANCE PROGRAM REQUIREMENTS (OPERATION)

1.9.33.1 Regulatory Guide 1.33 Position (HISTORICAL)

[The overall QAP requirements for the operation phase that are included in ANSI N18.7-1976/American Nuclear Society (ANS) 3.2 are acceptable to the NRC and provide an adequate basis for complying with the QAP requirements of Appendix B to 10 CFR 50, subject to the qualifications in the guide.]

1.9.33.2 VEGP Position

Regulatory Guide 1.33 provided NRC endorsement of ANSI N18.7 as an acceptable basis for a quality assurance program that meets the requirements of 10 CFR 50, Appendix B. In addition, Regulatory Guide 1.33 includes regulatory positions that qualify NRC endorsement of ANSI N18.7.

The SNC QATR is based on ASME NQA-1-1994 and incorporates the applicable requirements of ANSI N18.7-1976. Accordingly, SNC complies with the applicable requirements of ANSI N18.7-1976 via compliance with the QATR without an explicit (or implied) commitment to either Regulatory Guide 1.33 or ANSI N18.7-1976.

1.9.34 REGULATORY GUIDE 1.34, DECEMBER 1972, CONTROL OF ELECTROSLAG WELD PROPERTIES

1.9.34.1 Regulatory Guide 1.34 Position

This guide describes an acceptable method of implementing requirements with regard to the control of weld properties when fabricating electroslag welds for nuclear components made of ferritic or austenitic materials.

1.9.34.2 VEGP Position

Conform. Refer to paragraph 5.2.3.4.6.

1.9.35 REGULATORY GUIDE 1.35, REVISION 2, JANUARY 1976, INSERVICE INSPECTION OF UNGROUTED TENDONS IN PRESTRESSED CONCRETE CONTAINMENT STRUCTURES

1.9.35.1 Regulatory Guide 1.35 Position

This guide describes an acceptable basis for developing an appropriate inservice inspection and surveillance program for ungrouted tendons in prestressed concrete containment structures.

1.9.35.2 VEGP Position

Conform as discussed in subsection 3.8.1.

1.9.36 REGULATORY GUIDE 1.36, FEBRUARY 1973, NON-METALLIC THERMAL INSULATION FOR AUSTENITIC STAINLESS STEEL

1.9.36.1 Regulatory Guide 1.36 Position

This guide describes an acceptable method for implementing criteria for the selection and use of nonmetallic thermal insulation to minimize contamination that could promote stress-corrosion cracking in stainless steel components.

1.9.36.2 VEGP Position

Conform. Refer to paragraphs 5.2.3.2.3 and 6.1.1.1.3.

1.9.37 REGULATORY GUIDE 1.37, MARCH 1973, QUALITY ASSURANCE REQUIREMENTS FOR CLEANING OF FLUID SYSTEMS AND ASSOCIATED COMPONENTS OF WATER-COOLED NUCLEAR POWER PLANTS

1.9.37.1 Regulatory Guide 1.37 Position (HISTORICAL)

[The requirements and recommendations for onsite cleaning of materials and components, cleanness control, and preoperational cleaning and layup of water-cooled nuclear power plant fluid systems that are included in ANSI N45.2.1-1973, Cleaning of Fluid Systems and Associated Components During Construction Phase of Nuclear Power Plants, are generally acceptable and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the qualifications identified in the guide.]

1.9.37.2 VEGP Position

[(HISTORICAL) The VEGP QAP during design and construction conforms to ANSI N45.2.1-1973 with the following exceptions and clarifications.

Exceptions are as follows:

1. *The VEGP QAP during design and construction conforms to ANSI N45.2.1-1973 except in regard to installation cleaning. Carbon steel piping is stored with the end caps removed and without desiccants. The piping is stored to allow drainage and to prevent entry of rainwater. Prior to installation the piping is inspected and cleaned if necessary.*

Clarifications are as follows:

1. *This guide applies to onsite cleaning of materials and components and, therefore, not in the direct scope of NSSS supply. However, controls for cleaning processes during manufacture of NSSS equipment satisfy the objective of ANSI N45.2.1-1973, which is to assure that components delivered to the plant site require only water flushing or rinsing to render them ready for service.*

2. *The items shall be flushed at the design velocity or other flow velocity as specified in the procedures.]*

Regulatory Guide 1.37 provided NRC endorsement of ANSI N45.2.1. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N 45.2.1. Accordingly, quality assurance requirements for cleaning of fluid systems and associated components are described in the SNC QATR.

1.9.38 REGULATORY GUIDE 1.38, REVISION 2, MAY 1977, QUALITY ASSURANCE REQUIREMENTS FOR PACKAGING, SHIPPING, RECEIVING, STORAGE AND HANDLING OF ITEMS FOR WATER-COOLED NUCLEAR POWER PLANTS

1.9.38.1 Regulatory Guide 1.38 Position (HISTORICAL)

[The requirements for the packaging, shipping, receiving, storage, and handling of items for water-cooled nuclear power plants that are included in ANSI N45.2.2-1972, Packaging, Shipping, Receiving, Storage, and Handling of Items for Nuclear Power Plants During the Construction Phase, are acceptable to the NRC staff and, when supplemented by the guidelines identified in regulatory position 2, provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the qualifications identified in this guide.]

1.9.38.2 VEGP Position

[(HISTORICAL) The VEGP QAP during design and construction as described in section 17.1, conforms with Regulatory Guide 1.38 (March 16, 1973) which also endorses ANSI N45.2.2-1972, with the following exceptions and clarifications.

Exceptions are as follows:

1. *Exception is taken to the requirements for use of nonhalogenated wrappings.*

Clarifications are as follows:

1. *Brightly or specially colored tape will not be used due to the rigorous flushing program scheduled prior to preoperation. Tapes and vapor barriers used in packaging processes for NSSS equipment contrast with the material being packaged when such packing materials are commercially available.*

2. *Caps and plugs are used only when required by the specification. See Regulatory Guide 1.37 comparison. Tape near a weld may be removed to clean, setup, and inspect surface.*

3. *The contact preservative used on the main condenser is not water flushable; it will be chemically cleaned.*

4. *Quality assurance for packaging, shipping, receiving, storage, and handling of NSSS equipment is described in WCAP-8370/7800, Table 17-1. Refer to section 17.1 for further discussion.]*

Regulatory Guide 1.38 provided NRC endorsement of ANSI N45.2.2. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.2. Accordingly, quality assurance requirements for packaging, shipping, receiving, storage, and handling are described in the SNC QATR.

1.9.39 REGULATORY GUIDE 1.39, REVISION 2, SEPTEMBER 1977, HOUSEKEEPING REQUIREMENTS FOR WATER-COOLED NUCLEAR POWER PLANTS

1.9.39.1 Regulatory Guide 1.39 Position (HISTORICAL)

[This guide describes an acceptable method of complying with regulations with regard to housekeeping requirements for the control of work activities, conditions, and environments at water-cooled nuclear power plant sites.]

1.9.39.2 VEGP Position

Regulatory Guide 1.39 provided NRC endorsement of ANSI N45.2.3. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.3. Accordingly, housekeeping requirements are described in the SNC QATR.

1.9.40 REGULATORY GUIDE 1.40, MARCH 1973, QUALIFICATION TESTS OF CONTINUOUS-DUTY MOTORS INSTALLED INSIDE THE CONTAINMENT OF WATER-COOLED NUCLEAR POWER PLANTS

1.9.40.1 Regulatory Guide 1.40 Position

The procedures for conducting qualification tests of continuous-duty motors installed inside the containment of water-cooled nuclear power plants which are specified by IEEE Std. 334-1971, IEEE Trial-Use Guide for Type Tests of Continuous-Duty Class I Motors Installed Inside the Containment of Nuclear Power Generating Stations, are generally acceptable and provide an adequate basis for complying with the qualification testing requirements of Criterion III of Appendix B to 10 CFR 50, to verify adequacy of design for service under the most adverse design conditions, subject to the qualifications identified in the guide.

1.9.40.2 VEGP Position

To the extent practicable, the procedures for conducting qualification tests specified by IEEE Std. 334-1974 are used to supplement the requirements of IEEE 323-1974 for Class 1E motors inside the containment. Refer to Regulatory Guide comparison 1.100 and section 3.11.

1.9.41 REGULATORY GUIDE 1.41, MARCH 1973, PREOPERATIONAL TESTING OF REDUNDANT ONSITE ELECTRIC POWER SYSTEMS TO VERIFY PROPER LOAD GROUP ASSIGNMENTS

1.9.41.1 Regulatory Guide 1.41 Position

As part of the initial preoperational testing program, and also after major modifications or repairs to a facility, those onsite electric power systems designed in accordance with Regulatory Guides 1.6 and 1.32 (Safety Guides 6 and 32) should be tested as follows to verify the existence of independence among redundant onsite power sources and their load groups.

1. C.1 The plant electric power distribution system, not necessarily including the switchyard and the startup and auxiliary transformers, should be isolated from the offsite transmission network. Preferably, this isolation should be effected by direct actuation of the undervoltage-sensing relays within the onsite system.

2. C.2 Under the conditions of C.1 above, the onsite electric power system should be functionally tested successively in the various possible combinations of power sources and load groups with all dc and onsite ac power sources for one load group at a time completely disconnected. Each test should include injection of simulated accident signals, startup of the onsite power source(s) and load group(s) under test, sequencing of loads, and the functional performance of the loads. Each test should be of sufficient duration to achieve stable operating conditions and thus permit the onset and detection of adverse conditions which could result from improper assignment of loads; e.g., the lack of forced cooling of a vital device.

3. C.3 During each test, the dc and onsite ac buses and related loads not under test should be monitored to verify absence of voltage at these buses and loads.

1.9.41.2 VEGP Position

VEGP is committed to follow this regulatory guide. Refer to section 14.2 for further discussion.

1.9.42 REGULATORY GUIDE 1.42

Withdrawn.

1.9.43 REGULATORY GUIDE 1.43, MAY 1973, CONTROL OF STAINLESS STEEL WELD CLADDING OF LOW-ALLOY STEEL COMPONENTS

1.9.43.1 Regulatory Guide 1.43 Position

This guide describes acceptable methods for implementing requirements with regard to the selection and control of the welding process used for cladding ferritic steel components with austenitic stainless steel to restrict practices that could result in underclad cracking.

1.9.43.2 VEGP Position

Qualification testing is performed on any high-heat input welding process (such as the submerged-arc wide-strip welding process or the submerged arc 6-wire process) used to clad coarse or fine grained SA-508 Class 2 material. This test follows the recommendations of this guide. Production welding is monitored by the fabricator to ensure that essential variables remain within the limits established by the qualification. If the essential variables exceed the qualification limits, an evaluation is performed to determine if the cladding is acceptable for use. Where Westinghouse permits the use of submerged-arc strip process on SA-508 Class 2 material, a two-layer technique is used to minimize intergranular cracking. Refer to paragraph 5.2.3.3.2.

1.9.44 REGULATORY GUIDE 1.44, MAY 1973, CONTROL OF THE USE OF SENSITIZED STAINLESS STEEL

1.9.44.1 Regulatory Guide 1.44 Position

This guide describes acceptable methods for controlling the application and processing of stainless steel to avoid severe sensitization that could lead to stress corrosion cracking.

1.9.44.2 VEGP Position

Conforms to the basic concept of controlling the use of sensitized stainless steel. Pipe and piping products are solution annealed by heating the material to the manufacturer's recommended solution annealing temperature and holding at this temperature for the required period, followed by water quenching or spraying rapidly enough to prevent carbide precipitation. Verification of nonsensitization of casting material is not specified, however, because sensitization is controlled by requiring the above solution annealing procedure and control of delta ferrite content.

Control of the use of sensitized stainless steel for NSSS equipment is discussed in paragraph 5.2.3.4.

1.9.45 REGULATORY GUIDE 1.45, MAY 1973, REACTOR COOLANT PRESSURE BOUNDARY LEAKAGE DETECTION SYSTEM

1.9.45.1 Regulatory Guide 1.45 Position

This guide describes acceptable methods for implementing requirements with regard to the selection of leakage detection systems for the RCPB.

1.9.45.2 VEGP Position

VEGP conforms as described in subsection 5.2.5.

1.9.46 REGULATORY GUIDE 1.46, MAY 1973, PROTECTION AGAINST PIPE BREAK INSIDE CONTAINMENT

1.9.46.1 Regulatory Guide 1.46 Position

This guide presents guidelines acceptable to the NRC staff for implementing General Design Criterion 4 of Appendix A to 10 CFR 50 in the protection of safety-related structures, systems, and components from the effects of pipe ruptures.

1.9.46.2 VEGP Position

In lieu of Regulatory Guide 1.46, the basis for conformance with General Design Criterion 4 of Appendix A to 10 CFR 50 is the implementation of NRC Branch Technical Position (BTP) MEB 3-1 and NRC BTP ASB 3-1. Refer to table 3.6.1-2.

The criteria originally implemented in the evaluation of the main reactor coolant loop was based on draft ANSI Standard 20.2, Design Basis for Protection Against Pipe Whip, and is documented in WCAP-8172-A, Pipe Breaks for the LOCA Analysis of the Westinghouse

Primary Coolant Loop. WCAP-8172-A has received NRC approval as providing an equivalent degree of protection as would be obtained by applying the criteria of Regulatory Guide 1.46. By specific exemption (50 FR 5454 February 8, 1985), the elimination of eight of these postulated large pipe breaks in the RC loop was authorized. (See section 3.6.) Elimination of the dynamic effects of pipe rupture in the NSSS Class 1 branch lines for Unit 2 is also discussed in section 3.6.

1.9.47 REGULATORY GUIDE 1.47, MAY 1973, BYPASSED AND INOPERABLE STATUS INDICATION FOR NUCLEAR POWER PLANT SAFETY SYSTEMS

1.9.47.1 Regulatory Guide 1.47 Position

Describes an acceptable method for implementing the requirements of Section 4.13 of IEEE Std. 279-1971 and Criterion XIV of Appendix B to 10 CFR 50 with respect to indicating the bypass or inoperable status of portions of the protection system, systems actuated or controlled by the protection system, and auxiliary or supporting systems that must be operable for the protection system and the system it actuates to perform their safety-related functions.

1.9.47.2 VEGP Position

VEGP conforms as described in subsection 7.5.5 and paragraph 10.4.9.5.

1.9.48 REGULATORY GUIDE 1.48, MAY 1973, DESIGN LIMITS AND LOADING COMBINATIONS FOR SEISMIC CATEGORY 1 FLUID SYSTEM COMPONENTS

1.9.48.1 Regulatory Guide 1.48 Position

This guide describes acceptable design limits and appropriate combinations of loadings associated with normal operation, postulated accidents, and specified seismic events for the design of Seismic Category 1 fluid system components.

1.9.48.2 VEGP Position

VEGP conforms as described in subsection 3.9.N.3 and tables 3.9.B.3-3 through -10.

1.9.49 REGULATORY GUIDE 1.49, REVISION 1, DECEMBER 1973, POWER LEVELS OF NUCLEAR POWER PLANTS

1.9.49.1 Regulatory Guide 1.49 Position

This guide describes acceptable maximum power levels for nuclear power plants.

1.9.49.2 VEGP Position

Conform. VEGP has a maximum licensed core thermal power. Refer to chapter 4 for further discussion.

1.9.50 REGULATORY GUIDE 1.50, MAY 1973, CONTROL OF PREHEAT TEMPERATURES FOR WELDING OF LOW-ALLOY STEEL

1.9.50.1 Regulatory Guide 1.50 Position

Weld fabrication for low-alloy steel components should comply with the fabrication requirements specified in Section III and Section IX of the ASME Boiler and Pressure Vessel Code supplemented by the following:

1. C.1 The procedure qualification should require that:

   a. A minimum preheat and a maximum interpass temperature be specified.

   b. The welding procedures be qualified at the minimum preheat temperature.

2. C.2 For production welds, the preheat temperature should be maintained until a post-weld heat treatment has been performed.

1.9.50.2 VEGP Position

1. C.1 Paragraph 1.a is conformed with when impact testing, in accordance with ASME Boiler and Pressure Vessel Code, Section III, Subarticle 2300, is required. When impact testing is not required, specification of a maximum interpass temperature in the welding procedures is not necessary in order to assure that the other required mechanical properties of the weld are met.

   Paragraph 1.b conforms.

2. C.2 Conforms for pressure vessels with nominal thicknesses greater than 1 in. Maintenance of preheat beyond completion of welding until postweld.

The NSSS Class 1 components are in conformance with Regulatory Guide 1.50 except for regulatory positions 1.b and 2. For Class 2 and 3 components, Westinghouse does not apply Regulatory Guide 1.50 recommendations.

In the case of regulatory position 1.b, the welding procedures are qualified within the preheat temperature ranges required by Section IX of the ASME Code. Experience has shown excellent quality of welds using the ASME qualification procedures.

In the case of regulatory position 2, it is felt that this position is both unnecessary and impractical. Code acceptance low-alloy steel welds have been and are being made under present Westinghouse specified procedures. It is not necessary to maintain the preheat

temperature until a post-weld heat treatment has been performed as required by the guide, in the case of large components. In the case of reactor vessel main structural welds, the practice of maintaining preheat until the intermediate or final post-weld heat treatment has been followed by Westinghouse. In either case, the welds have shown high integrity. Westinghouse practices are documented in WCAP-8577, The Application of Preheat Temperature After Welding of Pressure Vessel Steel, which has been accepted by the NRC. Refer to subsection 5.2.3 for further discussion.

1.9.51 REGULATORY GUIDE 1.51

Withdrawn.

1.9.52 REGULATORY GUIDE 1.52, REVISION 2, MARCH 1978, DESIGN, TESTING, AND MAINTENANCE CRITERIA FOR POST-ACCIDENT ENGINEERED SAFETY FEATURE ATMOSPHERE CLEANUP SYSTEM AIR FILTRATION AND ADSORPTION UNITS OF LIGHT-WATER-COOLED NUCLEAR POWER PLANTS

1.9.52.1 Regulatory Guide 1.52 Position

This guide describes an acceptable method for the design, testing, and maintenance of post-accident ESF atmosphere cleanup systems designed to mitigate the consequences of postulated accidents.

1.9.52.2 VEGP Position

Conform, except for the following exceptions and clarifications:

1. C.2.a The atmospheric cleanup systems consist of moisture eliminators (demisters), high-efficiency particulate air (HEPA) filters before the adsorber, iodine adsorbers, HEPA filters after the adsorber, valves, fans, and related instrumentation. The fan systems are integrated with the filter housing. The filtration units and associated instrumentation and valving are redundant; however, the air distribution system is common for the redundant filtration units.

2. C.2.g No recording of filter pressure drops or flowrates is available in the control room.

3. C.2.j The design and installation of the ESF atmosphere cleanup systems do not permit the replacement of a train as an intact unit. The design of the filtration unit consists of an all-welded housing construction to minimize outleakage of potentially radioactive contaminants. Missile protection and shielding are provided for the units. Component servicing and maintenance provisions are provided consistent with Regulatory Guide 8.8.

4. C.3.k Anticipated charcoal bed loading for the design basis accident is not sufficient to raise bed temperature to the desorption and adsorbent automatic ignition range.

However, a water spray system for the adsorber has been included in the design to prevent excessive heating, if required.

5. C.5.c Fuel handling post accident filter satisfies 99.0 percent retention of DOP on HEPA filters instead of 99.95 percent.

6. C.5.d Fuel handling post accident filter satisfies 99.0 percent retention of gaseous halogenated hydrocarbon refrigerant on the adsorber instead of 99.95 percent.

7. C.6.b Carbon samples taken for laboratory tests shall meet the following acceptance criteria as specified in the VEGP Technical Specifications.

    a. Control room emergency filtration system criterion is greater than or equal to 99.8 percent when tested with methyl iodide at 30°C and 70-percent relative humidity.

    b. Piping penetration area filtration and exhaust system and fuel handling building post accident ventilation system criteria are greater than or equal to 90.0 percent when tested with methyl iodide at 30°C and 95-percent relative humidity.

Wherever ANSI N509-1976 is referenced in the regulatory guide, conformance is with ANSI N509-1976, ANSI N509-1980, or ASME N509-1989 depending on the date of the applicable purchase order. Conformance may be with a particular revision when specifically called out in the corresponding specification.

Wherever ANSI N510-1975 is referenced in the regulatory guide, conformance is with ANSI N510-1975, ANSI N510-1980, or ASME N510-1989 depending on the date of the applicable purchase order. Conformance may be with a particular revision when specifically called out in the corresponding specification.

1.9.53 REGULATORY GUIDE 1.53, JUNE 1973, APPLICATION OF THE SINGLE-FAILURE CRITERION TO NUCLEAR POWER PLANT PROTECTION SYSTEMS

1.9.53.1 Regulatory Guide 1.53 Position

The guidance in trial-use IEEE Std. 379-1972 for applying the single-failure criterion to the design and analysis of nuclear power plant protection systems is generally acceptable and provides an adequate interim basis for complying with Section 4.2 of IEEE Std. 279-1971, subject to the qualifications identified in the guide.

1.9.53.2 VEGP Position

Conform. Refer to paragraph 7.1.2.6 and subsection 15.0.8.

1.9.54 REGULATORY GUIDE 1.54, JUNE 1973, QUALITY ASSURANCE REQUIREMENTS FOR PROTECTIVE COATINGS APPLIED TO WATER-COOLED NUCLEAR POWER PLANTS

1.9.54.1 Regulatory Guide 1.54 Position

The requirements and guidelines included in ANSI N101.4-1972, Quality Assurance for Protective Coatings Applied to Nuclear Facilities, for protective coatings applied to ferritic steels, aluminum, stainless steel, zinc-coated (galvanized) steel, concrete, or masonry surfaces of water-cooled nuclear power plants are generally acceptable and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50 subject to the qualifications identified in the guide.

1.9.54.2 VEGP Position

The VEGP conformance is discussed in table 6.1.2-1.

1.9.55 REGULATORY GUIDE 1.55, JUNE 1973, CONCRETE PLACEMENT IN CATEGORY 1 STRUCTURES

1.9.55.1 Regulatory Guide 1.55 Position

This guide describes acceptable bases for implementing the quality assurance requirements applicable to the placement of concrete in Category 1 structures.

1.9.55.2 VEGP Position

VEGP conforms to the following standards identified in the regulatory guide:

- ACI 305-72 Recommended Practice for Hot Weather Concreting.
- ACI 306-66 Recommended Practice for Cold Weather Concreting.
- ACI 308-71 Recommended Practice for Curing Concrete.
- ACI 347-68 Recommended Practice for Concrete Formwork.

VEGP also conforms to ACI 318-71 "Building Code Requirements for Reinforced Concrete" with the following exceptions:

- Section 6.2.2 - Forms not supporting the weight of concrete may be removed in less than 24-h to satisfy the curing requirements.

- Section 6.4.1 - Vertical construction joints are not coated with neat cement grout before placing new concrete. The use of neat cement grout in vertical construction joints is impractical and can be detrimental where deep forms and steel congestion prevent proper access.

In addition, the following standards not identified in the regulatory guide are used for concrete placement of Category 1 structures:

- ACI 304-73 Recommended Practice For Measuring, Mixing, Transporting, and Placing Concrete.

- ACI 309-72 Consolidation of Concrete.

Refer to section 3.8 for further discussion.

1.9.56 REGULATORY GUIDE 1.56, REVISION 1, JULY 1978, MAINTENANCE OF WATER PURITY IN BOILING WATER REACTORS

Not applicable to VEGP.

1.9.57 REGULATORY GUIDE 1.57, JUNE 1973, DESIGN LIMITS AND LOADING COMBINATIONS FOR METAL PRIMARY REACTOR CONTAINMENT SYSTEM COMPONENTS

1.9.57.1 Regulatory Guide 1.57 Position

This guide delineates acceptable design limits and appropriate combinations of loadings associated with normal operation, postulated accidents, and specified seismic events for the design of components of metal primary reactor containment systems.

1.9.57.2 VEGP Position

Conformance is discussed in subsection 3.8.2.

1.9.58 REGULATORY GUIDE 1.58, REVISION 1, SEPTEMBER 1980, QUALIFICATION OF NUCLEAR POWER PLANT INSPECTION, EXAMINATION, AND TESTING PERSONNEL

1.9.58.1 Regulatory Guide 1.58 Position (HISTORICAL)

[The requirements and recommendations for qualification of nuclear power inspection, examination, and testing personnel that are included in ANSI N45.2.6-1978 are generally acceptable and provide an

adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the qualifications identified in the guide.]

1.9.58.2 VEGP Position - Prelicense (HISTORICAL)

[VEGP shall conform with this guide, which endorses ANSI N45.2.6-1978, with the following exceptions and clarifications:

Exceptions are as follows:

1. *Position C.8 of Regulatory Guide 1.58. Exposure during construction is restricted to radiography, and training in radiation protection is supplied in the radiography certification course. Construction personnel entering restricted areas of the operative plant will receive training in radiation protection from power generation prior to being allowed access.*

Clarifications are as follows:

1. *Paragraph 1.2 of ANSI N45.2.6-1978, Applicability. VEGP personnel who approve preoperational, startup, and test results and who direct or supervise the conduct of individual preoperational, startup, and operational tasks shall be qualified in accordance with the VEGP position to Regulatory Guide 1.8 in lieu of being qualified to ANSI N45.2.6 as allowed by regulatory position C.1 of Regulatory Guide 1.58, Rev. 1. For nuclear operations, VEGP elects to apply the requirements of this guide to quality control inspection personnel.*

   Regulatory position C.2 of Regulatory Guide 1.58, Rev. 1, states that the 1975 version of SNT-TC-1A is acceptable for the qualification of personnel performing nondestructive examinations. In lieu of this, VEGP Construction personnel (SNC only) and Nuclear Operations personnel shall use the requirements of the 1980 version of SNT-TC-1A for qualifying personnel performing nondestructive inspection, examination, or testing, and in addition VEGP shall supplement these requirements by replacing the "shoulds" contained in the 1980 version with "shalls" except as follows:

   a. *The "should" in the last sentence of Section 4.3 (3) of SNT-TC-1A - 1980, which refers to a NDE level III being qualified to instruct and examine NDT personnel, is not replaced with "shall." For the VEGP, a NDE level III individual shall be responsible for the training and examination of NDT personnel; while conducting of actual training and examination may be delegated to qualified individuals appointed by the level III individual.*

   b. *The "should" in the second sentence of Section 8.4.3 of SNT-TC-1A - 1980 remains as a "should," as allowed by Section 8.4.2. VEGP elects to use a composite grade based on the simple average of the examinations; this shall be prescribed in VEGP's written practice.*

   Contractors may use the 1975 or later version of SNT-TC-1A, as allowed, for qualifying their personnel.

   For personnel performing calibration, installation, checkouts, or routine surveillances the requirements of this guide will not be applied, as allowed by Section 1.2 of ANSI N45.2-6-1978; personnel performing these functions shall either meet the minimum education and experience recommendations of ANSI N18.1-1971 or will complete a qualification program which will demonstrate their ability to perform their job functions.

FSAR table 13.1.3-1 designates the minimum education and experience recommendations for plant personnel, while FSAR subsection 13.2.2 describes the training programs which demonstrate the ability of plant personnel to perform their job functions.

2. *Paragraph 2.5 of ANSI N45.2.6-1978, Physical. VEGP will implement the requirements of this section with the stipulation that, where no special physical characteristics are required, none will be specified. The converse is also true; if no special physical requirements are stipulated by VEGP, none are considered necessary. SNC employees receive an initial physical examination to assure satisfactory physical condition; SNC management shall determine which personnel are required to receive an annual examination.*

3. *Paragraph 3 of ANSI N45.2.6-1978, Qualification. Same clarification as 1.*

4. *Personnel performing visual examinations (VT-2, 3, and 4) per the requirements of ASME Section XI will either meet the experience requirements in sections 3.5.1, 3.5.2, or 3.5.3 of ANSI N45.2.6-1978 or alternately, a program based on experience, comprehensive training, and testing as allowed in section 3.5 of ANSI N45.2.6-1978. Detailed training and testing will be accomplished either through the visual examination training program at the Electric Power Research Institute's NDE Center or an equivalent program.*

5. *The Georgia Power Company (GPC) Construction Department Inspector Certification Program utilizes level III administrators to coordinate training, certify qualified level I & II personnel, and maintain records. Level III administrators must be high school graduates as a minimum. Level III administrators are appointed by the manager of quality control. They must be members of the quality control staff, and must have extensive inspection, technical or engineering construction experience that has been appropriately documented. The primary responsibility of the level III administrator is to evaluate the adequacy of specific programs used to train and test inspection, examination and testing personnel. The level III administrator does not plan, implement, record, evaluate, or report inspections.]*

1.9.58.3 VEGP Position - Post-License

Regulatory Guide 1.58 provided NRC endorsement of ANSI N45.2.6. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.6. Accordingly, the requirements for qualification of inspection, examination, and testing personnel are described in the SNC QATR.

1.9.59 REGULATORY GUIDE 1.59, REVISION 2, AUGUST 1977, DESIGN BASIS FLOODS FOR NUCLEAR POWER PLANTS

1.9.59.1 Regulatory Guide 1.59 Position

This guide describes the conditions resulting from the worst site-related flood probable at a nuclear power plant that safety-related structures, systems, and components must be designed to withstand and retain capability for cold shutdown and maintenance thereof.

1.9.59.2 VEGP Position

Conform. See subsections 2.4.3, 2.4.4, and 3.4.1 for a detailed discussion on flood protection.

1.9.60 REGULATORY GUIDE 1.60, REVISION 1, DECEMBER 1973, DESIGN RESPONSE SPECTRA FOR SEISMIC DESIGN OF NUCLEAR POWER PLANTS

1.9.60.1 Regulatory Guide 1.60 Position

This guide describes an acceptable procedure for defining response spectra for seismic design of nuclear power plants.

The Newmark-Blume-Kapur design spectra curves for free field ground accelerations are endorsed.

1.9.60.2 VEGP Position

Conform. Refer to sections 3.7.B and 3.7.N for discussion on this subject.

1.9.61 REGULATORY GUIDE 1.61, OCTOBER 1973, DAMPING VALUES FOR SEISMIC DESIGN OF NUCLEAR POWER PLANTS

1.9.61.1 Regulatory Guide 1.61 Position

This guide delineates acceptable damping values to be used in the elastic model dynamic seismic analysis of Seismic Category 1 structures, systems, and components.

1.9.61.2 VEGP Position

Conformance is discussed in subsections 3.7.B.1 and 3.7.N.1.

1.9.62 REGULATORY GUIDE 1.62, OCTOBER 1973, MANUAL INITIATION OF PROTECTIVE ACTIONS

1.9.62.1 Regulatory Guide 1.62 Position

This guide describes an acceptable method for complying with the requirements of Section 4.17 of IEEE Std. 279-1971 for including the means for manual initiation of protective actions.

1.9.62.2 VEGP Position

Conform as described in paragraph 7.3.1.2.2.7.

1.9.63 REGULATORY GUIDE 1.63, REVISION 2, JULY 1978, ELECTRIC PENETRATION ASSEMBLIES IN CONTAINMENT STRUCTURES FOR LIGHT-WATER-COOLED NUCLEAR POWER PLANTS

1.9.63.1 Regulatory Guide 1.63 Position

Conformance with the requirements of IEEE Std. 317-1976, IEEE Standard for Electric Penetration Assemblies in Containment Structures for Nuclear Power Generating Stations, provides an acceptable method of complying with General Design Criterion 50 of Appendix A and with Appendix B to 10 CFR 50 with respect to mechanical, electrical, and test requirements for the design, qualification, construction, installation, and testing of electric penetration assemblies in containment structures for light-water-cooled nuclear power plants, subject to the following:

1. C.1 Section 4.2.4 should be supplemented by the following: The electric penetration assembly should be designed to withstand, without loss of mechanical integrity, the maximum short circuit current vs. time conditions that could occur given single random failures of circuit overload protection devices. The circuit overload protection system should conform to the criteria of IEEE Std. 279-1971, Criteria for Protection Systems for Nuclear Power Generating Stations (also designated ANSI N42.7-1972).

2. C.2 The maximum short circuit current assessed at the penetration assembly should be consistent with the criteria used in establishing the interrupting capability of the protective device associated with the penetration assembly conductors. Accordingly, the provisions of Section 4.2.4 pertaining to the rated short circuit current for ac circuits should be modified as follows:

Rated Voltage (V)	Ratio
300 600 1000	>8
5000 8000 15,000	>15

3. C.3 The provisions of Section 4.2.4 pertaining to the duration of the maximum short circuit current are representative of circuits protected by molded-case circuit breakers but are not representative of circuits using other air circuit breakers. The provisions pertaining to the duration of the maximum short-circuit current should be modified as follows:

Service Classification	Duration(s)

Low-voltage power and control	0.033 (for molded-case circuit breakers) 0.066 (for other air circuit breakers)

4. C.4 Section 6.4.4, Dielectric-Strength Test, should be supplemented, for qualification testing only, by the following:

   (3) Each medium-voltage power conductor shall be given an impulse withstand test by applying a 1.2 x 50 μs impulse voltage test series consisting of three positive and three negative impulse voltages. If flashover occurs on only one test during any group of three consecutive tests, three more shall be made. If no flashover occurs in the second group of tests, the flashover in the first group shall be considered as a random flashover and the equipment shall be considered as having passed the test.

   The test voltages for the above shall be based on the voltage rating of the conductor in accordance with the following table:

Conductor-Rated Voltage (V)	Impulse Voltage (V)
300 and 600	.....
1000	.....
5000	60,000
8000	95,000
15,000	95,000

5. C.5 The 500-h aging time at minimum aging temperature of Section 6.3.3 is a printing error and should be changed to 5000 h.

6. C.6 The definition of "Double Aperture Seal" in Section 2 is a printing error and should be changed as follows: "Two single aperture seals in series."

7. C.7 The specific applicability or acceptability of the codes, standards, and guides referenced in Section 3 will be covered separately in other regulatory guides, where appropriate.

1.9.63.2 VEGP Position

1. C.1 Protection against single random failure of circuit overload protection devices are as follows:

   a. For medium-voltage circuits, the circuit breaker associated with the load is backed up by a second load breaker in series. The second breaker is Class 1E.

   b. For 480-V loads fed from load centers, the circuit breaker associated with the load is backed up by series fuses. Primary protection is provided by the individual load circuit breaker.

c. For 480-V loads fed from motor control centers, a second breaker in series with the primary breaker to each load is used. A thermal overload relay may also be used in conjunction with these breakers in some cases.

d. For control circuits with sufficient capacity to potentially damage a penetration, backup overload protection is provided. The fault current in other low-energy level control circuits and instrument circuits is limited and does not need backup overload protection.

2. C.2 The X/R ratios used in calculating faults at the penetrations are consistent with the X/R ratios used in establishing the interrupting capability of the protective device associated with the penetration assembly conductors. Fault currents are based on: the full interrupting capability of the associated switchgear or X/R ratios as recommended by Regulatory Guide 1.63; X/R ratios as provided by industry standard for motor fault contribution; X/R ratios from actual transformer impedances and conservative cable impedances.

3. C.3 Clearing time of the circuit breakers supplying the penetrations is used to determine the penetration conductor capability.

4. C.4 Conform.

5. C.5 Conform.

6. C.6 Conform.

7. C.7 Not applicable.

Refer to subsection 8.3.1 for further discussion.

1.9.64 REGULATORY GUIDE 1.64, REVISION 2, JUNE 1976, QUALITY ASSURANCE FOR THE DESIGN OF NUCLEAR POWER PLANTS

1.9.64.1 Regulatory Guide 1.64 Position (HISTORICAL)

[The requirements and recommendations for establishing and executing a QAP during the design phase of nuclear power plants that are included in ANSI N45.2.11-1974 are acceptable to the NRC staff and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the following:

1. *C.1 Subdivision 1.5 of ANSI N45.2.11-1974 states that other documents that are required to be included as a part of this standard will be identified at the point of reference and described in Section 12 of the standard. The specific acceptability of these listed documents has been or will be covered separately in other regulatory guides or in Commission regulations where appropriate.*

2. *C.2 Instead of the second sentence of the second paragraph of Section 6.1 on design verification, the following should be used: "The duties of a supervisor and the relationship with subordinates vary widely in different organizations. Regardless of their title, individuals*

performing design verification should not (1) have immediate supervisory responsibility of the individual performing the design, (2) have specified a singular design approach, (3) have ruled out certain design considerations, or (4) have established the design inputs for the particular design aspect being verified. While design verification by the designer's immediate supervisor is encouraged, it should not be construed that such verification constitutes the required independent design verification, nor should the independent design verification be construed to dilute or replace the clear responsibility of supervisors for the quality of work performed under their supervision."

3. *C.3 In the first sentence of Section 8 of N45.2.11-1974, the word "effecting" should be inserted before "design changes" for clarification. Further, the term "approved design document" should be construed to mean "design output" (Section 1.4) approved by the organization performing the design.*

4. *C.4 Sections 4.3, 4.4, and 4.5 of N45.2.11-1974 concern the establishment of procedures for the preparation and control of drawings, specifications, and other design documents. These sections list typical subjects to be covered by such procedures. One of the subjects to be covered is "nonconformances." The NRC staff considers the "nonconformances" listed in these sections to be nonconformances with procedural requirements. Thus in Section 4.3, item (11), "Nonconformance with drawing requirements," should be construed to mean "Nonconformance with procedures for the preparation and control of drawings;" in Section 4.4, item (7), "Nonconformance with specification requirements," should be construed to mean "Nonconformance with procedures for the preparation and control of specifications;" and in Section 4.5, item (7), "Nonconformance with design document requirements," should be construed to mean "Nonconformance with procedures for the preparation and control of design documents."]*

1.9.64.2 VEGP Position

[(HISTORICAL) The VEGP QAP during design and construction phase conforms to ANSI N45.2.11, Quality Assurance Requirements for the Design of Nuclear Power Plants (Draft 2, Revision 2, May 1973) as discussed in section 17.1 and appendix 17A. With regard to the design of NSSS equipment, alternatives and clarification to the text of ANSI N45.2.11-1974 are contained in WCAP-8370/7800 Table 17-1.]

Regulatory Guide 1.64 provides NRC endorsement of ANSI N45.2.11. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.11. Accordingly, quality assurance requirements applicable to design activities are described in the SNC QATR.

1.9.65 REGULATORY GUIDE 1.65, OCTOBER 1973, MATERIALS AND INSPECTIONS FOR REACTOR VESSEL CLOSURE STUDS

1.9.65.1 Regulatory Guide 1.65 Position

This guide defines acceptable materials and testing procedures for implementing criteria with regard to reactor vessel closure stud bolting for light-water-cooled reactors.

1.9.65.2 VEGP Position

VEGP conforms with this guide except as noted below:

1. The use of modified SA-540 Grade B24 material as specified in ASME Boiler and Pressure Vessel Code Case 1605 is not specified in the guide but is used by Westinghouse. The use of this Code Case has been approved by the NRC via Regulatory Guide 1.85.

2. The maximum limit of 170 ksi ultimate tensile strength is not explicitly specified by Westinghouse as required by the guide. Westinghouse does specify fracture toughness of 45 ft/lb and 25 mils lateral expansion as required by the ASME Code and 10 CFR 50, Appendix G. These requirements also result in strength levels below the maximum limit, as demonstrated by the actual stud material properties for VEGP which are listed in tables 5.3.1-4 and 5.3.1-5.

3. Inservice inspection of the bolting will be performed using the following guidelines.

    a. All surface examinations will be performed in accordance with ASME Section XI in lieu of paragraph NB-2545 or NB-2546 of ASME Section III.

    b. Washers will be examined using the visual techniques as required by ASME Section XI in lieu of the surface examination in Section C.4.

1.9.66 REGULATORY GUIDE 1.66

Withdrawn.

1.9.67 REGULATORY GUIDE 1.67, OCTOBER 1973, INSTALLATION OF OVERPRESSURE PROTECTION DEVICES

1.9.67.1 Regulatory Guide 1.67 Position

This guide describes an acceptable method for the design of piping for safety valve and relief valve stations which have open discharge systems with limited discharge pipes, and which have inlet piping that neither contains a water seal nor is subject to slug flow of water upon discharge of the valves.

1.9.67.2 VEGP Position

Conform. Refer to paragraph 3.9.B.3.

1.9.68 REGULATORY GUIDE 1.68, REVISION 2, AUGUST 1978, INITIAL TEST PROGRAMS FOR WATER-COOLED NUCLEAR POWER PLANTS

1.9.68.1 Regulatory Guide 1.68 Position

This guide describes the general scope and depth of initial test programs acceptable to the NRC for light-water-cooled nuclear power plants. The guide provides a representative listing of the plant structures, systems, components, and the design features and performance capability tests that should be demonstrated during the initial test program. The guide also provides information on inspections that will be performed by the NRC and provides guidance on the preparation of procedures for the conduct of initial test programs.

1.9.68.2 VEGP Position

Conform as follows, except for Appendix A, Section 5, Subsections F, I, M, U, CC, HH, II, and KK, MM, Section 1, Subsections 0(1) and M(5), and Section 4, Subsections B, C, and T. Reverse flow through idle RCS loops will not be measured as required by paragraph 5(m). VEGP will not be licensed to operate with idle loops; therefore, these measurements are not applicable.

Tests U and MM will not be performed as the results obtained will be similar to the results obtained during a turbine trip from 100-percent power which will be performed. The closure times for the MSIVs will be verified during hot functional and preoperational testing.

The loss of or bypass of feedwater heaters test (test KK) will not be performed as results will be similar, but less severe than those obtained during the load swing test, paragraph 14.2.8.2.27.

The gaseous and liquid radwaste systems (test CC) will be tested as described in the gaseous waste processing system preoperational test abstract (paragraph 14.2.8.1.48) and the liquid waste processing system preoperational test abstract (paragraph 14.2.8.1.49). Performance of these tests during the power ascension test phase would produce the same results as testing during the preoperational test phase.

The complete loss of flow at full-power test (test II) will not be performed. Results for reactor coolant system (RCS) flowrates obtained in the flow coastdown test, paragraph 14.2.8.2.5, will verify that the RCS flowrates assumed in section 15.3.2 are conservative.

The load swing test (test HH) will be performed at 30 percent, 75 percent, and 100 percent. A load swing at 50-percent power will not be performed. This conforms with the standard Westinghouse startup program. The load swings at 30-percent, 75-percent and 100-percent power and are adequate to demonstrate the dynamic response of the facility.

The dropped control test (test F) will not be performed. The design has been verified and documented by testing at a prototype plant. In addition, difficulties identified in a recent INPO report in performing the test have caused significant transients in the reactor plant resulting in peaking problems.

The pseudo-rod ejection test (test 4.C) will not be performed on Unit 2. The test was successfully performed on Unit 1 and prototype plants, and no significant information would be

obtained by performing the test on Unit 2. The pseudo-rod ejection test at greater than 10-percent power will not be performed. Tests have been performed to validate the rod ejection accident analysis at prototype plants.

Testing of the reactor vessel head lifting rig and internal lifting rig (test 0 (1)) shall be in accordance with paragraph 9.1.5.2.3.4 "Special Lifting Devices" as delineated in table 9.1.5-7.

The worth of all control rods, with the greatest worth control rod stuck out (test 4.B), will not be performed. This test was performed on the essentially identical Unit 1 core and no significant additional information would be obtained from a test on Unit 2. Unit 2 will be treated as a restart and low-power testing will determine the total worth of the control rod banks (not including shutdown rod banks).

Demonstration of incore and excore nuclear instrumentation to detect control rod misalignment (test I) will not be performed because the individual rod position indication system is the primary means of determining control rod misalignments.

The natural circulation test (4.T) will not be performed on Unit 2 because the test was successfully performed on Unit 1 and satisfied all requirements. The Unit 1 test was performed during power ascension using decay heat instead of nuclear heat. A decay heat test is preferable because it eliminates the need to determine actual reactor power level and avoids unrelated trips.

All Fuel Transfer System normal operating features will be tested, including interlocks and bypasses (test M (5)). Emergency handwheel and emergency pull-out cable testing will not be performed due to the amount of equipment disassembly required and the destructive impact it would have on the equipment and dummy fuel assembly.

1.9.68.3 Regulatory Guide 1.68.2, Revision 1, July 1978, Initial Startup Test Program to Demonstrate Remote Shutdown Capability for Water-Cooled Nuclear Power Plants

1.9.68.3.1 Regulatory Guide 1.68.2 Position

This guide describes an initial startup test program acceptable to the NRC for demonstrating hot shutdown capability and the potential for cold shutdown from outside the control room.

1.9.68.3.2 VEGP Position

Conform; the initial startup test program is described in chapter 14.

1.9.68.4 Regulatory Guide 1.68.3, April 1982, Preoperational Testing of Instrument and Control Air Systems

1.9.68.4.1 Regulatory Guide 1.68.3 Position

This guide describes a method acceptable to the NRC for verifying that instrument and control air systems and the loads they supply will operate properly at normal system pressures and to assure the operability of functions important to safety in the event that system pressure is lost, reduced below normal operating level, or increased above the design pressure of the air system components to the upstream safety valve accumulation pressure.

1.9.68.4.2 VEGP Position

The instrument air system has no safety design basis as discussed in subsection 9.3.1. The ability of the instrument air system to perform its design function will be demonstrated during the instrument air preoperational test described in chapter 14. VEGP conforms with this guide with the following clarifications:

1. The provisions of position C.8 are satisfied as follows:

   Monitoring of the response of each safety-related pneumatic valve upon loss of air occurs during construction acceptance tests for each valve and is a prerequisite test for the preoperational test of the system. In performing this testing, the air pressure that will be supplied will be equivalent to the air pressure supplied by the instrument air system during normal plant operation, and it will be demonstrated that each valve responds properly (assumes its fail-safe position) for both a simulated sudden loss of air and for a gradual loss of air pressure. Since it is verified, on an individual basis, that each safety-related pneumatically operated valve will assume its fail-safe position, performance of a large-scale loss-of-air test encompassing several branches of the instrument air system is not necessary to verify correct valve response.

2. Position C.6 states that the "....ability of the system to meet the quality requirements of the system design should be verified," and references ANSI/ISA S7.3-1975 as an acceptable standard with respect to oil, water, and particulate matter contained in the product air. It further states that the quality should be verified by analyzing the air at the end of each feeder line. The VEGP instrument air system design is such that instrument air is filtered at the dehumidifier, and at each instrument (by a local filter/regulator) in accordance with individual instrument manufacturer's requirements. This ensures that particulate will be filtered at the individual instrument to prevent possible plugging and/or erosion of the individual air passages. Because of this individual filtration, verification of the particulate size at the end of each feeder is not necessary. The oil and moisture content are verified in conformance with the Regulatory Guide position.

1.9.69 REGULATORY GUIDE 1.69, DECEMBER 1973, CONCRETE RADIATION SHIELDS FOR NUCLEAR POWER PLANTS

1.9.69.1 Regulatory Guide 1.69 Position

This guide endorses ANSI N101.6-1972 which addresses the design and construction of concrete radiation shields.

1.9.69.2 VEGP Position

Not applicable since VEGP uses conventional concrete for shielding, not concrete shields addressed in ANSI N101.6-1972.

1.9.70 REGULATORY GUIDE 1.70, REVISION 3, NOVEMBER 1978, STANDARD FORMAT AND CONTENT OF SAFETY ANALYSIS REPORTS FOR NUCLEAR POWER PLANTS

1.9.70.1 Regulatory Guide 1.70 Position

The purpose of the FSAR is to inform the NRC of the nature of the plant, the plans for its use, and the safety evaluations that have been performed to evaluate whether the plant can be operated without undue risk to the health and safety of the public. The FSAR is the principal document for the applicant to provide this information. The purpose of this guide is to indicate the information to be provided in the FSAR and to establish a uniform format acceptable to the NRC for presenting this information.

1.9.70.2 VEGP Position

Conform as discussed in subsection 1.1.6.

1.9.71 REGULATORY GUIDE 1.71, DECEMBER 1973, WELDER QUALIFICATION FOR AREAS OF LIMITED ACCESSIBILITY

1.9.71.1 Regulatory Guide 1.71 Position

This guide describes a method acceptable to the NRC for implementing requirements with regard to the control of welding for nuclear components.

1.9.71.2 VEGP Position

This guide provides guidelines above and beyond requirements of ASME Section IX. All welder qualification at VEGP is in conformance with ASME Section IX. Few welds of limited accessibility are expected to be encountered. For field application, the type of qualification should be considered on a case-by-case basis due to the great variety of circumstances encountered. Reasonable engineering judgment will be used to determine if performance qualification is necessary under simulated access conditions for any specific case.

Westinghouse practice does not require qualification or requalification of welders for areas of limited accessibility as described by the guide. Experience shows that the current shop practice produces high quality welds. Limited accessibility qualification or requalification, which are additional to ASME Section III and IX requirements, is an unduly restrictive requirement for shop fabrication, where the welders' physical position relative to the welds is controlled and does not present any significant problems. In addition, shop welds of limited accessibility are repetitive due to multiple production of similar components, and such welding is closely supervised.

Refer to subsection 5.2.3 for further discussion.

1.9.72 REGULATORY GUIDE 1.72, REVISION 2, NOVEMBER 1978, SPRAY POND PIPING MADE FROM FIBERGLASS-REINFORCED THERMO-SETTING RESIN

1.9.72.1 Regulatory Guide 1.72 Position

This guide describes a method acceptable to the NRC for designing, fabricating, and testing fiberglass-reinforced thermo-setting resin piping for spray pond applications.

1.9.72.2 VEGP Position

Spray pond piping is not used at VEGP; therefore, this guide is not applicable.

1.9.73 REGULATORY GUIDE 1.73, JANUARY 1974, QUALIFICATION TESTS OF ELECTRIC VALVE OPERATORS INSTALLED INSIDE THE CONTAINMENT OF NUCLEAR POWER PLANTS

1.9.73.1 Regulatory Guide 1.73 Position

The procedures specified by IEEE Std. 382-1972, IEEE Trial-Use Guide for Type Test of Class I Electric Valve Operators for Nuclear Power Generating Stations, dated April 10, 1973, for conducting qualification tests of electric valve operators for service inside the containment vessel of water-cooled and gas-cooled nuclear power plants are generally acceptable and provide an adequate basis for complying with the qualification testing requirements of Section III of Appendix B to 10 CFR 50 to verify adequacy of design for service under design basis event conditions, subject to the qualifications in the guide.

1.9.73.2 VEGP Position

Qualification is discussed in paragraph 3.9.B.3.2.2, subsection 3.11.B.2, and WCAP-8587. The NSSS components are qualified to meet IEEE Std. 382-1972 using the qualification program described in WCAP-8587. See Regulatory Guide comparisons 1.40, 1.89, and 1.100 for additional information.

1.9.74 REGULATORY GUIDE 1.74, FEBRUARY 1974, QUALITY ASSURANCE TERMS AND DEFINITIONS

1.9.74.1 Regulatory Guide 1.74 Position (HISTORICAL)

[The quality assurance terms and definitions contained in ANSI N45.2.10-1973 are generally acceptable for use in describing and implementing quality assurance programs for the design, construction, and operation of nuclear power plant structures, systems, and components subject to the following:

The definition of "procurement documents" should be considered to include such documents as contracts, letters of intent, work orders, purchase orders, or proposals and their acceptances which authorize the seller to perform services or supply equipment, material, or facilities on behalf of the purchaser.]

1.9.74.2 VEGP Position

Regulatory Guide 1.74 provided NRC endorsement of ANSI N45.2.10. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.10. Accordingly, terms and definitions used in the quality assurance program are provided in the SNC QATR.

1.9.75 REGULATORY GUIDE 1.75, REVISION 2, SEPTEMBER 1978, PHYSICAL INDEPENDENCE OF ELECTRIC SYSTEMS

1.9.75.1 Regulatory Guide 1.75 Position

The IEEE Std. 384-1974 sets forth criteria for the separation of circuits and equipment that are redundant.

The guidance in IEEE Std. 384-1974, IEEE Trial-Use Standard Criteria for Separation of Class 1E Equipment and Circuits, dated March 15, 1974, is generally acceptable to the NRC staff and provides an adequate basis for complying with IEEE Std. 279-1971 and the Commission's General Design Criteria 3, 17, and 21 of Appendix A to 10 CFR 50 with respect to the physical independence of the circuits and electric equipment comprising or associated with the Class 1E power system, the protection system, systems actuated or controlled by the protection system, and auxiliary or supporting systems that must be operable for the protection system and the systems it actuates to perform their safety-related functions, subject to the qualifications in the guide.

1.9.75.2 VEGP Position

Effective April 1990, VEGP has adopted IEEE 384-1981 as the "Standard Criteria for Independence of Class 1E Equipment and Circuits," effectively replacing IEEE 384-1974. The 1981 version of IEEE 384 is generally consistent with the 1974 version and provides clearer, more concise criteria. IEEE 628-1987, "Standard Criteria for the Design, Installation and Qualification of Raceway Systems for Class 1E circuits for Nuclear Power Generating Station" requires VEGP to meet the criteria of IEEE 384-1981. Independence of Class 1E equipment and circuits has not been adversely affected by the adoption of IEEE 384-1981.

Conform as discussed below and in paragraph 7.1.2.2.

Where isolation devices are used to isolate Class 1E circuits from non-Class 1E circuits, the circuits within or from the Class 1E equipment or devices to the isolation device(s) are identified as Class 1E and are treated as such. Beyond the isolation device(s) these circuits are identified as non-Class 1E and are separated from Class 1E circuits in accordance with paragraph 8.3.1.4.3. Power supply circuits from the transmission network which are connected to the Class 1E switchgear are identified as non-Class 1E.

All Class 1E and non-Class 1E circuits are separated in accordance with paragraph 8.3.1.4.3.

Splicing or repairing cable in raceway is prohibited except:

A. Replacing splices in equipment is permitted where such splices have been included in vendor qualification testing or analysis.

B. Cable manufacturers' splices and repairs are permitted provided they are qualified by the manufacturer to the same requirements and standards as specified in the cable specification.

C. Cable splices in raceway are permitted where specifically justified by analysis or testing. Proposed splices in raceway will be evaluated on a case-by-case basis and documented in engineering calculations. The engineering evaluation and analysis for splicing in raceway must assure that:

    1. The splice would not be overstressed by potential future overpulls or other mechanical tensioning, including cable creep.

    2. Administrative controls include documentation of the location of the splice and constraints on any future activity that could place additional cable in the same raceway.

    3. Normal and fault currents are evaluated to determine the potential for heating effects assuming, under worst case condition, that the splice will introduce some additional circuit resistance.

All cable splices and repairs must utilize qualified materials and methods, and their location shall be specifically described in design documents.

All raceways and circuits are uniquely identified (by permanent numbering markers and color coding) to distinguish between redundant Class 1E and non-Class 1E systems at intervals stipulated in Section 5.1.2 of IEEE Std. 384-1974.

Circuits and raceways of safety-related groups A and C are routed in the lower cable spreading room. Circuits and raceways of safety-related groups B and D are routed in the upper cable spreading room. Group N raceways and circuits are routed in both upper and lower cable spreading rooms.

In general, VEGP complies with the separation requirements of IEEE 384-1974. However, as allowed by Sections 5.1.1.2 and 5.6.2 of IEEE Std. 384-1974 and by Regulatory Guide 1.75, a series of tests has been performed to establish alternate reduced separation distances where the separation distances specified in IEEE 384 are not met. The test results are documented in Wyle Laboratories Test Report No. 48141-02 which has been submitted to the NRC under separate cover.

Refer to paragraph 8.3.1.4.3 and table 8.3.1-4 for additional information.

1.9.76 REGULATORY GUIDE 1.76, APRIL 1974, DESIGN BASIS TORNADO FOR NUCLEAR POWER PLANTS

1.9.76.1 Regulatory Guide 1.76 Position

This guide designates values of the design basis tornado and locations within the United States.

1.9.76.2 VEGP Position

Conform. The tornado design parameters for Region 1 are used for VEGP. Refer to section 3.3 for discussion on this subject.

1.9.77 REGULATORY GUIDE 1.77, MAY 1974, ASSUMPTIONS USED FOR EVALUATING A CONTROL ROD EJECTION ACCIDENT FOR PRESSURIZED WATER REACTORS

1.9.77.1 Regulatory Guide 1.77 Position

1. C.1 Reactivity excursions will not result in a radial average fuel enthalpy greater than 280 cal/g at any axial location in any fuel rod.

2. C.2 Maximum reactor pressure during any portion of the assumed transient will be less than the value that will cause stresses to exceed the emergency condition stress limits as defined in Section III of the ASME Boiler and Pressure Vessel Code.

3. C.3 Offsite dose consequences will be well within the guidelines of 10 CFR 100, Reactor Site Criteria.

1.9.77.2 VEGP Position

1. C.1 Conform.

2. C.2 Exception is taken to this position which implies that the rod ejection accident should be considered as an emergency condition. This is considered a faulted condition as stated in ANSI N18.2. Faulted condition stress limits will be applied for this accident.

3. C.3 Conform.

Refer to section 15.4 for further discussion.

1.9.78 REGULATORY GUIDE 1.78, ASSUMPTIONS FOR EVALUATING THE HABITABILITY OF A NUCLEAR POWER PLANT CONTROL ROOM DURING A POSTULATED HAZARDOUS CHEMICAL RELEASE

1.9.78.1 Regulatory Guide 1.78 Position

This guide describes acceptable assumptions to be used in assessing the habitability of the control room during and after a postulated external release of hazardous chemicals and describes acceptable criteria for the protection of control room operators.

1.9.78.2 VEGP Position

Conform. Refer to sections 2.2.3 and 6.4.

In addition, with respect to positions C-13 and C-15, VEGP will train control room operators annually. Operators will be able to distinguish the smell of toxic chemicals. This will be covered in the hazardous waste class for operators. Control room personnel will be trained to don breathing apparatus in 2 minutes.

1.9.79 REGULATORY GUIDE 1.79, REVISION 1, SEPTEMBER 1975, PREOPERATIONAL TESTING OF EMERGENCY CORE COOLING SYSTEMS FOR PRESSURIZED WATER REACTORS

1.9.79.1 Regulatory Guide 1.79 Position

This guide describes a preoperational test program acceptable to the NRC specifically for emergency core cooling systems in PWRs.

1.9.79.2 VEGP Position

Conform, with the following exceptions:

Position C.1C (2)

This position requires that the accumulator isolation be opened against maximum differential pressure using both normal and emergency power supplies. Conditions at the valve motor are independent of the power source for this test; therefore, opening the valve using emergency power would provide no additional meaningful data. For this reason, the valves will be cycled using the normal power sources only.

Position C.1B (2)

This position requires RHR system testing "include taking a suction on the RCS containment sump to verify vortex control and acceptable pressure drops across screening and suction lines and valves."

The in-plant testing requirement is unnecessary for the screens installed to satisfy requirements of the NRC Generic Letter, (GL) 2004-02, for the following reasons:

- Head loss (pressure drop) for the screen was developed using the laboratory test results conducted by General Electric Company (GE), and applied to the full-scale plant design conditions.

- Analysis and testing was conducted to provide assurance that vortexing will not result in degraded pump performance.

Preoperational testing conducted during installation of the previous screens already verified pressure drop through the suction lines and valves for the same operating conditions used for the new screen. The suction line is not modified.

1.9.80 REGULATORY GUIDE 1.80, JUNE 1974, PREOPERATIONAL TESTING OF INSTRUMENT AIR SYSTEMS

1.9.80.1 Regulatory Guide 1.80 Position

Withdrawn. Refer to paragraph 1.9.68.3.

1.9.81 REGULATORY GUIDE 1.81, REVISION 1, JANUARY 1975, SHARED EMERGENCY AND SHUTDOWN ELECTRIC SYSTEMS FOR MULTI-UNIT NUCLEAR POWER PLANTS

1.9.81.1 Regulatory Guide 1.81 Position

This guide describes an acceptable method for complying with NRC requirements with respect to the sharing of onsite emergency and shutdown electric systems for multiunit nuclear power plants.

1.9.81.2 VEGP Position

Conform. Refer to paragraph 8.3.1.1.2M for further discussion.

1.9.82 REGULATORY GUIDE 1.82, REVISION 3, WATER SOURCES FOR LONG-TERM RECIRCULATION COOLING FOLLOWING A LOSS-OF-COOLANT ACCIDENT

1.9.82.1 Regulatory Guide 1.82 Position

This guide describes an acceptable method for designing, fabricating, and testing of sump or suction inlet conditions for pumps in the emergency core cooling and containment spray systems.

1.9.82.2 VEGP Position

Conform except for the following exceptions:

1. The subject guide describes requirements for a trash rack to perform the following:

    a) Protect the inner screen from missiles that may be generated by a LOCA or by trash

    b) Prevent large debris from entering the screen.

VEGP is taking exception to the above requirements based on the following reasons:

There are no high-energy line breaks postulated to occur near the screens, and there are no missiles generated in the vicinity of the suction strainers; therefore, there are no jet loads, no pipe whip restraint loads, nor missiles applicable to the screens. The screens are designed to withstand the loading for the largest postulated debris quantity, pieces, and types. The design of the stacked disk screen prevents large debris from reaching the perforated inner area of the screens due to small slots between the strainer disks.

2. The subject guide suggests that a vertically mounted screen be provided:

VEGP takes exception to the vertically mounted screen for the following reasons:

The vertical modular stacked disk screen does not allow gravitationally-influenced settling on the perforated flow area; the top surface is a solid stainless steel plate and protects the perforated plates below it. Therefore, the horizontal screens are functionally equivalent to vertical screens.

See subsections 6.2.2 and 6.3.2 for further discussion. Note that some nuclear fuel used at the VEGP units may contain design features that provide for flow passage dimensions smaller than those of the containment sump screen. This condition could lead to flow blockage downstream of the containment sump. The fuel flow passage dimensions have been evaluated and the conclusion has been reached that excessive flow blockage in the fuel will not occur. Therefore, long-term core cooling requirements and 10 CFR 50.46 emergency core cooling system (ECCS) acceptance criteria will continue to be met.

1.9.83 REGULATORY GUIDE 1.83, REVISION 1, JULY 1975, INSERVICE INSPECTION OF PRESSURIZED WATER REACTOR STEAM GENERATOR TUBES

1.9.83.1 Regulatory Guide 1.83 Position

This guide describes a method acceptable to the NRC for implementing the applicable general design criteria by reducing the probability and consequences of steam generator tube failures through periodic inservice inspection for early detection of defects and deterioration.

1.9.83.2 VEGP Position

The steam generators are designed to permit access to tubes for inspection and/or repair or plugging (if necessary). Plugging will be accomplished by either welded plugs or nonwelded mechanical plugs. The inservice inspection program is discussed in subsection 5.4.2 and the Technical Specifications.

1.9.84 REGULATORY GUIDE 1.84, REVISION 20, NOVEMBER 1982, DESIGN AND FABRICATION CODE CASE ACCEPTABILITY - ASME SECTION III, DIVISION 1

1.9.84.1 Regulatory Guide 1.84 Position

This guide lists those Section III ASME Code Cases oriented to design and fabrication that are generally acceptable to the NRC for implementation in the licensing of light-water-cooled nuclear power plants.

1.9.84.2 VEGP Position

Conform, except as noted in table 1.9-1 for Code Case N-31.

The NSSS Code Cases are discussed in table 1.9-3 and paragraph 5.2.1.2. Other design and fabrication code cases used for VEGP are identified in and table 1.9-1.

1.9.85 REGULATORY GUIDE 1.85, REVISION 20, NOVEMBER 1982, MATERIALS CODE CASE ACCEPTABILITY - ASME SECTION III DIVISION 1

1.9.85.1 Regulatory Guide 1.85 Position

This guide lists those Section III ASME Code cases oriented to materials and testing that are generally acceptable to the NRC for implementation in the licensing of light-water-cooled nuclear power plants.

1.9.85.2 VEGP Position

Conform.

The NSSS Code Cases are discussed in table 1.9-3 and paragraph 5.2.1.2. Other materials code cases used for VEGP are identified in table 1.9-2.

1.9.86 REGULATORY GUIDE 1.86, JUNE 1974, TERMINATION OF OPERATING LICENSES FOR NUCLEAR REACTORS

1.9.86.1 Regulatory Guide 1.86 Position

This guide describes methods and procedures considered acceptable by the NRC for the termination of operating licenses for nuclear reactors.

1.9.86.2 VEGP Position

The termination of the operating license and subsequent decommissioning of VEGP will be performed in accordance with the regulations applicable at that time.

1.9.87 REGULATORY GUIDE 1.87, REVISION 1, JUNE 1975, GUIDANCE FOR CONSTRUCTION OF CLASS 1 COMPONENTS IN ELEVATED TEMPERATURE REACTORS

Not applicable to VEGP.

1.9.88 REGULATORY GUIDE 1.88, REVISION 2, OCTOBER 1976, COLLECTION, STORAGE, AND MAINTENANCE OF NUCLEAR POWER PLANT QUALITY ASSURANCE RECORDS

1.9.88.1 Regulatory Guide 1.88 Position (HISTORICAL)

[The requirements and guidelines for collection, storage, and maintenance of nuclear power plant quality assurance records that are included in ANSI N45.2.9-1974 are acceptable to the NRC staff and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the following:

1. *C.1 Subdivision 1.5 of ANSI N45.2.9-1974 states, "Other documents that are required to be included as part of this standard are either identified at the point of reference or described in Section 8 of this standard." The specific applicability or acceptability of these documents has been or will be covered separately in other regulatory guides or in Commission regulations where appropriate.*

2. *C.2 Two methods for protection of quality assurance records from the hazards of fire are described in Subdivision 5.6 of ANSI N45.2.9-1974. The National Fire Protection Association (NFPA) No. 232-1975, "Standard for the Protection of Records," also contains provisions for records protection equipment and records handling techniques that provide protection from the hazards of fire. This standard, within its scope of coverage, is considered by the NRC staff to provide an acceptable alternative to the fire protection provisions listed in Subdivision 5.6 of N45.2.9-1974. When NFPA No. 232-1975 is used, quality assurance records should be classified as NFPA Class 1 records (NFPA No. 23-1975, Chapter 5, Section 5222).]*

1.9.88.2 VEGP Position

[(HISTORICAL) The VEGP QAP for the design and construction phase is designed to conform with ANSI N45.2.9 (Draft 11, Revision 0, 1/17/73) as described in chapter 17.1. With regard to quality assurance, records for NSSS equipment, alternatives, clarifications, and exceptions to ANSI N45.2.9-1974 are contained in WCAP 8370/7800 table 17-1.

The VEGP QAP conforms for the operations phase with the requirements of ANSI N45.2.9-1974 as endorsed by Regulatory Guide 1.88, Rev. 2.

The VEGP QAP for both the design and construction phase and the operations phase conform with the revision to ANSI N45.2.9 referenced above with the following clarification.

1. *Subdivision 3.2.1 states that quality assurance records "shall be legible, completely filled out, and adequately identifiable to the item involved." Completely filled out is considered to mean "completed as appropriate to the work accomplished," as clarified in the 1979 Revision.]*

Regulatory Guide 1.88 provided NRC endorsement of ANSI N45.2.9. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.9. Accordingly, the requirements for collection, storage, and maintenance of quality assurance records are described in the SNC QATR.

1.9.89 REGULATORY GUIDE 1.89, NOVEMBER 1974, QUALIFICATION OF CLASS 1E EQUIPMENT FOR NUCLEAR POWER PLANTS

1.9.89.1 Regulatory Guide 1.89 Position

The procedures described in IEEE Std. 323-1974, IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations, dated February 28, 1974, for qualifying Class 1E equipment for service in light-water-cooled and gas-cooled nuclear power plants are generally acceptable and provide an adequate basis for complying with design verification requirements of Criterion III of Appendix B to 10 CFR 50 to verify adequacy of design under the most adverse design conditions subject to the following:

1. C.1 Reference is made in IEEE Std. 323-1974, Section 2, 6.3.2(5), and 6.3.5, to IEEE Std. 344-1971, Guide for Seismic Qualification of Class 1 Electric Equipment for Nuclear Power Generating Stations. The specific applicability or acceptability of IEEE Std. 344 will be covered separately in other regulatory guides, where appropriate.

2. C.2 The radiological source term for qualification tests in a nuclear radiation environment should be based on the same source term as that used in Regulatory Guide 1.7 (Safety Guide 7, March 10, 1971) for boiling water reactors (BWRs) and PWRs. An equivalent source term (i.e., 100 percent of the noble gases, 50 percent of the halogens, and 1 percent of the remaining solids developed from maximum full-power operation of the core) should be used for high temperature gas-cooled reactors (HTGRs). The containment size should be taken into account in each case. For exposed organic materials, calculations should take into account both beta and gamma radiation.

1.9.89.2 VEGP Position

Conform. See section 3.11.B for information on environmental conditions and design bases for mechanical, instrumentation, and electrical safety-related equipment. For NSSS equipment, Westinghouse conforms to IEEE Std. 323-1974 by implementation of the final NRC approved version of WCAP-8587.

1. C.1 See Regulatory Guide 1.100 comparison.

2. C.2 Conform.

1.9.90 REGULATORY GUIDE 1.90, REVISION 1, AUGUST 1977, INSERVICE INSPECTION OF PRESTRESSED CONCRETE CONTAINMENT STRUCTURES WITH GROUTED TENDONS

1.9.90.1 Regulatory Guide 1.90 Position

This guide describes bases acceptable to the NRC for developing an appropriate surveillance program for prestressed concrete containment structures with grouted tendons.

1.9.90.2 VEGP Position

This guide is not applicable since VEGP does not use grouted tendons.

1.9.91 REGULATORY GUIDE 1.91, REVISION 1, FEBRUARY 1978, EVALUATIONS OF EXPLOSIONS POSTULATED TO OCCUR ON TRANSPORTATION ROUTES NEAR NUCLEAR POWER PLANTS

1.9.91.1 Regulatory Guide 1.91 Position

This guide describes methods acceptable to the NRC for determining whether the risk of damage due to an explosion on a nearby transportation route is sufficiently high to warrant a detailed investigation.

1.9.91.2 VEGP Position

Conform. Refer to subsection 2.2.3 for discussion on this subject.

1.9.92 REGULATORY GUIDE 1.92, REVISION 1, FEBRUARY 1976, COMBINING MODAL RESPONSES AND SPATIAL COMPONENTS IN SEISMIC RESPONSE ANALYSIS

1.9.92.1 Regulatory Guide 1.92 Position

This guide describes the procedures to be used for combining modal responses of individual modes and the combination of effects due to the three independent spatial components of an earthquake in seismic analyses of nuclear power plant structures, systems, and components.

1.9.92.2 VEGP Position

Conform with the exception that Westinghouse uses an alternative method of combining modal responses to satisfy Regulatory Guide 1.92, Revision 1, as described in paragraph 3.7.N.2.7.

Refer to sections 3.7.B and 3.7.N for discussion on this subject.

1.9.93 REGULATORY GUIDE 1.93, DECEMBER 1974, AVAILABILITY OF ELECTRIC POWER SUPPLIES

1.9.93.1 Regulatory Guide 1.93 Position

This guide describes operating procedures and restrictions acceptable to the NRC which should be implemented if the available electric power sources are less than the limiting conditions for operation (LCO).

1.9.93.2 VEGP Position

VEGP will conform with this guide by implementing the appropriate NRC approved Technical Specifications.

Refer to the Technical Specifications for further discussion.

1.9.94 REGULATORY GUIDE 1.94, REVISION 1, APRIL 1976, QUALITY ASSURANCE REQUIREMENTS FOR INSTALLATION, INSPECTION, AND TESTING OF STRUCTURAL CONCRETE AND STRUCTURAL STEEL DURING THE CONSTRUCTION PHASE OF NUCLEAR POWER PLANTS

1.9.94.1 Regulatory Guide 1.94 Position (HISTORICAL)

[This guide describes a method acceptable to the NRC for complying with the quality assurance requirements for installation, inspection, and testing of structural concrete and structural steel during the construction phase of nuclear power plants. This guide endorses ANSI N45.2.5-1974 as generally acceptable to the NRC as a basis for complying with Appendix B to 10 CFR 50.]

1.9.94.2 VEGP Position

[(HISTORICAL) The VEGP QAP is described in section 17.1 for the design and construction phase and in section 17.2 for the operations phase. The VEGP QAP does not conform to Regulatory Guide 1.94, Revision 1, which endorses ANSI N45.2.5-1974. Instead, VEGP QAP for both the design and construction and the operations phases conforms to the requirements of ANSI/ASME N45.2.5-1978, Supplementary Quality Assurance Requirements for Installation, Inspection, and Testing of Structural

Concrete, Structural Steel, Soils, and Foundations during the Construction Phase of Nuclear Power Plants with the following clarifications:

1. *Paragraph 3.2, Materials Suitability (3.2.1)*

   This paragraph, Table A, states reinforcement shall be tested for physical properties per (ASTM A615) ASTM A370. VEGP conforms to this guidance in that the certification of compliance provided by the vendor ensures materials have been tested for physical properties (ASTM A615) per ASTM A370. In such a case, the vendor is subject to QA/QC provisions in accordance with 10 CFR 50, Appendix B.

2. *Paragraph 4.6, In-Process Tests on Compacted Fill*

   This paragraph, Table B, requires one grain-size test in accordance with ASTM D-422 (hydrometer or sieve as appropriate) for each compaction test. VEGP specifications require one grain-size test per ASTM D-422 without hydrometer test for every 26,250 yd^3 (i.e., one for every 5 compaction tests).

3. *Paragraph 4.6, In-Process Tests on Compacted Fill*

   This paragraph, Table B, requires borrow moisture test per ASTM D-1556, 2167, 3017, or 2937 for each soil type, each work shift, and when moisture content changes or is questionable. VEGP specifications require fill to be within an acceptable range of optimum moisture at the time of placement. Testing per ASTM D-2216, or rapid method correlation, is performed each time soil is compacted.

4. *Paragraph 4.6, In-Process Tests on Compacted Fill*

   This paragraph, Table B, requires field density test per ASTM D-1556, 2167, 2922, or 2937 as specified at a minimum of every 10,000 ft^2. VEGP specifications require one test per ASTM D-1556 for every 20,000 ft^2 per ft of depth.

5. *Paragraph 6.11, Inprocess Tests on Concrete and Reinforcing Steel*

   This paragraph, Table B, requires in-process compressive strength tests shall be performed daily on grout in accordance with ASTM C109. VEGP tests batch plant grout daily in accordance with ASTM C109, while nonshrink grout is tested in accordance with CRD621-82A prior to its initial use on the jobsite.

6. *Paragraph 6.11, Inprocess Tests on Concrete and Reinforcing Steel*

   This paragraph, Table B, requires that fly ash and pozzolans be checked for physical properties (ASTM C618) in accordance with ASTM C311 every 400 tons. VEGP specifications requires this testing to be performed every 1000 tons except for loss of ignition and sieve no. 325 which are performed every 200 tons.

7. *Paragraph 6.12, Mechanical (Cadweld) Splice Testing (6.12.2)*

   This paragraph requires visual inspection on completed splices shall be performed only after the splices have cooled to ambient temperatures. The VEGP specification does not require splices to be cooled to ambient temperature prior to inspection. Cadwelds are inspected after

they have cooled such that it will allow the inspection to be performed without any danger of burns.

8. *Paragraph 6.12, Mechanical (Cadweld) Splice Testing (6.12.4)*

   This paragraph, Item 2.C, requires that one splice, either production or sister splice, be tested for the next and subsequent units of 33 splices. VEGP specifications require three test splices for the next and subsequent units of 100 splices.

9. *Paragraph 7.5, Welding*

   This paragraph requires inspection of structural steel welding to be performed in accordance with the provisions of Section 6 of the AWS D1.1. Visual welding inspection performed on or after December 2, 1985, is in accordance with the Visual Weld Acceptance Criteria (VWAC) for Structural Welding at Nuclear Power Plants, NCIG-01, Revision 2, prepared by the Nuclear Construction Issues Group (NCIG) and accepted by the NRC in their letter to the NCIG dated June 26, 1985.

   For inspection performed prior to December 2, 1985, the visual acceptance criteria is in accordance with AWS D1.1-75 with the clarifications and modifications described below and in NRC Inspection Report Nos. 50-424/86-03 and 50-425/86-02.

10. *Paragraph 8.2 Evaluation of Aggregate Test Results (8.2.3)*

    This paragraph states that, when aggregate tests specified fail to meet specified requirements, two additional tests shall be made from samples of the same lot of aggregate. If one or both of the two additional tests fail to meet the specified requirements, the data shall be submitted to the responsible engineering organization for evaluation and corrective action. The VEGP specifications require that, when a running average of five gradation tests fails to meet the specified requirements, the data shall be submitted to the responsible engineering organization for evaluation and corrective action. In addition, the VEGP construction procedures require that all failing aggregate production tests be reported to construction engineering for evaluation.

    In order to designate the specific set of visual acceptance criteria applicable for a weld, structural steel weld joints within the jurisdiction of AWS D1.1 are classified into the following categories:

    - *Category A - Structural steel joints which are a part of the main building frame and those joints which connect miscellaneous structural steel with the main building frame.*

    - *Category B - Miscellaneous structural steel joints, not covered in Category A, but provide auxiliary support or framing for systems, components, and equipment (e.g., supports for cable tray and heating, ventilation, and air-conditioning (HVAC) ductwork, miscellaneous stiffeners and bracing, etc.).*

The base metal adjacent to a weld joint of dissimilar categories shall meet the visual inspection acceptance criteria of the category under which that base metal falls.

A. AWS D1.1, paragraph 3.1.4, is clarified as follows:

For Category A and B joints, the fillet leg dimension may underrun the nominal fillet size by 1/16 in. provided the undersize length does not exceed 10 percent of the weld length. For flange to web joints, the undersize may not be within two flange thicknesses of the weld end.

For Category A and B joints, where an intermittent weld is specified, a continuous weld of the same size is acceptable.

B. AWS D1.1, paragraph 3.6.1, is modified as follows:

For Category A and B joints, the faces of fillet welds may be slightly convex, flat, or slightly concave as shown in figure 3.6A, B, and C, with none of the unacceptable profiles shown in Figure 3.6.D. Convexity height may not exceed 1/8 in.

C. AWS D1.1, paragraph 3.6.4, is modified as follows:

For Category A joints, undercut not exceeding 1/32 in. for the full length of the weld is acceptable. For members welded from both sides, the cumulative undercut depth includes the sum of both sides if the undercuts on both sides are directly opposite to each other.

For Category B joints, undercut not exceeding 1/32 in. for the full length of the weld is acceptable.

Undercut greater than 1/32 in. but not exceeding 1/16 in. is acceptable provided the width is greater than the depth, and the undercut does not exhibit an acute intersection at its root. The cumulative length of 1/16 in. undercut shall not exceed 50 percent of the total weld length. For members welded from sides, the cumulative undercut depth or length includes the sum of both sides if the undercuts on both sides are directly opposite to each other.

D. AWS D1.1, paragraph 3.6.6, is modified as follows:

For Category A and B joints, overlap/rollover may not exceed 1/8 in.

E. AWS D1.1, paragraph 3.10.1, is clarified as follows:

For Category A and B joints, "trap" slag on the root and/or end of fillet and partial penetration welds is acceptable.

For Category A joints, loose spatter is not acceptable. Tightly adhering spatter is acceptable, if it will not interfere with subsequent NDE (when required).

For Category B joints, spatter is not a cause for rejection.

F. AWS D1.1, paragraph 4.4, is modified as follows:

For Category A joints, arc strikes shall be blended with the surface.

For Category B joints, arc strikes are acceptable provided the craters do not contain cracks and the material specification minimum gauge thickness is not violated.

G. AWS D1.1, paragraph 8.15.1.3, is modified as follows:

Underfill of a weld is not acceptable. A crater is not considered as an underfill.

For Category A joints, underfilled groove weld craters are acceptable provided the depth of underfill is 1/32 in. or less. Underfilled single-pass fillet weld craters are acceptable provided the crater length is less than 10 percent of the weld length. On multipass fillet weld, crater depth of 1/32 in. or less is acceptable.

For Category B joints, underfilled groove weld craters are acceptable provided the depth of underfill is 1/16 in. or less. Underfilled single-pass fillet weld craters are acceptable provided the crater length is less than 10 percent of the weld length. On multipass fillet welds, crater depth of 1/16 in. or less are acceptable.

H. AWS D1.1 paragraph 8.15.1.5, is clarified as follows:

For Category A joints, the sum diameters of porosity shall not exceed 3/8 in. in one linear in. of weld and 3/4 in. in 12 linear in. of weld.

For Category B joints, piping porosity is not a cause for rejection if the major axis is equal to or less than 1/16 in. For porosity greater than 1/16 in., the sum of the diameters shall not exceed 3/8 in. in any linear in. of weld nor 3/4 in. in any 12 in. of weld.

Refer to Regulatory Guide 1.55 comparison for a discussion of the standards being used in the placement of concrete in Category 1 structures.]

Regulatory Guide 1.94 provided NRC endorsement of ANSI N45.2.5. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.5. Accordingly, the quality assurance requirements for installation, inspection, and testing of structural concrete and structural steel applicable during the operation phase of nuclear power plants are described in the SNC QATR.

1.9.95 REGULATORY GUIDE 1.95, REVISION 1, JANUARY 1977, PROTECTION OF NUCLEAR POWER PLANT CONTROL ROOM OPERATORS AGAINST AN ACCIDENTAL CHLORINE RELEASE

1.9.95.1 Regulatory Guide 1.95 Position

This guide describes design features and procedures that are acceptable to the NRC for the protection of nuclear plant control room operators against an accidental chlorine release.

1.9.95.2 VEGP Position

VEGP does not store liquefied gaseous chlorine in excess of 20 lbs onsite.

1.9.96 REGULATORY GUIDE 1.96, REVISION 1, JUNE 1976, DESIGN OF MAIN STEAM ISOLATION VALVE LEAKAGE CONTROL SYSTEMS FOR BOILING WATER REACTOR NUCLEAR POWER PLANTS

Not applicable to VEGP.

1.9.97 REGULATORY GUIDE 1.97, REVISION 2, DECEMBER 1980, INSTRUMENTATION FOR LIGHT-WATER-COOLED NUCLEAR POWER PLANTS TO ASSESS PLANT CONDITIONS DURING AND FOLLOWING AN ACCIDENT

1.9.97.1 Regulatory Guide 1.97 Position

This guide describes an acceptable method for complying with NRC regulations to provide instrumentation to monitor plant variables and systems during and following an accident in a light-water-cooled nuclear power plant. Refer to section 7.5.

1.9.97.2 VEGP Position

VEGP conformance is as described in section 7.5.

1.9.98 REGULATORY GUIDE 1.98, MARCH 1976, ASSUMPTIONS USED FOR EVALUATING THE POTENTIAL RADIOLOGICAL CONSEQUENCES OF A RADIOACTIVE OFFGAS SYSTEM FAILURE IN A BOILING WATER REACTOR

Not applicable to VEGP.

1.9.99 REGULATORY GUIDE 1.99, REVISION 2, MAY 1988, RADIATION EMBRITTLEMENT OF REACTOR VESSEL MATERIALS

1.9.99.1 Regulatory Guide 1.99 Position

This guide describes general procedures acceptable to the NRC for predicting the effects of neutron radiation damage to the low-alloy steels currently used for light-water-cooled reactor vessels.

1.9.99.2 VEGP Position

There are two primary issues with the guide:

1. The guide provides a procedure and curves for predicting radiation damage (as relating to the shift of the reference temperature, RT_{NDT}), in terms of chemistry (Cu and Ni) and fluence.

   Since the adjustments in reference temperature obtained from the radiation damage curves are used in developing heatup and cooldown limits for plant operation, the use of the curves in the guide could result in conservative heatup and cooldown limits during plant life.

2. The guide restricts the end of life transition temperature to 200°F maximum at the 1/4 T position in the vessel wall. Control of residual elements such as copper, nickel, sulfur, and vanadium in the reactor vessel beltline materials of new plants to levels that result in a predicted adjusted reference temperature of less than 200°F at end of life is considered technically unnecessary and could lead to unnecessary changes in chemistry (Cu and Ni) requirements with corresponding adverse impact on cost and materials availability.

One additional feature of the guide constitutes a lesser but nevertheless important issue:

1. Figure 2 of the guide presents a curve which gives the decrease of upper shelf impact energy with fluence as a function of Cu content. Although it appears that the prescribed relationship does not predict unacceptable drops in upper shelf toughness for vessels with controlled chemistry, the curves are nevertheless overly conservative.

The VEGP position, with respect to each of the guide positions, is as follows:

1. The basis, as well as the scope of the guide for predicting adjustment of reference temperature as given in regulatory position C.1, are inappropriate since use of the Westinghouse trend curves yield more conservative heatup and cooldown curves for VEGP than those that would have been obtained using the Regulatory Guide 1.99, Revision 2 trend curves.

2. The VEGP is in agreement with the guide position C.2a. However, with respect to guide position C.2b, Westinghouse believes that figure 2 of the guide is incorrect since the upper shelf energy for 6-in.-thick American Society of Testing Materials (ASTM) A302B reference correlation monitor material reported by Hawthorne indicates essentially a constant upper shelf at fluences above ~1 x 10^{19} n/cm^2.[(a)]

3. The VEGP position with reference to the guide position C.3, controlling residual elements to levels that result in a predicted adjusted reference temperature of less than 200°F at end of life, is that the stresses in the vessel can be limited during operation in order to comply with the requirements of Appendix G to 10 CFR 50 even though the end of life adjusted reference temperature may exceed 200°F. By

[a] Hawthorne, J. R., "Radiation Effects Information Generated on the ASTM Reference. Correlation-Monitor Steels," ASTM, Philadelphia, 1974.

applying the procedures of Appendix G to ASME Section III, the stress limits including appropriate Code safety margin can be met.

4. Recent surveillance capsule data indicate a steady-state condition of radiation damage well below that predicted by current trend curves.[a] This effect is believed to be due to the annealing of the vessels at the operating temperature. As an alternative to Regulatory Guide1.99 Rev 2, operating limits will be determined using the current radiation damage curves developed by Westinghouse.[b] It is expected that as future surveillance specimens are evaluated, it will become increasingly evident that both the Regulatory Guide 1.99 Rev 2 and Westinghouse trend curves are very conservative.

Refer to section 5.3 for further discussion.

1.9.100 REGULATORY GUIDE 1.100, REVISION 1, AUGUST 1977, SEISMIC QUALIFICATION OF ELECTRIC EQUIPMENT FOR NUCLEAR POWER PLANTS

1.9.100.1 Regulatory Guide 1.100 Position

Conformance with the requirements and recommendations specified by IEEE Std. 344-1975 for conducting seismic qualification of Class 1E equipment, when such qualification is performed in conjunction with Regulatory Guide 1.89, provides an adequate basis for complying with design verification requirements of Criterion III of Appendix B to 10 CFR 50 with respect to verifying the seismic adequacy of electric equipment, subject to the qualifications identified in the guide.

1.9.100.2 VEGP Position

Conform for Seismic Category 1 electrical equipment. Conformance for instrumentation and electrical equipment is discussed in subsection 3.10.B.2. Conformance for pumps and valves to assure operability is discussed in paragraph 3.9.B.3.2 and section 3.10.

The Westinghouse program for seismic qualification of safety-related electrical equipment is delineated in the latest revision of WCAP-8587. The procedures utilized in performing seismic qualification will be in accordance with IEEE Std. 344-1975.

1.9.101 REGULATORY GUIDE 1.101, REVISION 3, AUGUST 1992, EMERGENCY PLANNING FOR NUCLEAR POWER PLANTS

1.9.101.1 Regulatory Guide 1.101 Position

This guide describes a method acceptable to the NRC for complying with regulations for emergency response plans and preparedness at nuclear power plants.

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[a] Letter NS-TMA-1843 to the Secretary of the Commission, T. M. Anderson, June 23, 1978
[b] Westinghouse RESAR-3S, chapter 16, figure B/3/4.2, page B3/4 4-8

1.9.101.2 VEGP Position

Conform. The VEGP emergency plan, which has been submitted to the NRC, describes how SNC implements NUREG-0654, Criteria for Preparation and Evaluation of Radiological Emergency Response Plans and Preparedness in Support of Nuclear Power Plants. Refer to section 13.3 for further discussion.

1.9.102 REGULATORY GUIDE 1.102, SEPTEMBER 1976, FLOOD PROTECTION FOR NUCLEAR POWER PLANTS

1.9.102.1 Regulatory Guide 1.102 Position

This guide defines acceptable methods of flood protection for nuclear power plants.

1.9.102.2 VEGP Position

The roofs of all safety-related structures are designed to pass probable maximum precipitation (PMP) generated runoff through the use of overflow scuppers as described below. Reservoir routing using the modified Puls technique develops the discharge hydrograph from each roof taking into account inflows from other roofs. Roof drains and 100-year scuppers with downdrains are designed to drain the 100-year storm. For the local intense precipitation analysis, roof drains and 100-year scuppers are considered plugged. Overflow scuppers are considered half plugged to account for possible blockage by ice or debris.

Various combinations of fully opened or half-plugged scuppers are used to develop the maximum ponded depth on each roof and the maximum flow to the yard.

The VEGP is a dry site and conforms with this regulatory guide. Refer to section 3.4 and subsection 2.4.3 for further discussion.

1.9.103 REGULATORY GUIDE 1.103, REVISION 1, OCTOBER 1976, POST-TENSIONED PRESTRESSING SYSTEMS FOR CONCRETE REACTOR VESSELS AND CONTAINMENTS

1.9.103.1 Regulatory Guide 1.103 Position

Defines post-tensioning schemes for containments. The 55-strand VSL tendon system is approved by the NRC.

1.9.103.2 VEGP Position

VEGP utilizes the VSL system and is in conformance with this regulatory guide. Refer to subsection 3.8.1 for discussion on this subject.

1.9.104 REGULATORY GUIDE 1.104, OVERHEAD CRANE HANDLING FOR NUCLEAR POWER PLANTS

Withdrawn.

1.9.105 REGULATORY GUIDE 1.105, REVISION 1, NOVEMBER 1976, INSTRUMENT SETPOINTS

1.9.105.1 Regulatory Guide 1.105 Position

This guide describes a method acceptable to the NRC for complying with regulations with regard to ensuring that the instrumentation setpoints in systems important to safety initially are within and remain within the specified limits.

1.9.105.2 VEGP Position

Conformance is discussed in section 7.1.

1.9.106 REGULATORY GUIDE 1.106, REVISION 1, MARCH 1977, THERMAL OVERLOAD PROTECTION FOR ELECTRIC MOTORS ON MOTOR-OPERATED VALVES

1.9.106.1 Regulatory Guide 1.106 Position

To ensure that safety-related, motor-operated valves, whose motors are equipped with thermal overload protection devices integral with the motor starter, will perform their function, one of the two alternatives described in regulatory position 1 or the one described in regulatory position 2 should be implemented:

1. C.1 Provided that the completion of the safety function is not jeopardized or that other safety systems are not degraded, (a) the thermal overload protection devices should be continuously bypassed and temporarily placed in force only when the valve motors are undergoing periodic or maintenance testing, or (b) those thermal overload protection devices that are normally in force during plant operation should be bypassed under accident conditions.

   The bypass initiation system circuitry should conform to the criteria of Sections 4.1, 4.2, 4.3, 4.4, 4.5, 4.10, and 4.13 of IEEE Std. 279-1971, Criteria for Protection Systems for Nuclear Power Generating Stations, and should be periodically tested.

2. C.2 The trip setpoint of the thermal overload protection devices should be established with all uncertainties resolved in favor of completing the safety-related action. With respect to those uncertainties, consideration should be given to (a) variations in the ambient temperature at the installed location of the overload protection devices and the valve motors, (b) inaccuracies in motor heating data,

overload protection device trip characteristics, and the matching of these two items, and (c) setpoint drift. In order to ensure continued functional reliability and the accuracy of the trip point, the thermal overload protection device should be periodically tested.

1.9.106.2 VEGP Position

1. C.1 Conform. Refer to paragraph 8.3.1.1.2, item K.5.

2. C.2 Not applicable to VEGP.

1.9.107 REGULATORY GUIDE 1.107, REVISION 1, FEBRUARY 1977, QUALIFICATIONS FOR CEMENT GROUTING FOR PRESTRESSING TENDONS IN CONTAINMENT STRUCTURES

1.9.107.1 Regulatory Guide 1.107 Position

This guide describes quality standards acceptable to the NRC for the use of portland cement grout as the corrosion inhibitor for prestressing tendons in prestressed concrete containment structures.

1.9.107.2 VEGP Position

As described in Regulatory Guide 1.90 comparison, VEGP does not use grouted tendons; therefore, this guide is not applicable.

1.9.108 REGULATORY GUIDE 1.108, REVISION 1, AUGUST 1977, PERIODIC TESTING OF DIESEL GENERATOR UNITS USED AS ONSITE ELECTRIC POWER SYSTEMS AT NUCLEAR POWER PLANTS

1.9.108.1 Regulatory Guide 1.108 Position

This guide describes a method acceptable to the NRC for complying with regulations with regard to periodic testing of diesel electric power units to ensure that the diesel electric power systems will meet their availability requirements.

1.9.108.2 VEGP Position

This guide has been withdrawn and superseded by Regulatory Guide 1.9, Revision 3, Selection, Design, and Qualification of Diesel-Generator Units Used as Standby (Onsite) Electrical Power Systems at Nuclear Power Plants. Regulatory Guide 1.108 was the original licensing basis for diesel generator testing; however, since the time that Regulatory Guide 1.108 was withdrawn,

Regulatory Guide 1.9, Revision 3, has been used in specific instances to revise diesel generator testing requirements for VEGP. Therefore, diesel generator testing requirements at VEGP represent a combination of Regulatory Guide 1.108 and Regulatory Guide 1.9, Revision 3. Diesel generator testing requirements are discussed in paragraph 8.3.1.1.3 and are specified in the Technical Specifications.

1.9.109 REGULATORY GUIDE 1.109, REVISION 1, OCTOBER 1977, CALCULATION OF ANNUAL DOSES TO MAN FROM ROUTINE RELEASES OF REACTOR EFFLUENTS FOR THE PURPOSE OF EVALUATING COMPLIANCE WITH 10 CFR 50, APPENDIX I

1.9.109.1 Regulatory Guide 1.109 Position

This guide describes basic features of calculational models and parameters acceptable to the NRC for the estimation of radiation doses to man from effluent releases.

1.9.109.2 VEGP Position

The VEGP ODCM contains equations and constants for estimation of radiation doses to man from effluent releases. These equations and constants are in conformance with this guide. Refer to subsections 11.2.3 and 11.3.3 for discussion of preoperational calculations.

1.9.110 REGULATORY GUIDE 1.110, MARCH 1976, COST-BENEFIT ANALYSIS FOR RADWASTE SYSTEMS FOR LIGHT-WATER-COOLED NUCLEAR POWER REACTORS

1.9.110.1 Regulatory Guide 1.110 Position

This guide describes a method acceptable to the NRC for performing a cost-benefit analysis for liquid and gaseous radwaste system components.

1.9.110.2 VEGP Position

Cost-benefit evaluations were not used in the design of VEGP. Therefore, this guide is not applicable in the design of the radwaste system.

1.9.111 REGULATORY GUIDE 1.111, REVISION 1, JULY 1977, METHODS FOR ESTIMATING ATMOSPHERIC TRANSPORT AND DISPERSION OF GASEOUS EFFLUENTS IN ROUTINE RELEASES FROM LIGHT-WATER-COOLED REACTORS

1.9.111.1 Regulatory Guide 1.111 Position

This guide identifies types of atmospheric transport and diffusion models, source configuration and removal mechanism modifications, and input data that are acceptable to the NRC for use in providing assessments of potential annual radiation doses to the public resulting from routine releases of radioactive materials in gaseous effluents.

1.9.111.2 VEGP Position

The dispersion models used for VEGP follow those described in this guide as discussed in subsection 2.3.5.

1.9.112 REGULATORY GUIDE 1.112, REVISION O-R, APRIL 1976, CALCULATION OF RELEASES OF RADIOACTIVE MATERIALS IN GASEOUS AND LIQUID EFFLUENTS FROM LIGHT-WATER-COOLED POWER REACTORS

1.9.112.1 Regulatory Guide 1.112 Position

This guide references two NUREG reports (NUREG-0016 and -0017) that provide methods acceptable to the NRC for calculating annual average expected releases of radioactive material in liquid and gaseous effluents from light-water-cooled nuclear power reactors.

1.9.112.2 VEGP Position

The calculation of radioactive releases in gaseous and liquid effluents are made using the GALE computer code which is in accordance with this guide and NUREG-0017. Refer to sections 11.2 and 11.3 for further discussion.

1.9.113 REGULATORY GUIDE 1.113, REVISION 1, APRIL 1977, ESTIMATING AQUATIC DISPERSION OF EFFLUENTS FROM ACCIDENTAL AND ROUTINE REACTOR RELEASES FOR THE PURPOSE OF IMPLEMENTING APPENDIX I

1.9.113.1 Regulatory Guide 1.113 Position

This guide describes basic features of calculational models and suggests methods of determining values of model parameters acceptable to the NRC for the estimation of aquatic dispersion of both routine and accidental releases of liquid effluents.

1.9.113.2 VEGP Position

Analysis of aquatic dispersion of radioactive releases is performed using the LADTAP computer code which is in accordance with the provisions of this guide. Refer to subsections 2.4.13, 11.2.3, and 15.7.3 for further discussion.

1.9.114 REGULATORY GUIDE 1.114, REVISION 1, NOVEMBER 1976, GUIDANCE ON BEING OPERATOR AT THE CONTROLS OF A NUCLEAR POWER PLANT

1.9.114.1 Regulatory Guide 1.114 Position

This guide describes a method acceptable to the NRC for complying with regulations that require an operator to be present at the controls of a nuclear power plant.

1.9.114.2 VEGP Position

Conform. Refer to chapter 18 and section 13.5.

1.9.115 REGULATORY GUIDE 1.115, REVISION 1, JULY 1977, PROTECTION AGAINST LOW TRAJECTORY TURBINE MISSILES

1.9.115.1 Regulatory Guide 1.115 Position

This guide describes methods acceptable to the NRC for protecting safety-related structures, systems, and components against low-trajectory missiles resulting from turbine failure.

1.9.115.2 VEGP Position

Conformance is discussed in paragraph 3.5.1.3.

1.9.116 REGULATORY GUIDE 1.116, REVISION O-R, JUNE 1976, QUALITY ASSURANCE REQUIREMENTS FOR INSTALLATION, INSPECTION, AND TESTING OF MECHANICAL EQUIPMENT AND SYSTEMS

1.9.116.1 Regulatory Guide 1.116 Position (HISTORICAL)

[This guide endorses ANSI N45.2.8-1975 which describes a method acceptable to the NRC for complying with regulations with regard to quality assurance requirements for installation, inspection, and testing of mechanical equipment and systems for water-cooled nuclear power plants.]

1.9.116.2 VEGP Position

[(HISTORICAL) The VEGP QAP for design and construction as described in section 17.1 does not address this Regulatory Guide nor ANSI N45.2.8. The VEGP operations QAP is described in section 17.2.]

Regulatory Guide 1.116 provided NRC endorsement of ANSI N45.2.8. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.8. Accordingly, the quality assurance requirements for installation, inspection, and testing of mechanical equipment and systems applicable to operation phase activities are described in the SNC QATR.

1.9.117 REGULATORY GUIDE 1.117, REVISION 1, APRIL 1978, TORNADO DESIGN CLASSIFICATION

1.9.117.1 Regulatory Guide 1.117 Position

This guide describes a method acceptable to the NRC for identifying those structures, systems, and components of light-water-cooled reactors that should be protected from the effects of the design basis tornado (including tornado missiles) and remain functional.

1.9.117.2 VEGP Position

Conform, with exception of the nuclear service cooling water tower fans which are not missile protected. Refer to section 3.5 for discussion on this subject.

1.9.118 REGULATORY GUIDE 1.118, REVISION 2, JUNE 1978, PERIODIC TESTING OF ELECTRIC POWER AND PROTECTION SYSTEMS

1.9.118.1 Regulatory Guide 1.118 Position

The requirements and recommendations contained in IEEE Std. 338-1977 are considered acceptable methods for the periodic testing of electric power and protection systems, subject to the qualifications identified in the guide.

1.9.118.2 VEGP Position

Conform, with the following clarifications.

The definition of auxiliary or supporting system is interpreted to mean the first auxiliary or supporting system to prevent expanding the safety system to encompass all plant systems.

Overlap testing is considered to have occurred if a relay coil is included in one test and the contacts are included in another test, or if a pressure switch is included in one test and its contacts included in a different test. If contacts are shown to function with one test, they may be replaced with jumper wires for other tests. Complete systems tests is interpreted to mean completion of all overlapping tests where overlapping tests are used. Each overlapping test will normally be completed on its own interval.

Test equipment, portable or installed, that must be connected to the safety system by cable and/or wire to perform the test, is not considered part of the safety system.

Regulatory position C.8 is not considered at this time since the NRC has not formally endorsed the referenced sections.

Westinghouse defines "protective action systems" to mean the electric, instrumentation, and control portions of those protection systems and equipment actuated and controlled by the protection system. Equipment performing control functions but actuated from protection system sensors is not part of the safety system and will not be tested for time response. Status, annunciating, display, and monitoring functions, except those related to the post-accident monitoring system (PAMS) are considered by Westinghouse to be control functions. Reasonability checks; i.e., comparison between or among similar such display functions, will be made.

Response time testing for control functions operated from system sensors will not be performed. Moreover, nuclear instrumentation system detectors will not be tested for time response. Also, nuclear instrumentation sensors are exempt from testing since "their worst-case response time is not a significant fraction of the total overall system response (i.e., less than 5 percent)." This exemption is permitted by IEEE-338.

Verification of trip set points that are continuously calculated over a range of input variables is not required during functional testing if the setpoints are displayed and subject to routine instrument checks. The functional test should verify the trip setpoints for at least one value of each input variable.

1.9.119 REGULATORY GUIDE 1.119

Withdrawn.

1.9.120 REGULATORY GUIDE 1.120, REVISION 1, NOVEMBER 1977, FIRE PROTECTION GUIDELINES FOR NUCLEAR POWER PLANTS

1.9.120.1 Regulatory Guide 1.120 Position

This guide presents guidelines acceptable to the NRC staff for implementing General Design Criterion 3 of Appendix A to 10 CFR 50 in the development of a fire protection program for nuclear power plants.

1.9.120.2 VEGP Position

Conformance with BTP CMEB 9.5-1 is addressed in appendix 9B.

1.9.121 REGULATORY GUIDE 1.121, AUGUST 1976, BASES FOR PLUGGING DEGRADED PWR STEAM GENERATOR TUBES

1.9.121.1 Regulatory Guide 1.121 Position

This guide describes an acceptable method for establishing the limiting safe conditions of tube degradation of steam generator tubing, beyond which defective tubes as established by inservice inspection should be removed from service by plugging at each end of the tube. Plugging will be accomplished by either welded plugs or nonwelded mechanical plugs.

1.9.121.2 VEGP Position

Conform, with the following exceptions:

1. C.1 Westinghouse interprets the term "unacceptable defects" to apply to those imperfections resulting from service induced mechanical or chemical degradation of the tube walls which have penetrated to a depth in excess of the plugging limit.

2. C.2a(2) and C.2.a(4) Westinghouse will use a 200-percent margin of safety based on the following definition of tube failure. Westinghouse defines tube failure as plastic deformation of a crack to the extent that the sides of the crack open to a nonparallel, elliptical configuration. This 200-percent margin of safety compares favorably with the 300 percent margin requested by the NRC against gross failure.

3. C.2.b In cases where sufficient inspection data exist to establish degradation allowance, the rate used will be an average time-rate determined from the mean of the test data.

Where requirements for minimum wall are markedly different for different areas of the tube bundle; e.g., U-bend area versus straight length in Westinghouse designs, two plugging limits may be established to address the varying requirements in a manner which will not require unnecessary plugging of tubes.

4. C.3.d(1) and C.3.d(3) The combined effect of these requirements would be to establish a maximum permissible primary-to-secondary leak rate which may be below the threshold of detection with current methods of measurement. Westinghouse has determined the maximum acceptable length of a through-wall crack based on secondary pipe break accident loadings which are typically twice the magnitude of normal operating pressure loads. Westinghouse will use a leak rate associated with the crack size determined on the basis of accident loadings.

5. C.3.e(6) Westinghouse will supply computer code names and references rather than the actual codes.

6. C.3.f(1) Westinghouse will establish a minimum acceptable tube wall thickness (plugging limit) based on structural requirements and consideration of loadings, measurement accuracy and, where applicable, a degradation allowance as discussed in this position and in accordance with the general intent of this guide. Analyses to determine the maximum acceptable number of tube failures during a postulated condition are normally done to entirely different bases and criteria are not within the scope of this guide.

Refer to subsection 5.4.2 for further discussion.

1.9.122 REGULATORY GUIDE 1.122, REVISION 1, FEBRUARY 1978, DEVELOPMENT OF FLOOR DESIGN RESPONSE SPECTRA FOR SEISMIC DESIGN OF FLOOR-SUPPORTED EQUIPMENT OR COMPONENTS

1.9.122.1 Regulatory Guide 1.122 Position

This guide describes the procedures acceptable to the NRC for combining and smoothing the floor response spectra, with peaks broadened, to obtain the floor design response spectra.

1.9.122.2 VEGP Position

Conform. Refer to subsection 3.7.B.2 for discussion on this subject.

1.9.123 REGULATORY GUIDE 1.123, REVISION 1, JULY 1977, QUALITY ASSURANCE REQUIREMENTS FOR CONTROL OF PROCUREMENT OF ITEMS AND SERVICES FOR NUCLEAR POWER PLANTS

1.9.123.1 Regulatory Guide 1.123 Position (HISTORICAL)

The requirements that are included in ANSI N45.2.13-1976 for control of procurement of items and services for nuclear power plants are acceptable to the NRC staff and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the qualifications identified in the guide.

1.9.123.2 VEGP Position

[(HISTORICAL) The VEGP QAP during the design and construction phase (except for NSSS components) conforms to the AEC Extracts from ANSI N45.2.13, Supplementary Quality Assurance Requirements for Control of Procurement of Equipment, Materials, and Services for Nuclear Power Plants (draft dated 5/31/73), as discussed in section 17.1.

VEGP QAP during the design and construction and the operation phases for components purchased from the NSSS vendor conform with ANSI N45.2.13-1976, except for regulatory position C.6b. The NSSS vendor routinely identifies notification points in procurement documents when applicable. Such points are not always identified in pre- and post-award meetings. However, the required notification/hold points are specified by changes to the procurement documents in a reasonable time prior to their being accomplished to allow the purchaser the opportunity to witness the event.

Alternatives and clarifications to the text of ANSI N45.2.13-1976 are contained in the text of WCAP-8370/7800, table 17-1. The VEGP QAP for NSSS components is described in chapter 17, appendix 17B.]

Regulatory Guide 1.123 provided NRC endorsement of ANSI N45.4.13. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.13. Accordingly, quality assurance requirements for control of procurement of items and services during the operation phase are described in the SNC QATR.

1.9.124 REGULATORY GUIDE 1.124, REVISION 1, JANUARY 1978, SERVICE LIMITS AND LOADING COMBINATIONS FOR CLASS 1 LINEAR-TYPE COMPONENT SUPPORTS

1.9.124.1 Regulatory Guide 1.124 Position

This guide delineates acceptable levels of service limits and appropriate combinations of loadings associated with normal operation, postulated accidents, and specified seismic events for the design of Class 1 linear-type component supports as defined in Subsection NF of Section III of the ASME Code.

1.9.124.2 VEGP Position

In general, VEGP conforms. Additional information is provided in subsection 3.8.3. For the NSSS scope, the following exceptions are taken.

Paragraph C.2 of the regulatory guide presents two methods of estimating the ultimate tensile strength S, at temperature. It is believed that method No. 2 is not conservative at elevated metal temperature (in excess of 800°F). In Westinghouse's judgment, values of S at these elevated temperatures should be determined by test rather than via the method given in C.2(b).

Paragraph C.4 of the regulatory guide states: "However, all increases; i.e., those allowed by NF-3231.1(a), XVII-2110(a), and F-1370(a), should always be limited by XVII-2110(b) of Section III." Paragraph XVII-2110(b) specifies that member compressive axial loads shall be limited to two-thirds of critical buckling. Satisfaction of this criteria for the faulted condition is unnecessarily restrictive.

The most significant faulted condition loads on equipment supports result from seismic disturbances and postulated LOCAs, both of which are dynamic events. The allowable faulted condition compressive load should not be limited to two-thirds of critical buckling because these faulted dynamic loads are of extremely short duration, and support members can take impulsive loads that exceed static critical buckling load. Westinghouse will use a compressive axial load of 0.9 of critical buckling since the dynamic buckling capacity of the member is greater than the static buckling capacity.

Paragraph C.6(a) of the regulatory guide appears to erroneously allow the use of faulted stress limits for the emergency condition. Westinghouse will interpret this paragraph as follows: "The stress limits of XVII-2000 of Section III and regulatory position 3, increased according to the provisions of XVII-2100(a) of Section III, should not be exceeded for component supports designed by the linear elastic analysis method."

Westinghouse will use the provisions of F-1370(d) to determine faulted condition allowable loads for supports designed by the load rating method. The method described in paragraph C.7(b) of the regulatory guide is very conservative and inconsistent with the remainder of the faulted stress limits.

In paragraphs B.5 and C.8 of the regulatory guide, Westinghouse takes exception to the requirement that systems whose safety- related function occurs during emergency or faulted plant conditions must meet upset limits. The reduction of allowable stress to no greater than upset limits (which in reality are only design limits since design, normal, and upset limits are the same for linear supports) for support structures in those systems with normal safety-related functions occurring during emergency or faulted plant conditions is overly conservative for components which are not required to mechanically function (inactive components). In addition, Westinghouse believes that emergency and faulted condition criteria are acceptable for active components. However, when these criteria are invoked for active components, any significant deformation that might occur is considered in the evaluation of equipment operability.

1.9.125 REGULATORY GUIDE 1.125, REVISION 1, OCTOBER 1978, PHYSICAL MODELS FOR DESIGN AND OPERATION OF HYDRAULIC STRUCTURES AND SYSTEMS FOR NUCLEAR POWER PLANTS

1.9.125.1 Regulatory Guide 1.125 Position

This guide describes coordination between VEGP and NRC relating to the use of physical hydraulic model testing for predicting performance of safety-related hydraulic structures.

1.9.125.2 VEGP Position

Since there are no safety-related hydraulic structures at VEGP, this regulatory guide is considered not applicable.

1.9.126 REGULATORY GUIDE 1.126, REVISION 1, MARCH 1978, AN ACCEPTABLE MODEL AND RELATED STATISTICAL METHODS FOR THE ANALYSIS OF FUEL DENSIFICATION

1.9.126.1 Regulatory Guide 1.126 Position

This guide provides an analytical model and related assumptions and procedures acceptable to the NRC for predicting the effects of fuel densification in light-water-cooled nuclear power reactors.

1.9.126.2 VEGP Position

This guide states that the model presented in this guide is not intended to supersede NRC approved vendor models. VEGP uses the Westinghouse model which has been approved by the NRC. Refer to paragraph 4.2.3.2 for further discussion.

1.9.127 REGULATORY GUIDE 1.127, REVISION 1, MARCH 1978, INSPECTION OF WATER-CONTROL STRUCTURES ASSOCIATED WITH NUCLEAR POWER PLANTS

1.9.127.1 Regulatory Guide 1.127 Position

This guide describes a basis acceptable to the NRC for developing an appropriate inservice inspection and surveillance program for dams, slopes, canals, and other water-control structures associated with emergency cooling water systems or flood protection of nuclear power plants.

1.9.127.2 VEGP Position

This guide applies only to water-control structures specifically built for use in conjunction with a nuclear power plant and whose failure could cause radiological consequences adversely affecting the public health and safety. There are no such structures in the VEGP design.

1.9.128 REGULATORY GUIDE 1.128, REVISION 1, OCTOBER 1978, INSTALLATION DESIGN AND INSTALLATION OF LARGE LEAD STORAGE BATTERIES FOR NUCLEAR POWER PLANTS

1.9.128.1 Regulatory Guide 1.128 Position

This guide describes a method acceptable to the NRC for performing the installation design and installation of large lead storage batteries for nuclear power plants.

1.9.128.2 VEGP Position

Conform-except as noted below.

Paragraph C.6(i) of the Regulatory Guide states that upon completion of a freshening charge (after battery installation), ". . . a hydrogen survey should be performed to verify that the design criteria required by Position 1 are met." A hydrogen survey of the battery rooms was performed during preoperational testing to verify that the ventilation system limits hydrogen concentration to this level in accordance with Regulatory Guide 1.128. The monitoring equipment locations were judged to be boundary locations for determination of hydrogen concentrations at any location within the battery area. As stated in paragraph 8.3.2.2, for future battery replacements, hydrogen evolution for the new battery will be calculated and compared to the battery being replaced, at which time an engineering evaluation will be performed to determine if a new hydrogen survey is necessary. Additionally, a new survey may be required if the battery room configuration or battery room ventilation system is modified in a manner that reduces air flow or creates a new dead air space in the battery room.

Refer to sections 8.1 and 8.3 for further discussion.

1.9.129 REGULATORY GUIDE 1.129, REVISION 1, FEBRUARY 1978, MAINTENANCE, TESTING, AND REPLACEMENT OF LARGE LEAD STORAGE BATTERIES FOR NUCLEAR POWER PLANTS

1.9.129.1 Regulatory Guide 1.129 Position

This guide endorses IEEE Std. 450-1975, which describes a method acceptable to the NRC for performing the maintenance, testing, and replacement of large lead storage batteries for all types of nuclear power plants subject to the qualifications identified in the guide.

1.9.129.2 VEGP Position

Conform, as discussed in Technical Specifications and subsection 8.3.2, with the following clarification:

> The requirement for an equalizing charge other than for voltage or specific gravity criteria is based upon an analysis of the records of operation and maintenance inspections and is not based upon a specific time interval.
>
> The safety-related batteries will be tested periodically in accordance with the Technical Specifications and the version of IEEE 450 as described in the Bases for the Technical Specifications. A temperature correction methodology approved by the battery manufacturer may be utilized in lieu of the electrolyte temperature correction methodology of IEEE-450.

1.9.130 REGULATORY GUIDE 1.130, REVISION 1, OCTOBER 1978, SERVICE LIMITS AND LOADING COMBINATIONS FOR CLASS 1 PLATE- AND SHELL-TYPE COMPONENT SUPPORTS

1.9.130.1 Regulatory Guide 1.130 Position

This guide delineates acceptable levels of service limits and appropriate combinations of loadings associated with normal operation, postulated accidents, and specified seismic events for the design of Class 1 plate- and shell-type component supports as defined in subsection NF of Section III of the ASME Code.

1.9.130.2 VEGP Position

Conform, except as indicated below. Refer to subsection 3.9.B.3.

Paragraphs C.3, C.4(a), and C.6(a) of the regulatory guide state that the allowable buckling strength should be calculated using a design margin of 2 for flat plates and 3 for shells for normal, upset, and emergency conditions.

In the design of plate- and shell-type supports, member compressive axial loads shall be limited per the requirements of paragraph C.3 for normal, upset, and emergency conditions.

In paragraph C.6 of the regulatory guide, inclusion of the upset plant condition is inappropriate in the load combination under discussion. Westinghouse does not include the upset plant condition in this combination.

In paragraphs C.6(a) and B.1 of the regulatory guide, the stress limits of F-1370(c) are discussed. The criterion states in F-1370(c), "...loads should not exceed 0.67 times the critical buckling strength of the support...." In the design of plate- and shell-type component supports, member compressive axial loads shall be limited to 0.67 times the critical buckling strength. If, as a result of a more detailed evaluation of the supports the member compressive axial loads can be shown to safely exceed 0.67 times the critical buckling for the faulted condition,

verification of the support function adequacy will be documented and submitted to the NRC for review. The member compressive axial loads will not exceed 0.67 times the critical buckling strength without NRC acceptance.

In paragraph C.6(b) of the regulatory guide, the limit based on the test load given in the regulatory guide T.L. x 0.7 S'/S , is overly conservative and is inconsistent with ASME Code requirements presented in Appendix F.

The provisions of F-1370(c) to determine service level D allowable loads are used for supports designed by the load rating method.

1.9.131 REGULATORY GUIDE 1.131, AUGUST 1977, QUALIFICATION TESTS OF ELECTRICAL CABLES, FIELD SPLICES, AND CONNECTIONS FOR LIGHT-WATER-COOLED NUCLEAR POWER PLANTS

1.9.131.1 Regulatory Guide 1.131 Position

Conformance with the requirements of IEEE Std. 383-1974, IEEE Standard for Type Test of Class 1E Electric Cables, Field Splices, and Connections for Nuclear Power Generating Stations, is acceptable for qualifying electric cables, field splices, and connections as components (fire test provisions do not apply to qualification for an installed configuration) for service in light-water-cooled nuclear power plants to ensure that the cables, field splices, and connections can perform their safety-related function subject to the qualifications identified in the guide.

1.9.131.2 VEGP Position

Conform for Class 1E cables except as noted below. Design criteria, regulatory guides, and IEEE standards for electrical equipment are discussed in paragraph 8.1.4.3.

The burner location for flame tests is in accordance with IEEE 383-1974 in lieu of this guide.

The flame source is developed per IEEE 383-1974 in lieu of this guide.

1.9.132 REGULATORY GUIDE 1.132, REVISION 1, MARCH 1979, SITE INVESTIGATIONS FOR FOUNDATIONS OF NUCLEAR POWER PLANTS

1.9.132.1 Regulatory Guide 1.132 Position

Paragraph C of the guide addresses site investigations for foundations.

1.9.132.2 VEGP Position

VEGP site investigation conforms with the requirements of this regulatory guide. Refer to section 2.5 for discussion on this subject.

1.9.133 REGULATORY GUIDE 1.133, REVISION 1, MAY 1981, LOOSE PART DETECTION PROGRAM FOR THE PRIMARY SYSTEM OF LIGHT-WATER-COOLED REACTORS

1.9.133.1 Regulatory Guide 1.133 Position

This guide describes a method acceptable to the NRC for implementing requirements with respect to detecting a potentially safety-related loose part in light-water-cooled reactors during normal operation.

1.9.133.2 VEGP Position

Conforms except as noted below. Refer to paragraph 4.4.6.4 for a discussion of the digital metal impact monitoring system (DMIMS) which is the VEGP loose part monitoring system.

Position C.5.a. states that the sensor location should be noted in the Technical Specifications. The VEGP Technical Specifications do not note sensor locations. This is in conformance with the improved Standard Technical Specifications, NUREG-1431.

Positions C.3.a.(3) and C.5.c. recommend a channel calibration be performed at least once per 18 months. In lieu of this recommendation, the DMIMS will be calibrated at the frequency stated in subsection 16.3, requirement 3.

Positions C.3.a.(2) (a) and (e) state that the alert levels for startup and power operation be submitted to the Commission within 90 days (60 days for subsection (e)) following the completion of the startup test program or when there is a change to the preexisting alert levels for power operation. VEGP will report changes in the alert level alarm to the Commission when they exceed the setpoint determination criteria described in paragraph 4.4.6.4

1.9.134 REGULATORY GUIDE 1.134, APRIL 1987, MEDICAL CERTIFICATION AND MONITORING OF PERSONNEL REQUIRING OPERATOR LICENSES

1.9.134.1 Regulatory Guide 1.134 Position

The requirements contained in ANSI/ANS-3.4-1983, Medical Certification and Monitoring of Personnel Requiring Operator Licenses for Nuclear Power Plants, provide a method acceptable to the NRC staff for determining the medical qualifications of applicants for initial or renewal operator or senior operator licenses, subject to the following:

Section 3.2 of ANSI/ANS-3.4-1983 requires that the facility operator forward to the designated medical examiner a report on each employee referred for a reactor operator medical examination prior to the examination and that the report include information specified by the designated medical examiner.

ANSI/ANS-3.4-1983 further presents recommendations as to what type of information is to be addressed. To preclude the possibility that the facility operator report information that potentially could be biased and not pertinent to an individual's medical qualifications, the facility operator should forward only the information that has been specified or requested by the designated medical examiner for each employee referred for a reactor operator medical examination for the purpose of completing Form NRC-396, Certificate of Medical Examination.

1.9.134.2 VEGP Position

Conform. Medical certification and monitoring of personnel requiring operator licenses is in accordance with Regulatory Guide 1.134.

1.9.135 REGULATORY GUIDE 1.135, SEPTEMBER 1977, NORMAL WATER LEVEL AND DISCHARGE AT NUCLEAR POWER PLANTS

1.9.135.1 Regulatory Guide 1.135 Position

Safety-related structures resisting combinations of loading conditions require identification of normal water levels or discharges. The normal water level is generally defined as that water level either equal to or exceeded, 50 percent of the time.

1.9.135.2 VEGP Position

The level of the Savannah River does not affect the safety-related structures. The groundwater level has been firmly established at the site by an over-200 well canvas.

Groundwater levels at the site have been monitored for a period of greater than 10 years. Historic groundwater levels in the surrounding region have been monitored by others for a much longer period. The requirements of the guide are therefore satisfied for VEGP. Refer to section 2.4 for further discussion.

1.9.136 REGULATORY GUIDE 1.136, REVISION 2, JUNE 1981, MATERIALS, CONSTRUCTION, AND TESTING OF CONCRETE CONTAINMENTS

1.9.136.1 Regulatory Guide 1.136 Position

This guide describes bases acceptable to the NRC for implementing requirements with regard to the materials, construction, and testing of concrete containments. This guide endorses the

requirements specified in Articles CC-1000, CC-2000 and CC-4000 through CC-6000 of the Code for Concrete Reactor Vessels and Containments, ASME Section III, Division 2, 1980 Edition (also known as ACI Standard 359-80), subject to the qualifications provided in this guide.

1.9.136.2 VEGP Position

The VEGP containment design generally conforms to Article CC-3000 of the ASME Code, Section III, Division 2, 1974 through Winter 1975 addenda. Refer to subsection 3.8.1 for further discussion on this subject.

1.9.137 REGULATORY GUIDE 1.137, REVISION 1, OCTOBER 1979, FUEL-OIL SYSTEMS FOR STANDBY DIESEL GENERATORS

1.9.137.1 Regulatory Guide 1.137 Position

The requirements for the design of fuel-oil systems for diesel generators that provide standby electrical power for a nuclear power plant that are included in ANSI N195-1976, Fuel Oil Systems for Standby Diesel Generators, provide a method acceptable to the NRC staff for complying with the pertinent requirements of General Design Criterion 17 of Appendix A to 10 CFR 50, subject to the qualifications identified in the guide.

1.9.137.2 VEGP Position

Conform to Regulatory Guide 1.137 which endorses ANSI N195-1976 with the following clarifications and exceptions.

Clarification to regulatory position C.1.e is as follows:

1. “Section 7.3 of ANSI N195-1976 states that the arrangement of the fuel-oil system “shall provide for inservice inspection and testing in accordance with ASME Boiler and Pressure Vessel Code, Section XI, *Rules for Inservice of Nuclear Power Plant Components*.” For those portions of the fuel-oil system for standby diesel generators that are designed to Section III, Subsection ND of the Code, an acceptable method of meeting the requirements of Section 7.3 is to ensure that the system arrangement would allow:

    (1) Pressure testing of the fuel-oil system to a pressure 1.10 times the system design pressure at 10-year intervals...

    (2) A visual examination to be conducted during the pressure test for evidence of component leakages, structural distress, or corrosion....”

The diesel fuel oil transfer system is designed, fabricated, erected, and tested to quality standards commensurate with the safety function to be performed. VEGP will perform inservice inspection and testing commensurate with the safety function of the system in accordance with the above defined acceptable methods (1) and (2), not per ASME Section XI. Specifically, the

operational readiness of the pumps and valves to meet their intended safety function is maintained in accordance with the VEGP Technical Specifications.

VEGP utilizes Regulatory Guide 1.26 (reference UFSAR paragraph 1.9.26.1) to determine the safety classification of systems and components, and then applies ASME Section XI inspection and testing requirements. ASME Section XI requires that Subsection IWD rules for inservice inspection and testing requirements be applied to those components classified as Class 3, except where the system has been optionally upgraded.

Regulatory Guide 1.26 provides a method for determining acceptable quality standards for safety-related components containing radioactive material, water, or steam but does not require that fuel oil transfer system components be classified as Class 3. Although the fuel oil transfer system was constructed to ASME Section III, Subsection ND (Class 3) per ANSI N195-1976, the original classifications of this system as Class 3 is considered to be an optional upgrade for Section XI purposes, since it is not required to be Class 3 per Regulatory Guide 1.26. Therefore, the diesel fuel oil transfer system is not included in the scope of ASME Section XI.

Exception to regulatory position C.2 is as follows:

1. Regarding Appendix B to ANSI N195-1976, *Recommended Fuel Oil Practices*. Testing of the diesel generator fuel oil will be addressed in the Technical Specifications.

1.9.138 REGULATORY GUIDE 1.138, APRIL 1978, LABORATORY INVESTIGATIONS OF SOILS FOR ENGINEERING ANALYSIS AND DESIGN OF NUCLEAR POWER PLANTS

1.9.138.1 Regulatory Guide 1.138 Position

This guide describes laboratory investigations and testing practices acceptable to the NRC for determining soil and rock properties and characteristics needed for engineering analysis and design for foundations and earthworks for nuclear power plants.

1.9.138.2 VEGP Position

The soil investigations for VEGP were performed prior to the issuance of this guide. Soils testing is described in section 2.5.

1.9.139 REGULATORY GUIDE 1.139, MAY 1978, GUIDANCE FOR RESIDUAL HEAT REMOVAL

1.9.139.1 Regulatory Guide 1.139 Position

This guide describes a method acceptable to the NRC for complying with regulations with regard to the removal of decay heat and sensible heat after a reactor shutdown.

1.9.139.2 VEGP Position

VEGP design was reviewed and modified to provide safety-grade cold shutdown in accordance with this guide, as described in subsection 5.4.7.

RHR outlet Instrument Society of America (ISA) check valves are in conformance with regulatory position C.2 except that they will be tested once every refueling.

1.9.140 REGULATORY GUIDE 1.140, REVISION 1, OCTOBER 1979, DESIGN, TESTING, AND MAINTENANCE CRITERIA FOR NORMAL VENTILATION EXHAUST SYSTEM AIR FILTRATION AND ADSORPTION UNITS OF LIGHT-WATER-COOLED NUCLEAR POWER PLANTS

1.9.140.1 Regulatory Guide 1.140 Position

This guide describes acceptable methods of implementing regulations with regard to the design, testing, and maintenance criteria for air filtration and adsorption units designed to collect airborne radioactive materials during normal operation, including anticipated operational occurrences.

1.9.140.2 VEGP Position

Conform, except as discussed below.

C.3.f Duct stiffener angles are attached to the duct by mechanical fasteners (huck bolts) or by welding.

C.4.b The steam packing exhauster filtration system does not meet Regulatory Guide 1.140 spacing criteria between filter components. Since the system is operated only when high radiation is present, maintenance will only be required on a minimal basis. The physical location in which the filtration unit is installed has limited, but adequate, space availability.

The in-place testing of the HEPA filter and charcoal adsorber for the containment preaccess filter system is performed initially at acceptance testing and after subsequent filter or charcoal replacement. As discussed in paragraph 9.4.6.2.3, the preaccess filter system is a recirculating, nonducted system inside containment that reduces airborne contamination inside containment prior to personnel entry. Leakage through the HEPA or charcoal adsorber would not contribute to any increased radiological release to the environment.

C.5.d The activated carbon adsorber section of the post-LOCA containment hydrogen purge filter is leak-tested with a gaseous halogenated hydrocarbon refrigerant in accordance with Section 12 of ANSI N510-1975 to ensure that bypass leakage through the adsorber section is less than 1 percent as shown in table 6.2.5-3.

C.6.a New, activated, impregnated carbon is tested per Table 5-1 of ANSI N509-1980, except that the performance requirements shall be 0.5-percent penetration maximum when tested with molecular iodine per ASTM D3803 at 30°C and 95-percent relative humidity, and 50-percent penetration maximum when tested with methyl iodide per ASTM 3803 at 30°C and 95-percent relative humidity.

C.6.b Used carbon samples taken for laboratory testing shall meet the acceptance criteria of greater than or equal to 90 percent when tested with methyl iodide at 30°C and 70-percent relative humidity.

C.6.c Condenser vacuum exhaust and steam packing exhaust filtration systems carbon samples taken for laboratory testing shall meet the acceptance criteria of greater than or equal to 90 percent when tested with methyl iodide at 30°C and 95-percent relative humidity.

Wherever ANSI N509-1976 is referenced in the regulatory guide, conformance is with ANSI N509-1976, ANSI N509-1980, or ASME N509-1989 depending on the date of the applicable purchase order. Conformance may be with a particular revision when specifically called out in the corresponding specification. In addition, the fan peak pressure test delineated in paragraph 5.10.8.2 of ANSI N509-1980 will not be performed since two separate interlocks are provided to preclude overpressurization of the ducting.

Wherever ANSI N510-1975 is referenced in the regulatory guide, conformance is with ANSI N510-1975, ANSI N510-1980, or ASME N510-1989 depending on the date of the applicable purchase order.

Conformance may be with a particular revision when specifically called out in the corresponding specification.

The radwaste processing facility HEPA filtration unit was designed in accordance with ASME AG-1 (1997) and subject to testing described in paragraph 9.4.3.4.

1.9.141 REGULATORY GUIDE 1.141, APRIL 1978, CONTAINMENT ISOLATION PROVISIONS FOR FLUID SYSTEMS

1.9.141.1 Regulatory Guide 1.141 Position

The requirements and recommendations for containment isolation of fluid systems that penetrate the primary containment of light-water-cooled reactors as specified in ANSI N271-1976, Containment Isolation Provisions for Fluid Systems, are generally acceptable and provide an adequate basis for complying with the pertinent containment isolation requirements of Appendix A to 10 CFR 50, subject to the qualifications identified in the guide.

1.9.141.2 VEGP Position

VEGP conforms as discussed in subsection 6.2.4.

1.9.142 REGULATORY GUIDE 1.142, OCTOBER 1981, REVISION 1, SAFETY-RELATED CONCRETE STRUCTURES FOR NUCLEAR POWER PLANTS (OTHER THAN REACTOR VESSELS AND CONTAINMENTS)

1.9.142.1 Regulatory Guide 1.142 Position

This guide endorses the procedures and requirements described in American Concrete Institute (ACI) 349-76 subject to the qualifications provided in this guide.

1.9.142.2 VEGP Position

ACI 318-71 is used in lieu of ACI 349-76; but ACI 349-76, including Appendix B, may be utilized subject to the qualifications provided in Regulatory Guide 1.142.

Refer to subsections 3.8.3, 3.8.4, and 3.8.5 for discussion on this subject.

1.9.143 REGULATORY GUIDE 1.143, REVISION 1, OCTOBER 1979, DESIGN GUIDANCE FOR RADIOACTIVE WASTE MANAGEMENT SYSTEMS, STRUCTURES, AND COMPONENTS INSTALLED IN LIGHT-WATER-COOLED NUCLEAR POWER PLANTS

1.9.143.1 Regulatory Guide 1.143 Position

This guide furnishes design guidance acceptable to the NRC regarding seismic and quality group classification and quality assurance provisions for radioactive waste management systems, structures, and components.

1.9.143.2 VEGP Position

Conform, with the following clarifications:

- Radioactive waste management systems, structures, and components are classified in table 3.2.2-1.
- ACI 318-71 is used for design of concrete structures in lieu of ACI 318-77.
- Flexible, nonmetallic hoses with couplings are utilized in the radwaste processing facility for connecting various portable processing equipment.
- Portable washer/dryer and associated connections utilized in health physics laundry room.
- Regulatory Guide 1.143 stipulates that radwaste piping is to be hydro tested to 1.5 times the design pressure. Some butt weld connections on the radwaste process

piping in the north end of the radwaste transfer tunnel are not hydro tested as stipulated. In lieu of the hydro test, a full volumetric radiographic or ultrasonic NDE is performed on these welds in conjunction with an inservice leak test at nominal operating pressure and temperature. This ensures VEGP radwaste process piping continues to meet or exceed the inspection requirements of Regulatory Guide 1.143.

- Regulatory Guide 1.143 stipulates that radwaste piping is to be hydro tested to 1.5 times design pressure and held for 30 minutes. In lieu of hydrostatic testing following repair replacement activities (by welding) of ASME Class 1, 2, and 3 components, the NRC has endorsed the use of Code Case N-416-2 with a stipulation that hold time for use of this Code Case is conditionally acceptable provided the provisions of IWA-5213, "Test Condition Holding Times," 1989 Edition, are to be used (Ref. RG 1.147, Rev 13, January 2004, Inservice Inspection Code Case Acceptability). IWA 5213 requires a 10 minute hold time unless the system has been in operation for at least 10 minutes for an uninsulated system; 4 hours for an insulated system. Additionally, this Code Case stipulates that NDE shall be performed on welded repairs and fabrication and installation joints in accordance with the methods and acceptance criteria of the applicable subsection of the 1992 Edition of Section III. This requirement applies to piping and safety systems (ASME Section III, Class 1, 2, and 3). Apply the Code Case requirements for Class 3 piping to radwaste system piping. For applications related to radwaste systems, the requirements in the Code Case to complete an NIS-2 form are not applicable. Also, there are no size exemptions that can be utilized when applying this Code Case to radwaste systems. Additionally, as stated above, the surface examination requirements of Class 3 will also apply, i.e., welds 2-1/2 in. and greater will require a magnetic particle or liquid penetrant examination. All welds 2 in. and under will require a visual examination.

See section 11.4 for further discussion.

1.9.144 REGULATORY GUIDE 1.144, REVISION 1, SEPTEMBER 1980, AUDITING OF QUALITY ASSURANCE PROGRAMS FOR NUCLEAR POWER PLANTS

1.9.144.1 Regulatory Guide 1.144 Position (HISTORICAL)

[The requirements that are included in ANSI/ASME N45.2.12-1977 for auditing QAPs for nuclear power plants are acceptable to the NRC staff and provide an adequate basis for complying with the pertinent quality assurance requirements of Appendix B to 10 CFR 50, subject to the qualifications identified in the guide.]

1.9.144.2 VEGP Position

[(HISTORICAL) The VEGP QAP for the design and construction phase conforms with Regulatory Guide 1.144, January 1979, which also endorses ANSI/ASME N45.2.12-1977, subject to qualification identified in the guide, with the following clarifications. VEGP does not conform to the latest revision of the following ANSI standards: ANSI N45.2, ANSI N45.2.9, and ANSI N45.2.10. VEGP conforms to ANSI N45.2-1971, ANSI N45.2.9 (Draft 11, Revision 0, January 17, 1973), and ANSI N45.2.10-1973. The VEGP quality assurance program is described in section 17.1, appendix 17A, and appendix 17C.]

Regulatory Guide 1.144 provided NRC endorsement of ANSI N45.2.12. The SNC QATR is based on ASME NQA-1-1994 which incorporates the requirements of ANSI N45.2.12. Accordingly, requirements for auditing quality assurance programs applicable to operations phase activities are described in the QATR.

1.9.145 REGULATORY GUIDE 1.145, AUGUST 1979, ATMOSPHERIC DISPERSION MODELS FOR POTENTIAL ACCIDENT CONSEQUENCE ASSESSMENTS AT NUCLEAR POWER PLANTS

1.9.145.1 Regulatory Guide 1.145 Position

This guide identifies acceptable methods for:

- Calculating atmospheric relative concentration (χ/Q) values.
- Determining χ/Q values on an overall site basis.
- Determining χ/Q values on a directional basis.
- Choosing χ/Q values to be used in evaluations of the types of events described in Regulatory Guides 1.3 and 1.4.

1.9.145.2 VEGP Position

Conform. Refer to subsection 2.3.4.

1.9.146 REGULATORY GUIDE 1.146, AUGUST 1980, QUALIFICATION OF QUALITY ASSURANCE PROGRAM AUDIT PERSONNEL FOR NUCLEAR POWER PLANTS

1.9.146.1 Regulatory Guide 1.146 Position (HISTORICAL)

[This guide describes a method acceptable to the NRC for complying with regulations with regard to qualification of QAP audit personnel for nuclear power plants.]

1.9.146.2 VEGP Position

[(HISTORICAL) Conform except as discussed below.

ANSI N45.2.23-1978, Section 2.3.4 states that the prospective lead auditor shall have participated in a minimum of five quality assurance audits within a period of time not to exceed 3 years prior to the date of qualification, one audit of which shall be a nuclear QA audit within the year prior to his or her qualification.

In lieu of the requirements of Section 2.3.4 of ANSI N45.2.23-1978, the prospective lead auditor shall demonstrate their ability to effectively implement the audit process and effectively lead an audit team. The demonstration process will be described in written procedures and shall evaluate and document the results of the demonstration. Regardless of the methods used for the demonstration, the prospective lead auditor shall have participated in at least one nuclear QA audit within the year preceding the individual's effective date of qualification. Upon successful demonstration of the ability to effectively implement the audit process and lead audits, and having met the other provisions of Section 2.3 of ANSI N45.2.23-1978, the individual may be certified as being qualified to lead audits.

The QAP is discussed in Chapter 17.]

Regulatory Guide 1.146 provided NRC endorsement of ANSI N45.2.23. The SNC QATR is based on ASME NQA-1-1994 which contains qualification requirements applicable to quality assurance audit personnel in NQA-1 Basic Requirement 2 and Supplement 2S-3. Accordingly, requirements for qualification of quality assurance program audit personnel applicable to operation phase activities are described in the SNC QATR.

1.9.147 REGULATORY GUIDE 1.147, INSERVICE INSPECTION CODE CASE ACCEPTABILITY, ASME SECTION XI, DIVISION 1

1.9.147.1 Regulatory Guide 1.147 Position

This regulatory guide lists those Section XI ASME code cases that are generally acceptable to the NRC for implementation in the ISI of light-water-cooled nuclear power plants.

1.9.147.2 VEGP Position

Conform. Refer to section 6.6 for further discussion.

1.9.148 REGULATORY GUIDE 1.148, MARCH 1981, FUNCTIONAL SPECIFICATION FOR ACTIVE VALVE ASSEMBLIES IN SYSTEMS IMPORTANT TO SAFETY IN NUCLEAR POWER PLANTS

1.9.148.1 Regulatory Guide 1.148 Position

This guide delineates a procedure acceptable to the NRC for implementing regulations with respect to the detailed specification of information pertinent to defining operating requirements for active valve assemblies in light-water-cooled nuclear power plants.

1.9.148.2 VEGP Position

Conformance is addressed in table 3.9.B.3-10.

1.9.149 REGULATORY GUIDE 1.149, APRIL 1987, NUCLEAR POWER PLANT SIMULATION FACILITIES FOR USE IN OPERATOR LICENSE EXAMINATIONS

1.9.149.1 Regulatory Guide 1.149 Position

This regulatory guide describes a method acceptable to the NRC for specifying the functional requirements of a nuclear power plant simulator to be used for operator training.

1.9.149.2 VEGP Position

Conform with exceptions as described by subsection 13.2.1.

1.9.150 REGULATORY GUIDE 1.150, FEBRUARY 1983, ULTRASONIC TESTING OF REACTOR VESSEL WELDS DURING PRESERVICE AND INSERVICE EXAMINATIONS

1.9.150.1 Regulatory Guide 1.150 Position

This guide describes a method for volumetric examination of reactor vessel welds.

1.9.150.2 VEGP Position

Conform to the extent practicable. See subsection 5.2.4.

1.9.155 REGULATORY GUIDE 1.155, AUGUST 1988, STATION BLACKOUT

1.9.155.1 Regulatory Guide 1.155 Position

This regulatory guide describes a means acceptable to the NRC staff for meeting the requirements of 50.63 of 10 CFR Part 50 for station blackout coping. NUMARC 87-00 also provides guidance acceptable to the staff for meeting these requirements.

1.9.155.2 VEGP Position

Conform. See section 8.4.

1.9.163 REGULATORY GUIDE 1.163, SEPTEMBER 1995, PERFORMANCE-BASED CONTAINMENT LEAK-TESTING PROGRAM

1.9.163.1 Regulatory Guide 1.163 Position

This regulatory guide provides guidance on an acceptable performance-based leak-test program, leakage-rate test methods, procedures, and analysis that may be used to comply with the performance-based Option B in 10 CFR Part 50, Appendix J.

1.9.163.2 VEGP Position

Conform with the following exceptions:

1. Leakage rate testing for the containment purge valves with resilient seals is performed once per 18 months in accordance with the Limiting Condition of Operation of Technical Specification 3.6.3, Surveillance Requirements 3.6.3.6 and 3.0.2.

2. Containment personnel air lock door seals will be tested prior to reestablishing containment integrity when the air lock has been used for containment entry. When containment integrity is required and the air lock has been used for containment entry, door seals will be tested at least once per 30 days during the period that containment entry(ies) is (are) being made.

3. The visual examination of containment concrete surfaces intended to fulfill the requirements of 10 CFR 50, Appendix J, Option B testing, will be performed in accordance with the requirements of and frequency specified by ASME Section XI Code, Subsection IWL, except where relief has been authorized by the NRC. At the discretion of the licensee, the containment concrete visual examinations may be performed during either power operation, e.g., performed concurrently with other containment inspection-related activities such as tendon testing, or during a maintenance/refueling outage.

1.9.166 REGULATORY GUIDE 1.166, REVISION 0, MARCH 1997 PRE-EARTHQUAKE PLANNING AND IMMEDIATE NUCLEAR POWER PLANT OPERATOR POST-EARTHQUAKE ACTIONS

1.9.166.1 Regulatory Guide 1.166 Position

This guide provides guidance acceptable to the NRC for a timely evaluation of the recorded instrumentation data after an earthquake and for determining whether plant shutdown is required by 10 CFR 50.

1.9.166.2 VEGP Position

VEGP conforms to this guide.

1.9.167 REGULATORY GUIDE 1.167, REVISION 0, MARCH 1997, RESTART OF A NUCLEAR POWER PLANT SHUTDOWN BY A SEISMIC EVENT

1.9.167.1.1 Regulatory Guide 1.167 Position

This guide provides guidance acceptable to the NRC for performing inspections and tests of nuclear power plant equipment and structures prior to restart of a plant that has been shutdown by a seismic event.

1.9.167.2 VEGP Position

VEGP conforms to this guide.

TABLE 1.9-1 (SHEET 1 OF 8)

DESIGN AND FABRICATION CODE CASES

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-101 (1712)	Reg. Guide 1.84	None
N-31[(a)(b)] (1540-2)	Reg. Guide 1.84	Elastomer Diaphragm Valves, Section III, Class 2 and 3 Code Case 1540-2 is acceptable subject to the following conditions in addition to those conditions specified in the code case: (1) Each applicant who applies the code case should indicate in the referencing Safety Analysis Report that the service life of the elastomer diaphragm should not exceed the manufacturer's recommended service life. This recommended service life should not exceed one-third of the minimum cycle life as established by the requirements of paragraph 3 of the code case. (2) The service life of the elastomer diaphragm should not exceed 5 years. (3) The combined service and storage life of the elastomer diaphragm should not exceed 10 years.
N-142 (1774)	Reg. Guide 1.84	None
N-100 (1711)	Reg. Guide 1.84	Pressure Relief Valve Design Rules, Section III, Division 1, Class 1, 2, and 3.

TABLE 1.9-1 (SHEET 2 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
		Code Case 1711 is acceptable subject to the following conditions in addition to those conditions specified in the code case. The following information should be provided in the Safety Analysis Report: (1) If stress limits are used in excess of those for the upset operating condition, it should be demonstrated how the pressure relief function is assured. Refer to paragraph 3.1, Section I, of the case for Class 1 and paragraph 3.2, Section II, of the case for Class 2 and 3 pressure relief valves. (2) If Case 1660 is to be used in conjunction with this case, it should be stated that the stress limits of Case 1660 supersede those of paragraph 3.2(6), Section I, of Case 1711. Functional assurance of(1) above is required in all situations.
N-154	Reg. Guide 1.84	None
N-32-3 (1541-3)	Reg. Guide 1.84	None
N-237	Reg. Guide 1.84	None
N-86 (1686)	Reg. Guide 1.84	None
N-30	Reg. Guide 1.84	None

TABLE 1.9-1 (SHEET 3 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
1606 1606-1	Reg. Guide 1.84 Reg. Guide 1.84R.24	Stress Criteria Section III, Classes 2 and 3, Piping Subject to Upset, Emergency, and Faulted Operating Conditions Code Cases 1606 and 1606-1 are acceptable subject to the interpretation that the stress limit designations of "upset," "emergency," and "faulted" do not necessarily imply agreement with specified plant conditions applicable to ASME Code Class 2 and 3 components for fluid systems. These designations should be established and justified in the design specifications.
N-40 (1569)	Reg. Guide 1.84	Design of Piping for Pressure Relief Valve Station, Section III Code Case 1569 (N-40) was acceptable subject to compliance with the recommendations contained in Regulatory Guide 1.67, Installation of Overpressure Protection Devices.
N-275	Reg. Guide 1.84	Repair of Welds, Section III, Division 1 Code Case N-275 is acceptable subject to the following condition in addition to those conditions specified in the code case. Use of the code case is applicable only when the removal of an indication requires that the full weld thickness be removed and, in addition, the backside of the weld assembly joint is not accessible for the removal of examination material. If an indication is removed and weld-metal layers still remain, it is not acceptable to gouge through the wall in order to qualify for use of the code case. Instead, examination of the cavity is required when such an indication has been removed.

TABLE 1.9-1 (SHEET 4 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-111	Reg. Guide 1.84	None
N-55	Reg. Guide 1.84	None
N-82	Reg. Guide 1.84	None
N-175 (1818)	Reg. Guide 1.84	Welded Joints in Component Standard Supports, Section III, Division 1 Code Case 1818 is acceptable subject to the following condition in addition to those conditions specified in the code case: That portion of the unwelded housing that is limited to 90° maximum should include a minimum of two sectors that are uniform in length.
N-24 (1516-2)	Reg. Guide 1.84	None
N-69 (1635-1)		Stress Criteria for Section III, Class 2 and 3 Valves Subjected to Upset, Emergency, and Faulted Operating Conditions Code Case 1635-1 is acceptable subject to the interpretation that the stress limit designations of "upset," "emergency," and "faulted" do not necessarily imply agreement with specified plant conditions applicable to ASME Code Class 2 and 3 components for fluid systems. These designations should be established and justified in the design specifications.

TABLE 1.9-1 (SHEET 5 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-240	Reg. Guide 1.84	None
N-174 (1812)	Reg. Guide 1.84	None
N-302	Reg. Guide 1.84	None
N-316	Reg. Guide 1.84	None
N-318-2	Reg. Guide 1.84	Code Case N-318 is acceptable subject to the following conditions in addition to those conditions specified in the Code Case. Applicants should identify in their SAR: (1) method of lug attachment, (2) piping system involved; and (3) location in the system where the case is to be applied.
N-411	NRC letter, Novak to D. O. Foster March 3, 1985.	To apply to piping systems analyzed by the response spectra method. Increased pipe deflections due to greater piping flexibility will not violate project separation criteria.

TABLE 1.9-1 (SHEET 6 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
		Criteria outlined in Regulatory Guide 1.61 will not mix with the criteria of Code Case N-411 for a given piping analysis. As part of the integrated piping analysis/as-built reconciliation program, increased piping displacements and clearances will be reviewed for acceptance. With the exception of certain stress calculations described in GPC Letter GN-1257, J. A. Bailey to B. J. Youngblood, dated December 22, 1986, Code Case N-411 damping values are not used in conjunction with multiple response spectrum methodology piping analysis.
N-62-2 (1621-2)	Reg. Guide 1.84	Code Case 1621-2 is acceptable subject to the following condition: In addition to those conditions specified in the code case, the Code requires that Class 1 and Class 2 valve manufacturers meet the provisions of NCA-4000, "Quality Assurance," and in addition, Class 3 valve manufacturers should also meet the provisions of NCA-4000.
N-272	Reg. Guide 1.84	None
N-412	NRC letter, Denton to D. O. Foster, dated, November 18, 1985	None

TABLE 1.9-1 (SHEET 7 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-397	Reg. Guide 1.84 and NRC letter, Youngblood to D. O. Foster, dated December 26, 1985	Code Case N-397 is acceptable subject to the following condition in addition to those conditions specified in the Code Case. The Code Case is acceptable for specific plant applications on a case by case basis pending revision of Regulatory Guide 1.122, "Development of Floor Design Response Spectra for Seismic Design of Floor-Supported Equipment or Components."
N-316	Reg. Guide 1.84	None
N-411	NRC letter, Novak to D. O. Foster dated, March 18, 1985	
N-392	Reg. Guide 1.84	None
N-315	Reg. Guide 1.84	Code Case N-315 to ASME III Division 1 is acceptable subject to the following conditions specified in the Code Case: Prior to implementation of the Code Case, the applicant should present a description of the repair and a justification why the bellows should be repaired rather than replaced. Following receipt of approval for the repair, the applicant should present the results of the qualification of the full-scale facsimile bellows, including the design requirements, to ensure that the repair meets the design specification.

TABLE 1.9-1 (SHEET 8 OF 8)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-82 (1677)	Reg. Guide 1.84	None
N-368	Reg. Guide 1.84	Pressure Testing of Pump Discharge, Section III, Division 1, Class 2 and 3. Code case is acceptable subject to the following conditions in addition to those specified in the code case: Applicants using this code case should provide information to demonstrate that the length of discharge piping is reasonably short.
N-192-1 N-192-2	Reg. Guide 1.84 Reg. Guide 1.84	Use of Flexible Hose for Section III, Division 1. Class 1, 2, and 3 Construction. Code cases N-192-1 and N-192-2 are acceptable subject to the following conditions in addition to those specified in the code case: The applicant should indicate system application, design and operating pressure, and pressure temperature rating of the flexible hose. Data to demonstrate compliance of the flexible hose with NC/ND-3649, particularly NC/ND-3649.4(e) are required to be furnished with the application.
N-247	Reg. Guide 1.84	None

a. The service life of elastomer diaphragm material used for zero leakage valves does not exceed the manufacturer's recommended service life. The recommended service life does not exceed one-third of the minimum cycle life established in paragraph 3 of the code case.

b. The combined storage and service life on elastomer diaphragms may be greater than 10 years, and the service life may exceed 5 years, but neither is the environmental qualified life exceeded nor does the service life exceed one-third of the minimum established cycle life of paragraph 3 of the code case. Evaluation of the environmental qualified life considers process fluid and location, including temperature and radiation environments encountered by the valves.

TABLE 1.9-2 (SHEET 1 OF 7)

MATERIALS CODE CASES

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
1713	Reg. Guide 1.85	None
N-71-8	Reg. Guide 1.85	Additional Materials for Component Supports and Alternate Design Requirements for Bolted Joints, Section III, Division 1,Subsection NF, Class 1, 2, 3 and MC Construction Code Case 1644-8 is acceptable subject to the following conditions in addition to those specified in the code case: The maximum measured ultimate tensile strength (UTS) of the component support material should not exceed 170 ksi in view of the susceptibility of high-strength materials to brittleness and stress corrosion cracking. Certain applications may exist where a UTS value of up to 190 ksi could be considered acceptable for a material and, under this condition, the design specification should specify impact testing for the material. For these cases, it should be demonstrated by the applicant that: (1) The impact test results for the material meet code requirements. (2) The material is not subject to stress corrosion cracking by virtue of the fact that: (a) A corrosive environment is not present. (b) The component that contains the material has essentially no residual stresses or assembly stresses, and it does not experience frequent sustained loads in service.

TABLE 1.9-2 (SHEET 2 OF 7)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-71-7	Reg. Guide 1.85	Additional Materials for Component Supports, Section III, Division 1, Subsection NF, Class 1, 2, 3 and MC Component Supports Code Case 1644-7 was acceptable subject to the following conditions in addition to those specified in the code case: The maximum measured UTS of the component support material should not exceed 170 ksi in view of the susceptibility of high-strength materials to brittleness and stress corrosion cracking. Certain applications may exist where a UTS value of up to 190 ksi could be considered acceptable for a material and, under this condition, the design specification should specify impact testing for the material. For these cases, it should be demonstrated by the applicant that: (1) The impact test results for the material meet code requirements. (2) The material is not subject to stress corrosion cracking by virtue of the fact that: (a) A corrosive environment is not present. (b) The component that contains the material has essentially no residual stresses or assembly stresses, and it does not experience frequent sustained loads of service.

TABLE 1.9-2 (SHEET 3 OF 7)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-225	Reg. Guide 1.85	None
N-71-5	Reg. Guide 1.85	Additional Materials for Component Supports and Alternate Design Requirements for Bolted Joints, Section III, Division 1, 2, 3 and MC Construction Code Case 1644-5 was acceptable subject to the following conditions in addition to those specified in the code case: The maximum measured UTS of the component support material should not exceed 170 ksi in view of the susceptibility of high-strength materials to brittleness and stress corrosion cracking. Certain applications may exist where a UTS value of up to 190 ksi could be considered acceptable for a material and, under this condition, the design specification should specify impact testing for the material. For these cases, it should be demonstrated by the applicant that: (1) The impact test results for the material meet code requirements. (2) The material is not subject to stress corrosion cracking by virtue of the fact that: (a) Corrosive environment is not present. (b) The component that contains the material has essentially no residual stresses or assembly stresses, and it does not experience frequent sustained loads in service.

TABLE 1.9-2 (SHEET 4 OF 7)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-38	Reg. Guide 1.85	None
1649	Reg. Guide 1.85	None
N-188-1	Reg. Guide 1.85	None
N-147 (1781)	Reg. Guide 1.85	None
N-294	Reg. Guide 1.85	None
N-3-10	Reg. Guide 1.85	None
N-249	Reg. Guide 1.85	Additional Materials for Component Supports Fabricated Without Welding, Section III, Division 1, Subsection NF, Class 1, 2, and 3 MC Component Supports Code Case N-249 is acceptable subject to the following conditions in addition to those conditions specified in the code case. Footnote 2 of the code case should apply to all materials listed in tables 1 through 5 of the code case and should be so indicated in line 5 of the "Reply."
N-249-1	Reg. Guide 1.85	Code Case N-249-1 is acceptable subject to the following condition in addition to those conditions specified in the code case. Paragraph 7 of the "Reply" should reference the requirements of NF-2600 instead of NF-2800. This is a typographical error in that NF-2800 does not exist.
N-249-2	Reg. Guide 1.85	None

TABLE 1.9-2 (SHEET 5 OF 7)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-249-4 through N-249-5	Reg. Guide 1.85	Code Cases N-249-4 and N-249-5 are acceptable subject to the following: The fracture toughness requirements as listed in Code Cases N-249-4 and N-249-5 apply only to piping supports and not to Classes 1, 2, and 3 component supports. The fracture toughness of Classes 1, 2, and 3 should be characterized in accordance with paragraph 5.3.4 of the USNRC Standard Review Plan (NUREG-0800) or on a case-by-case basis.
N-249-6 through N-249-12	Reg. Guide 1.85	Code Cases N-249-6 through N-249-12 are acceptable subject to the following conditions in addition to those conditions specified in the Code Case: the fracture toughness requirements apply only to piping supports and not to Classes 1, 2, and 3 component supports. The fracture toughness of Classes 1, 2, and 3 component supports should be characterized in accordance with paragraph 5.3.4 of the USNRC Standard Review Plan (NUREG – 0800) or on a case-by-case basis. The following is to be added to paragraph (e) of the Case: For these cases, it should be demonstrated by the owner that: (1) The impact test results for the material meet Code Requirements and (2) The material is not subject to stress corrosion cracking by virtue of the fact that: (a) A corrosive environment is not present and (b) The component that contains the material has essentially no residual stresses or assembly stresses, and it does not experience frequent sustained loads in service.

TABLE 1.9-2 (SHEET 6 OF 7)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
N-71-10	Reg. Guide 1.85	Code Case N-71-10 is acceptable subject to the following condition in addition to those conditions specified in the code case. In the last sentence of paragraph 5-3, reference should be made to paragraph 4.5.2.2, Alternate Atmosphere Exposure Time Periods Established by Test, of the American Welding Society D.1.1 Code for the evidence presented to and accepted by the authorized inspector concerning exposure of electrodes for longer periods of time.
N-242-1	Reg. Guide 1.85	Code Case N-242-1 is acceptable subject to the following condition in addition to those conditions specified in the code case: applicants should identify in their safety analysis reports the components and supports for which the code case is being applied and should specify the respective paragraphs of the code case. For VEGP, this code case is being used for the reactor coolant pumps, control rod drive mechanism housing, and safety injection system valves. The applicable code case paragraphs are 1.0 through 4.0.
N-274	Reg. Guide 1.85	Code Case N-274 is acceptable subject to the following: In addition to those conditions specified in the Code Case, paragraph 6 should be expanded as follows: The ultrasonic examination procedure shall be proven by actual demonstration, to the satisfaction of the authorized nuclear inspector, that the procedures are capable of detecting cracks according to Section XI requirements.

TABLE 1.9-2 (SHEET 7 OF 7)

Code Case No.	NRC Acceptance of Code Case	NRC Contingency for Use of Code Case
		The reason for the conditional acceptance of paragraph 6 is to make sure that there is a qualified procedure capable of detecting small flaws and differentiating the small flaws from geometric reflectors. This paragraph does not in any way alter the acceptance as specified in paragraph 3.
N-71-15	Reg. Guide 1.85	Additional Materials for Subsection NF, Classes 1, 2, and 3 and MC Component Supports Fabricated by Welding, Section III, Division 1 Code Case N-71-15 is acceptable subject to the following condition in addition to those conditions specified in the Code Case: The maximum measured ultimate tensile strength (UTS) of the component support material should not exceed 170 Ksi in view of the susceptibility of high-strength materials to brittleness and stress corrosion cracking. Certain applications may exist where a UTS value of up to 190 Ksi could be considered acceptable for a material and, under this condition, the Design Specification should specify impact testing for the material. For these cases, it should be demonstrated by the applicant that (1) the impact test results for the material meet code requirements and (2) the material is not subject to stress corrosion cracking by virtue of the fact that (a) a corrosive environment is not present and (b) the component that contains the material has essentially no residual stresses or assembly stresses, and it does not experience frequent sustained loads in service.

TABLE 1.9-3 (SHEET 1 OF 4)

NSSS ASME CODE CASES

Code Case No.	Council Approval	Title (NRC Comments)	Regulatory Guide
1335-10 or N-3-10	8/28/78	Requirements for Bolting Materials, Section III	1.85-15
1423-2[a]	3/9/72	Wrought Type 304 and 316 with Nitrogen Added, Section I, III, VIII, Divisions 1 and 2	1.85-0
1484-3 (N-20)[b]	08/13/76	SB-163 Nickel-Chromium-Iron Tubing (Alloy 600 and 690) and Nickel-Iron-Chromium Alloy 800 at a Specified Minimum Yield Strength of 40.0 ksi, Section III, Division 1, Class 1	1.85-19
1528-3[c]	11/3/75	High Strength Steel SA-508, Class 2, and SA-541, Class 2 Forgings, Section III, Class 1 Components	1.85-7
1540-1	3/3/73	Elastomer Diaphragm Valves Section III, Classes 2 and 3	1.84-0
1553-1	3/3/75	Upset Heading and Roll Threading of SA-453 for Bolting in Section III	1.84-2
1567	3/3/73	Testing Lots of Carbon and Low-Alloy Steel-Covered Electrodes, Section III	1.85-0
1607	11/5/73	Stress Criteria for Section III Classes 2 and 3 Vessels Subjected to Upset, Emergency, and Faulted Operating Conditions	1.84-0
1612	12/17/73	Use of Type 308 Stainless Steel Rod and Bar for Section III, Class 1, 2, 3, and CS Construction	1.85-1

TABLE 1.9-3 (SHEET 2 OF 4)

Code Case No.	Council Approval	Title (NRC Comments)	Regulatory Guide
1644 (1644-4)[d]	8/12/74	Additional Materials for Component Supports Section III, Subsection NF, Class 1, 2, 3, and MC Construction	1.85-1
1649	8/12/74	Modified SA 453-GR 660 for Class 1, 2, 3 and CS Construction	1.85-1
N-242-1[e]	04/10/80	Materials Certification Section III, Division 1, Classes 1, 2, 3, MC, and CS Construction	1.85-18
N-71-9[f]	12/80	Additional Materials for Component Supports Fabricated by Welding	1.85-17
N-249-1[g]	04/82	Additional Materials for Component Supports Fabricated Without Welding	1.85-19
N-249-2	6/17/82	Additional Materials for subsection NF Class 1, 2, 3, and MC Component Supports Fabricated Without Welding Section III, Division I	1.85-23
N-62-2 (1621-2)	05/15/80	Internal and External Valve Items Section III, Division 1, Class 1, 2, and 3	1.84-12
N-272	05/15/80	Compiling Data Report Records Section III, Division I	1.84-23
N-274	03/17/80	Alternative Rules for Examination of Weld Repairs for Section III, Division 1 Construction	1.85-17
N-316	12/11/81	Alternative Rules for Fillet Weld Dimensions for Socket Welded Fittings Section III, Division 1, Class 1, 2, and 3	1.84-20

TABLE 1.9-3 (SHEET 3 OF 4)

Code Case No.	Council Approval	Title (NRC Comments)	Regulatory Guide
N-397	02/20/84	Alternative Rules to the Spectral Broadening Procedures of N-1226.3 for Classes 1, 2, and 3 Piping Section III, Division 1	1.84-23
N-411[(h)]	09/17/84	Alternative Damping Values for Seismic Analysis of Classes 1, 2, and 3 Piping Sections Section III, Division 1	-
N-412	04/15/85	Alternative Rules for Witnessing Piping System Pressure Tests of Classes 1, 2, and 3 Piping Systems Section III, Division 1	-
1948	11/28/83	Alternative Rules for Selection of Penetrameters and Shims for Welds with Reinforcement Section V	-
N-237	07/09/79	Hydrostatic Testing of Internal Piping Section III, Division 1	1.84-16
N-240	03/19/79	Hydrostatic Testing of Open Ended Piping Section III, Division 1	1.84-16
N-174	03/23/77	Size of Fillet Welds for Socket Welding of Piping Section III, Division 1	1.84-11
N-275	03/07/80	Repair of Welds Section III, Division 1	1.84-14
N-388	07/25/83	Component Support Bolting, Section III, Division 1, Classes 2, 3, and MC	1.85-22
N-392	11/28/83	Procedure for Evaluation of the Design of Hollow Circular Cross-Section Welded Attachments on Classes 2 and 3 Piping, Section III, Division 1	1.84-23

TABLE 1.9-3 (SHEET 4 OF 4)

a. Code Case 1423-2 is acceptable subject to the conditions established in Regulatory Guides 1.31, Control of Stainless Steel Welding, and 1.44, Control of the Use of Sensitized Stainless Steel.

b. Code Case 1484-3 is acceptable subject to the following condition in addition to those conditions specified in the code case: Alloy 690 is not acceptable on a generic basis.

c. Code Case 1528-3 is acceptable subject to the following condition in addition to those conditions specified in the code case: The information required to be developed by note 1 in the code case should be provided in each referencing Safety Analysis Report.

d. Code Case 1644 is acceptable subject to the following condition, in addition to those conditions specified in the code case: The maximum measured UTS of the component support material should not exceed 170 ksi.

e. Code Case N-242-1 is acceptable subject to the following condition in addition to those conditions specified in the code case: Applicants should identify in their Safety Analysis Reports the components and supports for which the code case is being applied and should specify the respective paragraphs of the code case. For VEGP, this code case is being used for the reactor coolant pumps, CRDM housing, and safety injection system valves. The applicable code case paragraphs are 1.0 through 4.0.

f. Code Case N-71-9 is acceptable subject to the contingency delineated in Regulatory Guide 1.85, Revision 17, dated December 1980.

g. Code Case N-249-1 is acceptable subject to the contingency delineated in Regulatory Guide 1.85, Revision 19, dated April 1982.

h. With the exception of certain stress calculations described in GPC letter GN-1257, J. A. Bailey to B. J. Youngblood dated December 22, 1986, Code Case N-411 damping values are not used in conjunction with multiple response spectrum methodology piping analysis.